FIFTH EDITION

Formulas and Calculations for Drilling, Production, and Workover

All the Formulas You Need to Solve Drilling and Production Problems



Thomas S. Carter
William C. Lyons
Norton J. Lapeyrouse

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THOMAS S. CARTER
WILLIAM C. LYONS
NORTON J. LAPEYROUSE†

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There are practical examples included in all chapters. The examples require the use of Excel spreadsheets that can be found at https://www.elsevier.com/books-and-journals/book-companion/9780323905497.

Preface

The coauthors for this edition are Thomas (Tom) S. Carter and William (Bill) C. Lyons. The first edition of this book written by Norton J. Lapeyrouse was published in 1992, and the second edition was released in 2008. With the permission of the Lapeyrouse family, Elsevier asked Tom and Bill to continue the publication of this very popular oil and gas "in the field" operations managerial and crew engineering calculation support tool. The coauthors have continued this book through three more editions, the third, fourth, and this very new fifth edition. This newest edition will provide example tables within the published book that individually can be downloaded from Elsevier's website into a field manager's personal analog cell phone as well as other field analog cell phones. Each individual table, when downloaded, will enable the user to simply change the input data (i.e., the "givens") and obtain the new output data (i.e., the "solution"). This type of analog cell phone calculations for field operations was recommended by two younger oil field engineering managers and owners of small oil and gas production companies, Earl Berkenhiemer and Glenn Lyle from the Houston, Texas, area. Tom developed most of the active downloadable example mud drilling fluid tables that are presented in the analog MS Excel software. Bill, at 84 and more familiar with MathCad analog software, needed a little help from Salim Sator, a drilling manager with Sonatrach in the Algerian Sahara Desert. Salim, an air and gas analysis expert with Sonatrach, was asked to help convert the air and gas tables in the last chapter to an active analog MS Excel table similar in design and operation to the mud drilling fluid tables developed by Tom, as mentioned previously. The coauthors are also indebted to Ann Gardner and Trish Richardson who are the longtime on-call artists. They were called on to create two important new graphs for this book. This is not the last edition of this important book. Younger engineers will take over the tasks of preparing new editions in the future. Lastly, the coauthors wish to thank Elsevier editor Katie Hammon and all those other supporting editors in the system who have guided Tom and Bill very skillfully through the ever more complicated world of book publication.

Thank you all.

Tom Carter and Bill Lyons

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Prologue

We have greatly appreciated the patience of our families over the past 12 years as we have devoted much of our spare time to these three new editions. During these periods, our initial work would start with calculating different examples. From the resulting solutions of those examples, we would make hand-drawn plots that in turn were made into publication printable drawings artfully created by Ann Gardner and Trish Richardson, our on-call artists. After these efforts came the organizing of chapters and the writing of the text sections for each chapter within each edition. The third and fourth edition books progressively "hardened" to exposure to the typical field fluids and heavy equipment greases (e.g., drilling, production pumping, and workover sites). These additional book protection efforts resulted in the fourth edition being published with water- and oil-resistant pages and a spiral binding that would not corrode in formation salt water solutions. The publishers introduced e-book versions for both the third and fourth editions. This allowed the individual engineer and technician user to download edition contents directly to their laptops. By the fourth edition, the field managers were doing calculations in the field at on-site locations on their personal analog smart phones. Soon after we started our fifth edition, two of Bill's old undergraduate engineering students from NM Tech in the 1980s, Earl Berkenhiemer and Glenn Lyle, suggested that each of the example tables be capable of making new active calculations that would allow a new reader to input new field input data in those tables and obtain a new solution particular to that reader's field situation. Both of these engineers have their own small petroleum-producing companies and have had extensive experiences managing field drilling, production, and workover operation crews both at the field site and from their offices in the Houston, Texas, area. Our interest in pursuing such field-friendly calculations led Tom to search for how we might prepare and transmit these new fifth edition tables to the engineers and technicians who would be using this new edition. This led Tom to another Elsevier-published petroleum engineering book, *Petroleum Production Engineering: A Computer-Assisted Approach*, by Boyun Guo, Xinghui "Lou" Liu, and Xuehoa Tan, 2nd Edition, 2017. In this production book, the authors had offered downloadable MS Excel spreadsheet example tables that would allow the user to alter the input and obtain new resultant solutions. Our new fifth edition has utilized similar MS Excel spreadsheet

example tables that Tom has developed for most of the book tables. Bill, being the oldest in our coauthor partnership, needed a little help with the air and gas MS Excel spreadsheet table designs in the book's last chapter. Fortunately, Bill obtained some spreadsheet table development help for those air and gas calculations from Salim Sator. Bill had taught Salim at an extensive all summer program in 2000 for Sonatrach in the Algerian Sahara Desert. Salim was a drilling engineer student in that summer program and has since risen in Sonatrach's structure to a Senior Drilling Advisor with additional expertise in MS Excel spreadsheet design.

That's the full story of this very important book.

Tom Carter and Bill Lyons

CHAPTER

1

Basic equations

1.1 Terminology

Density: The term "density" is the mass per unit volume. In the System International (SI), this is kg/m^3, or kg/L, gr/cm^3. In the British Imperial System (BIS) and United States Customary System (USCS), the mechanical properties of a fluid are not published in mass per unit volume units (the BIS and USCS are basically the same). In the USCS, the mass per unit volume must be calculated from the published weight per unit volume (this latter term is denoted as specific weight). For decades, the oil and gas industry in the West has used the "density" name as a form of an oil field slang term for the USCS weight per unit volume published fluid mechanical properties usually published as lb/ft^3 or lb/gal (the latter also written as *ppg*). The weight per unit volume of fresh water is $62.4\,lb/ft^3$ or $8.34\,lb/gal$. To obtain the USCS density terms (equivalent to the SI density terms) from the published specific weight values, both terms must be divided by the USCS acceleration of gravity constant, namely $32.2\,ft/s^2$. This would give density values of

$$\rho_{fw} = \frac{\gamma_{fw}}{g} = \frac{62.4}{32.2} = 1.94\,\frac{lb - s^2}{ft^4} = 1.94\,\frac{slug}{ft^3}$$

$$\rho_{fw} = \frac{\gamma_{fw}}{g} = \frac{8.34}{32.2} = 0.258\,\frac{lb - s^2}{ft - gal} = 0.258\,\frac{slug}{gal}$$

where

ρ_{fw} = density of fresh water
^{g} = gravity constant

The slug term is not used often in engineering practice. Basically, it is the USCS equivalent to the SI kilogram. The USCS slug is

$$1\ slug = 1\ \frac{lb - s^2}{ft}$$

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Likewise, the SI kilogram is

$$1\,\text{kg} = 1\,\frac{\text{N} - \text{s}^2}{\text{m}}$$

where N is the Newton which is the force unit equivalent to the lb force unit in the USCS (the conversion is $4.445N = 1\,\text{lb}$). As the slug is not often written in the technical literature and the kilogram is very rarely written in the terms its basic terms of

$$\frac{\text{N} - \text{s}^2}{\text{m}}$$

Specific weight: Since the SI mechanical properties of a fluid are listed in density units, then these density terms must be used to calculate the specific weight so that practical engineering calculations can be made. Therefore, the density of fresh water can be written in SI units as $1000\,\text{kg/m}^3$, or $1\,\text{kg/L}$. To carry these calculations out, we must **multiply** these density terms by the SI acceleration of gravity constant, namely $9.81\,\text{m/s}^2$. This would give the specific weight values of

$$\gamma_{\text{fw}} = \rho_{\text{fw}} g = 1000\ (9.81) = 9810\ \frac{\text{N}}{\text{m}^{3.}}$$

or

$$\gamma_{\text{fw}} = \rho_{\text{fw}} g = 1(9.81) = 9.810\ \frac{\text{N}}{\text{L}}$$

where

γ_{fw} = specific weight

Specific gravity: The above is a complicated unit situation especially for engineers who may have worked in one part of the world where either the SI or the USCS was being used and later is assigned to work in another part of the world where the other unit system dominates. Fortunately, instead of dealing with such complicated units situations, the specific gravity of a fluid can be determined from either the published SI density data or from the published USCS specific weight data. The specific gravity term can be defined as

$$\begin{aligned}\text{Specific gravity} &= \frac{\text{Density of a fluid}}{\text{Density of fresh water}} \\ &= \frac{\text{Specific weight of a fluid}}{\text{Specific weight of fresh water}}\end{aligned}$$

For example, if an engineer from Germany is working temporarily for his operating company at a drilling location in the Gulf Coast region of the United States and wants to convert his SI density calculation result for a new cement slurry to a USCS-specific weight value for service company staff working at the location. His calculation result for the new cement slurry is 1.88 kg/L. Therefore, the specific gravity of this new cement slurry is

$$\text{Specific gravity} = \frac{1.88}{1.00} = 1.88$$

Therefore, the new cement slurry-specific weight is

$$\text{Specific weight} = 1.88\ (8.34) = 15.7\ \frac{\text{lb}}{\text{gal}}$$

or

$$\text{Specific gravity} = 1.88\ (62.4) = 117.3\ \frac{\text{lb}}{\text{ft}^3}$$

Spreadsheet 1.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Specific gravity.

1.2 Mud weight MW (lb/ft^3), mud weight MW (ppg), and specific gravity (SG)

Definition: Mud weight of fresh water in lb/ft^3:

$$MW_{\text{fw}} = 62.4\ \text{lb/ft}^3 \tag{1.1}$$

where

MW_{fw} = fresh water mud weight in lb/ft^3.

Example: Mud weight of fresh water in ppg:

$$MW_{\text{fw}} = \left(\frac{62.4}{(12^3)}\right)(231) \tag{1.2}$$

$$MW_{\text{fw}} = 8.34\,\text{ppg}$$

where

1 gal = 231 in.^{3}
1 ft = 12 in.

Example: Specific gravity of fresh water SG:

$$\text{SG}_{\text{fw}} = \left(\frac{62.4}{62.4}\right) = 1.0 \tag{1.3}$$

TABLE 1.1 Mud weight and density conversion factors summary.

From		To		
Density	**Mud wt**	**Mud wt**	**SG**	**Multiply by**
	lb/ft^3	lb/gal		0.134
	lb/ft^3		SG	(1/62.4)
	lb/gal		SG	(1/8.34)
	lb/gal	kg/m^3		119.83
kg/m^3		N/m^3		9.81
kg/L		N/L		9.81
kg/m^3		N/L		(9.81/1000)
	N/L		SG	(1/9.81)
	N/m^3		SG	(1/9810)

or

$$SG_{fw} = \left(\frac{8.34}{8.34}\right) = 1.0 \tag{1.4}$$

Example: SG of a mud weight of 12.0 ppg (Table 1.1):

$$SG_{fw} = \left(\frac{12.0}{8.34}\right) = 1.44 \tag{1.5}$$

Spreadsheet 1.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 MW-specific gravity.

1.3 Hydrostatic pressure (*P*) and (*p*)

Definition: The Conversion Factor used to convert the mud weight to a pressure gradient in psi/ft is

$$F_c = \left(\rho_w, \frac{lb}{ft^3}\right)\left(\frac{1\ ft^2}{A,\ in.^2}\right)\left(\frac{1,\ gal}{\rho_w,\ lb}\right) \tag{1.6}$$

where

F_c = Conversion Factor in gal/ft in.^{2}
ρ_w = weight of water in lb/ft^3
A = area in in.^{2}

Example: Determine the conversion factor using

$\rho_w = 62.4\,\text{lb/ft}^3$
$A = 144\,\text{in.}^2$

$$F_c = \left(\frac{62.4\text{ lb}}{\text{ft}^3}\right)\left(\frac{1\text{ ft}^2}{144,\text{ in.}^2}\right)\left(\frac{1,\text{ gal}}{8.34,\text{ lb}}\right)$$

$$F_c = (62.4)(0.00694)(0.1199)$$

$$F_c = 0.0519\frac{\text{gal}}{\text{ft in.}^2}$$

or

$$F_c = 0.052\frac{\text{gal}}{\text{ft in.}^2}$$

Definition: Hydrostatic pressure P (lb/ft^2) at a depth H (ft) below surface is

$$p = (MW)(H) \tag{1.7}$$

where

P = hydrostatic pressure in lb/ft^2
MW = mud weight in lb/ft^3
H = True Vertical Depth (TVD) in ft

Note: The TVD is always used and *not* the measured depth even for an inclined wellbore.

Example: Pressure (lb/ft^2) in fresh water at a depth of 1000 ft:

$$p = (62.4)(1000) = 62,000\frac{\text{lb}}{\text{ft}^3}$$

Example: Pressure (lb/ft^2) in 12.0 ppg at a depth of 1000 ft:

$$p = (89.9)(1000) = 89,900\frac{\text{lb}}{\text{ft}^3}$$

Definition: Hydrostatic pressure p (psi) at a depth H (ft) below surface is (using Eq. 1.7):

$$p\left(12^2\right) = (MW)(H) \tag{1.8}$$

which reduces to

$$P = (MW)\left(\frac{1}{12^2}\right)(H)$$

or

$$p = (0.00694)(MW)(H)$$

where

p = hydrostatic pressure in psi

Example: Pressure (psi) in fresh water (lbft^3) at a depth of 1000 ft:

$$p = (0.00694)(62.4)(1000) = 434 \text{ psi}$$

Example: Pressure (psi) in 89.9 lb/ft^3 (12.0 ppg) at a depth of 1000 ft:

$$p = (0.00694)(89.9)(1000) = 624 \text{ psi}$$

Definition: Hydrostatic pressure p (psi) at a depth H (ft) below surface is (using Eq. 1.7):

$$p\left(12^2\right) = (MW)\left(\frac{12^3}{231}\right)(H)$$

which reduces to

$$p = (MW)\left(\frac{12}{231}\right)(H)$$

or

$$p = (MW)(0.052)(H) \tag{1.9}$$

Example: Pressure (psi) in fresh water at a depth of 1000 ft:

$$p = (8.34)(0.052)(1000) = 434 \text{ psi}$$

Example: Pressure (psi) in 12.0 ppg mud at a depth of 1000 ft:

$$p = (12.0)(0.052)(1000) = 624 \text{ psi}$$

Spreadsheet 1.3 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Hydrostatic pressure.

1.4 Pressure gradient ∇ (psi/ft), G (ppg)

Definition: Pressure gradient ∇ (psi/ft) is obtained from Eq. (1.8):

$$\nabla = \left(\frac{p}{H}\right) = (0.052)(MW)$$

$$\nabla = (0.052)(MW) \tag{1.10}$$

where

∇ = pressure gradient in psi/ft

Example: Pressure gradient ∇_{fw} (psi/ft) for fresh water:

$$\nabla = (0.052)(8.34) = 0.434 \frac{\text{psi}}{\text{ft}}$$

Example: Pressure gradient ∇_{m} (psi/ft) for 12.0 ppg mud

$$\nabla = (0.052)(12.0) = 0.624 \frac{\text{psi}}{\text{ft}}$$

Definition: Pressure gradient G (ppg) is also obtained from Eq. (1.8):

$$G = \left(\frac{p}{0.052}\right) = MW \text{ ppg} \tag{1.11}$$

Example: Pressure gradient G_{fw} (ppg) for fresh water:

$$G = 8.34 \text{ ppg}$$

Example: Pressure gradient G_{m} (ppg) for 12.0 ppg mud (Table 1.2):

$$G = 12.0 \text{ ppg}$$

TABLE 1.2 Pressure gradient conversion factors summary.

Pressure unit	Depth unit	Mud weight unit	Factor
psi	Feet	lb/ft^3	0.00694
psi	Feet	ppg	0.052
N/m^2	Meters	N/L	1000
N/m^2	Meters	SG	9810

Spreadsheet 1.4 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Pressure gradient.

1.5 Mud pump output q (bbl/stk) and Q (gpm)

1.5.1 Triplex pump

Formula 1: Pump output in gallons per stroke (gal/stk):

$$Q_{gs} = (0.0102)(D_l^2)(S)(e_v) \tag{1.12}$$

where

Q_{gs} = pump output in gallons per stroke (gal/stk)
D_l = liner diameter in inches
S = stroke length in inches
e_v = volumetric efficiency in percent (%)

Example: Q_{gs} (gal/stk) at 100% volumetric efficiency for a 7 in. by 12 in. triplex pump:

$$Q_{gs} = (0.0102)(7^2)(12)(1.0) = 6.0 \text{ gal/stk}$$

The above assumes 100% volumetric efficiency of the pump.

Note: Most published information on pump output per stroke assumes 100% volumetric efficiency. The pump manufacturers can be contacted to get actual pump volumetric efficiencies. These efficiencies can vary from 0.85 to 0.98. Published data can be checked by assuming $e_v = 1.0$.

Example: Adjust the above result for a pump with a volumetric efficiency of 0.90:

$$Q_a = 6.0 \times 0.90 = 5.4 \text{ gal/stk}$$

where

Q_a = actual pump output with a reduced volumetric efficiency in gal/stk

Formula 2: Pump output in bbl/stk:

$$Q_{bs} = \frac{(0.0102)(D_l^2)(S)(e_v)}{42} \tag{1.13}$$

where

Q_{bs} = pump output in bbl/stk

Example: Determine the pump output in bbl/stk.

$$Q_{bs} = \left(\frac{(0.0102)(7^2)(12)(1.0)}{42} = 0.1428 \frac{\text{bbl}}{\text{stk}}\right)$$

Formula 3: Pump output in gpm:

$$Q = (0.0102)(D_l^2)(S)(N)(e_v) \tag{1.14}$$

where

Q = pump output in gpm
N = strokes per minute (also rpm of pump flywheel)

Example: Determine the pump output q (gpm) at 100% volumetric efficiency, for a 7 in. by 12 in. triplex pump at 60 SPM.

$$Q = (0.0102)\left(7^2\right)(12)(60)(1.0) = 360 \text{ gpm}$$

Formula 4: Pump output in bbl/min:

$$Q_{\text{bm}} = \frac{(0.0102)\left(D_{\text{l}}^2\right)(S)(N)(e_{\text{v}})}{42} \tag{1.15}$$

where

Q_{bm} = pump output in bbl/min

Example: Determine the pump output in bbl/min.

$$Q_{\text{bm}} = \frac{(0.0102)\left(7^2\right)(12)(60)(1.0)}{42} = 8.57 \frac{\text{bbl}}{\text{min}}$$

Spreadsheet 1.5.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Triplex pump.

1.5.2 Duplex pump

Formula 1: Pump output in gal/stk:

$$Q_{\text{gs}} = (0.0068)\left(2\left({D_{\text{l}}}^2 - {D_{\text{r}}}^2\right)\right)(S)(e_{\text{v}}) \tag{1.16}$$

where

Q_{gs} = pump output in gal/stk
D_{l} = liner diameter in inches
D_{r} = rod diameter in inches
S = stroke length in inches
e_{v} = volumetric efficiency in percent (%)

Example: Determine the output Q_{bs} (bbl/stk) of a 5½ in. by 14 in. duplex pump at 100% efficiency. Pump has a rod diameter = 2.0 in.

$$Q_{\text{gs}} = (0.0068) \times \left(2\left(5.5^2 - 2^2\right)\right)(14)(1.0) = 5.0 \frac{\text{gal}}{\text{stk}}$$

Example: Adjust the above result for a pump with a volumetric efficiency of 0.88:

$$Q_{\text{a}} = (5.00)(0.88) = 4.4 \frac{\text{gal}}{\text{stk}}$$

Formula 2: Pump output in bbl/stk

$$Q_{bs} = \frac{(0.0068)\left(2\left(D_l^2 - D_r^2\right)\right)(S)(e_v)}{42} \tag{1.17}$$

where

Q_{bs} = pump output in bbl/stk

Example: Determine the pump output in bbl/stk:

$$Q_{bs} = \frac{(0.0068)\left(2\left(5.5^2 - 2.0^2\right)\right)(14)(1.0)}{42} = 0.119 \frac{\text{bbl}}{\text{stk}}$$

Formula 3: Pump output in gpm:

$$Q = (0.0068)\left(2\left(D_l{}^2 - D_r{}^2\right)\right)(S)(N)(e_v) \tag{1.18}$$

where

Q = pump output in gpm
D_l = liner diameter in inches
D_r = rod diameter in inches
S = stroke length in inches
N = strokes per minute (also rpm of pump flywheel)
e_v = volumetric efficiency in percent (%)

Example: Determine the output Q (gpm) of a 5½ in. by 14 in. duplex pump at 100% efficiency. Pump has a rod diameter = 2.0 in. N is 50 spm.

$$Q = (0.0068)\left(2\left(5.5^2 - 2.0^2\right)\right)(14)(50)(1.0) = 249.9 \cong 250 \text{ gpm}$$

Spreadsheet 1.5.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Duplex pump.

1.6 Hydraulic horsepower

Formula 1: (Circulating)

$$\text{HHP} = \frac{(P)(Q)}{1714} \tag{1.19}$$

where

HHP = hydraulic horsepower
P = circulating pressure in psi
Q = circulating rate in gpm

$$\text{HHP} = \frac{(2950)(520)}{1714} = 894.9 \cong 895$$

Formula 2: (Input)

$$\mathrm{HHP_i} = \frac{(P)(Q)}{(1714)(e_v)(e_m)} \tag{1.20}$$

where

$\mathrm{HHP_i}$ = input hydraulic horsepower
e_v = volumetric efficiency (~0.85 to 0.98)
e_m = mechanical efficiency (~0.80 for continuous operations and ~0.9 for intermittent operations)

Example: Determine the hydraulic horsepower of a pump that has a volumetric output of 480 gpm at a pressure of 1800 psig.

$$\mathrm{HHP} = \frac{(1800)(480)}{1714} = 504$$

This represents the horsepower that the pump must apply to move the drilling mud within the pump.

Example: Determine the input horsepower that must be applied to the pumping unit by a prime mover to pump the drilling mud in the above example with continuous operations ($e_m \sim 0.80$) and the pump has a volumetric efficiency of 0.96.

$$\mathrm{HHP_i} = \frac{(1800)(480)}{(\mathbf{1714})(0.96)(0.80)} = 656$$

Spreadsheet 1.6 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Hydraulic horsepower.

1.7 Estimated weight of drill collars in AIR

Formula 1: REGULAR Drill Collar Weight in lb/ft:

$$W_{\mathrm{Rdc}} = (2.66)\left(OD^2 - ID^2\right) \tag{1.21}$$

where

W_{Rdc} = weight of REGULAR drill collar in lb/ft
OD = outside diameter of drill collar in inches
ID = inside diameter of drill collar in inches

Example: Determine the weight of an 8×2 13/16 in. regular drill collar in lb/ft:

OD = 8.0 in.
ID = 2 13/16 in. $\left(\frac{13}{16} = 0.81125\right)$

$$W_{\text{Rdc}} = (2.66)\left(8^2 - 2.8125^2\right) = 149.2\,\frac{\text{lb}}{\text{ft}}$$

Formula 2: SPIRAL Drill Collar Weight in lb/ft:

$$W_{\text{Sdc}} = (2.56\ )\left(OD^2 - ID^2\right) \tag{1.22}$$

where

W_{Sdc} = weight of SPIRAL drill collar in lb/ft

Example: Determine the weight of an $8 \times 2\,13/16$ in. spiral drill collar in lb/ft

$OD = 8.0$ in.
$ID = 2\ 13/16$ in. $\left(\frac{13}{16} = 0.81125\right)$

$$W_{\text{Sdc}} = (2.56)\left(8^2 - 2.8125^2\right) = 143.6\ \text{lb/ft}$$

Spreadsheet 1.7 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 DC weight.

1.8 Open hole and tubular capacity and displacement formulas

1.8.1 Capacity of open hole or tubulars

Formula 1: (Open Hole or Tubular Capacity in bbl/ft):

$$C = \frac{ID^2}{1029.4} \tag{1.23}$$

where

C = capacity of open hole, casing, drill pipe or drill collars in bbl/ft
ID = internal diameter of open hole, casing, drill pipe or drill collars in inches

Note: For hole "washout," increase the DIAMETER of the hole size (not volume) with the estimated percent increase.

Example: Determine the capacity in bbl/ft, of a 12¼ in. hole.

$$C = \frac{12.25^2}{1029.4} = 0.1458\ \text{bbl per ft}$$

Determine the capacity in bbl/ft, of a 12¼ in. hole with a 10% washout in hole DIAMETER. (That is about equal to a ⅝ in. increase in hole size on either side of the bit).

$$C = \frac{(12.25 \times 1.10)^2}{1029.4} = 0.1764 \text{ bbl per ft}$$

For 8×2 13/16 in. Drill Collars, the capacity in bbl/ft would be:

Convert $^{13}/_{16}$ to decimal equivalent: $13 \div 16 = 0.8125$

$$C = \frac{2.8125^2}{1029.4} = 0.0077 \text{ bbl per ft}$$

Formula 2: (Open Hole or Tubular Capacity in ft/bbl):

$$C = \frac{1029.4}{ID^2} \tag{1.24}$$

Example: Determine the capacity of a 12¼ in. hole in ft/bbl.

$$C_a = \frac{1029.4}{12.25^2} = 6.8598 \text{ ft per bbl}$$

Formula 3: (Open Hole or Tubular Capacity in gal/ft):

$$C_a = \frac{ID^2}{24.51} \tag{1.25}$$

Example: Determine the capacity, gal/ft, of an 8½ in. hole.

$$C_a = \frac{8.5^2}{24.51} = 2.9478 \text{ gal per ft}$$

Formula 4: (Open Hole or Tubular Capacity in ft/gal):

$$C_a = \frac{24.51}{ID^2} \tag{1.26}$$

Example: Determine the capacity, ft/gal, of an 8½ in. hole.

$$C_a = \frac{24.51}{8.5^2} = 0.3392 \text{ ft per gal}$$

Formula 5: (Open Hole or Tubular Capacity in ft^3/linear ft):

$$C_a = \frac{ID^2}{183.35} \tag{1.27}$$

Example: Determine the capacity, ft^3/linear ft, for a 6.0 in. hole.

$$C_a = \frac{6.0^2}{183.35} = 0.1963 \text{ ft}^3 \text{ per linear ft}$$

Formula 6: (Open Hole or Tubular Capacity in linear ft/ft^3):

$$C_a = \frac{183.35}{ID^2} \tag{1.28}$$

Example: Determine the capacity, linear ft/ft^3, for a 6.0 in. hole.

$$C_a = \frac{183.35}{6.0^2} = 5.0931 \text{ linear ft per ft}^3$$

Spreadsheet 1.8.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 OH tubular capacity.

1.8.2 Displacement of tubulars

Formula 1: (Tubular Displacement in bbl/ft):

$$D = \frac{(OD^2 - ID^2)}{1029.4} \tag{1.29}$$

where

D = displacement of tubular in bbl/ft
OD = outside diameter of tubular in inches
ID = internal diameter of tubular in inches

Example: Determine the displacement for 5 × 4.276 in. drill pipe in bbl/ft

$$D = \frac{(5.0^2 - 4.276^2)}{1029.4} = 0.0065 \text{ bbl per ft}$$

Formula 2: (Tubular Displacement in ft/bbl):

$$D = \frac{1029.4}{(OD^2 - ID^2)} \tag{1.30}$$

where

D = displacement of tubular in ft/bbl

Example: Determine the displacement for 5 × 4.276 in. drill pipe in ft/bbl.

$$D = \frac{1029.4}{(5^2 - 4.276^2)} = 153.2798 \text{ ft per bbl}$$

Formula 3: (Tubular Displacement in gal/ft):

$$D = \frac{(OD^2 - ID^2)}{24.51} \tag{1.31}$$

where

D = displacement of tubular in gal/ft

Example: Determine the displacement for 5×4.276 in. drill pipe in gal/ft

$$D = \frac{(5^2 - 4.276^2)}{24.51} = 0.2740 \text{ gal per ft}$$

Formula 4: (Tubular Displacement in ft/gal):

$$D = \frac{24.51}{(OD^2 - ID^2)} \tag{1.32}$$

where

D = displacement of tubular in ft/gal

Example: Determine the displacement for 5×4.276 in. drill pipe in ft/gal.

$$D = \frac{24.51}{(5^2 - 4.276^2)} = 3.6496 \text{ ft per gal}$$

Formula 5: (Tubular Displacement in ft^3/linear ft):

$$D = \frac{(OD^2 - ID^2)}{183.35} \tag{1.33}$$

where

D = displacement of tubular in ft^3/linear ft

Example: Determine the displacement for 5×4.276 in. drill pipe in ft^3/linear ft

$$D = \frac{(5^2 - 4.276^2)}{183.35} = 0.0366 \text{ ft}^3 \text{ per linear ft}$$

Formula 6: (Tubular Displacement in linear ft/ft^3):

$$D = \frac{183.35}{(OD^2 - ID^2)} \tag{1.34}$$

Example: Determine the displacement for 5×4.276 in. drill pipe in linear ft/ft^3.

$$D = \frac{183.35}{(OD^2 - ID^2)} = 27.3013 \text{ linear ft per ft}^3$$

Spreadsheet 1.8.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Tubular displacement.

1.8.3 Annular capacity between casing or hole and drill pipe, tubing, or casing

Formula 1: (Capacity of annulus in bbl/ft):

$$C_a = \frac{\left(D_h^2 - D_p^2\right)}{1029.4} \tag{1.35}$$

where

C_a = capacity of annulus in bbl/ft
D_h = diameter of hole in inches

Note: The bit size may be used if the actual hole size is not known. Larger sizes may be used based on estimated hole "washout."

D_p = diameter of drill pipe in inches

Example: Bit or hole size (D_h) = 12¼ in.

Drill pipe OD (D_p) = 5.0 in.

$$C_a = \frac{12.25^2 - 5^2}{1029.4} = 0.1215 \text{ bbl per ft}$$

Formula 2: (Capacity of annulus in ft/bbl):

$$C_a = \frac{1029.4}{\left(D_h^2 - D_p^2\right)} \tag{1.36}$$

where

C_a = capacity of annulus in ft/bbl

Example: Bit or hole size (D_h) = 12¼ in.

Drill pipe OD (D_p) = 5.0 in.

$$C_a = \frac{1029.4}{\left(12.25^2 - 5^2\right)} = 8.2311 \text{ ft per bbl}$$

Formula 3: (Capacity of annulus in gal/ft):

$$C_a = \frac{\left(D_h^2 - D_p^2\right)}{24.51} \tag{1.37}$$

where

C_a = capacity of annulus in gal/ft

Example: Bit or hole size (D_h) = 12¼ in.

Drill pipe OD (D_p) = 5.0 in.

$$C_a = \frac{12.25^2 - 5^2}{24.51} = 5.1025 \text{ gal per ft}$$

Formula 4: (Capacity of annulus in ft/gal):

$$C_a = \frac{24.51}{\left(D_h^2 - D_p^2\right)} \tag{1.38}$$

where

C_a = capacity of annulus in ft/gal

Example: Bit or hole size (D_h) = 12¼ in.

Drill pipe OD (D_p) = 5.0 in.

$$C_a = \frac{24.51}{\left(12.25^2 - 5^2\right)} = 0.1960 \text{ ft per gal}$$

Formula 5: (Capacity of annulus in ft^3/linear ft):

$$C_a = \frac{\left(D_h^2 - D_p^2\right)}{183.35} \tag{1.39}$$

where

C_a = capacity of annulus in ft^3/linear ft

Example: Bit or hole size (D_h) = 12¼ in.

Drill pipe OD (D_p) = 5.0 in.

$$C_a = \frac{12.25^2 - 5^2}{183.35} = 0.6821 \text{ ft}^3 \text{ per linear ft}$$

Formula 6: (Capacity of annulus in linear ft/ft^3):

$$C_a = \frac{183.35}{\left(D_h^2 - D_p^2\right)} \tag{1.40}$$

where

C_a = capacity of annulus in linear ft/ft^3

Example: Bit or hole size (D_h) = 12¼ in:

Drill pipe OD (D_p) = 5.0 in.

$$C_a = \frac{183.35}{\left(12.25^2 - 5^2\right)} = 1.4661 \text{ linear ft per ft}^3$$

Spreadsheet 1.8.3 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Capacity between OH and tubulars.

1.8.4 Annular capacity between casing and multiple strings of tubing

Formula 1: (Annular capacity between casing and multiple strings of tubing in bbl/ft):

$$C_a = \frac{D_i^2 - \left[(T_1)^2 + (T_2)^2\right]}{1029.4} \tag{1.41}$$

where

C_a = capacity of annulus in bbl/ft or appropriate units
D_i = ID of casing in inches
T_1 = tubing no. 1 OD size in inches
T_2 = tubing no. 2 OD size in inches

Example: Using two strings of tubing of same size:

D_i = casing—7.0 in., 29 lb/ft ID = 6.184 in.
T_1 = tubing no. 1—OD = 2.375 in.
T_2 = tubing no. 2—OD = 2.375 in.

$$C_a = \frac{6.184^2 - \left[(2.375)^2 + (2.375)^2\right]}{1029.4} = 0.0262 \text{ bbl per ft}$$

Formula 2: (Annular capacity between casing and multiple strings of tubing in ft/bbl):

$$C_a = \frac{1029.4}{D_i^2 - \left[(T_1)^2 + (T_2)^2\right]} \tag{1.42}$$

Example: Using two strings of tubing of same size:

D_i = casing—7.0 in., 29 lb/ft ID = 6.184 in.
T_1 = tubing no. 1—OD = 2.375 in.
T_2 = tubing no. 2—OD = 2.375 in.

$$C_a = \frac{1029.4}{6.184^2 - \left[(2.375)^2 + (2.375)^2\right]} = 38.1816 \text{ ft per bbl}$$

Formula 3: (Annular capacity between casing and multiple strings of tubing in gal/ft):

$$C_a = \frac{D_i^2 - \left[(T_1)^2 + (T_2)^2\right]}{24.51} \tag{1.43}$$

Example: Using two strings of tubing of same size:

D_i = casing—7.0 in., 29 lb/ft ID = 6.184 in.
T_1 = tubing no. 1—OD = 2.375 in.
T_2 = tubing no. 2—OD = 3.5 in.

$$C_a = \frac{6.184^2 - \left[(2.375)^2 + (3.5)^2\right]}{24.51} = 0.8303 \text{ gal per ft}$$

Formula 4: (Annular capacity between casing and multiple strings of tubing in ft/gal):

$$C_a = \frac{24.51}{D_i^2 - \left[(T_1)^2 + (T_2)^2\right]} \tag{1.44}$$

Example: Using two strings of tubing of same size

D_i = casing—7.0 in., 29 lb/ft ID = 6.184 in.
T_1 = tubing no. 1—OD = 2.375 in.
T_2 = tubing no. 2—OD = 3.5 in.

$$C_a = \frac{24.51}{6.184^2 - \left[(2.375)^2 + (3.5)^2\right]} = 1.2043 \text{ ft per gal}$$

Formula 5: (Annular capacity between casing and multiple strings of tubing in ft^3/linear ft):

$$C_a = \frac{D_i^2 - \left[(T_1)^2 + (T_2)^2 + (T_3)^2\right]}{183.35} \tag{1.45}$$

Example: Using three strings of tubing:

D_i = casing—9⅝ in., 47 lb/ft ID = 8.681 in.
T_1 = tubing no. 1—OD = 3.5 in.
T_2 = tubing no. 2—OD = 3.5 in.
T_3 = tubing no. 3—OD = 3.5 in.

$$C_a = \frac{8.681^2 - \left[(3.5)^2 + (3.5)^2 + (3.5)^2\right]}{183.35} = 0.2106 \text{ ft}^3 \text{ per linear ft}$$

Formula 6: (Annular capacity between casing and multiple strings of tubing in linear ft/ft^3):

$$C_a = \frac{183.35}{D_i^2 - \left[(T_1)^2 + (T_2)^2 + (T_3)^2\right]} \tag{1.46}$$

Example: Using three strings of tubing:

D_i = casing—9⅝ in., 47 lb/ft ID = 8.681 in.
T_1 = tubing no. 1—OD = 3.5 in.
T_2 = tubing no. 2—OD = 3.5 in.
T_3 = tubing no. 3—OD = 3.5 in.

$$C_a = \frac{183.35}{8.681^2 - \left[(3.5)^2 + (3.5)^2 + (3.5)^2\right]} = 4.7488 \text{ linear ft per ft}^3$$

Spreadsheet 1.8.4 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Capacity between casing—multiple tubulars.

1.9 Amount of cuttings drilled per foot of hole

Formula 1: BARRELS of cuttings generated per foot of hole drilled:

$$V_c = \frac{{D_h}^2}{1029.4}\left(1 - \left(\frac{\phi}{100}\right)\right) \tag{1.47}$$

where

V_c = volume of cuttings in bbls/ft
D_h = diameter of bit or hole size (plus washout) in inches
ϕ = formation porosity in percent (%)

Example: Determine the number of barrels of cuttings drilled for 1 ft of 12¼ in. a hole drilled with 20% porosity.

$$V_c = \frac{12.25^2}{1029.4}\left(1 - \left(\frac{20}{100}\right)\right) = 0.1166 \text{ bbl per ft}$$

Formula 2: (Cuttings generated per foot of hole drilled in **CUBIC FEET**):

$$V_c = \left(\frac{{D_h}^2}{144}\right)(0.7854)\left(1 - \left(\frac{\phi}{100}\right)\right) \tag{1.48}$$

Example: Determine the cubic feet of cuttings drilled for 1 ft of 12¼ in. hole with 20% porosity:

$$V_c = \left(\frac{12.25^2}{144}\right)(0.7854)\left(1 - \left(\frac{20}{100}\right)\right) = 0.6548 \text{ ft}^3$$

Formula 3: (Cuttings generated per foot of hole drilled in **POUNDS**):

$$W_{cg} = 350((C_a)(L_d))\left(1 - \left(\frac{\phi}{100}\right)\right)(SG) \tag{1.49}$$

where

W_{cg} = solids generated in lbs
C_a = capacity of hole in bbl/ft
L_d = footage drilled in ft
ϕ = percent porosity of cuttings in percent (%)
SG = specific gravity of cuttings in g/cm^3

Example: Determine the total pounds of solids generated in. drilling 100 ft of a 12¼ in. hole (0.1458 bbl/ft).

Specific gravity (average bulk density) of cuttings = 2.40 g/cc.
Porosity = 20%.

$$W_{cg} = 350((0.1458)(100))\left(1 - \left(\frac{20}{100}\right)\right)(2.40) = 9797.8 \text{ lb}$$

Spreadsheet 1.9 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Cuttings drilled per foot of hole.

1.10 Annular velocity (AV)

Formula 1: (Annular velocity in ft/min):

$$AV = \frac{O_p}{C_a} \tag{1.50}$$

where

AV = annular velocity in ft/min
O_p = pump output in bbl/min
C_a = capacity of annulus in bbl/ft

Example: Pump output = 12.6 bbl/min:

Capacity of annulus = 0.1261 bbl/ft

$$AV = \frac{12.6}{0.1261} = 99.9 \text{ ft per min}$$

Formula 2: (Annular velocity in ft/min):

$$AV = \frac{24.5(Q)}{(D_h{}^2 - D_p{}^2)} \tag{1.51}$$

where

Q = circulation rate in gpm
D_h = inside diameter of casing or hole size in inches
D_p = outside diameter of pipe, tubing, or collars in inches

Example:

$Q = 530$ gpm
$D_h = 12¼$ in.
$D_p = 4½$ in.

$$\mathrm{AV} = \frac{24.5(530)}{\left(12.25^2 - 4.5^2\right)} = 100.0 \text{ ft per min}$$

Formula 3: (Annular velocity in ft/min):

$$\mathrm{AV} = \frac{O_p(1029.4)}{\left({D_h}^2 - {D_p}^2\right)} \tag{1.52}$$

Example: Pump output = 12.6 bbl/min:

Hole size = 12¼ in.
Pipe OD = 4½ in.

$$\mathrm{AV} = \frac{12.6(1029.4)}{\left(12.25^2 - 4.5^2\right)} = 99.9 \text{ ft per min}$$

Formula 4: (Annular velocity in ft/s):

$$\mathrm{AV} = \frac{(17.16)\left(O_p\right)}{\left({D_h}^2 - {D_p}^2\right)} \tag{1.53}$$

Example: Pump output = 12.6 bbl/min:

Hole size = 12¼ in.
Pipe OD = 4½ in.

$$\mathrm{AV} = \frac{(17.16)(12.6)}{\left(12.25^2 - 4.5^2\right)} = 1.67 \text{ ft per second}$$

Spreadsheet 1.10 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 AV.

1.11 Pump output required in GPM for a desired annular velocity, ft/min

Formula 1: Calculate pump output in gpm:

$$O_p = \frac{AV\left(D_h{}^2 - D_p{}^2\right)}{24.5} \tag{1.54}$$

where

O_p = pump output in gpm
AV = annular velocity in ft/min
D_h = inside diameter of casing or hole size, in.
D_p = outside diameter of pipe, tubing, or collars, in.

Example: AV = Annular velocity 120 ft/min

Hole size = 12¼ in.
Pipe OD = 4½ in.

$$O_p = \frac{120\left(12.25^2 - 4.5^2\right)}{24.5} = 635.8 \text{ gpm}$$

Formula 2: Strokes per minute (**SPM**) required for a given annular velocity:

$$SPM = \frac{(AV)(C_a)}{O_p} \tag{1.55}$$

where

SPM = pump strokes per minute

Example: AV = 120 ft/min

C_a = 0.1261 bbl/ft
O_p = 0.136 bbl/stk

$$SPM = \frac{(120)(0.126)}{0.136} = 111.3 \text{ stks per min}$$

Spreadsheet 1.11 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Pump output for AV.

1.12 Bottoms-up formula

$$\mathrm{BU} = \frac{\left(\frac{\left(D_h^2 - D_p^2\right)}{1029.4}\right)(L)}{\left(\frac{Q}{42}\right)} \tag{1.56}$$

where

BU = bottoms-up circulating time in min
L = measured depth in ft

Example: Determine the bottoms up time for the following conditions:

Hole diameter = 12¼ in.
Drill pipe size = 4½ in.
Measured depth = 10,000 ft
Pump output = 530 gpm

$$\mathrm{BU} = \frac{\left(\frac{\left(12.25^2 - 4.5^2\right)}{1029.4}\right)(10,\,000)}{\left(\frac{530}{42}\right)} = 100 \text{ min}$$

Spreadsheet 1.12 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Bottoms-up time.

1.13 Pump pressure/pump stroke relationship (the Roughneck's formula)

Step 1: Basic formula:

$$P_N = (P_O)\left(\frac{\mathrm{SPM_N}}{\mathrm{SPM_O}}\right)^2 \tag{1.57}$$

where

P_N = new circulating pressure in psi
P_O = old circulating pressure in psi
$\mathrm{SPM_N}$ = new pump rate in strokes per minute
$\mathrm{SPM_O}$ = old pump rate in strokes per minute

Example: Determine the new circulating pressure, psi using the following data.

Present circulating pressure = 1800 psi
Old pump rate = 60 spm
New pump rate = 30 spm

$$P_N = (1800)\left(\frac{30}{60}\right)^2 = 450 \text{ psi}$$

Step 2: Determination of exact factor in above equation.

The above formula is an approximation because the factor "2" is a rounded-off number. To determine the exact factor, obtain two pressure readings at different pump rates and use the following formula.

$$F = \frac{log\left(\frac{P_N}{P_O}\right)}{log\left(\frac{Q_N}{Q_O}\right)} \tag{1.58}$$

where

F = factor used to calculate a new pump pressure with a new pump rate
Q_N = new pump output in gpm
Q_O = old pump output in gpm

Example:

$P_N = 2500$ psi
$P_O = 450$ psi
$Q_N = 315$ gpm
$Q_O = 120$ gpm

$$F = \frac{log\left(\frac{2500}{450}\right)}{log\left(\frac{315}{120}\right)} = 1.7768$$

Example: Same example as above but with the correct factor:

$$P_N = (1800)\left(\frac{30}{60}\right)^{1.7768} = 525 \text{ psi}$$

Spreadsheet 1.13 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Roughneck formula.

1.14 Buoyancy factor (BF)

Formula 1: (Buoyancy factor using mud weight in ppg):

$$BF = \frac{65.5 - MW}{65.5} \tag{1.59}$$

where

BF = buoyancy factor
65.5 = weight of steel in ppg (plain carbon steel AISI SAE 1020 = 7.86 g/cm^3)
MW = mud weight in ppg

Example: Determine the buoyancy factor for a 15.0 ppg fluid.

$$\mathrm{BF} = \frac{65.5 - 15.0}{65.5} = 0.7710$$

Formula 2: (Buoyancy factor using mud weight in lb/ft^3):

$$(65.5)(7.48) = 489.9 \cong 490 \text{ lb/ft}^3$$

$$\mathrm{BF} = \frac{490 - \mathrm{MW}}{490} \tag{1.60}$$

where

490 = weight of steel in lb/ft^3

Example: Determine the buoyancy factor for a 112.2 lb/ft^3 (15.0 ppg) fluid.

$$\mathrm{BF} = \frac{490 - 112.2}{490} = 0.7710$$

Spreadsheet 1.14 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Buoyancy factor.

1.15 Formation temperature (T_f)

$$T_f = T_s + \left(T_G\left(\frac{\mathrm{TVD}}{100}\right)\right) \tag{1.61}$$

where

T_f = estimated temperature of formation at a specific depth in °F
T_s = ambient temperature at the surface in °F
T_G = geothermal gradient in °F per 100 ft of depth
TVD = total vertical depth in feet

Example: Determine the formation temperature with the following:

$T_s = 70$°F
$T_G = 1.2$°F/100 ft
TVD = 15,000 ft

$$T_f = 70 + \left(1.2\left(\frac{15,000}{100}\right)\right) = 250^{\circ}\text{F}$$

Spreadsheet 1.15 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Formation temperature.

1.16 Temperature conversion formulas

Formula 1a: Convert temperature, Fahrenheit (°F) to Centigrade or Celsius (°C):

$$C = \frac{5(F-32)}{9} \tag{1.62}$$

where

C = centigrade or Celsius temperature in degrees (°)
F = Fahrenheit temperature in degrees (°)

Example: Convert 95°F to °C.

$$C = \frac{5(95-32)}{9} = 35^{\circ}\text{C}$$

Formula 1b: Convert temperature, Fahrenheit (°F) to centigrade or Celsius (°C):

$$C = (F-32)(0.5556) \tag{1.63}$$

Example: Convert 95°F to °C.

$$C = (95-32)(0.5556) = 35^{\circ}\text{C}$$

Formula 2a: Convert temperature, centigrade or Celsius (°C) to Fahrenheit (°F):

$$F = \frac{(9(C))}{5} + 32 \tag{1.64}$$

Example: Convert 24°C to °F.

$$F = \frac{(9(24))}{5} + 32 = 75.2^{\circ}\text{F}$$

Formula 2b: Convert temperature, centigrade or Celsius (C) to Fahrenheit (°F):

$$F = (1.8(C)) + 32 \tag{1.65}$$

Example: Convert 24°C to °F.

$$F = (1.8(24)) + 32 = 75.2^{\circ}\text{F}$$

Formula 3a: Convert temperature, centigrade or Celsius (C) to Kelvin (°K):

$$K = C + 273.16 \tag{1.66}$$

Example: Convert 35°C to °K.

$$K = 35 + 273.16 = 308.16^{\circ}\text{K}$$

Formula 4a: Convert temperature, Fahrenheit (°F) to Rankin (°R):

$$R = F + 459.69 \tag{1.67}$$

Example: Convert 95°F to °R.

$$R = 95 + 459.69 = 554.69^{\circ}\text{R}$$

Rules of thumb formulas for temperature conversion
Convert °F to °C.

$$C = \frac{(F - 30)}{2} \tag{1.68}$$

Example: Convert 95°F to °C.

$$C = \frac{(95 - 30)}{2} = 32.5^{\circ}\text{C}$$

Convert °C to °F.

$$F = C + C + 30 \tag{1.69}$$

Example: Convert 24°C to °F.

$$F = 24 + 24 + 30 = 78^{\circ}\text{F}$$

Spreadsheet 1.16 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Temperature conversions.

Appendix: Supplementary material

Supplementary material related to this chapter can be found online at https://www.elsevier.com/books-and-journals/book-companion/9780323905497.

CHAPTER

2

RIG calculations

2.1 Accumulator capacity

2.1.1 Usable volume per bottle

Note: Thefollowing will be used as guidelines:

Volume per bottle	= 10 gal
Precharge pressure	= 1000 psi
Minimum pressure (remaining after activation)	= 1200 psi
Pressure gradient of hydraulic fluid	= 0.445 psi/ft
Maximum pressure	= 3000 psi

Boyle's law for ideal gases will be adjusted and used as follows:

$$(P_1)(V_1) = (P_2)(V_2) \tag{2.1}$$

where

P = pressure in psi
V = volume in gal

Spreadsheet 2.1.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Usable accumulator volume.

2.1.2 Surface application

Step 1: Calculate the hydraulic fluid necessary to increase the pressure from the precharge to the minimum:

$(1000)(10) = (1200)(V_2)$

$V_2 = \frac{(1000)(10)}{(1200)} = 8.33$ gal

The nitrogen has been compressed from 10.0 to 8.33 gal.

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Next, calculate the volume of the hydraulic fluid in the bottle.
$10.0 - 8.33 = 1.67$ gal of hydraulic fluid per bottle

Note: This is dead hydraulic fluid. The pressure must not drop below the minimum value of 1200 psi.

Step 2: Calculate the amount of hydraulic fluid necessary to increase the pressure from the precharge to the maximum:

$(1000)(10) = (1200)(V_2)$
$V_2 = \frac{(1000)(10)}{(3000)} = 3.33$ gal
The nitrogen has been compressed from 10 to 3.33 gal.
$10.0 - 3.33 = 6.67$ gal of hydraulic fluid per bottle

Step 3: Calculate the usable volume per bottle in gal:

$$V_u = (V_{tb} - V_d) \tag{2.2}$$

where

V_u = usable volume per bottle in gal
V_{tb} = total hydraulic volume per bottle in gal
V_d = dead hydraulic volume per bottle in gal
$V_u = (6.67 - 1.67) = 5.0$ gal

Spreadsheet 2.1.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Surface application.

2.1.3 Volume capacity of the accumulator bottle

$$V_u = (V_b)\left(\left(\frac{P_p}{P_f}\right) - \left(\frac{P_p}{P_m}\right)\right) \tag{2.3}$$

where

V_b = volume capacity of the bottle in gal
P_p = precharge pressure in psi
P_f = minimum pressure after activation in psi
P_m = maximum system pressure in psi

Example: Calculate the amount of usable hydraulic fluid delivered from a 20-gal bottle:

Precharge pressure	= 1000 psi (P_P)
Maximum system pressure	= 3000 psi (P_m)
Final pressure	= 1200 psi (P_f)

$$V_u = (20)\left(\left(\frac{1000}{1200}\right) = \left(\frac{1000}{3000}\right)\right) = 10 \text{ gal.}$$

Spreadsheet 2.1.3 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Volume capacity of the accumulator bottle.

2.1.4 Deepwater applications

In deepwater applications, the hydrostatic pressure exerted by the hydraulic fluid must be compensated for in the calculations due to the hydrostatic head of the seawater.

Example: The same guidelines as in surface applications:

Water depth = 2000 ft
Hydrostatic pressure on the hydraulic fluid = 889 psi or [(8.55)(0.052)(2000)]

Step 1: Adjust all hydraulic fluid pressures for the hydrostatic pressure of the seawater:

$P_p = (1000 + 889) = 1889$ psi
$P_m = (3000 + 889) = 3889$ psi
$P_f = (1200 + 889) = 2089$ psi

Step 2: Calculate the volume of the hydraulic fluid necessary to increase the pressure from the precharge to the minimum:

$(1889)(10) = (2089)(V_2)$
$V_2 = \frac{(1889)(10)}{(2089)} = 9.04\ gal$
$V_u = (10 - 9.04) = 0.96$ gal

Note: This is dead hydraulic fluid. The pressure must not drop below the minimum value of 2089 psi.

Step 3: Calculate the amount of hydraulic fluid necessary to increase the pressure from the precharge to the maximum:

$(1889)(10) = (3889)(V_2)$
$V_2 = \frac{(1889)(10)}{(3889)} = 4.86\ gal$
The nitrogen has been compressed from 10 to 4.86 gal.
$10 - 4.86 = 5.14$ gal of hydraulic fluid per bottle

Step 4: Calculate the usable hydraulic fluid per bottle:

$$V_u = (4.86 - 0.96) = 3.9\ gal$$

Spreadsheet 2.1.4 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Deepwater applications.

2.1.5 Accumulator precharge pressure

The following is a method for calculating the average accumulator precharge pressure by operating the unit with the charge pumps switched off:

$$P_p = \left(\frac{V_r}{V_t}\right)\left(\frac{((P_e)(P_s))}{(P_s - P_t)}\right) \tag{2.4}$$

where

P_p = precharge pressure in psi
V_r = volume of the hydraulic fluid removed in gal
V_t = total accumulator volume in gal
P_f = final accumulator pressure in psi
P_s = starting accumulator pressure in psi

Example: Calculate the average accumulator precharge pressure using the following data:

Starting accumulator pressure = 3000 psi
Final accumulator pressure = 2200 psi
Volume of the fluid removed = 20 gal
Total accumulator volume = 180 gal

$$P_p = \left(\frac{20}{180}\right)\left(\frac{((2200)(3000))}{(3000 - 2200)}\right) = (0.1111)\left(\frac{(6,600,000)}{800}\right) = 917 \text{ psi}$$

Spreadsheet 2.1.5 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Accumulator precharge pressure.

2.2 Slug calculations

2.2.1 Barrels of slug required for a desired length of dry pipe

Step 1: Calculate the hydrostatic pressure required to give a desired mud drop inside the drill pipe:

$$S_{HP} = (W_m)(0.052)(L_{pd}) \tag{2.5}$$

where

S_{HP} = slug hydrostatic pressure in psi
W_m = mud weight in ppg
L_{pd} = length of the dry pipe in ft

Step 2: Calculate the difference in pressure gradient between the slug weight and mud weight:

$$S_{PG} = (S_w - W_m)(0.052) \tag{2.6}$$

where

S_{PG} = slug pressure gradient in psi/ft
S_w = slug weight in ppg
W_m = mud weight in use in ppg

Step 3: Calculate the length of the slug in the drill pipe:

$$S_l = \left(\frac{S_{HP}}{S_{PG}}\right) \tag{2.7}$$

where

S_l = slug length in ft

Step 4: Calculate the volume of the slug in bbl:

$$S_v = (S_l)(V_{dp}) \tag{2.8}$$

where

S_v = volume of the slug in bbl
V_{dp} = capacity of the drill pipe in bbl/ft

Example: Calculate the barrels of slug required for the following:

Desired length of the dry pipe (2 stands)	= 184 ft
Mud weight	= 12.2 ppg
Slug weight	= 13.2 ppg
Drill pipe capacity (6⅝ in.)	= 0.03457 bbl/ft

Step 1: Hydrostatic pressure required:

$$S_{HP} = (12.2)(0.052)(184) = 117\,\text{psi}$$

Step 2: Difference in pressure gradient:

$$S_{PG} = (13.2 - 12.2)(0.052) = 0.052\,\text{psi/ft}$$

Step 3: Length of the slug in the drill pipe:

$$S_l = \left(\frac{117}{0.052}\right) = 2250\ \text{ft}$$

Step 4: Volume of the slug:

$$S_v = (2250)(0.03457) = 77.8\ \text{bbl} \cong 80\ \text{bbl}$$

Spreadsheet 2.2.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Barrels of slug required.

2.2.2 Weight of slug required for a desired length of dry pipe with a set volume of slug

Step 1: Calculate the length of the slug in the drill pipe in ft:

$$S_l = \left(\frac{S_v}{V_{dp}}\right) \tag{2.9}$$

where

S_l = slug length in ft

Step 2: Calculate the hydrostatic pressure required to give the desired drop of mud inside the drill pipe in psi (from Eq. (2.5)):

$$S_{HP} = (W_m)(0.052)(L_{pd})$$

where

S_{HP} = slug hydrostatic pressure in psi
W_m = mud weight in ppg
L_{pd} = length of the dry pipe in ft

Step 3: Calculate the weight of the desired slug:

$$S_w = \left(\frac{\left(\frac{S_{HP}}{0.052}\right)}{S_l}\right) + W_m \tag{2.10}$$

where

S_w = slug weight in ppg

Example: Calculate the weight of the slug required for the following:

Desired length of the dry pipe (2 stands)	= 184 ft
Mud weight	= 12.2 ppg
Volume of the slug	= 75 bbl
Drill pipe capacity (6⅝ in.)	= 0.03457 bbl/ft

Step 1: Length of the slug in the drill pipe:

$$S_l = \left(\frac{75}{0.03457}\right) = 2170 \text{ ft}$$

Step 2: Hydrostatic pressure required:

$$S_{HP} = (12.2)(0.052)(184) = 117 \text{ psi}$$

Step 3: Weight of the slug:

$$S_w = \left(\frac{\left(\frac{117}{0.052}\right)}{2170}\right) + 12.2 = 13.2 \text{ ppg}$$

Spreadsheet 2.2.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Weight of the slug required.

2.2.3 Volume, height, and pressure gained because of placement of the slug in the drill pipe

Step 1: Calculate the volume gained in mud pits after the slug is pumped, due to U-tubing:

$$S_{\text{pit}} = (S_{\text{l}})(V_{\text{dp}}) \tag{2.11}$$

where

S_{pit} = volume increase in the mud pit in bbl
V_{dp} = capacity of the drill pipe in bbl/ft

Step 2: Calculate the height of the slug in the annulus in ft:

$$S_{\text{h}} = (V_{\text{ac}})(S_{\text{v}}) \tag{2.12}$$

where

S_{h} = height of the slug in the annulus in ft
V_{ac} = volume of the open hole and drill pipe in ft/bbl

Note: If the slug is pumped out of the drill string that contains a float, then that will prevent U-tubing back into the string.

Step 3: Calculate the hydrostatic pressure in psi gained in the annulus because the slug has been pumped out of the drill string:

$$S_{\text{HP}} = (S_{\text{h}})(S_{\text{w}} - W_{\text{m}})(0.052) \tag{2.13}$$

where

S_{HP} = increase in hydrostatic pressure in the annulus due to the slug in psi

Example: Calculate the increase in hydrostatic pressure gained with the following:

Feet of the dry pipe (2 stands)	= 184 ft
Slug volume	= 75 bbl
Slug weight	= 13.2 ppg
Mud weight	= 12.2 ppg (after circulating the annulus clean of cuttings)
Drill pipe capacity (6⅝ in.)	= 0.03457 bbl/ft
Length of the drill collars	= 600 ft
Annulus volume (9⅞ × 8½ in.)	= 40.74 ft/bbl
Annulus volume (9⅞ × 6⅝ in.)	= 19.20 ft/bbl

Step 1: Volume gained in the mud pits after the slug is pumped, due to U-tubing:

$$S_{\text{pit}} = (184)(0.03457) = 6.4\,\text{bbl}$$

Step 2: Calculate the height of the slug in the annulus in ft:

Note: The slug volume will cover the drill collars and part of the drill pipe section.

$$S_{\text{h}} = \left[(19.2)\left(75 - \left(\frac{600}{40.74}\right)\right)\right] + (600) = 1757.2\ \text{ft}$$

Step 3: Calculate the hydrostatic pressure gained in the annulus because of the slug:

$$S_{\text{HP}} = (1757.2)(13.2 - 12.2)(0.052) = 91.4\ \text{psi}$$

Spreadsheet 2.2.3 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Volume, height, and pressure gained from the slug.

2.2.4 The amount of dry pipe in feet after pumping the slug

Step 1: Volume gained due to pumping the slug in bbl:

$$S_{\text{vg}} = \left((S_{\text{v}})\left(\frac{S_{\text{w}}}{W_{\text{m}}}\right)\right) - (S_{\text{v}}) \tag{2.14}$$

where

S_{vg} = volume gained due to pumping the slug in bbl

Step 2: Calculate the length of the dry pipe after pumping the slug in ft:

$$S_{\text{ldp}} = \left(\frac{S_{\text{vg}}}{V_{\text{dp}}}\right) \tag{2.15}$$

where

S_{ldp} = length of the dry pipe after pumping the slug in ft

Example: Calculate the number of barrels of mud gained due to pumping the slug, and calculate the feet of the dry pipe:

Mud weight	= 12.6 ppg
Slug weight	= 14.2 ppg
Barrels of slug pumped	= 50 barrels
Drill pipe capacity (6⅝ in.)	= 0.03457 bbl/ft

Step 1: Volume gained after pumping the slug due to U-tubing:

$$S_{vg} = \left((50)\left(\frac{14.2}{12.6}\right)\right) - (50) = 6.35\,\text{bbl}$$

Step 2: Calculate the number of feet of the dry pipe after pumping the slug:

$$S_{ldp} = \left(\frac{6.35}{0.03457}\right) = 184\,\text{ft}$$

Spreadsheet 2.2.4 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Amount of dry pipe after the slug.

2.3 Bulk density of cuttings using the mud balance

Procedure:

(1) Cuttings must be washed free of mud. In an oil mud, a base oil can be used instead of water.
(2) Set mud balance at 8.33 ppg.
(3) Fill the mud balance with the clean cuttings until a balance is obtained with the lid in place.
(4) Remove the lid, fill cup with fresh water (cuttings included), replace the lid, and dry outside of the mud balance.
(5) Move counterweight to obtain a new weight reading on the ppg scale.

$$SG_c = \left(\frac{1}{2 - ((0.12)(W_r))}\right) \tag{2.16}$$

where

SG_c = specific gravity (average bulk density) of cuttings in gm/cm^3
W_r = resulting mud weight with cuttings plus water in ppg

Example: Calculate the average bulk density of cuttings with a final weight of 13.0 ppg:

$$SG_c = \left(\frac{1}{2 - ((0.12)(13.0))}\right) = 2.27\,gm/cm^3$$

A graph may also be prepared to provide a quick direct reading of the average bulk density.

Spreadsheet 2.3 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Bulk density of cuttings.

2.4 Drill string design

2.4.1 Estimated weight of the drill collars in air

Formula 1: REGULAR drill collar weight in lb/ft:

$$W_{Rdc} = 2.674\left(\mathrm{OD}^2 - \mathrm{ID}^2\right) \tag{2.17}$$

where

W_{Rdc} = weight of the **REGULAR** drill collar in lb/ft
OD = outside diameter of the drill collar in in.
ID = inside diameter of the drill collar in in.

Example: Determine the weight of an $8 \times 2\frac{13}{16}$ in. **REGULAR** drill collar in lb/ft:

Drill collar OD = 8.0 in.
Drill collar ID = $2\frac{13}{16}$ in. $\left(\frac{13}{16} = 0.8125\right)$
$W_{\mathrm{Rdc}} = 2.674\ (8^2 - 2.8125^2)$
$W_{\mathrm{Rdc}} = 2.674\ (56.089844) = 149.98 \cong 150$ lb/ft

Formula 2: SPIRAL drill collar weight in lb/ft:

$$W_{Sdc} = 2.56\left(\mathrm{OD}^2 - \mathrm{ID}^2\right) \tag{2.18}$$

where

W_{Sdc} = weight of the **SPIRAL** drill collar in lb/ft

Example: Determine the weight of an $8 \times 2\frac{13}{16}$ in. **SPIRAL** drill collar in lb/ft:

Drill collar OD = 8.0 in.
Drill collar ID = $2\frac{13}{16}$ in. $\left(\frac{13}{16} = 0.8125\right)$
$W_{\mathrm{Sdc}} = 2.56\ (8.0^2 - 2.8125^2) = 143.6$ lb/ft

Spreadsheet 2.4.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Weight of the DCs in air.

2.4.2 Tensile strength of tubulars in lb (Ref.: API Spec 5D & 7)

Tensile strength values of the class 1 (new) drill pipe are listed in Table 2.1. These values should be reduced based on the class of the pipe as follows:

Class	Minimum weight%
1 (new)	87.5
Premium (used)	80.0
Class 2 (used)	70.0

TABLE 2.1 Pipe tensile requirements.

	Yield strength (psi)	
Drill pipe body grade	**Minimum**	**Maximum**
E	75,000	105,000
X	95,000	125,000
G	105,000	135,000
S	135,000	165,000
Tool joint	120,000	165,000

Ref: API Spec 5D, Table C.5.

Other dimensions may be found in Table C.1 of the API publication Spec 5D.

Non-API Z-140 and V-150 grades are covered by proprietary specifications and can be found in T. H. Hill's Standard DS-1™.

2.4.3 Reduced tensile yield strength of tubulars in lb

Step 1: Calculate the minimum weight wall thickness in in.:

$$P_t = W_{min}(W_t) \tag{2.19}$$

where

P_t = reduced pipe wall thickness in in.
W_{min} = wall thickness reduction according to pipe class in %
W_t = wall thickness of the new pipe in in.

Step 2: Calculate the reduction in the OD of the premium pipe in in.:

$$OD_r = ID + [2(P_t)] \tag{2.20}$$

where

OD_r = reduced pipe OD in in.

Step 3: Calculate the cross-sectional area of the reduced tube in in.^{2}:

$$A_r = \frac{\pi\left(OD_r^2 - ID^2\right)}{4} \tag{2.21}$$

where

A_r = reduced cross-sectional area in in.^{2}

Step 4: Calculate the reduced tensile yield strength in lb:

$$T_{Sr} = A_{\mathrm{r}}(\gamma_{\mathrm{m}}) \tag{2.22}$$

where

T_{sr} = reduced tensile yield strength in lb
γ_{m} = minimum yield strength in lb from Table 2.1.

Example: Calculate the reduced wall thickness of a premium 6⅝ in., ID—5.965 in., 25.5 lb/ft, S-135 drill pipe with a 0.330 in. wall.

Step 1: Calculate the minimum weight wall thickness in in.:

$$P_{\mathrm{t}} = 0.80(0.330) = 0.264\,\mathrm{in.}$$

Step 2: Calculate the reduction in the OD of the premium pipe in in.:

$$\mathrm{OD_r} = 5.965 + [2(0.264)] = 6.493\,\mathrm{in.}$$

Step 3: Calculate the cross-sectional area of the reduced tube in $\mathrm{in.}^2$:

$$A_{\mathrm{r}} = \frac{3.14\left(6.493^2 - 5.965^2\right)}{4} = 5.164\ \mathrm{in.}^2$$

Step 4: Calculate the reduced tensile yield strength in lb:

$$T_{\mathrm{sr}} = 5.164\ (135,000) = 697,140\,\mathrm{lb}$$

Spreadsheet 2.4.3 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Reduced tensile yield of tubulars.

2.4.4 Adjusted weight and length of the drill pipe and tool joints

Step 1: Calculate the approximate adjusted weight of the tube in lb/ft:

$$W_{\mathrm{ta}} = W_{\mathrm{pe}} + \left(\frac{\mathrm{e_w}}{29.4}\right) \tag{2.23}$$

where

W_{ta} = approximate adjusted weight of the tube in lb/ft
e_{w} = weight of the upset in API Spec 5D, Table C.13.
W_{pe} = plain-end pipe body unit mass (without upsets) in lb/ft

(Ref: API Spec 5-DP, Table C.14, p. 97).

Step 2: Calculate the approximate weight of the tool joint in lb:

$$W_{\mathrm{tja}} = \left(0.222{*}L_{\mathrm{tj}}{*}\left(\mathrm{OD_{tj}^2} - \mathrm{ID_{tj}^2}\right)\right) + \left(0.167{*}\left(\mathrm{OD_{tj}^3} - D_{\mathrm{te}}^3\right)\right) - \left(0.510{*}\mathrm{ID_{tj}^2}{*}\left(\mathrm{OD_{tj}} - D_{\mathrm{te}}\right)\right) \tag{2.24}$$

where

W_{tja} = approximate weight of the tool joint in lb
OD_{tj} = outside diameter of the tool joint in in.
ID_{tj} = inside diameter of the tool joint in in.
D_{te} = inside diameter of the pipe weld neck in in.
L_{tj} = length of the tool joint in in.

Step 3: Calculate the adjusted length of the tool joint in ft:

$$L_{tja} = \frac{L_{tj} + \left(2.253^{*}\left(OD_{tj} - D_{te}\right)\right)}{12} \tag{2.25}$$

where

L_{tja} = adjusted length of the tool joint in ft

Step 4: Calculate the adjusted weight of the drill pipe and tool joint in lb/ft:

$$W_{adj} = \frac{\left[W_{ta}(29.4) + W_{tja}\right]}{\left(W_{tja} + 29.4\right)} \tag{2.26}$$

where

W_{adj} = adjusted weight of the drill pipe and tool joint in lb/ft

Example: Calculate the adjusted weight of the drill pipe and length and tool joint of a 6⅝ in., 25.5 lb/ft, S-135 drill pipe. Calculate the adjusted length of the 6⅝ in. tool joint with an 8.5 in. OD, length—19 in., and a 6.938 weld neck.

Step 1: Calculate the approximate adjusted weight of the tube in lb/ft:

$$W_{ta} = 22.19 + \left(\frac{24.87}{29.4}\right) = 23.04\,\text{lb/ft}$$

Step 2: Calculate the approximate weight of the tool joint in lb:

$$W_{tja} = \left(0.222^{*}19^{*}\left(8.5^{2} - 4.25^{2}\right)\right) + \left(0.167^{*}\left(8.5^{3} - 6.938^{3}\right)\right) - \left(0.510^{*}4.25^{2}{}^{*}(8.5 - 6.938)\right) = 261.0\,\text{lb}$$

Step 3: Calculate the adjusted length of the tool joint in ft:

$$L_{tja} = \frac{19.0 + (2.253(8.5 - 6.938))}{12} = 1.877\ \text{ft}$$

Step 4: Calculate the adjusted weight of the drill pipe and tool joint in lb/ft:

$$W_{adj} = \frac{(23.04^{*}29.4) + 261.0}{(1.877 + 29.4)} = 30.1\ \text{lb/ft}$$

Spreadsheet 2.4.4 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Adjusted weight of the DP.

2.4.5 Length of the BHA necessary for a desired weight on bit

The following will be determined: the length of the bottom hole assembly (BHA) necessary for a desired weight on bit (WOB) and the feet of the premium drill pipe that can be used with a specific bottom hole assembly.

Step 1: Calculate the buoyancy factor (from Section 1.16):

$$\mathrm{BF} = \frac{65.5 - \mathrm{MW}}{65.5}$$

Step 2: Calculate the length of the BHA necessary for a desired weight on bit:

$$L_{\mathrm{BHA}} = \left(\frac{(W_{\mathrm{bit}})(1 + f_{\mathrm{dc}})}{(W_{\mathrm{dc}})(\mathrm{BF})}\right) \tag{2.27}$$

where

$\mathrm{LBM}_{\mathrm{HA}}$ = length of the BHA necessary for a desired WOB in ft
W_{bit} = desired weight on bit (WOB) in lb
f_{dc} = safety factor to place a neutral point in drill collars
W_{dc} = weight of the drill collar in lb/ft

Example: Calculate the BHA length necessary for a desired WOB:

Desired WOB while drilling	= 50,000 lb
Safety factor	= 15%
Mud weight	= 12.0 ppg
Drill collar weight (8×3 in.)	= 147 lb/ft

Step 1: Calculate the buoyancy factor (from Section 1.16):

$$\mathrm{BF} = \left(\frac{65.5 - 12.0}{65.5}\right) = 0.8168$$

Step 2: Calculate the length of the BHA necessary for this weight on bit:

$$L_{\mathrm{BHA}} = \left(\frac{(50,000)(1 + 0.15)}{(147)(0.8168)}\right) = 479\,\mathrm{ft}$$

Spreadsheet 2.4.5 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Length of the BHA for the desired bit weight.

2.4.6 Maximum length of the premium drill pipe that can be run into the hole with a specific BHA based on the margin of overpull

Note: Obtain the tensile strength of a new pipe from Table 2.2 and adjust for premium service.

Step 1: Calculate the buoyancy factor (from Section 1.16):

$$\mathrm{BF} = \frac{65.5 - \mathrm{MW}}{65.5}$$

Step 2: Calculate the maximum length of the premium drill pipe that can be run into the hole with a specific BHA based on the margin of overpull:

$$L_{\max} = \frac{\left(\left(T_{\mathrm{Sr}}\right)\left(\mathrm{f_{dp}}\right)\right) - \mathrm{MOP} - \left(\left(W_{\mathrm{BHA}}\right)(\mathrm{BF})\right)}{\left(W_{\mathrm{dpa}}\right)(\mathrm{BF})} \tag{2.28}$$

where

$L_{\max}$ = maximum length of the premium drill pipe that can be run into the hole with a specific BHA in ft
T_{Sr} = reduced tensile strength of the premium (used) drill pipe in lb
f_{dp} = safety factor
MOP = margin of overpull in lb
W_{BHA} = air weight of the BHA in lb
W_{dp} = adjusted weight of the premium drill pipe with tool joints in lb/ft

Step 3: Calculate the total depth that can be reached with a specific BHA in ft:

$$D_{\mathrm{T}} = L_{\max} + L_{\mathrm{BHA}} \tag{2.29}$$

where

D_{T} = total depth that can be reached with a specific BHA in ft
L_{BHA} = length of the BHA to be run in ft

Example:

Drill pipe (6⅝ in.)	= 25.20 lb/ft (S-135) (adjusted weight = 30.0 lb/ft)
Tensile strength	= 881,000 lb (Class 1—new)
Reduced tensile strength	= 697,410 lb (premium—used, calculated from Eq. (2.22))
BHA weight in air	= 50,000 lb
BHA length	= 500 ft
Desired overpull	= 100,000 lb
Mud weight	= 13.5 ppg
Safety factor	= 10%

TABLE 2.2 Drill pipe data.

Size OD (in.)	Size ID (in.)	Nominal weight (lb/ft)	Grade	Connection	Tensile yield strength (lb)	Pipe body section area (in.^{2})
3½	2.764	13.30	X 95	NC 38	344,000	3.621
			G 105	NC 38	380,000	
			S 135	NC 38	488,800	
			Z 140	HT 31	506,900	
			V 150	HT 31	543,100	
4	3.240	15.70	X 95	NC 40	410,000	4.322
			G 105	NC 40	453,000	
			S 135	NC 40	583,400	
			Z 140	HT 40	605,000	
			V 150	HT 40	648,200	
4½	3.826	16.60	X 95	NC 50	418,700	4.407
			G 105	NC 50	462,800	
			S 135	NC 50	595,000	
			Z 140	HT 50	617,000	
			V 150	HT 50	661,100	
5	4.276	19.5	X 95	NC 50	501,100	5.275
			G 105	NC 50	553,800	
			S 135	NC 50	712,100	
			Z 140	HT 50	738,400	
			V 150	HT 50	791,200	
6⅝	5.965	25.50	X 95	FH	620,000	6.526
			G 105	FH	685,200	
			S 135	FH	881,000	
			Z 140	FH	913,700	
			V 150	FH	978,900	

Ref: Grant Prideco, Drill Pipe Data Catalog, 2003.

Step 1: Buoyancy factor:

$$\mathrm{BF} = \left(\frac{65.5 - 13.5}{65.5}\right) = 0.7939$$

Step 2: Calculate the maximum length of the premium drill pipe that can be run into the hole based on a margin of overpull in ft:

$$L_{\max} = \frac{(((697,140)(0.9)) - 100,000 - ((50,000)(0.7939)))}{(30.1{*}0.7939)} = 20,410 \text{ ft}$$

Step 3: Calculate the total depth that can be reached with this BHA and this premium drill pipe in ft:

$$D_{\mathrm{T}} = 20,410 + 500 = 20,910 \text{ ft}$$

Spreadsheet 2.4.6 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Length of the DP with BHA overpull.

2.4.7 Length of the premium drill pipe based on the overpull and slip crushing

Step 1: Calculate the tensile strength of the drill pipe in psi:

$$W_{\mathrm{S}} = \gamma_{\mathrm{m}}(A_{\mathrm{r}}) \tag{2.30}$$

where

W_{S} = working strength in psi
γ_{m} = minimum yield strength in psi

Example: Calculate the tensile strength of the 6⅝ in. S-135 drill pipe with a cross-sectional area of 6.526 in.^{2}

$$W_{\mathrm{S}} = 135,000(6.526) = 881,010 \text{ psi}$$

Spreadsheet 2.4.7 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Length of the DP based on the overpull.

2.4.7.1 Slip crushing

Definition: The slip crushing relationship describes the possibility that the drill pipe can be crushed by the axial load due to high hoop stresses that exist in the cylindrical pipe body while hung in the rotary slips with a large string load.

Step 1: Calculate the minimum stress ratio $\left(\frac{\sigma_h}{\sigma_t}\right)$. (Also available in Table 2.3):

$$R_{ms} = \sqrt{1 + \left(\frac{D(k)}{2(L_s)}\right) + \left(\frac{D(k)}{2(L_s)}\right)^2} \tag{2.31}$$

where

R_{ms} = minimum stress ratio $\left(\frac{\sigma_h}{\sigma_t}\right)$. [$\sigma_h$ = pipe body hoop stress; σ_t = pipe body tensile axial stress]
D = OD of the pipe in in.
k = lateral load factor of the slips (refer to Table 2.3)
L_s = length of the slips in in.

Step 2: Calculate the axial tensile stress of the pipe body at the slips:

$$\sigma_t = \frac{W_s}{A_p} \tag{2.32}$$

where

σ_t = axial tensile stress of the tube in psi
W_s = weight of the string in lb
A_p = cross-sectional area of the pipe in in.^{2}

Step 3: Calculate the hoop stress of the pipe body in psi:

$$\sigma_h = R_{ms}(\sigma_t) \tag{2.33}$$

where

σ_h = pipe body hoop stress in psi
R_{ms} = minimum stress ratio from Step 1 or Table 2.3.

Step 4: Calculate the approximate safety factor for pipe body slip crushing:

$$SF_{sc} = \frac{\gamma_m}{\sigma_h} \tag{2.34}$$

where

SF_{SC} = safety factor for slip crushing
γ_m = minimum yield strength in psi (from Table 2.1).

Example: Calculate the safety factor for the following conditions:

DP	= 5.0 in. (G-105, 19.5 lb/ft, NC50, Class 1—new)
String weight	= 220,000 lb

TABLE 2.3 Minimum ratios $\left(\frac{\sigma_h}{\sigma_t}\right)$ to prevent slip crushing.

			Minimum ratio $\left(\frac{\sigma_h}{\sigma_t}\right)$ pipe size (in.)						
Slip length (in.)	**Coefficient of friction**	**Lateral load factor**	**2⅜**	**2⅞**	**3½**	**4**	**4½**	**5**	**5½**
12	0.06	4.36	1.27	1.34	1.43	1.50	1.58	1.65	1.73
	0.08	4.00	1.25	1.31	1.39	1.45	1.52	1.59	1.66
	0.10	3.68	1.22	1.28	1.35	1.41	1.47	1.54	1.60
	0.12	3.42	1.21	1.26	1.32	1.38	1.43	1.49	1.55
	0.14	3.18	1.19	1.24	1.30	1.34	1.40	1.45	1.50
16	0.06	4.36	1.20	1.24	1.30	1.36	1.41	1.47	1.52
	0.08	4.00	1.18	1.22	1.28	1.32	1.37	1.42	1.47
	0.10	3.68	1.16	1.20	1.25	1.29	1.34	1.38	1.43
	0.12	3.42	1.15	1.18	1.23	1.27	1.31	1.35	1.39
	0.14	3.18	1.14	1.17	1.21	1.25	1.28	1.32	1.36

Cross-sectional area $= 5.275\,\text{in.}^2$
Coefficient of friction $= 0.08$
Lateral load factor $= 4.0$
Slip length $= 16\,\text{in.}$

Step 1: Calculate the minimum stress ratio $\left(\frac{\sigma_h}{\sigma_t}\right)$:

$$R_{ms} = \sqrt{1 + \left(\frac{(5.0*4.0)}{(2*16)}\right) + \left(\frac{(5.0*4.0)}{(2*16)}\right)^2} = 1.4197$$

Step 2: Calculate the axial tensile stress of the pipe body at the slips:

$$\sigma_t = \frac{220,000}{5.275} = 41,706\,\text{psi}$$

Step 3: Calculate the hoop stress of the pipe body in psi:

$$\sigma_h = (1.4197)(41,706) = 59,210\ \text{psi}$$

Step 4: Calculate the approximate safety factor for pipe body slip crushing:

$$\text{SF}_{sc} = \frac{105,000}{59,210} = 1.77$$

Spreadsheet 2.4.7.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Slip crushing.

2.4.8 Design of a drill string for a specific set of well conditions

This design method will use two conditions to calculate the length of the various grades of the drill pipe: margin of overpull and slip crushing. The objective of this technique is to select the smallest length of the pipe calculated with the two methods.

$$L_{psc} = \left[\frac{\left(\frac{\gamma_m(f_{dp})}{K_s}\right) - [W_{DS}(\text{BF})]}{(W_{dpa})(\text{BF})}\right] \tag{2.35}$$

where

L_{psc} = length of the drill pipe based on slip crushing
γ_m = tensile strength from Table 2.4
f_{dp} = factor for the drill pipe (1—safety factor)

TABLE 2.4 Available string.

		Pipe size											
No.	**Item**	**OD (in.)**	**ID (in.)**	**Wt (lb/ft)**	**Grade**	**W_{min} (in.)**	**OD_r (in.)**	**A_r (in.)**	**TS_r (lb)**	**W_{ta} (lb/ft)**	**W_{tja} (lb)**	**L_{tja} (in.)**	**W_{adj} (lb/ft)**
1	DC	7	2.8125	110	NC50								
2	HW	5	3.0	42.72	NC50								49.77
3	DP	5	4.276	19.5	X-95	0.2896	4.855	4.152	394,440	18.22	137.8	1.748	21.62
4	DP	5	4.276	19.5	G-105	0.2896	4.855	4.152	435,960	18.22	145.7	1.748	21.88
5	DP	5	4.276	19.5	S-135	0.2896	4.855	4.152	560,520	18.22	159.7	1.748	22.32

K_s = constant for the minimum ratio $\left(\frac{\sigma_h}{\sigma_t}\right)$ of hoop stress to tensile stress that can be found in Table 2.3.
W_{DS} = air weight of the drill string, including the BHA, in lb
W_{dpa} = adjusted weight of the drill pipe with tool joints in lb/ft

Step 1: Calculate the length of the drill collars in ft:

$$L_{DCbj} = \frac{(W_b + T_j)}{(BF(W_{DC}))} \tag{2.36}$$

where

L_{DCbj} = length of the drill collars below jars in ft
W_b = weight on bit in lb
T_j = jarring tension in lb
W_{DC} = weight of the drill collar in lb/ft

Step 2: Calculate the total length of the drill collars required in ft:

$$L_{DCT} = L_{DCbj} + L_{DCaj} \tag{2.37}$$

where

L_{DCT} = total length of the drill collars required in ft
L_{DCaj} = length of the drill collars above the jars in ft

Step 3: Calculate the length of the heavy-weight drill pipe in ft:

$$L_{HW} = \frac{W_j - [(L_{HWj})(W_{DC})(BF)]}{(W_{HW})(BF)} \tag{2.38}$$

where

L_{HW} = length of the heavy-weight drill pipe in ft (round off heavy weight to full joints)
L_{HWj} = length of heavy weight to provide jarring weight in lb
W_{HW} = weight of the heavy-weight drill pipe in lb

Step 4: Calculate the buoyed weight of the BHA:

$$L_{BHA} = L_{DCT} + L_{HW} \tag{2.39}$$

Step 5: Calculate the air weight of the BHA in lb:

$$W_{BHAa} = [W_{DC}(L_{DCT})] + [W_{HW}(L_{HW})] \tag{2.40}$$

where

W_{BHAa} = air weight of the BHA in lb

Step 6: Calculate the buoyed weight of the BHA in lb:

$$W_{BHA} = W_{BHAa}(BF) \tag{2.41}$$

where

W_{BHA} = buoyed weight of the BHA in lb

Step 7: Design Section 1 of the drill string above the BHA:

Use Eq. (2.28) to calculate the L_{max} based on the margin of overpull in ft
Use Eq. (2.35) to calculate the L_{psc} based on slip crushing in ft Select the shortest length to use in Section 1.

Step 8: Add the BHA and the drill pipe selected for Section 1 to find the buoyed weight needed to design the next section (see example).

Step 9: Repeat Steps 7 and 8 until the top section of the drill string has been selected.

Step 10: Prepare a summary of the BHA and Sections 1, 2, and 3 of the drill pipe.

Step 11: Prepare a check of the margin of overpull for each grade of drill pipe selected.

Step 12: Prepare a summary of the string design.

Example: Design a drill string to drill to 18,000 ft using:

Weight on bit	= 30,000 lb
Jarring tension	= 7000 lb
DC for jarring wt.	= 62 ft
Average length of the DC	= 31 ft
HW to complete jarring wt.	= 12,000 lb (less 2 DCs)
Mud weight	= 16.0 ppg (BF = 0.7557)
Slip length	= 16 in.
Coefficient of friction	= 0.08
Safety factor	= 10%

Step 1: Calculate the length of the drill collars for bit weight and jarring tension:

$$L_{DCbj} = \frac{(30,000 + 7000)}{(0.7557)(110)} = 445\,\text{ft}$$

Step 2: Calculate the total length of the drill collars:

$$L_{DCT} = 445 + 62 = 507\,\text{ft}$$

Step 3: Calculate the length of the heavy-weight drill pipe:

$$L_{HW} = \frac{12,000 - [(62)(110)(0.7557)]}{(49.77)(0.7557)} = 182.4 \cong 186\,\text{ft}(2\,\text{stands})$$

Step 4: Summary: BHA.

$$\text{Length} = 507 + 186 = 693\,\text{ft}$$

$$\textit{Air}\ \text{wt.} = [507\ (110)] + [186\ (49.77)] = 65,027\,\text{lb}$$

$$\text{Buoyed wt.} = 65,027\ (0.7557) = 49,141\,\text{lb}$$

Step 5: Calculate the length of the first section (#3) of the drill pipe:

$$L_{\text{max}} = \frac{(((394,440)(0.9)) - 100,000 - (49,141))}{(21.62)(0.7557)} = 12,600\,\text{ft}$$

$$L_{\text{psc}} = \left[\frac{\left(\frac{(394,440(0.9))}{1.42}\right) - (65,027(0.7557))}{(21.62)(0.7557)}\right] = 12,294\,\text{ft}$$

Select the smallest length; therefore, the design is limited by slip crushing.

Step 6: Summary: BHA and Section 1 (#3) of the drill pipe:

BHA wt.	= 49,141 lb
Grade X air wt.	= 12,294 (21.62) = 265,796 lb
Grade X buoyed wt.	= 265,796 (0.7557) = 200,862 lb
Total buoyed wt.	= 49,141 + 200,862 lb = 250,003 lb

Step 7: Calculate the length of the second section (#4) of the drill pipe:

$$L_{\text{max}} = \frac{(((435,960)(0.9)) - 100,000 - 250,003)}{(21.62)(0.7557)} = 2562\ \text{ft}$$

$$L_{\text{psc}} = \left[\frac{\left(\frac{(435,960*0.9) - 250,003}{1.42}\right)}{(21.88*0.7557)}\right] = 1591\ \text{ft}$$

Select the smallest length; therefore, the design is limited by slip crushing.

Step 8: Summary: BHA and Sections 1 (#3) and 2 (#4) of the drill pipe:

BHA wt.	= 49,141 lb
Grade X	= 200,862 lb
Grade G air wt.	= 1591 (21.88) = 34,811 lb
Grade G buoyed wt.	= 34,811 (0.7557) = 26,307 lb
Total buoyed wt.	= 49,141 + 200,862 + 26,307 = 276,310 lb

Step 9: Calculate the length of the third section (#5) of the drill pipe:

$$L_{\text{max}} = \frac{(((560,520*0.9)) - 100,000 - 276,310)}{(22.32*0.7557)} = 7598\ \text{ft}$$

$$L_{\text{psc}} = \left[\frac{\left(\frac{(560,520*0.9)-276,310}{1.42}\right)}{(22.32*0.7557)}\right] = 4681 \text{ ft}$$

Select the smallest length; therefore, the design is limited by slip crushing.

Step 10: Summary: BHA and Sections 1 (#3), 2 (#4), and 3 (#5) of the drill pipe.

Drill collars	= 507 ft
Heavy-weight DP	= 186 ft
Grade X DP	= 12,294 ft
Grade G DP	= 1591 ft
Subtotal	= 14,578 ft
Grade S DP length	= 18,000 − 14,578 = 3422 ft
Total drill string	= 18,000 ft
Total buoyed wt.	= 276,310 + [(3422)(22.32)(0.7557)] = 334,030 lb
Total depth possible	= 14,578 + 4681 = 19,259 ft

Step 11: Margin of overpull check:

$$\text{Grade X} = [(394,440)(0.90)] - 250,003 = 104,993 \text{ lb}$$

$$\text{Grade G} = [(435,960)(0.90)] - 276,310 = 116,054 \text{ lb}$$

$$\text{Grade S} = [(560,520)(0.90)] - 334,030 = 170,438 \text{ lb}$$

Step 12: Summary of design.

Item description	Length (ft)	Air wt. (lb)	Buoyed wt. (lb)	Accumulated wt. (lb)	MOP (lb)
DC: 7 × 2 13/16 × 110, NC50	507	55,770	42,145	42,145	
HW: 5 × 3 × 49.77, NC50	186	9257	6996	49,141	
BHA summary	693	65,027	49,141	49,141	
DP:					
5,19.5, Grade X, NC50	12,294	265,796	200,862	250,003	104,993
5,19.5, Grade G, NC50	1591	34,811	26,307	276,310	116,054
5,19.5, Grade S, NC50	3422	76,379	57,720	334,030	170,438
Total	18,000	442,013	334,030	334,030	104,993 limited

Ref: Murchison (1982).

Spreadsheet 2.4.8 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Design of the drill string.

2.5 Depth of a washout in a drill pipe

Method 1: Pump soft line or other plugging material down the drill pipe, and note how many strokes are required before the pump pressure increases. Use a moderate pump rate to prevent forcing the plugging material through the washed-out pipe.

$$D_{\text{pwol}} = \left(\frac{(C_r)(O_p)}{V_{pc}}\right) \tag{2.42}$$

where

D_{pwo1} = depth of pipe washout in ft
C_r = strokes required for the pump pressure to increase
O_p = pump output in bbl
V_{pc} = capacity of the drill pipe in bbl/ft

Example: Calculate the depth of washout in the drill pipe:

Drill pipe = 3½ × 2.764 in. (13.3 lb/ft)
DP capacity = 0.00740 bbl/ft
Pump output = 0.112 bbl/stk (5½ × 14 in. duplex at 90% efficiency)

Note: A pressure increase was noted after 303 stk.

$$D_{\text{pwol}} = \left(\frac{(303*0.112)}{0.00740}\right) = 4586 \text{ ft}$$

Method 2: Pump some material that will go through the washout, up the annulus, and over the shale shaker. This material must be of the type that can be easily observed as it comes across the shaker. Examples: carbide, uncooked rice, corn starch, glass or plastic beads, brightly colored paint, and so on. In nonaqueous fluids, use a red dye designed for use to identify cement spacers.

$$D_{\text{pwo2}} = \left(\frac{(C_r)(O_p)}{(V_{pc} + V_{acb})}\right) \tag{2.43}$$

where

D_{pwo2} = depth of pipe washout in ft
V_{acb} = capacity of the annulus between the open hole or casing in bbl/ft

Example: Calculate the depth of washout in the drill pipe:

Drill pipe	= 3½ × 2.764 in. (13.3 lb/ft)
Drill pipe capacity	= 0.00740 bbl/ft
Pump output	= 0.112 bbl/stk (5½ × 14 in. duplex at 90% efficiency)
Casing size	= 8⅝ × 7.921 in. (32 lb/ft)
Casing/DP annulus capacity	= 0.0491 bbl/ft

Note: The material pumped down the drill pipe came over the shaker after 2310 stk.

$$D_{\text{pwo2}} = \left(\frac{(2310)(0.112)}{(0.00740 + 0.0491)}\right) = 4579 \text{ ft}$$

Spreadsheet 2.5 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Depth of a washout in a DP.

2.6 Stuck pipe calculations

2.6.1 Calculate the number of days to quit fishing

$$F_q = \frac{(V_F + C_S)}{(F_d + C_{rd})} \tag{2.44}$$

where

F_q = maximum number of days to fish
V_F = value of fish
C_S = cost of side-tracking original hole
F_d = fishing daily cost (op-ex)
C_{rd} = daily rig cost

Example: Calculate the number of fishing days with the following data:

Daily rig cost	= $25,000
Value of fish	= $215,000
Cost of side-tracking	= $250,000
Fishing daily cost	= $20,000

$$F_q = \frac{(215,000 + 250,000)}{(20,000 + 25,000)} = 10.3 \text{ days}$$

Spreadsheet 2.6.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Days to quit fishing.

2.6.2 Determine the length of the free pipe in feet and the free point constant

Method 1: The depth at which the pipe is stuck and the number of feet of free pipe can be estimated using the data in the drill pipe stretch (Table 2.5) and is shown below with the following formula.

$$P_{f1} = \frac{(P_{dps})(K_{fpt})}{F_p} \tag{2.45}$$

where

P_{f1} = length of the free pipe in ft
P_{dps} = drill pipe stretch in in.
K_{fpt} = free point constant from Table 2.5
F_p = pull force in 1000 lb

Example: Calculate the length of the free drill pipe with the following data:

Drill pipe = 6.625 × 5.965 in. (S-135, 25.2 lb/ft)
Stretch = 20 in.
Pull force = 35,000 lb

Step 1: Determine the drill pipe stretch from Table 2.5:

$$K_{fpt} = 16,315.0$$

Step 2: Calculate the length of the free pipe:

$$P_n = \frac{(20*16,315.0)}{35} = 9323 \text{ ft}$$

Method 2: Calculate the free point constant (K_{fpc}). The free point constant can be calculated for any type of steel drill pipe if the outside diameter (OD, in.) and the inside diameter (ID, in.) are known:

Step 1: Calculate the cross-sectional area of the drill pipe wall in in.^{2}:

$$A_s = \left(D_p^2 - D_i^2\right)(0.7854) \tag{2.46}$$

where

A_s = cross-sectional area of the pipe wall in in.^{2}
D_p = outside diameter of the drill pipe in in.
D_i = inside diameter of the drill pipe in in.

TABLE 2.5 Drill pipe stretch table.

ID (in.)	Nominal weight (lb/ft)	ID (in.)	Wall area (in.^{2})	Stretch constant in 1000 lb/1000 ft	Free point constant
2⅜	4.85	1.995	1.304	0.30675	3260.0
	6.65	1.815	1.843	0.21704	4607.7
2⅞	6.85	2.241	1.812	0.22075	4530.0
	10.40	2.151	2.858	0.13996	7145.0
3½	9.50	2.992	2.590	0.15444	6475.0
	13.30	2.764	3.621	0.11047	9052.5
	15.50	2.602	4.304	0.09294	10,760.0
4	11.85	3.476	3.077	0.13000	7692.5
	14.00	3.340	3.805	0.10512	9512.5
4½	13.75	3.958	3.600	0.11111	9000.0
	16.60	3.826	4.407	0.09076	11,017.5
	18.10	3.754	4.836	0.08271	12,090.0
	20.00	3.640	5.498	0.07275	13,745.0
5	16.25	4.408	4.374	0.09145	10,935.0
	19.50	4.276	5.275	0.07583	13,187.5
5½	21.90	4.778	5.828	0.06863	14,570.0
	24.70	4.670	6.630	0.06033	16,575.0
6⅝	25.20	5.965	6.526	0.06129	16,315.0

Step 2: Calculate the free point constant for the drill pipe:

$$K_{\text{fpc}} = (A_{\text{S}})(2500) \tag{2.47}$$

where

K_{fpc} = calculated free point constant

Example 1: Calculate the free point constant with the following data:

$$\text{Drill pipe size and weight} = 6.625 \times 5.965\,\text{in.}, 25.2\,\text{lb/ft}$$

Step 1: Calculate the cross-sectional area:

$$A_{\text{s}} = \left(6.625^2 - 5.965^2\right)(0.7854) = 6.53\,\text{in.}^2$$

Step 2: Calculate the free point constant:

$$K_{\text{fpc}} = (6.53*2500) = 16,325.0$$

Example 2: Calculate the free point constant and the depth at which the pipe is stuck using the following data:

Tubing size and weight = 2.375 × 2.441 in., 6.5 lb/ft (Table 2.5)
Stretch = 25 in.
Pull force = 20,000 lb

Step 1: Calculate the cross-sectional area:

$$A_{\text{s}} = \left(2.375^2 - 1.815^2\right)(0.7854) = 1.843\,\text{in.}^2$$

Step 2: Calculate the free point constant:

$$K_{\text{fpc}} = (1.843*2500) = 4607.5$$

Step 3: Calculate the depth of the stuck pipe:

$$P_{\text{fl}} = \frac{(25*4607.5)}{20} = 5759\ \text{ft}$$

Method 3: This method of calculating the length of the free pipe does not use the free point constant listed in Table 2.5.

Step 1: Calculate the weight of the drill pipe tube without the tool joints in lb/ft:

$$W_{\text{dpt}} = 2.674\left(D_{\text{p}}^2 - D_{\text{i}}^2\right) \tag{2.48}$$

where

W_{dpt} = weight of the drill pipe tube in lb/ft

Step 2: Calculate the length of the free pipe in ft:

$$P_{\text{f2}} = \frac{(735,294)\left(P_{\text{dps}}\right)\left(W_{\text{dpt}}\right)}{F_{\text{pd}}} \tag{2.49}$$

where

P_{f2} = length of the free pipe in ft
P_{dps} = pipe stretch in in.
W_{dpt} = weight of the drill pipe tube in lb/ft (excluding tool joints)
F_{pd} = differential pull force in lb

Note: The weight of the drill pipe tube without the tool joints may be used if known instead of the calculated value.

Example: Calculate the length of the free pipe using the following data:

Drill pipe size	$= 5.0 \times 4.276$ in. (19.5 lb/ft)
Stretch of the pipe	$= 24$ in.
Stretch of the pipe	$= 24$ in.
Differential pull force to obtain stretch	$= 30{,}000$ lb

Step 1: Calculate the weight of the drill pipe tube:

$$W_{dpt} = (2.674)\left(5.0^2 - 4.276^2\right) = 17.958\,\text{lb/ft}$$

Step 2: Calculate the length of the free pipe:

$$P_{f2} = \frac{(735,294)(24)(17.958)}{30,000} = 10,564\,\text{ft}$$

Spreadsheet 2.6.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Length of the free pipe.

2.6.3 Stuck pipe overbalance guidelines

$$P_{OPG} = \left(\frac{1500}{K}\right) - ((\sin \measuredangle)(1000)) \tag{2.50}$$

where

P_{OPG} = overbalance pressure guideline in psi (stuck pipe risk may be more than 90% above this value)
K = mud type factor (0.75 for OBM or SBM; 1.0 for WBM)
$\measuredangle$ = angle of the hole in degrees (decimal value) (Table 2.6)

Example 1: Determine the overbalance guideline for reducing the risk of the stuck pipe with SBM in psi:

$$\textbf{Data}: \text{Hole angle} = 30^{\circ}$$

$$\text{Mud type} = \text{synthetic base mud (SBM)}$$

$$P_{OPG} = \left(\frac{1500}{0.75}\right) - ((0.50)(1000)) = 1500\ \text{psi}$$

Example 2: Determine the overbalance guideline for reducing the risk of the stuck pipe with WBM in psi:

$$\textbf{Data}: \text{Hole angle} = 30^{\circ}$$

$$\text{Mud type} = \text{water base mud (WBM)}$$

TABLE 2.6 Angle sin values.

Angle	Sin	Angle	Sin	Angle	Sin
5	0.087156	35	0.573576	65	0.906308
10	0.173648	40	0.642788	70	0.939693
15	0.258819	45	0.707107	75	0.965926
20	0.342020	50	0.766044	80	0.984808
25	0.422618	55	0.819152	85	0.996195
30	0.500000	60	0.866025	90	1.000000

$$P_{OPG} = \left(\frac{1500}{1.0}\right) - ((0.50)(1000)) = 1000 \text{ psi}$$

Spreadsheet 2.6.3 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Stuck pipe overbalance.

2.6.4 Force required to pull the differently stuck pipe free

$$F_{PF} = (W_C)(Z_P{*}12)(P_d)(C_f) \tag{2.51}$$

where

F_{PF} = force required to pull the stuck pipe free in lb
W_C = contact width of the stuck pipe in in.
Z_P = permeable formation length in ft
P_d = differential pressure in psi
C_f = coefficient of friction (OBM/SBM = 0.1, WBM = 0.25). If known, use the actual coefficient of friction values measured on the mud system.

Example: determine the force required to pull the pipe free with these conditions:

Data: Contact width of the stuck pipe = 3.0 in.
Permeable formation length = 85 ft
Differential pressure = 200 psi
Type of mud = WBM

$$F_{PF} = (3.0)(85.0{*}12)(200)(0.25) = 153{,}000 \text{ lb}$$

Spreadsheet 2.6.4 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Force required to pull the differentially stuck pipe free.

2.7 Calculations required for placing spotting pills in an open hole annulus

2.7.1 Amount of spotting fluid pill in barrels required to cover the stuck point of the drill string or casing and the number of pump strokes required to spot the pill

Step 1: Calculate the hole "washout" size in in.:

$$D_{\text{hwo}} = \left((D_{\text{bit}}) \left(\frac{H_{\text{wo}}}{100} \right) \right) + (D_{\text{bit}}) \tag{2.52}$$

where

D_{hwo} = diameter of hole "washout" in in.
D_{bit} = diameter of bit in in.
H_{wo} = hole "washout" factor in percentage

Step 2: Calculate the internal capacity of the drill pipe, heavy-weight drill pipe, and drill collars in bbl/ft:

$$C_{\text{p}} = \frac{\left(\text{ID}_{\text{p}}^2 \right)}{1029.4} \tag{2.53}$$

where

C_{p} = capacity of the drill pipe, heavy-weight drill pipe, or drill collars in bbl/ft
ID_{p} = internal diameter of the drill pipe, heavy-weight drill pipe, or drill collars in in.

Step 3: Calculate the annular capacity of the open hole of the drill pipe, heavy-weight drill pipe, and drill collars in bbl/ft

$$C_{\text{a}} = \frac{\left(D_{\text{hwo}}^2 - \text{OD}_{\text{p}}^2 \right)}{1029.4} \tag{2.54}$$

where

C_{a} = annular capacity in bbl/ft
D_{hwo} = diameter of the washed-out hole in in.
OD_{p} = outside diameter of DP, HWDP, or DCs in in.

Step 4: Calculate the volume of the spotting fluid pill required for the open hole annulus in bbl:

$$V_{\text{sfpa}} = (C_{\text{a}})(L_{\text{sp}}) \tag{2.55}$$

where

V_{sfpa} = volume of the spotting fluid pill across the fish in the open hole annulus in bbl
L_{sp} = length of the excess spotting fluid pill required in the open hole annulus in ft

Step 5: Calculate the total volume of the spotting fluid pill required in bbl:

$$V_{sfpt} = V_{sfpa} + V_{sfpds} \tag{2.56}$$

where

V_{sfpt} = total volume of the spotting fluid pill required in bbl
V_{sfpds} = predetermined volume of the spotting fluid pill to be left inside the drill string in bbl

Step 6: Calculate the drill string volume for each pipe section in bbl:

$$V_p = (C_p)(L_s) \tag{2.57}$$

where

V_p = volume of the drill pipe, heavy-weight drill pipe, or drill collar section in bbl
L_s = length of the drill pipe, heavy-weight drill pipe, or drill collar section in ft

Step 7: Calculate the height of the pill volume to be left in the drill string (this allows selected pill volume to be lubricated out of the string at a selected time interval, i.e., 1 bbl every 30 min).

(a) Calculate the volume of the pill left in the drill pipe:

$$V_{PDP} = (V_{PDS}) - (V_{DC}) - (V_{HWDP}) \tag{2.58}$$

where

V_{PDP} = volume of the pill in the drill pipe section in bbl
V_{PDS} = volume of the drill collars in bbl
V_{HWDP} = volume of the heavy-weight drill pipe in bbl

(b) Calculate the depth of the top of the pill in the drill pipe in ft

$$P_{HDP} = (L_{DP}) - \left(\frac{V_{PDP}}{C_{DP}}\right) \tag{2.59}$$

where

P_{HDP} = depth of the top of the pill in the drill pipe in ft

L_{DP} = length of the drill pipe in ft
C_{DP} = capacity of the drill pipe in bbl/ft

Step 8: Calculate the total strokes required to spot the pill across the fish:

(a) Calculate the strokes required to pump the pill into the drill string:

$$S_{sfs} = \frac{(V_{pT})}{Q} + S_{SS} \tag{2.60}$$

where

S_{sfs} = strokes required to pump the pill into the drill string
Q = pump output in bbl/stk
S_{SS} = strokes required for the surface system

(b) Strokes required to spot the pill across the fish:

$$S_{sp} = \frac{(P_{HDP})(C_{DP})}{Q} + S_{sfs} \tag{2.61}$$

where

S_{sp} = strokes to spot the pill across the fish

Example: The drill collars are differentially stuck. Use the following data to spot a base oil pill around the drill collars plus 200 ft (volume optional) around the heavy-weight drill pipe and leave 30 barrels of pill in the drill string:

Well depth (MD)	= 10,240 ft
Hole diameter	= 8½ in.
Washout factor	= 20%
Casing size	= 9⅝ × 8.755 in. (N-80, 43.5 lb/ft)
Casing length	= 6000 ft
Drill pipe	= 5.0 × 4.276 in. (19.5 lb/ft)
DP length	= 9400 ft
HWDP	= 5.0 × 3.0 in.
HWDP length	= 240 ft
Drill collars	= 6½ × 2½ in.
DC length	= 600 ft
Pump output	= 0.117 bbl/stk
Surface system	= 80 stk (strokes required to pump the pill to the drill string).

Step 1: Calculate the hole "washout" size in in.:

$$D_{hwo} = ((8.5)(0.20)) + 8.5 = 10.2 \text{ in.}$$

Step 2: Calculate the internal capacity of the drill pipe, heavy-weight drill pipe, and drill collars:

(a) Capacity of the drill pipe:

$$C_{\text{dp}} = \frac{(4.276^2)}{1029.4} = 0.01776 \text{ bbl/ft}$$

(b) Capacity of the heavy-weight drill pipe:

$$C_{\text{hwdp}} = \frac{(3^2)}{1029.4} = 0.00874 \text{ bbl/ft}$$

(c) Capacity of the drill collars:

$$C_{\text{dc}} = \frac{(2.5^2)}{1029.4} = 0.00607 \text{ bbl/ft}$$

Step 3: Calculate the annular capacity of the open hole of the drill pipe, heavy-weight drill pipe, and drill collars in bbl/ft

(a) Annular capacity around the drill pipe in the open hole:

$$C_{\text{adp}} = \frac{(10.2^2 - 5.0^2)}{1029.4} = 0.0768 \text{ bbl/ft}$$

(b) Annular capacity around the heavy-weight drill pipe in the open hole:

$$C_{\text{ahwdp}} = \frac{(10.2^2 - 5.0^2)}{1029.4} = 0.0768 \text{ bbl/ft}$$

(c) Annular capacity around the drill collars in the open hole:

$$C_{\text{adc}} = \frac{(10.2^2 - 6.5^2)}{1029.4} = 0.0600 \text{ bbl/ft}$$

Step 4: Calculate the volume of the spotting fluid pill required for the open hole annulus in bbl:

(a) Volume opposite the drill collars:

$$V_{\text{PDC}} = (0.0600)(600) = 36.0 \text{ bbl}$$

where

V_{PDC} = volume of the pill across the drill collars in bbl.

(b) Volume opposite the heavy-weight drill pipe:

$$V_{\text{PHWDP}} = (0.0768)(200) = 15.4\,\text{bbl}$$

where

V_{PHWDP} = volume of the pill across the predetermined length of HWDP in bbl.

(c) Total pill volume required in the open hole annulus in bbl:

$$V_{\text{OH}} = 36.0 + 15.4 = 51.4\,\text{bbl}$$

where

V_{OH} = total volume of the pill in the open hole annulus in bbl.

Step 5: Calculate the total volume required for the spotting fluid pill:

$$V_{\text{pT}} = 51.4 + 30 = 81.4\,\text{bbl}$$

Step 6: Calculate the drill string volume for each pipe section in bbl:

(a) Drill pipe capacity in bbl:

$$V_{\text{dp}} = (0.01776)(9400) = 166.9\,\text{bbl}$$

(b) Heavy-weight drill pipe capacity in bbl:

$$V_{\text{hwdp}} = (0.00874)(240) = 2.1\,\text{bbl}$$

(c) Drill collar capacity in bbl:

$$V_{\text{dc}} = (0.00607)(600) = 3.6\,\text{bbl}$$

Step 7: Calculate the height of the pill volume to be left in the drill string:**(a)** Calculate the volume of the pill left in the drill pipe:

$$(30.0) - (3.6) - (2.1) = 24.3\,\text{bbl}$$

(b) Calculate the depth of the top of the pill in the drill pipe in ft

$$(9400) - \left(\frac{24.3}{0.01776}\right) = 8031.7\ \text{ft}$$

Step 8: Calculate the total strokes required to spot the pill across the fish:**(a)** Calculate the strokes required to pump the pill into the drill string:

$$\frac{(81.4)}{0.117} + 80 = 776 \text{ stks}$$

(b) Strokes required to spot the pill across the fish:

$$\frac{(8031.7)(0.01776)}{0.117} + 776 = 1995 \text{ stks}$$

Spreadsheet 2.7.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Amount of spotting pill required.

2.7.2 Determination of the length of an unweighted spotting fluid pill that will balance formation pressure in the annulus in ft

Step 1: Calculate the difference in pressure gradient between the mud weight and the spotting fluid pill in psi/ft:

$$G_{sfp} = (W_m - W_{sfp})(0.052) \tag{2.62}$$

where

G_{sfp} = difference in pressure gradient in psi/ft
W_m = weight of mud in lb/gal
W_{sfp} = weight of the spotting fluid pill in lb/gal

Step 2: Calculate the length of an unweighted spotting fluid pill that will balance formation pressure in the annulus:

$$L_{sfp} = \frac{OB}{G_{sfp}} \tag{2.63}$$

where

L_{sfp} = length of the unweighted spotting fluid pill in ft
OB = overbalance pressure needed to control pore pressure in psi

Example: Use the following data to determine the length of an unweighted spotting fluid pill that will balance formation pressure in the annulus:

Mud weight = 11.2 ppg (without cuttings)
Weight of spotting fluid pill = 6.7 ppg (diesel = 7.0 ppg/synthetic = 6.7 ppg)
Amount of overbalance = 250.0 psi

Step 1: Calculate the difference in pressure gradient in psi/ft:

$$G_{sfp} = (11.2 - 6.7)(0.052) = 0.234\,\text{psi/ft}$$

Step 2: Calculate the length of an unweighted spotting fluid pill that will balance formation pressure in the annulus:

$$L_{sfp} = \frac{250}{0.234} = 1068\,\text{ft}$$

Therefore: Less than 1068 ft of an unweighted spotting fluid pill should be used to maintain a safe balance of the formation pore pressure and prevent an influx that would cause a kick or blowout.

Spreadsheet 2.7.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Length of the unweighted spotting pill required.

2.8 Line size for a low pressure system

Determine the line size for a low pressure system.

(a) Calculate the fluid velocity in ft/s:

$$V_f = \frac{(0.4087)(Q)}{\left(\text{ID}_p^2\right)} \tag{2.64}$$

where

V_f = fluid velocity in ft/s
Q = centrifugal pump output in gpm
ID_p = internal diameter of the pipe in in.

The ideal range of fluid velocity is between 5 and 10 ft/s. This is based on achieving 300–600 ft/min velocity inside the pipe. Experience has shown that solid settlement in the pipe is minimized above a velocity of 300 ft/min.

Example: Calculate the fluid velocity in an 8.0″ pipe with 1100 gal of flow:

$$V_f = \frac{(0.4087)(1100)}{\left(8.0^2\right)} = 7.0\ \text{ft/s}.$$

Spreadsheet 2.8 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Line size for a low pressure system.

Appendix: Supplementary material

Supplementary material related to this chapter can be found online at https://www.elsevier.com/books-and-journals/book-companion/9780323905497.

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CHAPTER

3

Pressure control

3.1 Normal kill sheet

A normal kill sheet example form designed for a land or offshore well location can be accessed by selecting the "Spreadsheet 3.1" box shown below. This is a suggested form to record prerecorded data, well geometry and casing/hole and drill string volumes required for calculating information that will be used to control and remove a formation fluid influx from the wellbore.

Many operators, drilling contractors, and consultants have their own form for land or deep water wells that can be utilized to control kicks. The IADC also provides a suggested form that may be used to solve the influx problem. The same equations detailed in this section may be used in their kill sheet to calculate the information that will be needed to control the influx.

Spreadsheet 3.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Normal kill sheet.

(A) Determine the maximum allowable shut-in casing pressure (MASICP):

$$\text{MASICP} = \left((\text{LOT} - \text{MW}_c)(0.052)(\text{Csg}_d)\right) \tag{3.1}$$

where

MASICP = maximum allowable shut-in casing pressure in psi
MW_c = current mud weight in ppg
Csg_d = casing depth in ft
LOT = leak-off test in psi

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Example: Using data listed in the prerecorded data:

$$\text{MASICP} = ((14.1 - 9.6)(0.052)(4000)) = 936 \text{ psi}$$

Spreadsheet 3.1.A in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 MASCIP.

(B) Determine the capacity of the annular sections with the drill string in the wellbore in bbl/ft:

$$C_{\text{AwDS}} = \left(\frac{\left(D_{\text{h}}^2\right) - \left(D_{\text{p}}^2\right)}{1029.4}\right) \tag{3.2}$$

$$V_{\text{AwDS}} = \left(C_{\text{AwDS}}\right)\left(L_{\text{p}}\right) \tag{3.3}$$

where

C_{AwDS} = capacity of annulus with drill string in bbl/ft
V_{AwDS} = volume of annulus with drill string in bbl
D_{h} = diameter of casing or open hole in in.
D_{p} = diameter of drill string pipe in in.
L_{p} = length of pipe in ft

Example: Calculate the volume and capacity of the casing with the drill string pipe in bbl/ft and bbl:

$$C_{\text{AwDS}} = \left(\frac{\left(10.05^2\right) - \left(5.0^2\right)}{1029.4}\right) = 0.0738 \text{ bbl/ft}$$

$$V_{AwDS} = (0.0738)(4000) = 295.2 \text{ bbl}$$

Spreadsheet 3.1.B in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Capacity of annulus.

(C) Determine the capacity of the drill string in bbl/ft:

$$C_{\text{p}} = \left(\frac{\text{ID}_{\text{p}}^2}{1029.4}\right) \tag{3.4}$$

$$V_{\text{p}} = \left(C_{\text{p}}\right)\left(L_{\text{p}}\right) \tag{3.5}$$

where

C_{p} = capacity of drill string pipe in bbl/ft
ID_{p} = internal diameter of drill string pipe in in.
V_{p} = volume of drill string pipe in bbl

Example: Calculate the volume and capacity of the drill string pipe in bbl/ft and bbl:

$$C_p = \left(\frac{4.276^2}{1029.4}\right) = 0.0178 \text{ bbl/ft}$$

$$V_p = (0.0178)(9925) = 176.7 \text{ bbl}$$

Spreadsheet 3.1.C in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Capacity of drill string.

(D) Determine the number of strokes from the pump to the bit down the drill string:

$$S_{PB} = \left(\frac{V_{DS}}{P_O}\right) \tag{3.6}$$

where

S_{PB} = strokes from pump to the bit
V_{DS} = volume of the drill string in bbl
P_O = pump output in bbl/stk

Example: Calculate the number of strokes from the pump to the bit:

$$S_{PB} = \left(\frac{191.9}{0.136}\right) = 1411 \text{ stks}$$

Determine the strokes for the Bit to Casing Shoe and Casing Shoe to Surface sections using Eq. (3.7).

Spreadsheet 3.1.D in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Strokes from pump to bit.

(E) Determine the kill data with the following equations:

$$\text{KMW} = \left(\frac{((\text{SIDPP})(19.2))}{\text{TVD}}\right) + \text{MW}_c \tag{3.7}$$

$$\text{ICP} = (\text{SIDPP} + \text{KRP}_1) \tag{3.8}$$

$$\text{FCP} = \left(\frac{(\text{KMW})(\text{KRP}_1)}{\text{MW}_c}\right) \tag{3.9}$$

where

KMW = kill mud weight in ppg
SIDPP = shut-in drill pipe pressure in psi
TVD = total vertical depth in ft
MW_c = current mud weight in ppg

KRP_1 = kill rate pressure of pump #1 in psi
ICP = initial circulating pressure in psi

Example: Calculate the kill data:

$$KMW = \left(\frac{((480)(19.2))}{10{,}525}\right) + 9.6 = 10.47 \text{ ppg} \cong 10.5 \text{ ppg}$$

$$ICP = (480 + 1000) = 1480 \text{ psi}$$

$$FCP = \left(\frac{(10.5)(1000)}{9.6}\right) = 1094 \text{ psi}$$

Spreadsheet 3.1.E in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Kill data.

3.2 Pressure chart: Prepare a chart with pressure and strokes

3.2.1 Use this technique for rigs that have digital pressure gauges

Step 1: Calculate the number of strokes per line in the pressure chart:

$$S_{BL} = \left(\frac{S_{PB}}{10}\right) \tag{3.10}$$

where

S_{BL} = stokes to the bit per line in the pressure chart
S_{PB} = strokes from the pump to the bit

Note: You may select more than 10 lines to divide into the volume of the drill string if desired.

Step 2: Calculate the pressure decrease per line in the pressure chart:

$$P_{dL} = \left(\frac{(ICP - FCP)}{10}\right) \tag{3.11}$$

where

P_{dL} = pressure decrease per line in the pressure chart
ICP = initial circulating pressure in psi
FCP = final circulating pressure in psi

Example: Prepare pressure chart with the following data:

Strokes to the bit = 1441 stk
ICP = 1480 psi
FCP = 1094 psi

Step 1: Calculate the number of strokes per line in the pressure chart:

$$S_{BL} = \left(\frac{1441}{10}\right) = 144.1 \text{ strokes}$$

Step 2: Calculate the pressure decrease per line in the pressure chart:

$$P_{dL} = \left(\frac{(1480 - 1094)}{10}\right) = 38.6 \cong 39.0 \text{ psi/line}$$

Pressure Chart

	Strokes	Pressure	
	0	1480	
144.1 rounded to	144	1441	= –39.0
+ 144 =	288	1402	= –39.0
+ 144 =	432	1363	= –39.0
+ 144 =	576	1324	= –39.0
+ 144 =	720	1285	= –39.0
+ 144 =	864	1246	= –39.0
+ 144 =	1,008	1207	= –39.0
+ 144 =	1,152	1168	= –39.0
+ 144 =	1,296	1129	= –39.0
+ 144 =	1,440	1090	= –39.0

Spreadsheet 3.2.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Pressure chart with digital gauge.

3.2.2 Use this technique for rigs that have analog pressure gauges

Step 1: Calculate the amount of psi per stroke for the pressure chart:

$$T_{SP} = (\text{ICP} - P_S) \tag{3.12}$$

where

T_{SP} = table starting pressure in psi
P_S = pressure adjustment to reach 50 psi increments

Step 2: Calculate the amount of pressure per pump stroke:

$$P_{stk} = \frac{(\text{ICP} - \text{FCP})}{S_{TB}} \tag{3.13}$$

where

P_{stk} = pressure per pump stroke in psi
ICP = initial circulating pressure in psi
FCP = final circulating pressure in psi
S_{TB} = total strokes from the pump to the bit

Example: Using the kill sheet just completed, adjust the pressure chart to read in increments that are easy to read on pressure gauges. (Generally 50 psi.)

Data:

$$\begin{aligned}\text{Initial circulating pressure} &= 1480\,\text{psi}\\ \text{Final circulating pressure} &= 1094\,\text{psi}\\ \text{Strokes to bit} &= 1441\,\text{stk}\end{aligned}$$

$$P_{\text{stk}} = \frac{(1480 - 1094)}{1441} = 0.2679\ \text{psi/stk}$$

Step 3: Determine how many strokes will be required to decrease the pressure from 1480 to 1450 psi.

$$1480 - 1450 = 30\,\text{psi}$$

$$\frac{30}{0.2679} = 111.9\ \text{stks} \cong 112\ \text{stks}$$

How many strokes will be required to decrease the pressure by the 50 psi increments?

$$\frac{50}{0.2679} = 186.6\ \text{stks} \cong 187\ \text{stks}$$

The pressure side of the chart will appear as follows:

Pressure chart

Strokes	Pressure
0	1480
112	1450
299	1400
486	1350
673	1300
860	1250
1047	1200
1234	1150
1421	1100
1608	1050

Spreadsheet 3.2.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Pressure chart with analog gauge.

3.3 Kill sheet with a tapered string

When a kick is taken with a *tapered drill string* in the hole, interim pressures should be calculated for (a) the length of large drill pipe (DPL) and (b) the length of large drill pipe plus the length of small drill pipe.

Step 1: Determine the strokes to pump down the drill string:

$$S_{DS} = \frac{(L_{DS})(C_{DS})}{P_O} \tag{3.14}$$

where

S_{DS} = strokes to pump down the drill string
L_{DS} = length of pipe section in ft
C_{DS} = capacity of the pipe section in bbl/ft
P_O = pump output in bbl/stk

Step 2: Determine the pressure for the section of the pipe:

$$P_{dsps} = \text{ICP} - \left[\left(\frac{L_{DP}}{L_{DS}}\right)(\text{ICP} - \text{FCP})\right] \tag{3.15}$$

where

P_{dsps} = pressure for drill string pipe section in psi
ICP = initial circulating pressure in psi
FCP = final circulating pressure in psi

Step 3: Plot data on graph paper as shown in Fig. 3.1.

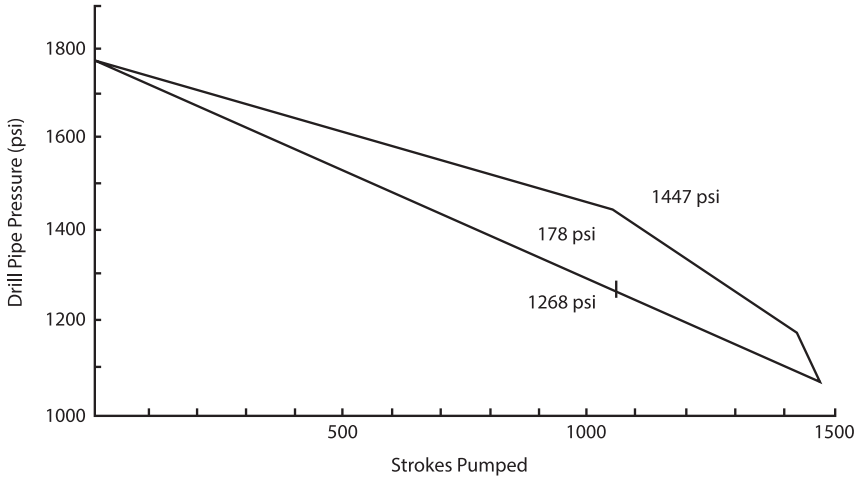


FIG. 3.1 Kill pressure graph for a tapered string.

Example: Plot the kill data for the following tapered string:

ICP	= 1780 psi
FCP	= 1067 psi
Section #1 Drill pipe	= 5.0 × 4.276 in. (19.5 lb/ft)
Section #1 DP capacity	= 0.0178 bbl/ft
Section #1 DP length	= 7000 ft
Section #2 Drill pipe	= 3½ × 2.764 in. (13.3 lb/ft)
Section #2 DP capacity	= 0.0073 bbl/ft
Section #2 Length	= 6000 ft
Section #3 Drill collars	= 4½ × 1½ in. (48.1 lb/ft)
Section #3 DC capacity	= 0.0022 bbl/ft
Section #3 DC length	= 2000 ft
Pump output	= 0.117 bbl/stk

Step 1: Determine the strokes to pump down the drill string:

(a) Section #1 DP:

$$S_{DS} = \frac{(7000)(0.0178)}{0.117} = 1065 \text{ stks}$$

(b) Section #2 DP:

$$S_{DS} = \frac{(6000)(0.0073)}{0.117} = 374 \text{ stks}$$

(c) Section #3 DC:

$$S_{DS} = \frac{(2000)(0.0022)}{0.117} = 38 \text{ stks}$$

Total strokes for tapered string = 1065 + 374 + 38 = 1477 stk.

Step 2: Determine the pressure for the section of the pipe:

(a) Section #1 DP:

$$P_{dsps} = 1780 - \left[\left(\frac{7000}{15{,}000}\right)(1780 - 1067)\right] = 1447 \text{ psi at } 1063 \text{ stks}$$

(b) Section #1 DP plus Section #2 DP:

$$P_{dsps} = 1780 - \left[\left(\frac{13{,}000}{15{,}000}\right)(1780 - 1067)\right] = 1162 \text{ psi at } 1444 \text{ stks}$$

Step 3: Plot data on graph paper.

Note: After pumping 1062 strokes, if a straight line would have been plotted, the well would have been under-balanced by 178 psi.

Spreadsheet 3.3 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Kill sheet with a tapered string.

3.4 Kill sheet for a highly deviated well

When a kick is taken in a highly deviated well, the circulating pressure can be excessive when the kill weight mud gets to the kick-off point (KOP). If the pressure is excessive, the pressure schedule should be divided into two sections: (1) from surface to KOP, and (2) from KOP to TD. The following calculations are used:

Step 1: Determine strokes from surface to KOP:

$$S_{\mathrm{KOP}} = \frac{(C_{\mathrm{dp}})(\mathrm{MD}_{\mathrm{KOP}})}{O_{\mathrm{p}}} \tag{3.16}$$

where

S_{KOP} = strokes to pump to KOP
C_{dp} = capacity of drill pipe in bbl/ft
$\mathrm{MD}_{\mathrm{KOP}}$ = length of drill pipe to KOP in ft
O_{p} = pump output in bbl/stk

Step 2: Determine strokes from KOP to TD:

$$S_{\mathrm{TD}} = \frac{(C_{\mathrm{dp}})(L_{\mathrm{MD}})}{O_{\mathrm{p}}} \tag{3.17}$$

where

S_{TD} = strokes from KOP to measured depth
L_{MD} = length of drill pipe to measured depth in ft

Step 3: Determine the total strokes from surface to the bit:

$$S_{\mathrm{B}} = S_{\mathrm{KOP}} + S_{\mathrm{TD}} \tag{3.18}$$

where

S_{B} = strokes from the surface to the bit

Step 4: Determine the kill weight mud:

$$\mathrm{KWM} = \left(\frac{\mathrm{SIDPP}}{(0.052)(\mathrm{TVD}_{\mathrm{TD}})}\right) + \mathrm{MW} \tag{3.19}$$

where

KWM = kill weight mud in ppg
SIDPP = shut-in drill pipe pressure in psi
$\mathrm{TVD}_{\mathrm{TD}}$ = total vertical depth in ft
MW = current mud weight in ppg

Step 5: Determine the initial circulating pressure:

$$\mathrm{ICP} = \mathrm{SIDPP} + \mathrm{KRP} \tag{3.20}$$

where

ICP = initial circulating pressure in psi
KRP = kill rate pressure in psi

Step 6: Determine the final circulating pressure:

$$\text{FCP} = \frac{(\text{KWM})(\text{KRP})}{\text{OMW}} \tag{3.21}$$

where

FCP = final circulating pressure in psi
OMW = old mud weight in ppg

Step 7: Determine the hydrostatic pressure increase from surface to the KOP:

$$\text{HP}_{\text{KOP}} = (\text{KWM} - \text{OMW})(0.052)(\text{MD}_{\text{KOP}}) \tag{3.22}$$

where

HP_{KOP} = hydrostatic pressure increase from surface to the KOP in psi
MD_{KOP} = total vertical depth at the KOP in ft

Step 8: Determine the friction pressure increase to KOP:

$$\text{FP}_{\text{KOP}} = (\text{FCP} - \text{KRP})\left(\frac{\text{MD}_{\text{KOP}}}{\text{MD}_{\text{TD}}}\right) \tag{3.23}$$

where

FP_{KOP} = friction pressure increase to KOP in psi
MD_{TD} = measured depth at TD in ft

Step 9: Circulating pressure when KWM gets to KOP:

$$\text{CP} = (\text{ICP} - \text{HP}_{\text{KOP}}) + \text{FP}_{\text{KOP}} \tag{3.24}$$

where

CP = circulating pressure when KWM gets to KOP in psi

Note: At this point, compare this circulating pressure to the value obtained when using a regular kill sheet.

Example:

Original mud weight (OMW)	= 9.6 ppg
Measured depth (MD)	= 15,000 ft
Measured depth @ KOP	= 5000 ft
True vertical depth @ KOP	= 5000 ft
Kill rate pressure (KRP) @ 30 spm	= 600 psi
Pump output	= 0.136 bbl/stk
Drill pipe capacity	= 0.01776 bbl/ft
Shut-in drill pipe pressure (SIDPP)	= 800 psi
True vertical depth (TVD)	= 10,000 ft

Step 1: Strokes from surface to KOP:

$$S_{\text{KOP}} = \frac{(0.0178)(5000)}{0.136} = 653 \text{ stks}$$

Step 2: Determine strokes from KOP to TD:

$$S_{\text{TD}} = \frac{(0.0178)(10{,}000)}{0.136} = 1306 \text{ stk}$$

Step 3: Determine the total strokes from the surface to the bit:

$$S_B = 653 + 1309 = 1959 \text{ stk}$$

Step 4: Determine the kill weight mud:

$$\text{KWM} = \left(\frac{800}{(0.052)(10{,}000)}\right) + 9.6 = 11.1 \text{ ppg}$$

Step 5: Determine the initial circulating pressure (ICP):

$$\text{ICP} = 800 + 600 = 1400 \text{ psi}$$

Step 6: Determine the final circulating pressure:

$$\text{FCP} = \frac{(11.1)(600)}{9.6} = 694 \text{ psi}$$

Step 7: Determine the hydrostatic pressure increase from surface to the KOP:

$$\text{HP}_{\text{KOP}} = (11.1 - 9.6)(0.052)(5000) = 390 \text{ psi}$$

Step 8: Determine the friction pressure increase to KOP:

$$\text{FP}_{\text{KOP}} = (694 - 600)\left(\frac{5000}{15{,}000}\right) = 31 \text{ psi}$$

Step 9: Circulating pressure when KWM gets to KOP:

$$\text{CP} = (1400 - 390) + 31 = 1041 \text{ psi}$$

Compare this circulating pressure to the value obtained when using a regular kill sheet:

(a) Calculate the psi/stk:

$$P_{\text{stk}} = \frac{(1400 - 694)}{1959} = 0.36 \text{ psi/stk}$$

(b) Calculate the pressure drop to the bit in psi:

$$(0.36)(653) = 235 \text{ psi}$$

(c) Calculate circulating pressure when KWM gets to KOP:

$$1400 - 235 = 1165 \text{ psi}$$

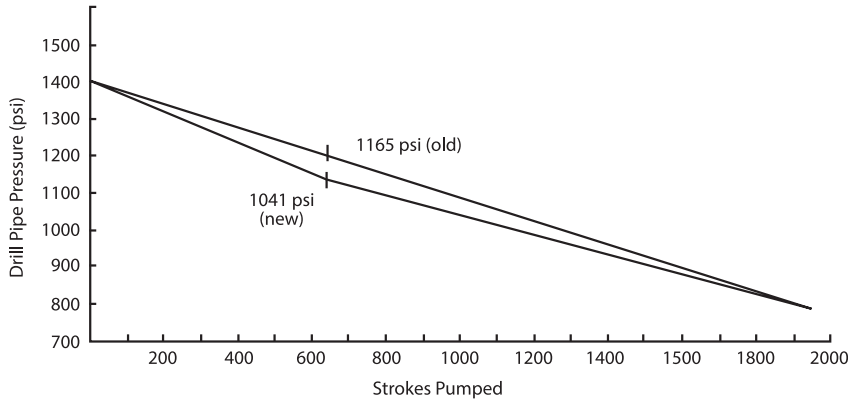


FIG. 3.2 Kill pressure graph for a deviated well.

Using a regular kill sheet, the circulating drill pipe pressure would be 1165 psi. The adjusted pressure chart would have 1041 psi on the drill pipe gauge. This represents 124 psi difference in pressure, which would also be observed on the annulus (casing) side. If the difference in pressure at the KOP is 100 psi or greater, then the adjusted pressure chart should be used to minimize the chances of losing circulation.

The chart (Fig. 3.2) below graphically illustrates the difference.

Spreadsheet 3.4 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Kill sheet for a deviated well.

3.5 Maximum anticipated surface pressure

Two methods are commonly used to determine maximum anticipated surface pressure:

Method 1: Use when assuming the maximum formation pressure is from TD:

Step 1: Determine maximum formation pressure:

$$P_{FM} = (MW_M + MW_{SF})(0.052)(TVD) \tag{3.25}$$

where

P_{FM} = maximum formation pressure in psi
MW_M = maximum mud weight in ppg
MW_{SF} = mud weight safety factor in ppg
TVD = total vertical depth in ft

Step 2: Assuming 100% of the mud is blown out of the hole, determine the hydrostatic pressure in the wellbore:

Note: 70% to 80% of mud being blown out is sometimes used instead of 100%.

$$\mathrm{HP_{gas}} = \left(G_{\mathrm{gas}}\right)(\mathrm{TVD}) \tag{3.26}$$

where

$\mathrm{HP_{gas}}$ = hydrostatic pressure of a gas column in psi
G_{gas} = gas gradient in psi/ft

Step 3: Determine maximum anticipated surface pressure (MASP):

$$\mathrm{MASP} = P_{\mathrm{FM}} - \mathrm{HP_{gas}} \tag{3.27}$$

where

MASP = maximum anticipated surface pressure in psi

Example:

Proposed total vertical depth = 12,000 ft
Maximum mud weight to be used in drilling the well = 12.0 ppg
Mud weight safety factor = 4.0 ppg
Gas gradient = 0.12 psi/ft
Assume that 100% of the mud is blown out of well.

Step 1: Determine maximum formation pressure:

$$P_{\mathrm{FM}} = (12.0 + 4.0)(0.052)(12{,}000) = 9984 \text{ psi}$$

Step 2: Determine the hydrostatic pressure of the gas in the evacuated wellbore:

$$\mathrm{HP_{gas}} = (0.12)(12{,}000) = 1440 \text{ psi}$$

Step 3: Determine maximum anticipated surface pressure (MASP):

$$\mathrm{MASP} = 9984 - 1440 = 8544 \text{ psi}$$

Method 2: Use when assuming the maximum pressure in the wellbore is attained when the formation at the shoe fractures:

Step 1: Determine fracture pressure in psi:

$$\mathrm{FP} = (\mathrm{FG} + \mathrm{MW_{SF}})(0.052)\left(\mathrm{TVD_{csg}}\right) \tag{3.28}$$

where

FP = fracture pressure of the formation at the casing shoe in psi
FG = fracture gradient at the casing shoe in ppg
$\mathrm{TVD_{csg}}$ = total vertical depth at the casing shoe in ft

Note: A safety factor is added to ensure the formation fractures before BOP pressure rating is exceeded.

Step 2: Determine the hydrostatic pressure of gas in the wellbore in psi:

$$\mathrm{HP_{gas}} = (G_{\mathrm{gas}})(\mathrm{TVD}) \tag{3.29}$$

Step 3: Determine the maximum anticipated surface pressure (MASP) in psi:

$$\mathrm{MASP} = P_{\mathrm{FM}} - \mathrm{HP_{gas}} \tag{3.30}$$

Example:

Proposed casing setting total vertical depth = 4000 ft
Estimated fracture gradient = 14.2 ppg
Mud weight safety factor = 1.0 ppg
Gas gradient = 0.12 psi/ft
Assume 100% of mud is blown out of the hole.

Step 1: Determine fracture pressure in psi:

$$\mathrm{FP} = (14.2 + 1.0)(0.052)(4000) = 3162 \text{ psi}$$

Step 2: Determine the hydrostatic pressure of gas in the wellbore in psi:

$$\mathrm{HP_{gas}} = (0.12)(4000) = 480 \text{ psi}$$

Step 3: Determine the maximum anticipated surface pressure (MASP) in psi:

$$\mathrm{MASP} = 3162 - 480 = 2682 \text{ psi}$$

Spreadsheet 3.5 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Maximum surface pressure.

3.6 Trip margin (TM)

This represents the mud weight required to prevent additional formation fluid from entering the wellbore when tripping the drill string out of the hole.

$$\mathrm{TM} = \frac{\overleftarrow{\mathrm{YP}}}{11.7(D_{\mathrm{h}} - D_{\mathrm{p}})} \tag{3.31}$$

where

TM = trip margin in ppg
YP = yield point in lb/100 ft^2
D_{h} = hole diameter in in.
D_{p} = drill pipe OD in in.

Example: Calculate the trip margin with the following data:

$$\begin{aligned}\text{Mud weight} &= 12.0\,\text{ppg}\\ \text{Yield point} &= 10\,\text{lb}/100\,\text{ft}^2\\ D_\text{h} &= 8.5\,\text{in.}\\ D_\text{p} &= 4.5\,\text{in.}\end{aligned}$$

$$\text{TM} = \frac{10}{11.7(8.5-4.5)} = 0.2\ \text{ppg}$$

$$\text{New mud weight} = 12.0 + 0.2 = 12.2\,\text{ppg}$$

Spreadsheet 3.6 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Trip margin.

3.7 Sizing the diverter line

Determine diverter line inside diameter in inches, equal to the area between the inside diameter of the casing and the outside diameter of drill pipe in use:

$$D_\text{DL} = \sqrt{\left(D_\text{h}^2 - D_\text{p}^2\right)} \tag{3.32}$$

where

D_DL = diameter of diverter line in in.
D_h = diameter of hole or casing in in.
D_p = diameter of pipe in in.

Example:

$$\begin{aligned}\text{Casing} &= 13\tfrac{3}{8}\ \text{in., J-55, 6 L lb/ft (ID} = 12.515\,\text{in.)}\\ \text{Drill pipe} &= 5.0\,\text{in., 19.5 lb/ft}\end{aligned}$$

$$D_\text{DL} = \sqrt{\left(12.515^2 - 5.0^2\right)} = 11.47 \cong 11.5\ \text{in.}$$

Spreadsheet 3.7 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Sizing the diverter line.

3.8 Fracture gradient (FG)

3.8.1 Fracture gradient determination—Surface application

Method 1: Matthews and Kelly method.
This technique assumes the overburden gradient is 1.0 psi/ft.

(A) Graphical method

$$FG_{MK} = \left(\frac{Fp_F}{D_{TVD}}\right) + K_i\left(\frac{\sigma}{D_{TVD}}\right) \tag{3.33}$$

where

FG_{MK} = fracture gradient by Matthews and Kelly method in psi/ft
Pp_F = formation pore pressure in psi
K_i = matrix stress coefficient from Fig. 3.3 and is dimensionless

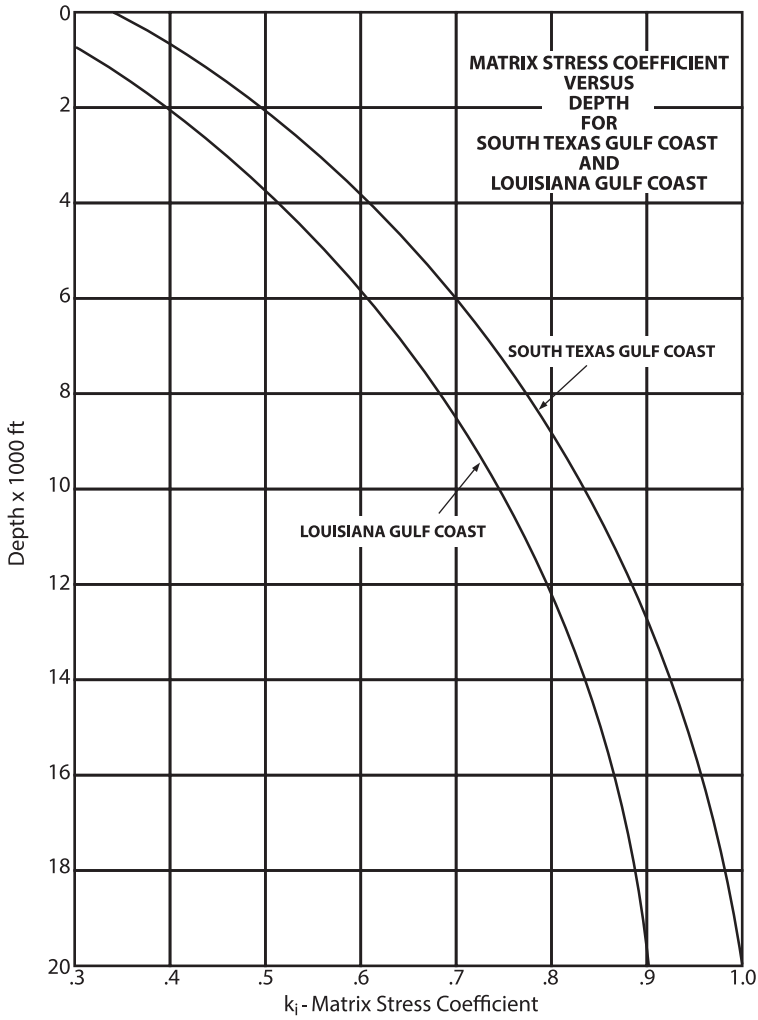


FIG. 3.3 Matrix stress coefficient.

σ = matrix stress at the point of interest in psi
D_{TVD} = true vertical depth at point of interest in ft

Procedure:

(1) Obtain formation pore pressure gradient (Fp_G) from electric logs, density measurements or mud logging data.
(2) Assume the overburden gradient is 1.0 psi/ft.
(3) Calculate the matrix stress at the point of interest:

$$\sigma = P_{OB} - Pp_F \tag{3.34}$$

where

σ = matrix stress in psi.
P_{OB} = overburden pressure in psi.
Pp_F = formation pore pressure in p

(4) Determine the K_i from Fig. 3.3 and the TVD in ft.
(5) Determine the fracture gradient in psi/ft with Eq. (3.33).
(6) Determine the fracture gradient in ppg:

$$FG_{MK} = \frac{FG_{MK}}{0.052} \tag{3.35}$$

Example: Use Louisiana Gulf Coast data in Fig. 3.3:

Casing TVD = 12,000 ft
Formation pore pressure = 12.0 ppg (0.624 psi/ft)
K_i = 0.79 (From Fig. 3.3)

$$Pp_F = (12.0)(0.052)(12{,}000) = 7488\,\text{psi}$$

$$\sigma = (12{,}000) - (7488) = 4512\,\text{psi}$$

$$FG_{MK} = \left(\frac{7488}{12{,}000}\right) + 0.79\left(\frac{4512}{12{,}000}\right) = 0.92\ \text{psi/ft}$$

$$FG_{MK} = \frac{0.92}{0.052} = 17.7\ \text{ppg}$$

Example: Use Texas Gulf Coast data in Fig. 3.3:

Casing TVD = 12,000 ft
Formation pore pressure = 12.0 ppg (0.624 psi/ft)
K_i = 0.89

$$Pp_F = (12.0)(0.052)(12{,}000) = 7488\,\text{psi}$$

$$\sigma = (12{,}000) - (7488) = 4512\,\text{psi}$$

$$FG_{MK} = \left(\frac{7488}{12{,}000}\right) + 0.89\left(\frac{4512}{12{,}000}\right) = 0.96\ \text{psi/ft}$$

$$FG_{MK} = \frac{0.96}{0.052} = 18.5 \text{ ppg}$$

Spreadsheet 3.8.1.A in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 FG—Graphical surface application.

(B) Calculations for Matthews and Kelly method:

(1) Louisiana Gulf Coast:

$$FG_L = (P_G) + (((\ln(TVD)(0.245)) - 1.516)(1.0 - P_G)) \quad (3.36)$$

where

FG_L = fracture gradient for Louisiana Gulf Coast in psi/ft
P_G = pore pressure gradient in psi/ft
TVD = true vertical depth in ft

$$FG_E = \frac{FG_L}{0.052} \quad (3.37)$$

(2) Texas Gulf Coast:

$$FG_T = (P_G) + (((\ln(TVD)(0.246)) - 1.428)(1.0 - P_G)) \quad (3.38)$$

where

FG_T = fracture gradient for Texas Gulf Coast in psi/ft

Example:

Casing TVD = 12,000 ft
Formation pore pressure = 12.0 ppg (0.624 psi/ft)

(1) Louisiana Gulf Coast:

$$FG_L = (0.624) + ((((9.39)(0.245)) - 1.516)(1.0 - 0.624)) = 0.92 \text{ psi/ft}$$

$$FG_E = \frac{0.92}{0.052} = 17.7 \text{ ppg}$$

(2) Texas Gulf Coast:

$$FG_T = (0.624) + ((((9.39)(0.246)) - 1.428)(1.0 - 0.624)) = 0.96 \text{ psi/ft}$$

$$FG_{MK} = \frac{0.96}{0.052} = 18.5 \text{ ppg}$$

Spreadsheet 3.8.1.B in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 FG—Calculation surface application.

Method 2: Ben Eaton method.

This technique assumes a **variable** overburden gradient.

(A) Graphical method

(1) Use Fig. 3.4 to determine the variable overburden stress gradient in psi/ft.

(2) Obtain formation pore pressure gradient from electric logs, density measurements or logging data.

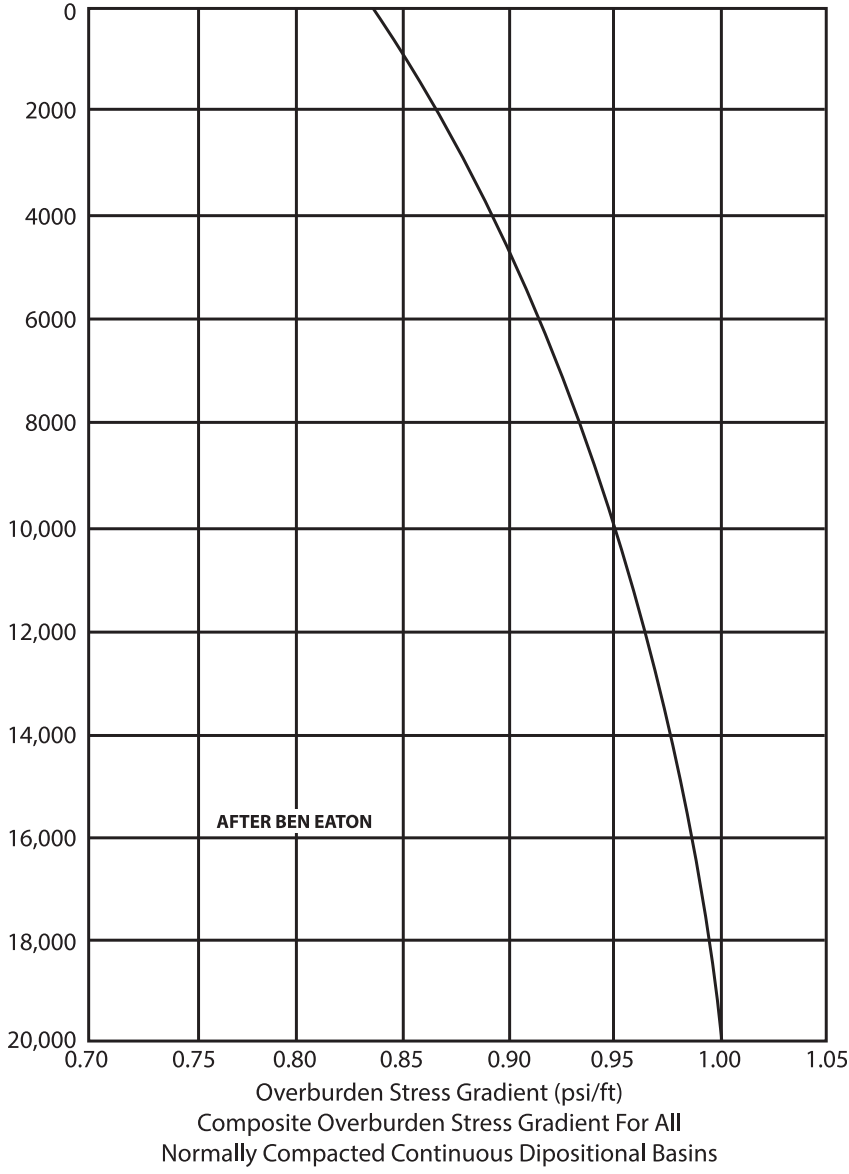


FIG. 3.4 Eaton's overburden stress graph.

(3) Use Fig. 3.5 to determine the Poisson's ratio at the depth of interest in ft.

(4) Determine the fracture gradient by the graphical method:

$$FG_E = \left((O_{VG} - Pp_G)\left(\frac{\mu}{(1.0-\mu)}\right)\right) + (Pp_G) \tag{3.39}$$

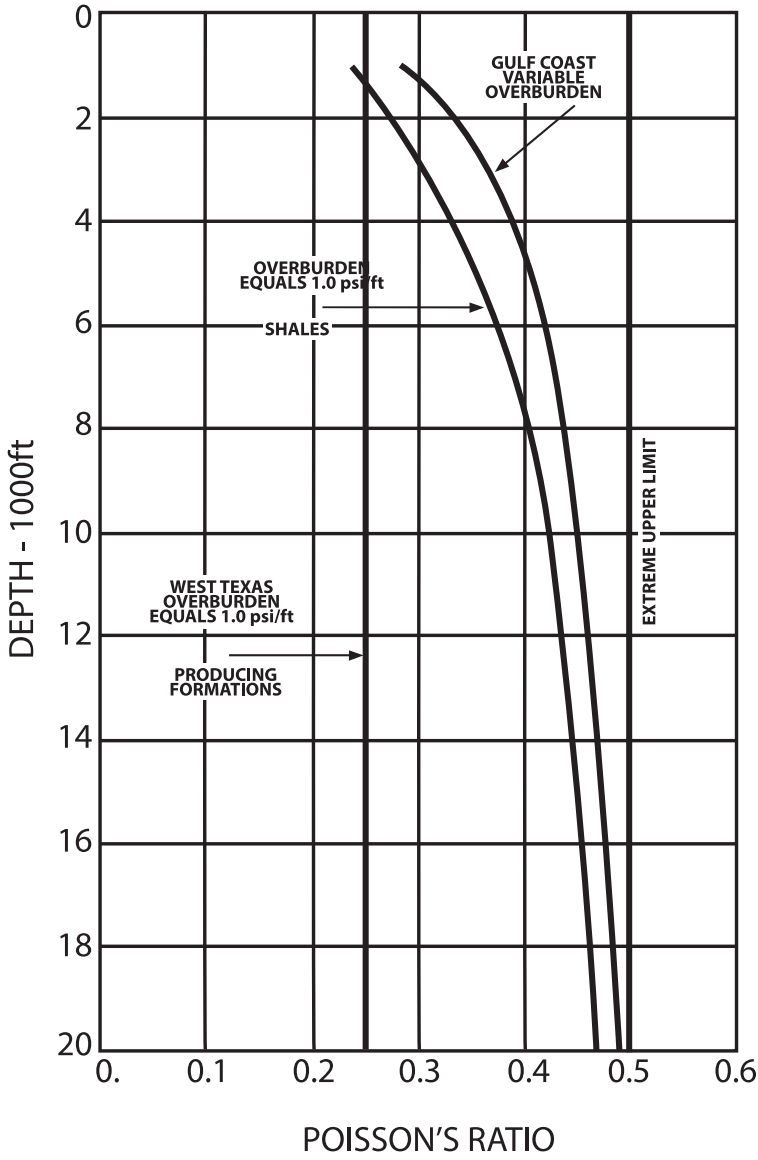


FIG. 3.5 Variation of Poisson's ratio with depth.

where

FG_E = fracture gradient by Eaton method in psi/ft
O_{VG} = variable overburden gradient in psi/ft
μ = Poisson's ratio
Pp_G = formation pore pressure in psi/ft

(5) Determine the FG_E in ppg (Eq. 3.7).

Example:

Casing TVD = 12,000 ft
Variable overburden gradient = 0.97 psi/ft (from Fig. 3.4)
Formation pore pressure = 0.624 psi/ft
Poisson's = 0.46 (from Fig. 3.5)

$$FG_E = \left((0.97 - 0.624)\left(\frac{0.46}{(1.0 - 0.46)}\right)\right) + (0.624) = 0.92 \text{ psi/ft}$$

$$FG_E = \frac{0.92}{0.052} = 17.7 \text{ ppg}$$

(B) Calculations for Eaton method:

(1) Determine the variable overburden gradient:

$$O_{VG} = ((\ln(\text{TVD})(0.072)) + (0.286)) \tag{3.40}$$

where

OVG = variable overburden gradient in psi/ft
TVD = true vertical depth in ft

(2) Determine the variable Poisson's ratio:

$$\mu = ((\ln(\text{TVD})(0.064)) - (0.140)) \tag{3.41}$$

where

μ = variable Poisson's ratio

(3) Use Eq. (3.39) to calculate the Fracture Gradient in psi/ft.
(4) Determine the F_{GE} in ppg (use Eq. 3.37).

Example:

Casing TVD = 12,000 ft
Formation pore pressure gradient = 0.624 psi/ft

(1) Determine the variable overburden gradient:

$$O_{VG} = (((9.39)(0.072)) + (0.286)) = 0.96 \text{ psi/ft}$$

(2) Determine the variable Poisson's ratio:

$$O_{VG} = (((9.39)(0.064)) - (0.140)) = 0.46$$

(3) Calculate the fracture gradient in psi/ft:

$$FG_E = \left((0.96 - 0.624)\left(\frac{0.46}{(1.0 - 0.46)}\right)\right) + (0.624) = 0.91 \text{ psi/ft}$$

$$FG_E = \frac{0.91}{0.052} = 17.5 \text{ ppg}$$

Spreadsheet 3.8.1.C in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 FG—Eaton variable overburden application.

3.8.2 Fracture gradient determination—Subsea application

During offshore deepwater drilling operations, it is necessary to correct the calculated fracture gradient for the effect of water depth and flowline height (air gap) above mean sea level. The following procedure can be used:

Graphical procedure:

(1) Convert water example to equivalent land area in ft:

(a) Determine the hydrostatic pressure of the seawater:

$$HP_{sw} = (MW_{sw})(0.052)(D_{sw}) \tag{3.42}$$

where

HP_{sw} = hydrostatic pressure of seawater in psi
MW_{sw} = mud weight of seawater in ppg
D_{sw} = depth of sea water in ft

(b) From Eaton's overburden stress graph, determine the overburden stress gradient from mean sea level to the casing setting depth. Using Fig. 3.4 enter the graph at 6000 ft on the left and intersect the curved line and read the Eaton overburden stress gradient (K_e) at the bottom of the graph.

(c) Determine the equivalent depth for an onshore land well in ft:

$$D_e = \frac{HP_{sw}}{K_e} \tag{3.43}$$

where

D_e = depth equivalent to onshore land well in ft
K_e = Eaton overburden stress gradient in psi/ft

(2) Determine the depth of interest for the fracture gradient calculation:

$$D_{FG} = Csg_D + D_e \tag{3.44}$$

where

D_{FG} = depth for the fracture gradient determination in ft
Csg_D = casing depth below the mud line in ft

(3) Use Eaton's fracture gradient graph in Fig. 3.6 to determine the fracture gradient at the depth of interest (D_{FG}). Enter the graph at the depth of interest (D_{FG}) and intersect the 9.0 ppg line, then proceed to the bottom of the graph to read the fracture gradient.

(4) Determine the fracture pressure in psi:

$$F_p = (FG_{MW})(0.052)(D_{FG}) \tag{3.45}$$

where

F_p = fracture pressure in psi

(5) Convert the fracture pressure relative to the flowline of the offshore well:

$$FG_{SS} = \left(\frac{F_p}{Csg_D + D_{sw} + L_{AG}}\right)(19.2) \tag{3.46}$$

where

FG_{SS} = fracture gradient subsea in ppg
Csg_D = casing depth below mud line in ft
L_{AG} = length of air gap in ft

Example:

Air gap	= 100 ft
Density of seawater	= 8.55 ppg
Water depth	= 2000 ft
Feet of casing below mudline	= 4000 ft

(1) (a) Determine the hydrostatic pressure of the seawater:

$$HP_{sw} = (8.55)(0.052)(4000) = 889 \text{ psi}$$

(b) From Fig. 3.4, the overburden stress gradient (K_e) is 0.92 psi/ft
(c) Determine the equivalent depth for an onshore land well in ft:

$$D_e = \frac{889}{0.92} = 966 \text{ ft}$$

(2) Determine the depth of interest for the fracture gradient calculation:

$$D_{FG} = 4000 + 966 = 4966 \text{ ft}$$

(3) The fracture gradient from Fig. 3.6 is estimated to be 14.6 ppg.

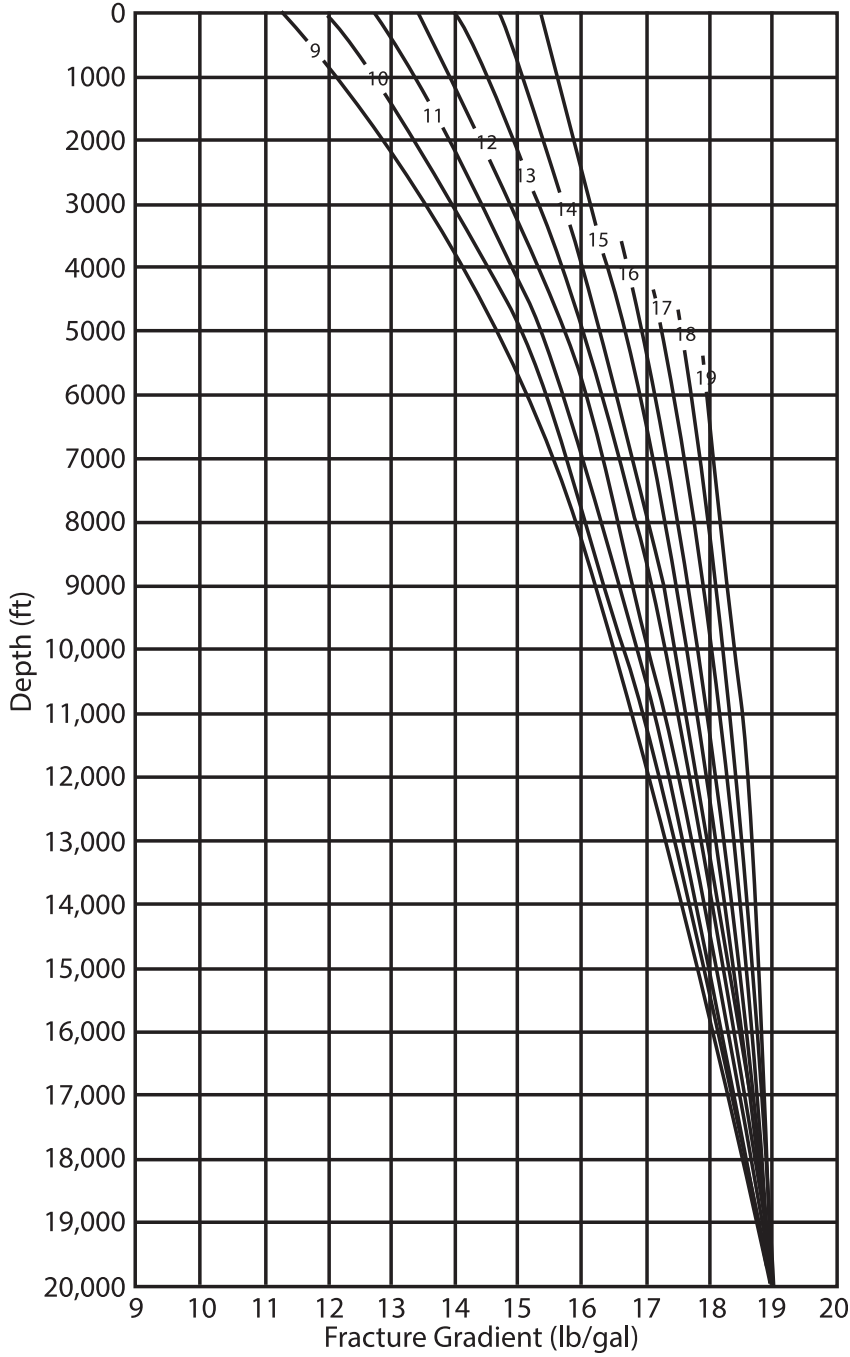


FIG. 3.6 Eaton's fracture gradient graph.

(4) Determine the fracture pressure in psi:

$$F_p = (14.6)(0.052)(4966) = 3770 \text{ psi}$$

(5) Convert the fracture pressure relative to the flowline of the offshore well:

$$FG_{SS} = \left(\frac{3770}{4000 + 2000 + 100}\right)(19.2) = 11.9 \text{ ppg}$$

Spreadsheet 3.8.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 FG—Subsea application.

3.9 Formation pressure tests

Two methods of testing:

- Equivalent mud weight test often referred to as the formation integrity test **(FIT)** normally used on development wells.
- Leak-off test **(LOT)** normally used on exploratory wells.

3.9.1 Precautions to be undertaken before testing

(1) Drill out of casing and penetrate at least 10 ft of new formation.
(2) Circulate and condition the mud to ensure the mud weight is consistent throughout the system.
(3) Change the pressure gauge (if possible) to a smaller increment gauge so a more accurate measure can be determined.
(4) Shut-in the well.
(5) Begin pumping at a very slow rate—¼ to ½ bbl/min.
(6) Monitor pressure, time, and barrels pumped.
(7) Some operators may use different procedures for running this test; others techniques may include:
  (a) Increasing the pressure by 100 psi increments, waiting for a few minutes, then increasing by another 100 psi, and so on, until either the equivalent mud weight or leak-off is achieved.
  (b) Some operators prefer not pumping against a closed system. They prefer to circulate through the choke and increase back pressure by slowly closing the choke. In this method, the annular pressure loss should be calculated and added to the test pressure results.

3.9.2 Testing to an equivalent mud weight (formation integrity test—FIT)

(1) This test is used primarily on development wells where the maximum mud weight that will be used to drill the next interval is known.
(2) Determine the equivalent test mud weight in ppg. Three methods can used to calculate the surface pressure for the test in psi.

Step 1: Determine the mud weight needed to calculate the surface pressure for the test in ppg.

Method 1: Use the maximum MW that is programed for the next hole interval with **NO** safety factor in ppg:

$$P_{S1} = (MW_M - MW_C)(0.052)\left(TVD_{csg}\right) \tag{3.47}$$

where

P_{S1} = surface pressure used for test in ppg
MW_M = maximum mud weight programed for next interval in ppg
MW_C = current mud weight in ppg
TVD_{csg} = true vertical depth of casing shoe in ft

Method 2: Add a safety factor to the maximum mud weight programed for the next internal to prevent taking a kick and exceeding the estimated fracture gradient at the casing shoe in ppg:

$$P_{S2} = ((MW_M + MW_{SF}) - MW_C)(0.052)\left(TVD_{csg}\right) \tag{3.48}$$

where

P_{S2} = surface pressure used for test with safety factor in ppg
MW_{SF} = safety factor mud weight in ppg

Method 3: Subtract a safety factor from the estimated fracture gradient at the casing shoe to prevent formation breakdown in ppg:

$$P_{S3} = ((FG - MW_{SF}) - MW_C)(0.052)\left(TVD_{csg}\right) \tag{3.49}$$

where

P_{S3} = surface pressure used for test with fracture gradient and safety factor in ppg
FG = estimated fracture gradient from the drilling program in ppg

Note: The pressure that would cause formation breakdown would be the estimated FG in psi:

$$P_{FG} = (FG - MW_C)(0.052)\left(TVD_{csg}\right) \tag{3.50}$$

where

P_{FG} = formation breakdown pressure at casing shoe in psi

Example:

Mud weight	= 10.0 ppg
Casing shoe TVD	= 4000 ft
Maximum mud weight for next interval	= 12.0 ppg
Fracture gradient at shoe	= 14.0 ppg
Safety factor	= 0.5 ppg

Method 1: Use the maximum mud weight that is programed for the next hole interval with **NO** safety factor in ppg:

$$P_{S1} = (12.0 - 10.0)(0.052)(4000) = 416 \text{ psi}$$

Method 2: Add a safety factor to the maximum mud weight programed for the next internal to prevent taking a kick and exceeding the estimated fracture gradient at the casing shoe in ppg:

$$P_{S2} = ((12.0 + 0.5) - 10.0)(0.052)(4000) = 520 \text{ psi}$$

Method 3: Subtract a safety factor from the estimated fracture gradient at the casing shoe to prevent formation breakdown in ppg:

$$P_{S3} = ((14.0 - 0.5) - 10.0)(0.052)(\text{TVD}_{csg}) = 728 \text{ psi}$$

The pressure that would cause formation breakdown would be the estimated FG in psi:

$$P_{FG} = (14.0 - 10.0)(0.052)(\text{TVD}_{csg}) = 832 \text{ psi}$$

Spreadsheet 3.9.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 FIT.

3.9.3 Testing to leak-off test pressure

(1) This test is used primarily on wildcat or exploratory wells or where the actual fracture is not known.
(2) Determine the estimated fracture gradient utilizing the "Matrix Stress Coefficient" in Fig. 3.3.
(3) Determine the estimated leak-off pressure.

3.9.4 Procedure to prepare the graph to record leak-off pressure data

On the linear chart drawn for the casing pressure test, draw three horizontal lines corresponding to:

(1) Maximum allowable pressure in psi (80% or 90% of the Overburden).
(2) Estimated LOT pressure in psi.
(3) Minimum acceptable leak-off test pressure in psi.
(4) V_{min} determined from intersection of minimum volume line from casing test with Estimated LOT line.
(5) Draw maximum volume line from zero through the intersection of $2 \times V_{min}$ with the Estimated LOT line (Fig. 3.7).

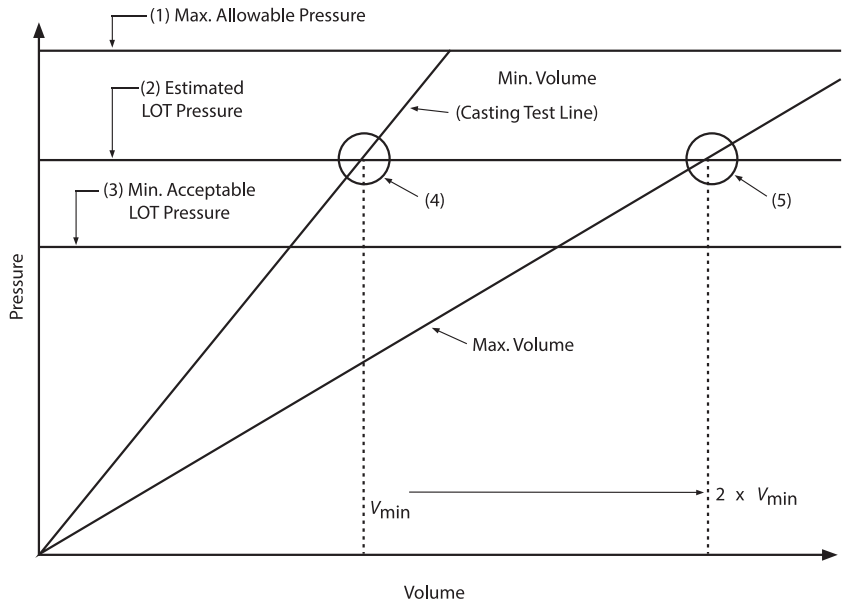


FIG. 3.7 Leak-off test graph.

Step 1: Determine the maximum allowable pressure line (1) in psi:

$$P_{MXL} = \left[(\sigma_{OB})(SF_{OB})(TVD_{csg})\right] - \left[(MW)(0.052)(TVD_{csg})\right] \quad (3.51)$$

where

P_{MXL} = maximum allowable pressure line in psi (option—use the casing pressure test line).
σ_{OB} = variable overburden in psi/ft
SF_{OB} = safety factor to prevent exceeding the overburden value
TVD_{csg} = true vertical depth of casing shoe in ft
MW = mud weight in ppg

Note: If the Variable Overburden is not known, 1.0 psi/ft can be used instead. In this case, the Safety Factor should be decreased to 85.0% or 80.0%.

Step 2: Determine the estimated leak-off test pressure line (2) in psi:

$$P_{LOTL} = (FG - MW)(0.052)(TVD_{csg}) \quad (3.52)$$

where

P_{LOTL} = estimated leak-off test pressure line (2) in psi
FG = estimated fracture gradient in ppg
MW = mud weight in use in ppg

Step 3: Determine the minimum acceptable leak-off test pressure line (3) in psi:

$$P_{\text{LOTmin}} = (\text{MW}_{\text{MXI}} - \text{MW})(0.052)(\text{TVD}_{\text{csg}}) \tag{3.53}$$

where

P_{LOTmin} = minimum acceptable leak-off test pressure line (3) in psi
MW_{MXI} = maximum mud weight for next interval in ppg

Step 4: Determine the frictional pressure loss in the system in psi:

(1) If only using the **drill pipe** to conduct the test, determine the frictional pressure loss in psi:

$$P_{\text{fdp}} = \frac{(\gamma_{\text{G}})(L_{\text{p}})}{300(\text{ID}_{\text{p}})} \tag{3.54}$$

where

P_{fdp} = frictional pressure loss down the drill pipe in psi
γ_{G} = gel strength of the mud in $\text{lb}/100\,\text{ft}^2$
L_{p} = length of the drill pipe in ft
ID_{p} = internal diameter of the drill pipe in in.

(2) When testing down the **casing annulus** only, determine the frictional pressure loss in psi:

$$P_{\text{fa}} = \frac{(\gamma_{\text{G}})(L_{\text{csg}})}{300(\text{ID}_{\text{csg}} - \text{ODp})} \tag{3.55}$$

where

P_{fa} = frictional pressure loss down the casing annulus in psi
γ_{G} = gel strength of the mud in $\text{lb}/100\,\text{ft}^2$
L_{csg} = length of the casing in ft
ID_{csg} = internal diameter of the casing in in.
OD_{p} = outside diameter of the drill pipe in in.

(3) When testing down **BOTH** the drill pipe and the casing annulus, determine the frictional pressure for each section and add those values together for the psi:

$$P_{\text{s}} = (P_{\text{fdp}}) + (P_{\text{fa}}) \tag{3.56}$$

where

P_{s} = frictional pressure loss for the system in psi

Step 5: Determine the slope for the minimum volume line (4), if the casing test line is **NOT** used:

(a) Determine the compressibility of the mud:

$$C_m = \left(\left[3.0 \times 10^{-6}\right](W_{\%})\right) + \left(\left[5.0 \times 10^{-6}\right](O_{\%})\right) + \left(\left[0.2 \times 10^{-6}\right](S_{\%})\right) \tag{3.57}$$

where

C_m = compressibility of the mud
$W_{\%}$ = water content of the mud in %
$O_{\%}$ = oil content of the mud in %
$S_{\%}$ = solids content of the mud in %

(b) Determine the volume of the mud system in bbl:

$$V_s = \left(\left(C_p\right)\left(L_p\right)\right) + \left((C_a)(L_a)\right) + \left((C_h)(L_h)\right) \tag{3.58}$$

where

V_s = volume of system in bbl
C_p = capacity of drill pipe in bbl/ft
L_p = length of drill pipe in ft
C_a = capacity of annulus in bbl/ft
L_a = length of annulus (casing) in ft
C_h = capacity of open hole in bbl/ft
L_h = length of open hole in ft

(c) Determine the slope of the minimum volume line (4) in psi/bbl:

$$S_{PVL} = \frac{1}{\left((C_m)(V_s)\right)} \tag{3.59}$$

where

S_{PVL} = slope of the minimum volume line in psi/bbl

(d) Record the volume as V_{ML} where the minimum volume line (4) crosses the estimated LOT pressure line in bbl.
where

V_{ML} = minimum volume line intersection at LOT pressure line in bbl

Step 6: Determine the maximum volume line (5) in bbl:

$$V_{MXL} = 2(V_{ML}) \tag{3.60}$$

where

V_{MXL} = maximum volume line at LOT pressure in bbl

Example:

Overburden	= 1.0 psi/ft
Overburden safety factor	= 80%
Casing TVD (also length of annulus)	= 4000 ft
Casing size	= 13⅜ in. (61 lb/ft, ID = 12.515 in.)
Casing annulus capacity w/DP	= 0.1279 bbl/ft
Drill pipe	= 5.0 in. × 4.276 in. (19.5 lb/ft)
Drill pipe length	= 4000 ft
Drill pipe capacity	= 0.0178 bbl/ft
Open hole size	= 12¼ in.
Open hole length	= 30 ft
Open hole capacity w/o DP	= 0.1458 bbl/ft
Mud weight	= 10.0 ppg
Oil content	= 72%
Water content	= 18%
Solids content	= 10%
Oil/water ratio	= 80/20
Gel strength	= 15 lb/100 ft^2
Maximum mud weight for next interval	= 12.0 ppg
Fracture gradient	= 14.0 ppg

Test performed down the drill pipe.

Step 1: Determine the maximum allowable pressure line (1) in psi:

$$P_{\mathrm{ML}} = [(1.0)(0.80)(4000)] - [(\mathrm{MW})(0.052)(4000)] = 1120 \text{ psi}$$

Step 2: Determine the estimated leak-off test pressure line (2) in psi:

$$P_{\mathrm{LOTL}} = (14.0 - 10.0)(0.052)(4000) = 832 \text{ psi}$$

Step 3: Determine the minimum acceptable leak-off test pressure line (3) in psi:

$$P_{\mathrm{LOTmin}} = (12.0 - 10.0)(0.052)(4000) = 416 \text{ psi}$$

Step 4: Determine the frictional pressure loss in the system in psi:

(1) If only using the **drill pipe** to conduct the test, determine the frictional pressure loss in psi:

$$P_{\mathrm{fdp}} = \frac{(15)(4000)}{300(4.276)} = 46.8 \approx 47 \text{ psi}$$

Step 5: Determine the slope for the minimum volume line (4), if the casing test line is **NOT** used:

(a) Determine the compressibility of the mud:

$$C_{\mathrm{m}} = \left(\left[3.0 \times 10^{-6}\right](0.18)\right) + \left(\left[5.0 \times 10^{-6}\right](0.72)\right) + \left(\left[0.2 \times 10^{-6}\right](0.10)\right) = 4.16 \times 10^{-6}$$

(b) Determine the volume of the mud system in bbl:

$$V_s = ((0.0178)(4000)) + ((0.1279)(4000)) + ((0.1458)(30)) = 587.1 \approx 587 \text{ bbl}$$

(c) Determine the slope of the minimum volume line (4) in psi/bbl:

$$S_{PVL} = \frac{1}{\left((4.16 \times 10^{-6})(587)\right)} = 409.5 \text{ psi/bbl}$$

This means that a 1000 psi casing test will require the following:

$$\text{Casing test} = \frac{1000}{409.5} = 2.44 \text{ bbl}$$

(d) Record the volume as V_{ML} where the minimum volume line (4) crosses the estimated LOT pressure line in bbl.

$$V_{ML} = 2.44 \text{ bbl@1000 psi}$$

Step 6: Determine the maximum volume line (5) in bbl:

$$V_{MXL} = 2(2.44) = 4.88 \text{ bbl@1000 psi}$$

Spreadsheet 3.9.4 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Prepare graph for LOT.

3.9.5 Maximum allowable mud weight from leak-off test data

$$\text{MW}_{MA} = \left(\frac{P_{LOT}}{(0.052)(\text{TVD}_{csg})}\right) + \text{MW} \tag{3.61}$$

where

MW_{MA} = maximum allowable mud weight in ppg

Example: Determine the maximum allowable mud weight in ppg using the following data:

Leak-off pressure = 1040 psi
Casing shoe TVD = 4000 ft
Mud weight in use = 10.0 ppg

$$\text{MW}_{MA} = \left(\frac{1040}{(0.052)(4000)}\right) + 10.0 = 15.0 \text{ ppg}$$

Spreadsheet 3.9.5 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Maximum allowable MW from LOT.

3.10 Kick tolerance (KT)

Definitions: The kick tolerance intensity (KT_I) is the amount of kick in equivalent mud weight that can be taken with the current mud weight in use at the time of the kick. The kick tolerance volume (KT_V) is the maximum kick volume for a specific kick intensity that can be taken when circulating out and not exceed the formation breakdown pressure (fracture gradient) at the casing shoe.

3.10.1 Kick tolerance intensity (KTI)

Formula 1: Determine the kick tolerance intensity at any vertical depth in ppg:

$$KT_I = (FG - MW - SF - TM)\left(\frac{TVD_{csg}}{TVD_{TD}}\right) \tag{3.62}$$

where

KT_I = kick tolerance intensity in ppg
FG = fracture gradient at the casing shoe in ppg
MW = mud weight in use at total vertical depth in ppg
SF = safety factor in ppg (optional, but recommended)
TM = trip margin in ppg
TVD_{csg} = true vertical depth of casing in ft
TVD_{TD} = true vertical depth in ft

Formula 2: Determine the maximum surface pressure in psi:

$$P_{MS} = (KT_I)(0.052)(TVD_{TD}) \tag{3.63}$$

where

P_{MS} = maximum surface pressure in psi

Formula 3: Determine the maximum formation pressure that can be controlled when shutting-in a well in psi:

$$P_{MF} = (KT_I + MW)(0.052)(TVD_{TD}) \tag{3.64}$$

where

P_{MF} = maximum formation pressure in psi

Formula 4: Determine the maximum influx length that can be taken with the gas at the casing shoe without breakdown in ft:

$$L_I = \frac{MASICP}{(G_M - G_I)} \tag{3.65}$$

where

L_I = length of influx in ft
G_M = gradient of mud in psi/ft
G_I = gradient of influx in psi/ft

Example Formula 1: Determine the kick tolerance intensity using the following data:

Maximum allowable mud weight	= 13.0 ppg (from leak-off test data)
Mud weight at TVD	= 12.0 ppg
Casing shoe TVD	= 5000 ft
Safety factor	= 0.25 ppg (optional, can be combined with trip margin)
Trip margin	= 0.10 ppg
Well depth TVD	= 10,000 ft

$$\mathrm{KT_I} = (13.0 - 12.0 - 0.25 - 0.1)\left(\frac{5000}{10{,}000}\right) = 0.325 \cong 0.3 \text{ ppg}$$

Example Formula 2: Determine the maximum surface pressure in psi:
Data:

MASICP at 10,000 ft	= 260 psi
Mud gradient	= 0.624 psi/ft (12.0 ppg)
Gradient of influx	= 0.12 psi/ft
Capacity of annulus	= 0.1215 bbl/ft (12¼ × 5.0 in.)

$$P_{\mathrm{MS}} = (0.3)(0.052)(10{,}000) = 156 \text{ psi}$$

Example Formula 3: Determine the maximum formation pressure that can be controlled when shutting-in a well, using the following data:

Kick tolerance intensity	= 0.3 ppg
Mud weight	= 12.0 ppg (0.624 psi/ft)
True vertical depth	= 10,000 ft

$$P_{\mathrm{MF}} = (0.3 + 12.0)(0.052)(10{,}000) = 6396 \text{ psi}$$

Example Formula 4: Determine the influx length necessary to equal the maximum allowable shut-in surface pressure in ft:

$$L_I = \frac{156}{(0.624 - 0.12)} = 309.5 \text{ ft}$$

Spreadsheet 3.10.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Kick tolerance intensity.

3.10.2 Kick tolerance volume (KTV)

Formula 1: Determine the maximum influx volume to equal maximum allowable shut-in surface pressure at the casing shoe in bbl:

$$KT_V = \frac{[((FG)(0.052)(TVD_{csg}))(L_I)(C_a)]}{[(MW)(0.052)(TVD_{TD})]} \tag{3.66}$$

where

KT_V = volume of kick at the casing shoe in bbl
C_a = capacity of the annulus with the drill pipe in bbl

$$KT_V = \frac{[((13.0)(0.052)(5000))(516)(0.1215)]}{[(12.0)(0.052)(10{,}000)]} = 33.95 \approx 34.0 \text{ bbl}$$

Spreadsheet 3.10.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Kick tolerance volume.

3.10.3 Summary

This summary shows how the kick tolerance changes with depth and casing set at 5000 ft.

TVD_{TD}, ft	MW, ppg	MASICP, psi	KT, ppg	Influx length @ Casing Shoe, ft	Kick volume, bbl
5000	11.0	520	2.0	1150.0	165.1
7500	11.5	390	1.0	815.8	74.7
10,000	12.0	260	0.5	515.9	34.0

3.11 Kick analysis

3.11.1 Formation pressure (FP) with the well shut-in on a kick

$$P_F = (SIDPP) + ((MW)(0.052)(TVD_{TD})) \tag{3.67}$$

where

P_F = formation pressure in psi

Example: Determine the formation pressure using the following data:

Shut-in drill pipe pressure = 500 psi
Mud weight = 9.6 ppg
True vertical depth = 10,000 ft

$$P_F = (500) + ((9.6)(0.052)(10{,}000)) = 5492 \text{ psi}$$

Spreadsheet 3.11.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 FP with well shut-in on a kick.

3.11.2 Bottom hole pressure (BHP) with the well shut-in on a kick

$$\text{BHP} = (\text{SIDPP}) + ((\text{MW})(0.052)(\text{TVD}_{\text{TD}})) \tag{3.68}$$

where

BHP = bottom hole pressure in psi

Example: Determine the bottom hole pressure (BHP) with the well shut-in on a kick:

Shut-in drill pipe pressure = 500 psi
Mud weight = 10.0 ppg
True vertical depth = 10,000 ft

$$\text{BHP} = (500) + ((10.0)(0.052)(10{,}000)) = 5700 \text{ psi}$$

Spreadsheet 3.11.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 BHP with a well shut-in on a kick.

3.11.3 Shut-in drill pipe pressure (SIDPP)

$$\text{SIDPP} = P_{\text{F}} - ((\text{MW})(0.052)(\text{TVD}_{\text{TD}})) \tag{3.69}$$

where

SIDPP = shut-in drill pipe pressure in psi

Example: Determine the shut-in drill pipe pressure using the following data:

Formation pressure = 12,480 psi
Mud weight in drill pipe = 15.0 ppg
True vertical depth = 15,000 ft

$$\text{SIDPP} = (12{,}480) - ((15.0)(0.052)(15{,}000)) = 780 \text{ psi}$$

Spreadsheet 3.11.3 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 SIDPP.

3.11.4 Shut-in casing pressure (SICP)

$$\text{SCIP} = (P_{\text{F}}) - [((\text{MW})(0.052)(\text{TVD}_{\text{TD}})) + ((G_{\text{I}})(L_{\text{I}}))] \tag{3.70}$$

where

SCIP = shut-in casing pressure in psi
G_I = gradient of influx in psi/ft
L_I = length of influx in ft

Example: Determine the shut-in casing pressure using the following data:

Formation pressure	= 12,480 psi
Mud weight in annulus	= 15.0 ppg
Feet of mud in annulus	= 14,600 ft
Influx gradient	= 0.12 psi/ft
Feet of influx in annulus	= 400 ft

$$\text{SCIP} = (12{,}480) - [((15.0)(0.052)(14{,}600)) + ((0.12)(400))] = 1044 \text{ psi}$$

Spreadsheet 3.11.4 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 SICP.

3.11.5 Length of influx in ft

$$L_I = \frac{V_{pg}}{C_a} \tag{3.71}$$

where

L_I = length of influx in ft
V_{pg} = volume of pit gain in bbl

Example 1: Determine the length of the influx in feet using the following data:

Pit gain = 20 bbl
Annular capacity – DC/OH = 0.02914 bbl/ft (D_h = 8½ in.; D_p = 6½ in.)

$$L_I = \frac{20}{0.02914} = 686 \text{ ft}$$

Example 2: Determine the length of the influx in feet using the following data:

Pit gain	= 20 bbl
Hole size	= 8½ in.
Drill collar OD	= 6½ in.
Drill collar length	= 450 ft
Drill pipe OD	= 5.0 in.

Step 1: Determine annular capacity for DC/OH in bbl/ft:

$$C_{DC/OH} = \frac{(8.5^2) - (6.5^2)}{1029.4} = 0.0291 \text{ bbl/ft}$$

Step 2: Determine the number of barrels opposite the drill collars:

$$V_{DC/OH} = (450)(0.0291) = 13.1 \text{ bbl}$$

Step 3: Determine annular capacity opposite the drill pipe/OH in bbl/ft:

$$C_{DP/OH} = \frac{(8.5^2) - (5.0^2)}{1029.4} = 0.0459 \text{ bbl/ft}$$

Step 4: Determine barrels of influx opposite drill pipe:

$$V_I = V_{pg} - V_{DC/OH} \tag{3.72}$$

where

V_I = volume of pit gain in bbl

$$V_I = 20 - 13.1 = 6.9 \text{ bbl}$$

Step 5: Determine length of influx opposite drill pipe in ft:

$$L_I = \frac{6.9}{0.0459} = 150 \text{ ft}$$

Step 6: Determine the total height of the influx in ft:

$$L_I = 450 + 150 = 600 \text{ ft}$$

Spreadsheet 3.11.5 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Length of influx.

3.11.6 Estimated type of influx

$$W_K = \text{MW} - \left(\frac{\text{SCIP} - \text{SIDPP}}{(L_I)(0.052)}\right) \tag{3.73}$$

where

W_K = weight of influx in ppg

Then:

1–3 ppg = Gas kick
4–6 ppg = Oil kick or combination
7–9 ppg = Saltwater kick

Example: Determine the type of the influx using the following data in ppg:

Shut-in casing pressure = 1044 psi
Shut-in drill pipe pressure = 780 psi
Length of influx = 400 ft
Mud weight = 15.0 ppg

$$W_K = 15.0 - \left(\frac{1044 - 780}{(400)(0.052)}\right)$$

$$W_K = 15.0 - \left(\frac{264}{20.8}\right) = 2.31 \text{ ppg}$$

Therefore, the influx is probably "**gas**."

Spreadsheet 3.11.6 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Type of influx.

3.12 Gas cut mud weight measurement calculations

3.12.1 Determine the original mud weight of a gas cut mud at the flowline

Procedure:

(1) Dilute a 250 mL sample of the gas cut mud with an equal amount of water or base oil.

(2) Be sure to keep the mud solids in suspension by stirring the slurry as the sample is poured into the mud balance cup.

(3) Measure the density of the slurry and use the following equation to determine the original mud weight in ppg:

$$MW_O = \frac{(MW_{GC})(MW_D)}{((MW_{GC} + MW_{BL}) - MW_D)} \tag{3.74}$$

where

MW_O = original uncut mud weight in ppg
MW_{GC} = mud weight of gas cut mud in ppg
MW_D = mud weight of diluted slurry in ppg
MW_{BL} = mud weight of base liquid used to dilute the gas cut mud in ppg

(4) Determine the gas content of the mud in percent:

$$C_{gas} = 100\left(\frac{(MW_O - MW_C)}{MW_O}\right) \tag{3.75}$$

where

C_{gas} = gas content in cut mud in %

Example: Determine the original mud weight in ppg and the percent of gas in the mud.

Data:

$$\begin{aligned}\text{Mud type} &= \text{Water base mud}\\ \text{MW}_\text{C} &= 9.2\,\text{ppg}\\ \text{MW}_\text{D} &= 10.0\,\text{ppg}\\ \text{MW}_\text{BL} &= 8.34\,\text{ppg}\end{aligned}$$

$$\text{MW}_\text{O} = \frac{(9.2)(10.0)}{((9.0 + 8.34) - 10.0)} = 12.5\ \text{ppg}$$

$$C_\text{gas} = 100\left(\frac{(12.5 - 9.2)}{12.5}\right) = 26.4\%$$

Spreadsheet 3.12.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Gas cut MW.

3.12.2 Determine the reduction in the mud weight at the flowline when a gas formation is drilled with no kick

Step 1: Determine the volume of cuttings generated by the ROP in ft^3/h:

$$V_\text{C} = \left[\left(5.45 \times 10^{-3}\right)\left(D_\text{h}{}^2\right)\right](\text{ROP}) \tag{3.76}$$

where

V_C = volume of cuttings from the ROP in ft^3/h
D_h = diameter of the hole in in. (Bit size if amount of washout is not known.)
ROP = rate of penetration in ft/h

Step 2: Determine the volume of the gas in the formation pores ft^3/ft^3:

$$V_\text{Gf} = (\varnothing_\text{f})(1 - W_\text{S}) \tag{3.77}$$

where

V_Gf = volume of gas in the formation pores in ft^3/ft^3
$\varnothing_\text{f}$ = formation porosity in %
W_S = water saturation in %

Step 3: Determine the volume of gas in the cuttings at the bottom of the hole in ft^3/h:

$$V_\text{GC} = (V_\text{C})(V_\text{Gf}) \tag{3.78}$$

where

V_Gf = volume of gas in the cuttings in ft^3/h

Step 4: Determine the bottom hole pressure in psi:

$$P_{BH} = (MW)(0.052)(TVD_{TD}) \tag{3.79}$$

where

P_{BH} = bottom hole pressure of gas in psi
MW = mud weight in ppg
TVD_{TD} = true vertical depth in ft

Step 5: Determine the volume of the gas at the flowline in ft^3/h:

$$V_{GFL} = (0.0292)(P_{BH}) \tag{3.80}$$

where

V_{GFL} = volume of gas at the flowline in gpm

Step 6: Determine the mud weight of the gas cut mud at the flowline in ppg:

$$MW_C = (MW)\left(\frac{Q}{(Q) + (V_{GFL})}\right) \tag{3.81}$$

where

MW_C = mud weight of gas cut mud at the flowline in ppg
Q = pump output in gpm

Example: Determine the mud weight of gas cut mud at the flowline with the following:

Data:

TVD depth of hole	= 10,000 ft
Bit size	= 12¼ in.
ROP	= 50 ft/h
Pump output	= 650 gpm
Formation porosity	= 18%
Water saturation	= 25%
Formation pressure	= 12.0 ppg
Mud weight	= 12.5 ppg

Step 1: Determine the volume of cuttings generated by the ROP in ft^3/h:

$$V_C = \left[\left(5.45 \times 10^{-3}\right)\left(12.25^2\right)\right](50) = 40.9\ ft^3/h$$

Step 2: Determine the volume of the gas in the formation pores ft^3/ft^3:

$$V_{Gf} = (0.18)(1 - 0.25) = 0.135\ ft^3/ft^3$$

Step 3: Determine the volume of gas in the cuttings at the bottom of the hole in ft^3/h:

$$V_{GC} = (40.9)(0.135) = 5.52\ ft^3/hr$$

Step 4: Determine the bottom hole pressure in psi:

$$P_{BH} = (12.0)(0.052)(10{,}000) = 6240\ psi$$

Step 5: Determine the volume of the gas at the flowline in gpm:

$$V_{GFL} = (0.0292)(6240) = 182.2\ gpm$$

Step 6: Determine the mud weight of the gas cut mud at the flowline in ppg:

$$MW_C = (12.5)\left(\frac{650}{(650) + (182.2)}\right) = 9.76 \approx 9.8\ ppg$$

Spreadsheet 3.12.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 MW of gas cut without a kick.

3.13 Gas migration in a shut-in well

3.13.1 Gas migration

Determining the rate of gas migration after a well has been shut-in on a kick in ft/h:

$$R_{GM} = \left(\frac{(SCIP_2) - (SCIP_1)}{((MW_G)(t))}\right) \tag{3.82}$$

where

R_{GM} = rate of gas migration in ft/h
$SCIP_1$ = initial shut-in casing pressure in psi
$SCIP_2$ = shut-in casing pressure at time 2 in psi
MW_G = pressure gradient of mud weight in psi/ft
t = time for pressure to change in hours

Example: Determine the rate of gas migration with the following data:

Initial shut-in casing pressure	= 500 psi
SICP at time 2	= 550 psi
Mud weight	= 12.0 ppg
Pressure gradient for 12.0 ppg mud	= 0.624 psi/ft
Time for pressure to change	= 1.0 h

$$R_{GM} = \left(\frac{(550) - (500)}{((0.624)(1.0))}\right) = 80.1 \text{ ft/h}$$

Spreadsheet 3.13.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Gas migration.

3.14 Hydrostatic pressure decrease at TD caused by formation fluid influx due to a kick

3.14.1 Hydrostatic pressure decrease

Method 1: Determine the hydrostatic pressure decrease caused by gas-cut mud in psi.

$$HP_D = ((TVD)(MW_G)) - \left((TVD)(MW_G) - \left(\left(\frac{V_{pg}}{C_a}\right)(MW_G)\right) + \left(\left(\frac{V_{pg}}{C_a}\right)(I_G)\right)\right) \quad (3.83)$$

where

HP_D = hydrostatic pressure decrease in psi
TVD = true vertical depth in ft
MW_G = mud weight gradient in psi/ft
V_{pg} = pit gain in bbl
C_a = capacity of annulus in bbl/ft
I_G = influx gradient in psi/ft

Method 2: Determine the hydrostatic pressure decrease using gradients:

$$HP_{DG} = \left(\frac{(MW_G - I_G)}{C_a}\right)(V_{pg}) \quad (3.84)$$

Example: Determine the hydrostatic pressure decrease caused by gas-cut mud using the following data:

TVD	= 10,000 ft
Mud weight gradient	= 0.624 psi/ft (12.0 ppg)
Capacity of annulus	= 0.0322 bbl/ft (8½ × 6¼ in. annulus)
Influx gradient	= 0.12 psi/ft
Pit gain	= 20 bbl

Method 1: Determine the hydrostatic pressure decrease:

$$\begin{aligned}\mathrm{HP_D} &= ((10{,}000)(0.624)) \\ &\quad - \left((10{,}000)(0.624) - \left(\left(\frac{20}{0.0322}\right)(0.624)\right) + \left(\left(\frac{20}{0.0322}\right)(0.12)\right)\right) \\ &= 313 \text{ psi}\end{aligned}$$

Method 2: Determine the hydrostatic pressure decrease:

$$\mathrm{HP_{DG}} = \left(\frac{(0.624 - 0.12)}{0.322}\right)(20) = 313 \text{ psi}$$

Spreadsheet 3.14.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 HP decrease due to kick influx.

3.14.2 Maximum surface pressure from a gas kick in a water-base mud

$$P_{MS} = (0.2)\sqrt{\frac{(P_F)(V_{PG})(\mathrm{KWM})}{C_a}} \tag{3.85}$$

where

P_{MS} = maximum surface pressure resulting from a gas kick in a water-base mud in psi
P_F = formation pressure in psi
V_{PG} = pit gain in bbl
KWM = kill weight mud in ppg
C_a = annular capacity in bbl/ft

Example:

Formation pressure = 12,480 psi
Pit gain = 20 bbl
KWM = 16.0 ppg
Annular capacity = 0.0505 bbl/ft (D_h = 8½ in.; D_p = 4½ in.)

$$P_{MS} = (0.2)\sqrt{\frac{(12{,}480)(20)(16.0)}{0.0505}} = 1778.6 \text{ psi} \cong 1779 \text{ psi}$$

Spreadsheet 3.14.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Maximum surface pressure due to influx in WBM.

3.14.3 Maximum pit gain from gas kick in a water-base mud

$$V_{MPG} = (4)\sqrt{\frac{(P_F)(V_{OPG})(C_a)}{KWM}} \tag{3.86}$$

where

V_{MPG} = maximum pit gain volume in bbl
V_{OPG} = original pit gain volume in bbl

Example:

Formation pressure = 12,480 psi
Pit gain = 20 bbl
KWM = 16.0 ppg
Annular capacity = 0.0505 bbl/ft (D_h = 8½ in.; D_p = 4½ in.)

$$V_{MPG} = (4)\sqrt{\frac{(12{,}480)(20)(0.0505)}{16.0}} = 112.3 \text{ bbl}$$

Spreadsheet 3.14.3 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Maximum pit gain due to influx in WBM.

3.15 Maximum pressures when circulating out a kick (Moore equations)

3.15.1 Maximum pressure calculations

Step 1: Determine formation pressure in psi:

$$P_F = (\text{SIDPP}) + [(\text{MW})(0.052)(\text{TVD})] \tag{3.87}$$

where

P_F = formation pressure in psi
SIDPP = shut-in drill pipe pressure in psi
MW = mud weight in ppg
TVD = true vertical depth in ft

Step 2: Determine the length of the influx in ft:

$$L_I = \left(\frac{V_{pg}}{C_a}\right) \tag{3.88}$$

where

L_I = length of influx in ft
V_{pg} = volume of pit gain in bbl
C_a = capacity of annulus in bbl/ft

Step 3: Determine hydrostatic pressure exerted by the influx in psi:

$$P_I = (P_F) - [MW_G(TVD - L_{DC}) + SICP] \tag{3.89}$$

where

P_I = hydrostatic pressure exerted by the influx in psi
MW_G = mud weight gradient in psi/ft
TVD = true vertical depth in ft
L_{DC} = length of drill collars in ft
SICP = shut-in casing pressure in psi

Step 4: Determine pressure gradient of influx in psi/ft:

$$P_{GI} = \left(\frac{P_I}{L_I}\right) \tag{3.90}$$

where

P_{GI} = pressure gradient of influx in psi/ft
L_I = length of influx in ft

Step 5: Determine temperature in Rankin (°R) at depth of interest:

$$T_{Di} = (T_A) + [(GT_G)(TVD)] + (460) \tag{3.91}$$

where

T_{Di} = temperature at depth of interest in °R
T_A = ambient temperature in °F
G_{TG} = geothermal gradient in °F/ft
TVD = true vertical depth in ft
460 = conversion of 0°F to °Rankin

Step 6: Determine the pressure at the top of the influx bubble (A) for an un-weighted mud in psi:

$$A = (P_F) - [MW_G(TVD - L_{DC}) + P_I] \tag{3.92}$$

where

A = pressure at the top of the gas bubble for an un-weighted mud in psi

Step 7: Determine pressure at depth of interest in psi:

$$P_{Di} = \left(\frac{A}{2}\right) + \left(\frac{A^2}{4} + \left(\frac{(MW_G)(P_F)(Z)(T_{csg})(L_I)}{(Z)(T_{TVD})}\right)\right)^{\frac{1}{2}} \tag{3.93}$$

where

P_{Di} = pressure at depth of interest in psi
Z = super compressibility factor (1.0)
T_{csg} = Rankin temperature at casing shoe in °R

L_I = length of influx in ft
T_{TVD} = Rankin temperature at true vertical depth in °R

Step 8: Determine kill weight mud in ppg:

$$\text{KWM} = \left(\frac{\text{SIDPP}}{(0.052)(\text{TVD})}\right) + \text{MW} \tag{3.94}$$

where

KWM = kill mud weight in ppg

Step 9: Determine gradient of kill weight mud in psi/ft:

$$\text{KWM}_G = \text{KWM}(0.052) \tag{3.95}$$

where

KWM_G = kill mud weight gradient in psi/ft

Step 10: Determine the internal volume of the components of the drill string in bbl:

$$V_P = (L_P)(C_P) \tag{3.96}$$

where

V_P = volume of pipe in bbl
L_P = length of pipe in ft
C_P = capacity of pipe in bbl/ft

Step 11: Determine length that the drill string volume will occupy in the annulus in ft:

$$L_{DSV} = \left(\frac{V_P}{C_a}\right) \tag{3.97}$$

where

L_{DSV} = length of drill string volume in the annulus in ft
V_P = pipe volume in bbl
C_a = capacity of the annulus with the drill string in bbl/ft

Step 12: Determine the length of the influx around the drill pipe in ft:

$$L_{IDP} = \left(\frac{V_{pg}}{C_{aDS}}\right) \tag{3.98}$$

where

L_{IDP} = length of influx around drill pipe in ft
V_{pg} = volume of pit gain in bbl
C_{aDS} = capacity of annulus around drill string in bbl/ft

Step 13: Determine the pressure of the influx around the drill pipe when the influx approaches the casing shoe in psi:

$$P_{IDP} = (P_{GI})(L_{IDP}) \tag{3.99}$$

where

P_{IDP} = pressure of influx around drill pipe when the influx approaches the casing shoe in psi

Step 14: Determine the pressure at the top of the influx bubble (A_W) for a weighted mud in psi:

$$A_W = (P_F) - \left[\left((KWM_G)(TVD - L_{csg})\right) + P_{IDP}\right] + \left[(L_{DSV})(KWM_G - MW_G)\right] \tag{3.100}$$

where

A_W = pressure at the top of the gas bubble for a weighted mud in psi
L_{csg} = depth of casing shoe in ft

Example: Assumed conditions in a vertical well:

Well depth	= 10,000 ft
Surface casing	= 9⅝ in. @ 2500 ft
Casing ID	= 8.921 in.
Casing capacity	= 0.077 bbl/ft
Hole size	= 8½ in.
Drill pipe	= 4½ in., 16.6 lb/ft
Drill collar OD	= 6¼ in.
Drill collar ID	= 2¼ in.
Drill collar length	= 625 ft
Mud weight	= 9.6 ppg (0.4992 psi/ft)
Fracture gradient @ 2500 ft	= 0.73 psi/ft (14.0 ppg)
Geothermal gradient	= 0.012°F/ft

Mud volumes:

8½ in. hole	= 0.0702 bbl/ft
8½ in. hole × 4½ in. drill pipe	= 0.0505 bbl/ft
8½ in. hole × 6¼ in. drill collars	= 0.0322 bbl/ft
8.921 in. casing × 4½ in. drill pipe	= 0.0576 bbl/ft
Drill pipe capacity (ID = 3.826 in.)	= 0.0142 bbl/ft
Drill collar capacity (ID = 2.25 in.)	= 0.0049 bbl/ft
Super compressibility factor (Z)	= 1.0

The well kicks and the following information is recorded:

SIDPP	= 260 psi
SICP	= 500 psi
Pit gain	= 20 bbl

Determine the following:
(A) Maximum pressure at the casing shoe with the drillers method.
(B) Maximum pressure at the surface with the drillers method.
(C) Maximum pressure at the casing shoe with the wait and weight method.
(D) Maximum pressure at the surface with the wait and weight method.

(A) Determine the maximum pressure at the shoe with the drillers method.

Step 1: Determine the formation pressure in psi:

$$P_F = (260) + [(9.6)(0.052)(10{,}000)] = 5252 \text{ psi}$$

Step 2: Determine the length of influx at TD in ft:

$$L_I = \left(\frac{20}{0.0322}\right) = 621 \text{ ft}$$

Step 3: Determine the hydrostatic pressure exerted by the influx at TVD in psi:

$$P_I = (5252) - [0.4992(10{,}000 - 621) + 500] = 70 \text{ psi}$$

Step 4: Determine pressure gradient of influx at TVD in psi/ft:

$$P_{GI} = \left(\frac{70}{621}\right) = 0.1127 \text{ psi/ft}$$

Step 5: Determine the length of the influx in the annulus in ft (as the influx approaches the casing shoe):
(a) In the open hole/drill pipe annulus:

$$L_{IDP} = \left(\frac{20}{0.0505}\right) = 396 \text{ ft}$$

(b) In the casing/drill pipe annulus:

$$L_{IDP} = \left(\frac{20}{0.0576}\right) = 347.2 \text{ ft} \cong 347.0 \text{ ft}$$

Step 6: Determine the pressure of the influx around the drill pipe in psi:

$$P_{IDP} = (0.1127)(396) = 44.6 \text{ psi} \cong 45 \text{ psi}$$

Step 7: Determine temperature in Rankin (°R) at the TVD:

$$T_{10{,}000} = (70) + [(0.012)(10{,}000)] + (460) = 650^{\circ}\text{R}$$

Step 8: Determine temperature in Rankin (°R) at the casing shoe:

$$T_{2500} = (70) + [(0.012)(2500)] + (460) = 560^{\circ}\text{R}$$

Step 9: Determine the pressure at the top of the influx bubble

$$A = (5252) - [0.4992(10{,}000 - 2500) + 45] = 1463.0 \text{ psi}$$

Step 10: Determine maximum pressure at shoe with Drillers Method:

$$P_{2500} = \left(\frac{1463}{2}\right) + \left(\frac{1463^2}{4} + \left(\frac{(0.4992)(5252)(1.0)(560)(396)}{(1.0)(650)}\right)\right)^{\frac{1}{2}}$$
$$= 1927.0 \text{ psi}$$

(B) Determine the maximum pressure at the surface with the drillers method.

Step 1: Determine A:

$$A = (5252) - [0.4992(10{,}000) + 45] = 215 \text{ psi}$$

Step 2: Determine maximum pressure at surface with drillers method:

$$P_{\text{Surface}} = \left(\frac{215}{2}\right) + \left(\frac{215^2}{4} + \left(\frac{(0.4992)(5252)(1.0)(530)(347)}{(1.0)(650)}\right)\right)^{\frac{1}{2}}$$
$$= 975.5 \cong 976.0 \text{ psi}$$

Note: The temperature in the numerator is calculated with the depth at 0 ft as follows:

$$T_{\text{Surface}} = (70) + (460) = 530^{\circ}\text{R}.$$

(C) Determine the maximum pressure at the casing shoe with the wait and weight method.

Step 1: Determine the kill weight mud in ppg:

$$\text{KWM} = \left(\frac{260}{(0.052)(10{,}000)}\right) + 9.6 = 10.1 \text{ ppg}$$

Step 2: Determine gradient of kill weight mud in psi/ft:

$$\text{KWM}_{\text{G}} = 10.1(0.052) = 0.5252 \text{ psi/ft}$$

Step 3: Determine internal volume of the drill string:

$$V_{\text{DP}} = (0.0142)(10{,}000 - 625) = 133.1 \text{ bbl}$$

$$V_{\text{DC}} = (0.0049)(625) = 3.1 \text{ bbl}$$

$$V_{\text{DS}} = (133.1) + (3.1) = 136.2 \text{ bbl}$$

Step 4: Determine length that the drill string volume will occupy in the annulus in ft:

$$L_{\text{DSV}} = \left(\frac{136.2 - ((625)(0.0322))}{0.0505}\right) + (625) = 2923.5 \cong 2924 \text{ ft}$$

Step 5: Determine the pressure at the top of the influx bubble (A_W) for a weighted mud in psi:

$$\begin{aligned} A_W &= (5252) - [((0.5252)(10{,}000 - 2500)) + 45] \\ &\quad + [(2924)(0.5252 - 0.4992)] \\ &= 1344.0 \text{ psi} \end{aligned}$$

Step 6: Determine the maximum pressure at shoe with wait and weight method in psi:

$$\begin{aligned} P_{2500} &= \left(\frac{1345}{2}\right) + \left(\frac{(1345^2)}{4} + \left(\frac{(0.5252)(5252)(1.0)(560)(396)}{(1.0)(650)}\right)\right)^{\frac{1}{2}} \\ &= 1852.9 \cong 1853.0 \text{ psi} \end{aligned}$$

(D) Determine the maximum pressure at the surface with the wait and weight method.

Step 1: Determine A:

$$\begin{aligned} A &= (5252) - [((0.5252)(10{,}000)) + 45] + [(2924)(0.5252 - 0.4992)] \\ &= 31.0 \text{ psi} \cong 32.0 \text{ psi} \end{aligned}$$

Step 2: Determine the maximum pressure at surface with the wait and weight method in psi:

$$P_{\text{Surface}} = \left(\frac{31}{2}\right) + \left(\frac{(31^2)}{4} + \frac{(0.5252)(5252)(1.0)(530)(347)}{(1.0)(650)}\right)^{\frac{1}{2}} = 899.0 \text{ psi}$$

3.15.2 Summary of maximum pressures when circulating out a kick (Moore equations)

Case	Method	Location	Pressure, psi
A	Drillers	Casing shoe	1927
B	Drillers	Surface	976
C	Wait and weight	Casing shoe	1853
D	Wait and weight	Surface	899

Spreadsheet 3.15.1 & 3.15.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Max pressure circulating out a kick (Moore's equations).

3.16 Gas flow into the wellbore

3.16.1 Gas flow into the wellbore

Flow rate into the wellbore increases as wellbore depth through a gas sand increases:

$$Q = \frac{(0.007)(k)(P_d)(L)}{(\mu)\left(\ln\left(\frac{R_e}{R_w}\right)\right)(1440)} \tag{3.101}$$

where

Q = flow rate in bbl/min
k = permeability in millidarcy (md)
P_d = pressure differential in psi
L = length of section open to wellbore in ft
μ = viscosity of intruding gas in centipoise (cP)
R_e = radius of drainage in ft
R_w = radius of wellbore in ft

Example:

$$\begin{aligned} k &= 200\,\text{md} \\ P_d &= 624\,\text{psi} \\ L &= 20\,\text{ft} \\ \mu &= 0.3\,\text{cP} \\ R_e &= 31.5\,\text{ft} \\ R_w &= 4.25\,\text{in. (8½ in. hole)} \end{aligned}$$

$$Q = \frac{(0.007)(200)(624)(20)}{(0.3)\left(\ln\left(\frac{31.5}{4.25}\right)\right)(1440)} = 20.2\,\text{bbl/min}$$

Therefore: If 1 min is required to shut-in the well, a pit gain of 20.2 bbl occurs in addition to the gain incurred while drilling the 20 ft section.

Spreadsheet 3.16.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Gas flow into the wellbore.

3.16.2 Gas flow through a choke

$$Q_G = \left(\frac{(\mu)(P_{FL})}{(G_{SG})(T_{RG})}\right) \tag{3.102}$$

where

Q_G = gas flow rate in MCF/day at 60°F and atmospheric pressure (1 atm = 14.7 psi)

μ = coefficient of friction
P_{FL} = pressure in flowline ahead of the choke (gage reading plus 15 psi)
G_{SG} = gas-specific gravity
T_{RG} = temperature ahead of choke, Rankin (gas temperature in °F plus 460)

Table of coefficients	
Choke size, in.	**Coefficient, μ**
1/8	6.25
3/16	14.44
¼	26.51
5/16	43.64
3/8	61.21
7/16	85.13
1/2	112.72
5/8	179.74
3/4	260.99

Example: Calculate the volume of gas through a choke with the following data:

Flowline pressure ahead of choke = 1215 psi
Choke size = ½ in.
Gas specific gravity = 0.7
Temperature of gas = 540°R

$$Q_G = \left(\frac{(112.72)(1215)}{(0.7)(540)}\right) = 362.3 \text{ MCF/day}$$

Spreadsheet 3.16.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Gas flow through a choke.

3.17 Pressure analysis

3.17.1 Gas expansion equation

The pressure (psi) of a gas bubble times its volume (bbl) in one part of the hole equals its pressure times its volume in another. Disregards effects of temperature (T) and gas compressibility (z).

$$((P_1)(V_1)) = ((P_2)(V_2)) \tag{3.103}$$

where

P_1 = pressure of gas downhole in psi
V_1 = volume of gas downhole in bbl
P_2 = pressure of gas at surface in psi
V_2 = volume of gas at surface in bbl

Example: Use the following data:

$$P_1 = 1000\,\text{psi}$$
$$V_1 = 20\,\text{bbl}$$
$$P_2 = 50\,\text{psi}$$

$$V_2 = \frac{(1000)(20)}{50} = 400\ \text{bbl}$$

$$P_2 = \frac{(1000)(20)}{400} = 50\ \text{psi}$$

Spreadsheet 3.17.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Gas expansion.

3.17.2 Hydrostatic pressure exerted by each barrel of mud in the casing

Formula 1: With pipe in the wellbore:

$$P = \left(\frac{1029.4}{(D_h{}^2 - D_p{}^2)}\right)(0.052)(\text{MW}) \tag{3.104}$$

where

P = pressure of mud with pipe in the wellbore in psi/bbl

Example:

D_h = 9⅝ in. casing (43.5 lb/ft, 8.755 in. ID)
D_p = 5.0 in.
Mud weight = 10.5 ppg

$$P = \left(\frac{1029.4}{(8.755^2 - 5.0^2)}\right)(0.052)(10.5) = 10.9\ \text{psi/bbl}$$

Formula 2: With **NO** pipe in the wellbore:

$$P = \left(\frac{1029.4}{\text{ID}^2}\right)(0.052)(\text{MW}) \tag{3.105}$$

Example:

D_h = 9⅝ in. casing (43.5 lb/ft, 8.755 in. ID)
Mud weight = 10.5 ppg

$$P = \left(\frac{1029.4}{8.755^2}\right)(0.052)(10.5) = 7.3 \text{ psi/bbl}$$

Spreadsheet 3.17.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Hydrostatic pressure of mud in casing.

3.17.3 Surface pressure during drill stem test (DST)

Step 1: Determine formation pressure in psi:

$$P_F = (P_{Fe})(0.052)(\text{TVD}) \tag{3.106}$$

where

P_F = formation pressure in psi
P_{Fe} = formation pressure equivalent mud weight in ppg

Step 2: Determine oil hydrostatic pressure in psi:

$$P_O = (F_F)(0.052)(\text{TVD}) \tag{3.107}$$

where

P_O = hydrostatic pressure of oil in psi
F_F = oil specific gravity (SG)

Step 3: Determine surface pressure in psi:

$$P_S = P_F - P_O \tag{3.108}$$

where

P_S = surface pressure in psi

Example: Oil bearing sand at 12,500 ft with a formation pressure equivalent to 13.5 ppg. If the specific gravity of the oil is 0.5, what will be the static surface pressure during a drill stem test?

Step 1: Determine formation pressure in psi:

$$P_F = (13.5)(0.052)(12{,}500) = 8775 \text{ psi}$$

Step 2: Determine oil hydrostatic pressure in psi:

$$P_O = (0.5(8.34))(0.052)(12{,}500) = 2711 \text{ psi}$$

Step 3: Determine the surface pressure in psi:

$$P_S = 8775 - 2711 = 6064 \text{ psi}$$

Spreadsheet 3.17.3 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Surface pressure during a DST.

3.18 Stripping/snubbing calculations

3.18.1 Breakover point between stripping and snubbing

Example: Use the following data to determine the breakover point:
Data:

Mud weight	= 12.5 ppg
Drill collars	= 6¼ in. (2–13/16 in., 83 lb/ft)
Length of drill collars	= 276 ft
Drill pipe	= 5.0 in.
Drill pipe weight	= 19.5 lb/ft
Shut-in casing pressure	= 2400 psi
Buoyancy factor	= 0.8092
Depth to kick	= 5000 ft
Depth of bit	= 2000 ft
Rate of gas migration	= 230.5 ft/h
Stripping speed	= 500 ft/h

Step 1: Determine the force created by wellbore pressure on 6¼ in. drill collars in lb:

$$F_{WP} = \left(P^2\right)(0.7854)(\text{SICP}) \tag{3.109}$$

where

F_{WP} = force created by wellbore pressure in lb
P = drill pipe or drill collars in in.
SICP = shut-in casing pressure in psi

$$F = \left(6.25^2\right)(0.7854)(2400) = 73{,}631 \text{ lb}$$

Step 2: Determine the weight of the drill collars in lb:

$$W_{DC} = (W_c)(L_{DC})(BF) \tag{3.110}$$

where

W_{DC} = weight of the drill collars in lb
W_c = drill collar weight in lb/ft
L_{DC} = length of the drill collars in ft

$$W_{DC} = (83)(276)(0.8092) = 18{,}537 \text{ lb}$$

Step 3: Additional weight required from drill pipe in lb:

$$W_{DP} = F_{WP} - W_{DC} \tag{3.111}$$

where

W_{DP} = weight of drill pipe in lb
F_{WP} = force created by wellbore pressure in lb

$$W_{DP} = 73{,}631 - 18{,}537 = 55{,}094 \text{ lb}$$

Step 4: Length of drill pipe required to reach breakover point in ft:

$$L_{DP} = \frac{W_{DP}}{((W_{dp})(BF))} \tag{3.112}$$

where

L_{DP} = length of drill pipe required to reach breakover point in ft
W_{dp} = drill pipe weight in lb/ft

$$L_{DP} = \frac{55{,}094}{((19.5)(0.8092))} = 3492 \text{ ft}$$

Step 5: Length of drill string required to reach breakover point in ft:

$$L_{DS} = L_{DC} + L_{DP} \tag{3.113}$$

where

L_{DS} = length of drill string in ft

$$L_{DS} = 276 + 3492 = 3768 \text{ ft}$$

Step 6: Time to kick penetration in hours:

$$T_{KP} = \frac{((D_K) - (D_B))}{((R_{GM}) + (S_S))} \tag{3.114}$$

where

T_{KP} = time to penetrate the kick when stripping in h
D_K = depth of kick in ft

D_B = depth of bit in ft
R_{GM} = rate of kick migration in ft/h
S_S = stripping speed in ft/h

$$T_{KP} = \frac{((5000) - (2000))}{((320.5) + (500))} = 3.7 \text{ h}$$

Spreadsheet 3.18.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Stripping—Snubbing calculations.

3.18.2 Minimum surface pressure before stripping is possible

$$P_{MS} = \frac{W_{DCS}}{A_{DC}} \tag{3.115}$$

where

P_{MS} = minimum surface pressure in psi
W_{DCS} = weight of one stand of drill collars in lb
A_{DC} = area of drill collar in in.^{2}

Example:

Drill collars = 8.0 in. OD × 3.0 in. ID (147 lb/ft)
Length of one stand = 92 ft

$$P_{MS} = \frac{(147)(92)}{(8^2)(0.7854)} = 269 \text{ psi}$$

Spreadsheet 3.18.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Minimum surface pressure before stripping.

3.18.3 Height gain from stripping into influx

$$H_G = \frac{\left((L_{SP})(C_{dp} + D_P)\right)}{C_a} \tag{3.116}$$

where

H_G = height gain from stripping into the influx in ft
L_{SP} = length of pipe stripped in ft

C_{dp} = capacity of drill pipe, drill collars, or tubing in bbl/ft
D_p = displacement of drill pipe, drill collars, or tubing in bbl/ft
C_a = annular capacity in bbl/ft

Example: If 300 ft of 5.0 in. drill pipe (19.5 lb/ft) is stripped into an influx in a 12¼ in. hole, determine the height gained in ft:

Data:

Drill pipe capacity	= 0.0178 bbl/ft
Drill pipe displacement	= 0.0065 bbl/ft
Length drill pipe stripped	= 300 ft
Annular capacity	= 0.1215 bbl/ft

$$H_G = \frac{((300)(0.0178 + 0.0065))}{0.1215} = 60.0 \text{ ft}$$

Spreadsheet 3.18.3 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Height gain due to stripping into influx.

3.18.4 Casing pressure increase from stripping into influx

$$P_{Ci} = (H_G)(MW_G - I_G) \tag{3.117}$$

where

P_{Ci} = casing pressure increase from stripping into influx in psi
I_G = influx gradient in psi/ft

Example:

Gain in height	= 60.0 ft
Mud weight gradient ((12.5 ppg)(0.052))	= 0.65 psi/ft
Influx gradient	= 0.12 psi/ft

$$P_{Ci} = (60.0)(0.65 - 0.12) = 31.8 \cong 32 \text{ psi}$$

Spreadsheet 3.18.4 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Casing pressure increase due to stripping into influx.

3.18.5 Volume of mud that must be bled to maintain constant bottom hole pressure with a gas bubble rising

Formula 1: With pipe in the hole:

$$V_{Mp} = \frac{(P_d)(C_a)}{MW_G} \tag{3.118}$$

where

V_{Mp} = volume of mud that must be bled to maintain CBHP with a gas bubble rising with pipe in the hole in bbl
P_d = incremental pressure steps that the casing pressure will be allowed to increase in psi
C_a = capacity of annulus in bbl/ft
MW_G = mud weight gradient in psi/ft

Example:

Casing pressure increase per step = 100 psi
Mud weight gradient = 0.702 psi/ft
((13.5 ppg) (0.052))
Annular capacity = 0.1215 bbl/ft
(D_h = 12¼ in.; D_p = 5.0 in.)

$$V_M = \frac{(100)(0.1215)}{0.702} = 17.3 \text{ bbl}$$

Formula 2: With no pipe in hole:

$$V_M = \frac{(P_d)(C_{h/p})}{MW_G} \tag{3.119}$$

where

V_M = volume of mud that must be bled to maintain CBHP with a gas bubble rising with **NO** pipe in the hole in bbl
$C_{h/p}$ = capacity of the hole or casing ID in bbl/ft

Example:

Casing pressure increase per step = 100 psi
Mud weight gradient (13.5 ppg) = 0.702 psi/ft
Hole capacity (12¼ in.) = 0.1458 bbl/ft

$$V_M = \frac{(100)(0.1458)}{0.702} = 20.8 \text{ bbl}$$

Spreadsheet 3.18.5 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Volume of mud to bleed to maintain CBHP.

3.18.6 Maximum allowable surface pressure (MASP) governed by the formation

$$\text{MASP} = (\text{MW}_\text{M} - \text{MW})(0.052)\left(\text{TVD}_\text{csg}\right) \tag{3.120}$$

where

MASP = maximum allowable surface pressure in psi
MW_M = maximum allowable mud weight in ppg
MW = mud weight in use in ppg

Example:

Maximum allowable mud weight = 15.0 ppg (from leak-off test data)
Mud weight = 12.0 ppg
Casing seat TVD = 8000 ft

$$\text{MASP} = (15.0 - 12.0)(0.052)(8000) = 1248.0 \text{ psi}$$

Spreadsheet 3.18.6 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 MASP controlled by formation pressure.

3.18.7 Maximum allowable surface pressure (MASP) governed by casing burst pressure

$$\text{MASP}_\text{CB} = ((P_\text{CB})(\text{SF})) - ((\text{MW}) - (\text{MW}_\text{OC}))(0.052)\left(\text{TVD}_\text{csg}\right) \tag{3.121}$$

where

MASP_CB = maximum allowable surface pressure governed by casing burst pressure in psi
P_CB = casing burst pressure in psi
SF = safety factor
MW_OC = mud weight outside of casing in ppg

Example:

Casing = 10¾ in. (51 lb/ft, N-80)
Casing burst pressure = 6070 psi
Casing setting depth = 8000 ft
Mud weight outside casing = 9.4 ppg
Mud weight = 12.0 ppg
Casing safety factor = 80%

$$\text{MASP}_{\text{CB}} = ((6070)(0.80)) - ((12.0) - (9.4))(0.052)(8000) = 3774.4 \text{ psi}$$
$$\cong 3774 \text{ psi}$$

Spreadsheet 3.18.7 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 MASP controlled by casing burst pressure.

3.19 Subsea considerations

3.19.1 Casing pressure decrease when bringing well on choke

When bringing the well on choke with a subsea stack, the casing pressure (annulus pressure) must be allowed to decrease by the amount of choke line pressure loss (friction pressure):

$$P_{\text{csgR}} = \text{SICP} - P_{\text{CL}} \tag{3.122}$$

where

P_{csgR} = reduced casing pressure in psi
SICP = shut-in casing pressure in psi
P_{CL} = choke line pressure loss in psi

Example:

Shut-in casing pressure (SICP) = 800 psi
Choke line pressure loss = 300 psi

$$P_{\text{csgR}} = 800 - 300 = 500 \text{ psi}$$

Spreadsheet 3.19.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Casing pressure decrease when bringing well on choke.

3.19.2 Pressure chart for bringing well on choke

The pressure/stroke relationship is not linear. When bringing the well on choke, to maintain a constant bottom hole pressure, the following chart should be used:

Pressure chart

	Strokes	Pressure
Line 1: Reset stroke counter to "0" =	0	
Line 2: 1/2 stroke rate = 50 × 0.5 =	25	
Line 3: 3/4 stroke rate = 50 × 0.75 =	38	
Line 4: 7/8 stroke rate = 50 × 0.875 =	44	
Line 5: Kill rate speed =	50	

Strokes side: Kill rate speed = 50 spm

Pressure side: Shut-in casing pressure (SICP) = 800 psi
Choke line pressure loss = 300 psi

Divide choke line pressure loss (P_{CL}) by 4, because there are four steps on the chart:

$$P_{Line} = \frac{P_{CL}}{4} \tag{3.123}$$

where

P_{Line} = choke line pressure loss for pressure chart in psi

$$P_{Line} = \frac{300}{4} = 75 \text{ psi/Line}$$

Pressure chart

	Strokes	Pressure
Line 1: Shut-in casing pressure, psi =	0	800
Line 2: Subtract 75 psi from Line 1 =	25	725
Line 3: Subtract 75 psi from Line 1 =	38	650
Line 4: Subtract 75 psi from Line 1 =	44	575
Line 5: Reduced casing pressure =	50	500

Spreadsheet 3.19.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Pressure chart for bringing well on choke.

3.19.3 Maximum allowable mud weight for subsea stack as derived from leak-off test data

$$\text{MAMW} = \frac{(P_{LOT})}{(0.052)(\text{TVD}_{csg})} + \text{MW} \tag{3.124}$$

where

MAMW = maximum allowable mud weight in ppg
P_{LOT} = leak-off test pressure in psi
TVD_{csg} = casing shoe depth from RKB in ft

Example:

Leak-off test pressure = 800 psi
TVD from rotary bushing to casing shoe = 4000 ft
Mud weight = 9.2 ppg

$$\text{MAMW} = \frac{(800)}{(0.052)(4000)} + 9.2 = 13.0 \text{ ppg}$$

Spreadsheet 3.19.3 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Max MW for subsea stack from LOT.

3.19.4 Casing burst pressure—Subsea stack

Step 1: Determine the internal yield pressure of the casing from the "Dimensions and Strengths" section in a cement company's service handbook.

Step 2: Correct internal yield pressure for safety factor in psi. Some operators use 80%; some use 75%; and others use 70%:

$$\gamma_{IC} = (\gamma_I)(SF) \tag{3.125}$$

where

γ_{IC} = correct internal yield pressure of casing in psi
γ_I = internal yield pressure of casing from reference book in psi
SF = safety factor

Step 3: Determine the hydrostatic pressure of the mud in use in psi:

Note: The depth is from the rotary Kelly bushing (RKB) to the mud line and includes the air gap plus the depth of seawater.

$$HP_M = (MW)(0.052)(TVD_{WD} + L_{AG}) \tag{3.126}$$

where

HP_M = hydrostatic pressure of the mud in psi
TVD_{ML} = total vertical depth from RKB to mud line in ft
L_{AG} = length of the air gap in ft

Step 4: Determine the hydrostatic pressure exerted by the seawater in psi:

$$HP_{SW} = (MW_{SW})(0.052)(TVD_{WD}) \tag{3.127}$$

where

HP_{SW} = hydrostatic pressure of seawater in psi
MW_{SW} = mud weight of seawater in ppg

TVD_{WD} = total vertical depth of seawater in ft

Step 5: Determine casing burst pressure in psi:

$$P_{CB} = (\gamma_{IC} - HP_M) + (HP_{SW}) \tag{3.128}$$

where

P_{CB} = casing burst pressure at the stack in psi

Example: Determine the casing burst pressure in psi with a subsea stack, using the following data:

Data:

Mud weight	= 10.0 ppg
Weight of seawater	= 8.7 ppg
Air gap	= 50 ft
Water depth	= 1500 ft
Safety factor	= 80%

Step 1: Determine the internal yield pressure of the casing from the "Dimension and Strengths" section of a cement company handbook:

$$9 - 5/8'' \text{ casing (C} - 75, 53.5 \text{ lb/ft)}$$

$$\text{Internal yield pressure} = 7430 \text{ psi}$$

Step 2: Correct internal yield pressure with the safety factor:

$$\gamma_{IC} = (7430)(0.80) = 5944 \text{ psi}$$

Step 3: Determine the hydrostatic pressure exerted by the mud in use in psi:

$$HP_M = (10.0)(0.052)(1500 + 50) = 806 \text{ psi}$$

Step 4: Determine the hydrostatic pressure exerted by the seawater in psi:

$$HP_{SW} = (8.7)(0.052)(1500) = 679 \text{ psi}$$

Step 5: Determine the casing burst pressure in psi:

$$P_{CB} = (5944 - 806) + (679) = 5817 \text{ psi}$$

Spreadsheet 3.19.4 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Casing burst pressure—Subsea stack.

3.19.5 Calculate choke line pressure loss in psi

$$P_{CL} = \frac{(0.000061)(MW)(L_{CL})\left(Q^{1.86}\right)}{\left(ID_{CL}{}^{4.86}\right)} \tag{3.129}$$

where

P_{CL} = choke line pressure loss in psi
L_{CL} = length of the choke line in ft
Q = flow rate in the choke line in gpm
ID_{CL} = internal diameter of the choke line in in.

Example: Determine the choke line pressure loss in psi, using the following data:

Data:

Mud weight = 14.0 ppg
Choke line length = 2000 ft
Circulation rate = 225 gpm
Choke line ID = 2.5 in.

$$P_{CL} = \frac{(0.000061)(14.0)(2000)\left(225^{1.86}\right)}{\left(2.5^{4.86}\right)} = 471.6 \text{ psi} \cong 472 \text{ psi}$$

Spreadsheet 3.19.5 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Choke line pressure loss.

3.19.6 Velocity through the choke line in ft/min

$$V_{CL} = \frac{24.5(Q)}{\left(ID_{CL}{}^2\right)} \tag{3.130}$$

where

V_{CL} = velocity through the choke line in ft/mi

Example: Determine the velocity through the choke line in ft/min using the following data:

Data:

Circulation rate = 225 gpm
Choke line ID = 2.5 in.

$$V_{CL} = \frac{24.5(225)}{\left(2.5^2\right)} = 882 \text{ ft/min}$$

Spreadsheet 3.19.6 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Velocity through the choke line.

3.19.7 Adjusting choke line pressure loss for a higher mud weight in ppg

$$P_{CLN} = \frac{(MW_N)(P_{CL})}{MW_O} \tag{3.131}$$

where

P_{CLN} = adjusted new choke line pressure loss in psi
MW_N = new mud weight in ppg
MW_O = old mud weight in ppg

Example: Use the following data to determine the new estimated choke line pressure loss in psi:

Data:

Old mud weight = 13. 5 ppg
New mud weight = 15.0 ppg
Old choke line pressure loss = 300 psi

$$P_{CLN} = \frac{(15.0)(300)}{13.5} = 333.3 \text{ psi} \cong 333 \text{ psi}$$

Spreadsheet 3.19.7 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Adjusting choke line pressure loss for a higher MW.

3.19.8 Minimum conductor casing setting depth in ft

$$D_C = \frac{((MW)(0.052)(D_W + L_{AG})) - ((D_W)((MW_{SW})(0.052)))}{((F_{FG}) - ((MW)(0.052)))} \tag{3.132}$$

where

D_C = depth of conductor below mud line in ft
MW = mud weight in ppg
D_W = depth of water in ft
L_{AG} = length of air gap in ft

MW_{SW} = mud weight of sea water in ppg
F_{FG} = formation fracture gradient in psi/ft

Example: Using the following data, determine the minimum setting depth of the conductor casing below the seabed:

Data:

Water depth	**= 450 ft**
Sea water mud weight	= 8.55 ppg (0.445 psi/ft)
Air gap	= 60 ft
Formation fracture gradient	= 0.68 psi/ft
Mud weight	= 9.0 ppg (to be used while drilling this interval)

$$D_C = \frac{((9.0)(0.052)(450 + 60)) - ((450)((8.55)(0.052)))}{((0.68) - ((9.0)(0.052)))} = 181.3 \text{ ft}$$

Spreadsheet 3.19.8 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Minimum conductor casing setting depth.

3.19.9 Maximum mud weight with returns back to rig floor in ppg

Step 1: Determine total pressure at casing seat in psi:

$$P_{csg} = (\text{FG})(L_C - \text{TVD}_{WD} - L_{AG}) + ((G_{SW})(\text{TVD}_{WD})) \tag{3.133}$$

where

P_{csg} = pressure at the casing shoe in psi

Step 2: Determine maximum mud weight in ppg:

$$\text{MW}_M = \frac{P_{csg}}{((0.052)(\text{TVD}_C))} \tag{3.134}$$

where

MW_M = maximum mud weight in ppg

Example: Using the following data, determine the maximum mud weight that can be used with returns back to the rig floor:

Data:

Air gap	= 75 ft
Water depth	= 600 ft
Conductor casing	= 1225 ft RKB
Seawater gradient	= 0.445 psi/ft
Fracture gradient	= 0.62 psi/ft

Step 1: Determine total pressure at casing seat in psi:

$$P_{csg} = (0.62)(1225 - 600 - 75) + ((0.445)(600)) = 608.0 \text{ psi}$$

Step 2: Determine maximum mud weight in ppg:

$$MW_M = \frac{608}{((0.052)(1225))} = 9.5 \text{ ppg}$$

Spreadsheet 3.19.9 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Max MW with returns back to rig floor.

3.19.10 Reduction in bottom hole pressure if riser is disconnected in psi

Step 1: Determine bottom hole pressure in psi:

$$P_{BH} = (MW)(0.052)(TVD_{TD}) \tag{3.135}$$

where

P_{BH} = bottom hole pressure in psi

Step 2: Determine bottom hole pressure with Riser disconnected in psi:

$$P_{BHRD} = ((G_{SW})(L_{WD})) + (MW)(0.052)((TVD_{TD}) - (L_{WD}) - (L_{AG})) \tag{3.136}$$

where

P_{BHRD} = bottom hole pressure with riser disconnected in psi

Step 3: Determine bottom hole pressure reduction in psi:

$$P_{BHR} = (P_{BH}) - (P_{BHRD}) \tag{3.137}$$

where

P_{BHR} = bottom hole pressure reduction in psi

Example: Use the following data and determine the reduction in bottom hole pressure if the riser is disconnected:

Data:

Air gap	= 75 ft
Water depth	= 700 ft
Seawater gradient	= 0.445 psi/ft
Well depth	= 2020 ft RKB
Mud weight	= 9.0 ppg

Step 1: Determine bottom hole pressure in psi:

$$P_{BH} = (9.0)(0.052)(2020) = 945.4 \text{ psi}$$

Step 2: Determine bottom hole pressure with Riser disconnected in psi:

$$P_{BHRD} = ((0.445)(700)) + (9.0)(0.052)((2020) - (700) - (75)) = 894.2 \text{ psi wi}$$

Step 3: Determine bottom hole pressure reduction in psi:

$$P_{BHR} = (945.4) - (894.2) = 51.2 \text{ psi}$$

Spreadsheet 3.19.10 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Reduction in BHP with riser disconnect.

3.19.11 Bottom hole pressure when circulating out a kick in psi

Example: Use the following data and determine the bottom hole pressure when circulating out a Kick in psi:

Data:

Total depth—RKB	= 13,500 ft
Length of gas kick in casing	= 1200 ft
Gas gradient	= 0.12 psi/ft
Original mud weight	= 12.0 ppg
Kill weight mud	= 12.7 ppg
Pressure loss in annulus	= 75 psi
Choke line pressure loss	= 200 psi
Air gap	= 75 ft
Water depth	= 1500 ft
Annulus (casing) pressure	= 630 psi
Original mud in casing below gas	= 5500 ft

Step 1: Determine the hydrostatic pressure in choke line in psi:

$$P_{\mathrm{HCL}} = (\mathrm{MW_O})(0.052)((\mathrm{TVD_{WD}}) + (L_{\mathrm{AG}})) \tag{3.138}$$

where

P_{HCL} = hydrostatic pressure in the choke line in psi
M_{WO} = original mud weight in ppg
$\mathrm{TVD_{WD}}$ = water depth in ft
L_{AG} = length of the air gap in ft

$$P_{\mathrm{HCL}} = (12.0)(0.052)((1500) + (75)) = 982.8 \text{ psi}$$

Step 2: Determine the hydrostatic pressure exerted by gas influx in psi:

$$P_{\mathrm{HI}} = (G_{\mathrm{I}})(L_{\mathrm{I}}) \tag{3.139}$$

where

P_{HI} = hydrostatic pressure of influx in psi
G_{I} = gas gradient in psi/ft
L_{I} = length of influx in ft

$$P_{\mathrm{HI}} = (0.12)(1200) = 144.0 \text{ psi}$$

Step 3: Determine the hydrostatic pressure of original mud below gas influx in psi:

$$P_{\mathrm{HOM}} = (\mathrm{MW_O})(0.052)(\mathrm{L_{BI}}) \tag{3.140}$$

where

P_{HOM} = hydrostatic pressure of original mud below gas influx in psi
L_{BI} = length of mud below influx in ft

$$P_{\mathrm{HOM}} = (12.0)(0.052)(5500) = 3432 \text{ ft}$$

Step 4: Determine the hydrostatic pressure of the kill weight mud in psi:

$$P_{\mathrm{HKWM}} = (\mathrm{KWM})(0.052)((\mathrm{TVD_{TD}}) - (L_{\mathrm{BI}}) - (L_{\mathrm{I}}) - (\mathrm{TVD_{WD}}) - (L_{\mathrm{AG}})) \tag{3.141}$$

where

P_{HKWM} = hydrostatic pressure of the kill weight mud in psi

$$P_{\mathrm{HKWM}} = (12.7)(0.052)((13{,}500) - (5500) - (1200) - (1500) - (75)) = 3450.6 \text{ psi}$$

Step 5: Summary

Bottom hole pressure while circulating out a kick:	
Pressure in choke line	982.8 psi
Pressure of gas influx	144.0 psi
Original mud below gas in casing	3432.0 psi
Kill weight mud	3450.6 psi
Annulus (casing) pressure	630.0 psi
Choke line pressure loss	200.0 psi
Annular pressure loss	+75.0 psi
Bottom hole pressure while circulating out a kick	8914.4 psi

Spreadsheet 3.19.11 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 BHP circulating out a kick.

3.20 Workover operations

Note: The following procedures and calculations are more commonly used in workover operations, but at times they are used in drilling operations.

3.20.1 Bullheading

Bullheading is a term used to describe killing the well by forcing formation fluids back into the formation. This involves pumping kill weight fluid down the tubing and, in some cases, down the casing.

The bullheading method of killing a well is primarily used in the following situations:

(a) Tubing in the well with a packer set. No communication exists between tubing and annulus.
(b) Tubing in the well, influx in the annulus, and for some reason, cannot circulate through the tubing.
(c) No tubing in the well. Influx in the casing. Bullheading is simplest, fastest, and safest method to use to kill the well.

Note: Tubing could be off bottom also.
(d) In drilling operations, bullheading has been used successfully in areas where hydrogen sulfide is a possibility.

Example: Calculations involved in bullheading operations:

Using the information given below, calculations will be performed to kill the well by bullheading. The example calculations will pertain to "a" above:

Data:

Depth of perforations	= 6480 ft
Fracture gradient	= 0.862 psi/ft
Formation pressure gradient	= 0.401 psi/ft
Tubing hydrostatic pressure	= 326 psi
Shut-in tubing pressure	= 2000 psi
Tubing size	= 2⅞ in. (6.5 lb/ft)
Tubing capacity	= 0.00579 bbl/ft
Tubing internal yield pressure	= 7260 psi
Kill fluid density	= 8.4 ppg

Note: Determine the best pump rate to use. The pump rate must exceed the rate of gas bubble migration up the tubing. The rate of gas bubble migration, ft/h, in a shut-in well can be determined by the following formula:

$$M_{\text{gas}} = \left(\frac{P_{\text{HI}}}{\text{CBF}_{\text{G}}}\right) \tag{3.142}$$

where

M_{gas} = rate of gas migration up hole in ft/h
P_{HI} = increase in surface pressure per hour in psi
CBF_{G} = completion brine fluid gradient in psi/ft

Step 1: Calculate the maximum allowable tubing (surface) pressure (MATP) for formation fracture in psi:

(a) Determine the initial maximum allowable tubing pressure initial with influx in the tubing in psi:

$$\text{MATP}_{\text{I}} = ((\text{FG}_{\text{G}})(\text{TVD}_{\text{P}})) - (\text{HP}_{\text{T}}) \tag{3.143}$$

where

MATP_{I} = maximum allowable tubing (surface) pressure in psi
FG_{G} = fracture gradient in psi/ft
HP_{T} = hydrostatic pressure of tubing in psi

Example: Use the data listed above:

$$\text{MATP}_{\text{I}} = ((0.862)(6480)) - (326) = 5260 \text{ psi}$$

(b) Determine the final maximum allowable tubing pressure with kill fluid in tubing in psi:

$$\mathrm{MATP_F} = ((\mathrm{FG_G})(\mathrm{TVD_P})) - (\mathrm{HP_T}) \tag{3.144}$$

where

$\mathrm{MATP_F}$ = final maximum allowable tubing pressure in psi

Example: Use the data listed above:

$$\mathrm{MATP_F} = ((0.862)(6480)) - ((8.4)(0.052)(6480)) = 2756 \text{ psi}$$

Step 2: Determine tubing capacity in bbl:

$$C_T = (L_T)(C_T) \tag{3.145}$$

where

C_T = tubing capacity in bbl
L_T = length of tubing in ft
C_T = capacity of tubing in bbl/ft

Example: Use the data listed above:

$$C_T = (6480)(0.00579) = 37.5 \text{ bbl}$$

Plot these values as shown in Fig. 3.8.

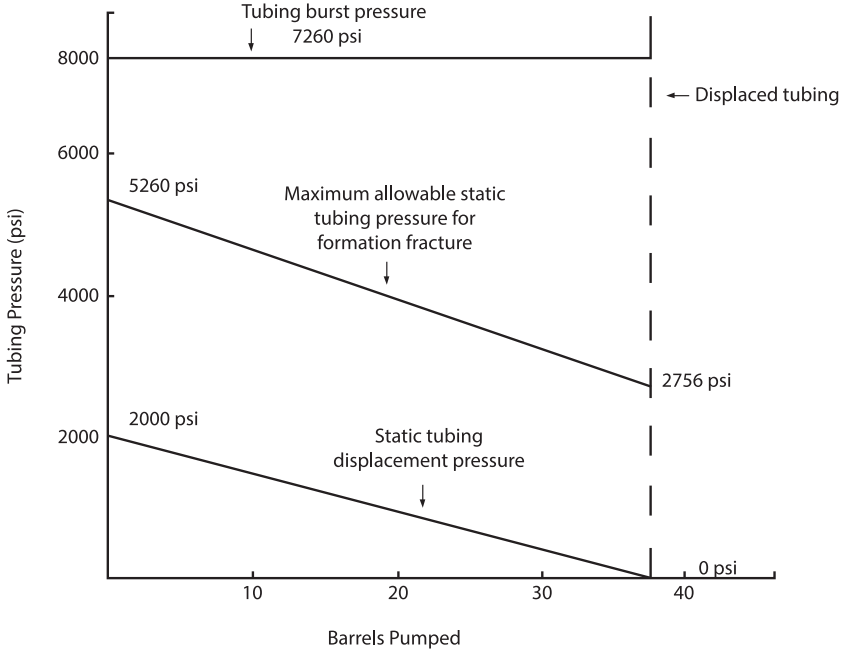


FIG. 3.8 Tubing pressure profile.

Spreadsheet 3.20.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Bullheading.

3.20.2 Lubricate and bleed

The lubricate and bleed method involves alternately pumping a kill fluid into the tubing (or into the casing if there is no tubing in the well), allowing the kill fluid to fall, then bleeding off a volume of gas until kill fluid reaches the choke. As each volume of kill fluid is pumped into the tubing, the SITP should decrease by a calculated value until the well is eventually killed.

This method is often used for two reasons: (1) shut-in pressures approach the rated working pressure of the wellhead or tubing and dynamic pumping pressure may exceed the limits, as in the case of bullheading, and (2) either to completely kill the well or lower the SITP to a value where other kill methods can be safely employed without exceeding rated limits.

This method can also be applied when the wellbore or perforations are plugged, rendering bullheading useless. In this case, the well can be killed without the use of tubing or snubbing small diameter tubing.

Users should be aware that the Lubricate and Bleed Method is often a very time consuming process, whereas another method may kill the well more quickly. The following is an example of a typical lubricate and bleed kill procedure.

Example: A workover is planned for a well where the SITP approaches the working pressure of the wellhead equipment. To minimize the possibility of equipment failure, the lubricate and bleed method will be used to reduce the SITP to a level at which bullheading can be safely conducted. The data below will be used to describe this procedure:

Step 1: Calculate the tubing capacity to the perforations in bbl:

$$C_{TP} = (C_T)(D_{perf}) \tag{3.146}$$

where

C_{TP} = capacity of the tubing to the perforations in bbl
C_T = capacity of the tubing in bbl/ft
D_{perf} = depth of the perforations in ft

Step 2: Calculate the expected pressure reduction for each barrel of kill fluid pumped in psi:

$$P_{RT} = (C_{TF})(0.052)(\text{KWF}) \tag{3.147}$$

where

P_{RT} = pressure reduction in the tubing for each barrel of kill fluid pumped in psi/bbl
C_{TF} = capacity of the tubing in ft/bbl
KWF = kill weight fluid in ppg

Step 3: Determine the pressure that can be bled off after lubricating the selected volume of fluid in psi:

$$P_B = (V_L)\left(\frac{(0.052)(MW)}{C_T}\right) \tag{3.148}$$

where

P_B = pressure to bleed off in psi
V_L = volume to lubricate into tubing in bbl
MW = fluid weight in ppg
C_T = capacity of tubing in bbl/ft

Example: Determine the amount of pressure that can be bled off the casing while lubricating the tubing:

Data:

TVD	= 6500 ft
Depth of perforations	= 6450 ft
SITP	= 2830 psi
Tubing size	= 2–7/8 in. (6.5 lb/ft, N-80)
Tubing capacity	= 0.0058 bbl/ft
Tubing capacity	= 172.8 ft/bbl
Tubing internal yield	= 10,570 psi
Wellhead working pressure	= 3000 psi
Kill weight fluid	= 9.0 ppg
Volume lubricated	= 0.5 bbl

Step 1: Calculate the tubing capacity to the perforations in bbl:

$$C_{TP} = (0.0058)(6450) = 37.4 \text{ bbl}$$

Step 2: Calculate the expected pressure reduction for each barrel of kill fluid pumped in psi:

$$P_{RT} = (172.8)(0.052)(9.0) = 80.85 \text{ psi/bbl} \cong 81.0 \text{ psi/bbl}$$

Step 3: Determine the pressure that can be bled off after lubricating the selected volume of fluid in psi:

$$P_B = (0.5)\left(\frac{(0.052)(9.0)}{0.0058}\right) = 40.3 \text{ psi}$$

Procedure: (for the example conditions).

(1) Rig up all surface equipment including pumps and gas flare lines.
(2) Record SITP and SICP.
(3) Open the choke to allow gas to escape from the well and momentarily reduce the SITP.
(4) Close the choke and pump in 9.0 ppg brine until the tubing pressure reaches 2830 psi.
(5) Wait for a period of time to allow the brine to fall in the tubing. This period will range from ¼ to 1 h depending on gas density, pressure, and tubing size.
(6) Open the choke and bleed gas until 9.0 ppg brine begins to escape.
(7) Close the choke and pump in 9.0 ppg brine water.
(8) Continue the process until a low level, safe working pressure is attained.

A certain amount of time is required for the kill fluid to fall down the tubing after the pumping stops. The actual waiting time is needed not to allow fluid to fall, but rather, for gas to migrate up through the kill fluid. Gas migrates at rates of 1000 to 2000 ft/h. Therefore, considerable time is required for fluid to fall or migrate to 6500 ft. Therefore, after pumping, it is important to wait several minutes before bleeding gas to prevent bleeding off kill fluid through the choke.

Spreadsheet 3.20.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Lubricate and bleed.

3.21 Controlling gas migration

Gas migration can occur when a well is shut in on a gas kick. It is indicated by a uniform increase in both the SIDPP and SICP. If the influx is allowed to migrate without expanding, pressures will increase everywhere in the well-bore. If it is ignored it can cause formation damage and mud losses. The worst-case scenario would be an underground blowout.

Gas migration can be controlled using two methods:

- Drill pipe pressure method
- Volumetric method

3.21.1 Drill pipe pressure method

This is a constant bottom hole pressure method of well control, and it is the simplest method. In order to use this method, the bit must be on bottom with no float in the string.

Procedure:

(1) Allow the SIDPP to increase by a safety margin: 50–100 psi. This is the lower limit. The SIDPP must not be allowed to decrease below this level.

(2) Next, allow the drill pipe pressure to further increase by another 50–100 psi. This is the upper limit.

(3) Open the choke and bleed fluid out of the well until the drill pipe pressure drops to the lower limit.

(4) Repeat steps 2 and 3 until circulation is possible or the gas migrates to the top of the well.

3.21.2 Volumetric method to control gas migration

Procedure:

(1) Select a safety margin, P_s, and a working pressure, P_w. (Recommended: $P_s = 100$ psi; $P_w = 100$ psi.)

(2) Calculate the hydrostatic pressure per barrel of mud, Hp/bbl:

$$P_H = \frac{(G_M)}{C_a} \tag{3.149}$$

where

P_H = hydrostatic pressure per barrel of mud in psi/bbl
G_M = mud gradient in psi/ft
C_a = capacity of the annulus in bbl/ft

(3) Calculate the volume to bleed each cycle: volume, bbl to bleed each cycle = P_w/Hp/bbl

(4) Allow the shut in casing pressure to increase by P_s without bleeding from the well.

(5) Allow the shut in casing pressure to further increase by P_w without bleeding from the well.

(6) Maintain casing pressure constant by bleeding small volumes of mud from the well until total mud bled equals the correct volume to bleed per cycle.

(7) Repeat steps 5 and 6 until another procedure is implemented or all gas is at the surface.

3.22 Gas lubrication

Gas lubrication is the process of removing gas from beneath the BOP stack while maintaining constant bottom hole pressure. Lubrication is best suited for surface stacks, but the dynamic gas lubrication procedure can be used to vent gas from beneath a subsea stack.

Lubrication can be used to reduce pressures or to remove gas from beneath a surface stack prior to stripping or after implementing the volumetric procedure for controlling gas migration. The volume of mud lubricated into the well must be accurately measured.

3.22.1 Gas lubrication—Volume method

Procedure:

(1) Select a range of working pressure, P_w in psi. (Recommended $P_w = 100–200$ psi.)

(2) Calculate the hydrostatic pressure increase in the upper annulus per bbl of lube mud:

$$P_H = \frac{G_M}{C_a} \tag{3.150}$$

where

P_H = hydrostatic pressure per barrel of mud in psi/bbl
G_M = mud gradient in psi/ft
C_a = capacity of the annulus in bbl/ft

(3) Pump lube mud through the kill line to increase the casing pressure by the working pressure range, P_w.

(4) Measure the trip tank and calculate the hydrostatic pressure increase of the mud lubricated for this cycle.

(5) Wait 10 to 30 min for the mud to lubricate through the gas.

(6) Bleed "dry" gas from the choke to reduce the casing pressure by the hydrostatic pressure increase plus the working pressure range.

(7) Repeat steps 3 through 6 until lubrication is complete.

3.22.2 Gas lubrication—Pressure method

Because of its simplicity, the pressure method is the preferred method of gas lubrication. However, it is only applicable when the mud weight being lubricated is sufficient to kill the well, as in the case of a swabbed influx. The pressure method is also the only accurate method whenever the formation is "taking" fluid, as is the case for most completed wellbores and whenever seepage loss is occurring.

The pressure method of gas lubrication utilizes the following formula:

$$P_3 = \frac{{P_1}^2}{P_2} \tag{3.151}$$

where

P_3 = pressure to bleed down after adding the hydrostatic of the lubricating fluid in psi
P_1 = original shut in pressure in psi
P_2 = pressure increase due to pumping lubricating fluid into the wellbore (increase is due to compression) in psi

Procedure:

(1) Select a working pressure range, P_w in psi. (Recommended $P_w = 50$–100 psi.)
(2) Pump lubricating fluid through the kill line to increase the casing pressure by the working pressure, P_w.
(3) Allow the pressure to stabilize. The pressure may drop by a substantial amount.
(4) Calculate the pressure to bleed down to by using the formula above.
(5) Repeat steps 2 through 4 until all the gas is lubricated out of the well.

3.23 Annular stripping procedures

3.23.1 Strip and bleed procedure

Application: Appropriate when stripping 30 stands or less or when gas migration is not a problem.

Procedure:

(1) Strip the first stand with the choke closed to allow the casing pressure to increase.

Note: Do not allow the casing pressure to rise above the maximum allowable surface pressure derived from the most recent leak-off test.

(2) Bleed enough volume to allow the casing pressure to decrease to a safety margin of 100 to 200 psi above the original shut in casing pressure.
(3) Continue to strip pipe with the choke closed unless the casing pressure approaches the maximum allowable surface pressure. If the casing pressure approaches the maximum allowable surface pressure, then bleed volume as the pipe is being stripped to minimize the casing pressure.
(4) Once the bit is back on bottom, utilize the Driller's method to circulate the influx out of the well.

3.23.2 Combined stripping/volumetric procedure

Application: Procedure to use when gas migration is a factor. Gas is allowed to expand while stripping. Mud is bled into a trip tank and then closed end displacement into a smaller tank. Trip tank measures gas expansion similar to volumetric method. Pressure is stepped up as in the volumetric method.

Procedure:

(1) Strip in the first stand with the choke closed until the casing pressure reaches $Pchoke_1$.
(2) As the driller strips the pipe, the choke operator should open the choke and bleed mud, being careful to hold the casing pressure at $Pchoke_1$.
(3) With the stand down, close the choke. Bleed the closed end displacement volume from the trip tank to the stripping tank.
(4) Repeat steps 2 and 3 above, stripping stands until V_M accumulates in the trip tank.
(5) Allow casing pressure to climb to the next Pchoke level.
(6) Continue stripping, repeating steps 2 through 4 at the new Pchoke value.
(7) When the bit is on the bottom, kill the well with the Driller's Method.

3.23.3 Worksheet

(1) Select a working pressure, P_w in psi. (Recommended $P_w = 50–100$ psi.)
(2) Calculate the hydrostatic pressure in psi/bbl:

$$P_H = \frac{G_M}{C_a} \tag{3.152}$$

where

P_H = hydrostatic pressure per barrel of mud in psi/bbl
G_M = mud gradient in psi/ft
C_a = capacity of the annulus in bbl/ft

(3) Calculate influx length in the open hole:

$$L_{Ioh} = \frac{V_I}{C_{oh}} \tag{3.153}$$

where

L_{Ioh} = length of influx in the open hole in ft
V_I = volume of influx in bbl
C_{oh} = open hole capacity in bbl/ft

(4) Calculate influx length after the BHA has penetrated the influx:

$$L_{IBHA} = \frac{V_I}{C_{ohDC}} \tag{3.154}$$

where

L_{IBHA} = length of influx after the BHA has penetrated the influx in ft

C_{ohDC} = annular capacity between the drill collars and the open hole in bbl/ft

(5) Calculate the pressure increase due to bubble penetration in psi:

$$P_b = (L_{IBHA} - L_{Ioh})(G_M - G_G) \tag{3.155}$$

where

P_b = pressure increase due to bubble in psi
G_M = mud gradient in psi/ft
G_G = gas gradient in psi/ft (Use 0.1 psi/ft if gas SG is not known).

(6) Calculate the Pchoke values in psi:

$$P_{C1} = \text{SCIP} + P_W = P_b \tag{3.156}$$

$$P_{C2} = P_{C1} + P_W \tag{3.157}$$

$$P_{C3} = P_{C2} + P_W \tag{3.158}$$

(7) Calculate the incremental volume (V_M) of hydrostatic equal to P_W in the upper annulus:

$$V_M = \frac{P_W}{HP} \tag{3.159}$$

where

V_M = volume of mud to bleed to maintain P_W in the annulus in bbl

Example: Prepare a worksheet with the following data:

Working pressure	= 100 psi
Shut-in casing pressure	= 75 psi
Mud weight	= 12.4 ppg (0.645 psi/ft)
Hole size	= 10.0 in.
Capacity of open hole	= 0.971 bbl/ft
Drill pipe OD	= 5.0 in.
Capacity of DP/OH	= 0.0729 bbl/ft
Drill collars OD	= 8.0 in.
Capacity of DC/OH	= 0.0350 bbl/ft
Volume of influx	= 22.0 bbl
Gas gradient	= 0.10 psi/ft

(1) Select a working pressure of 100.0 in psi.

(2) Calculate the hydrostatic pressure in psi/bbl:

$$P_H = \frac{0.645}{0.0971} = 6.6 \text{ psi/bbl for the open hole without the drill string}$$

$$P_H = \frac{0.645}{0.0729} = 8.8 \text{ psi/bbl for the open hole with the drill pipe}$$

$$P_H = \frac{0.645}{0.0350} = 18.4 \text{ psi/bbl for the open hole with the drill collars}$$

(3) Calculate influx length in the open hole:

$$L_{Ioh} = \frac{22.0}{0.0971} = 226.6 \text{ ft for the open hole without the drill string}$$

$$L_{Ioh} = \frac{22.0}{0.729} = 301.8 \text{ ft for the open hole with the drill pipe}$$

(4) Calculate influx length after the BHA has penetrated the influx:

$$L_{IBHA} = \frac{22.0}{0.0350} = 628.6 \text{ ft for the open hole with the drill collars}$$

(5) Calculate the pressure increase due to bubble penetration in psi:

$$P_b = (628.6 - 226.6)(0.645 - 0.100) = 219.1 \text{ psi}$$

(6) Calculate the Pchoke values in psi:

$$P_{C1} = 75.0 + 100.0 = 175.0 \text{ psi}$$

$$P_{C2} = 175.0 + 100.0 = 275.0 \text{ psi}$$

$$P_{C3} = 275.0 + 100.0 = 375.0 \text{ psi}$$

(7) Calculate the incremental volume (V_M) of hydrostatic equal to P_W in the upper annulus:

$$V_M = \frac{100.0}{18.4} = 5.4 \text{ bbl}$$

Spreadsheet 3.23.3 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Worksheet—Strip and bleed.

3.24 Barite plug

A barite plug is often used to control underground flows that have broken down the formation at the casing shoe. The technique involves placing a volume of barite and water or base oil that the solids settle out of suspension forming a solids plug of high density that can be drilled out easier than a cement plug. This also prevents the risk of sidetracking while drilling the plug.

Step 1: Determine the capacity of the open hole without drill pipe in bbl:

$$C_h = \frac{((D_B)(1 + (WO)))^2}{1029.4} \tag{3.160}$$

where

C_h = capacity of open hole without drill pipe in bbl/ft
D_B = diameter of the bit in in.
WO = washout of hole in % (optional)

Step 2: Determine the volume of barite to set a 100 ft plug of settle barite in bbl:

$$V_b = 100(C_h) \tag{3.161}$$

where

V_b = volume of barite for a 100 ft plug in bbl

Step 3: Determine the amount of base liquid needed to mix with the barite in bbl:

$$V_L = \frac{((V_b)((WM_{SG})(MW_L))) - ((V_b)(MW_p))}{(MW_p - MW_L)} \tag{3.162}$$

where

V_L = volume of base liquid to mix with barite in bbl
WM_{SG} = weight material specific gravity in g/cm^3
MW_L = mud weight of base liquid in ppg
MW_p = mud weight of plug slurry in ppg

Step 4: Determine the final volume of the plug slurry in bbl:

$$V_{ps} = V_L + V_b \tag{3.163}$$

where

V_{ps} = volume of plug slurry in bbl

Step 5: Determine the capacity of the annulus with drill pipe in bbl/ft:

$$C_{ap} = \frac{((D_B)(1 + (WO)))^2 - (D_P)^2}{1029.4} \tag{3.164}$$

where

C_{ap} = capacity of the annulus with drill pipe in bbl/ft
D_P = size of the drill pipe in in.

Step 6: Determine the length of the plug slurry in the annulus with the pipe in the hole in ft:

$$L_{pSP} = \frac{V_{pS}}{C_{ap}} \tag{3.165}$$

where

L_{pSP} = length of plug slurry in the annulus with the pipe in the hole in ft

Step 7: Determine the length of the plug slurry with **NO** pipe in the annulus in ft:

$$L_{pS} = \frac{V_{pS}}{C_a} \tag{3.166}$$

where

L_{pS} = length of the plug slurry in the annulus with **NO** pipe in ft

Step 8: Determine the length of the settled barite in the annulus in ft:

$$L_{bs} = \frac{V_b}{C_a} \tag{3.167}$$

where

L_{bs} = length of the settled barite in ft

Step 9: Determine the amount of barite required for the plug in sacks:

$$B_{ar} = (V_b)(14.7) \tag{3.168}$$

where

B_{ar} = number of 100 lb sacks of barite needed for the plug

Example: Calculate the materials required for an open hole, water-based, barite plug with the following:

Bit size	= 12¼ in.
Washout	= 15% (increase in diameter)
Drill collars	= 8.0 in.
Length of settled barite	= 100 ft
Weight material specific gravity	= 4.2 g/cm^3
Mud weight of water	= 8.34 ppg
Plug mud weight	= 18.0 ppg

Step 1: Determine the capacity of the open hole without the drill string in bbl:

$$C_h = \frac{((12.25)(1 + (0.15)))^2}{1029.4} = 0.1928 \text{ bbl/ft}$$

Step 2: Determine the volume of barite to set a 100 ft plug of settle barite in bbl:

$$V_b = 100(0.1928) = 19.3 \text{ bbl round up to } 20.0 \text{ bbl}$$

Step 3: Determine the amount of base liquid needed to mix with the barite in bbl:

$$V_{\mathrm{L}} = \frac{((20.0)((4.2)(8.34))) - ((20.0)(18.0))}{(18.0 - 8.34)} = 35.25 \cong 35.3 \text{ bbl}$$

Step 4: Determine the final volume of the plug slurry in bbl:

$$V_{\mathrm{ps}} = 35.3 + 20.0 = 55.3 \text{ bbl}$$

Step 5: Determine the capacity of the annulus with drill pipe in bbl/ft:

$$C_{\mathrm{ap}} = \frac{((12.25)(1 + (0.15)))^2 - (8.0)^2}{1029.4} = 0.1306 \text{ bbl/ft}$$

Step 6: Determine the length of the plug slurry in the annulus with the pipe in the hole in ft:

$$L_{\mathrm{pSP}} = \frac{55.3}{0.1306} = 423.4 \text{ ft}$$

Step 7: Determine the length of the plug slurry with **NO** pipe in the annulus in ft:

$$L_{\mathrm{pS}} = \frac{55.3}{0.1928} = 286.8 \text{ ft}$$

Step 8: Determine the length of the settled barite in the annulus in ft:

$$L_{\mathrm{bs}} = \frac{20.0}{0.1928} = 103.7 \text{ ft}$$

Step 9: Determine the amount of barite required for the plug in sacks:

$$B_{\mathrm{ar}} = (20.0)(14.7) = 294 \text{ sacks}$$

Spreadsheet 3.24 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Barite plug.

Appendix: Supplementary material

Supplementary material related to this chapter can be found online at https://www.elsevier.com/books-and-journals/book-companion/9780323905497.

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Petroleum Extension Service, Univ. of Texas, Houston, Texas..
Aberdeen Well Control School, Gene Wilson, Aberdeen, Scotland..

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CHAPTER

4

Drilling fluids

4.1 Mud density increase and volume change

4.1.1 Field procedure for determining the specific gravity of barite

Specific gravity of barite should be measured in a laboratory according to API procedures as outlined in API Specification A3. A quick field check to verify the average specific gravity of the weight material involves using the mud balance available at the rig.

Procedure:

(1) Place 100 mL of fresh water (8.34 lb/gal) in the mud balance cup.
(2) Add barite slowly to mud balance cup until the water level is ⅜ in. from the top of the cup. Make sure the cup is not so full that liquid overflows when the lid is placed on the cup.
(3) Stir the barite water mixture to ensure there are no trapped bubbles in the slurry.
(4) Place the lid on the cup and use a 10 mL plastic syringe with needle inserted through the lid and the end of the needle below the liquid level in the cup.
(5) Fill the cup up with water until it is level with the hole in the cup lid.
(6) The volume of the mud balance cup is 145 mL.
(7) Weight the slurry in the mud cop to the nearest 0.05 lb/gal.

$$SG_b = \frac{((145.0)(D_S)) - ((8.34)(V_W))}{8.34((145.0) - (V_W))} \tag{4.1}$$

where

SG_b = specific gravity of barite in g/cm^3
D_s = density of slurry in cup in lb/gal
V_W = volume of water in cup in ml.

Formulas and Calculations for Drilling, Production, and Workover
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Example: Use the following data to calculate specific gravity for a field sample of barite:

$$D_s = 13.3\,\text{lb/gal}$$

$$V_W = 118\,\text{mL}\ (100\,\text{mL plus } 18\,\text{mL volume from the syringe})$$

$$SG_b = \frac{((145)(13.3)) - ((8.34)(118))}{8.34((145) - (118))} = 4.2\ \text{SG gm/cm}^3$$

Spreadsheet 4.1.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Field procedure to determine the SG of barite.

4.1.2 Increase mud density—*No* base liquid added and *no* volume limit

Various dry materials may be used to increase the density of drilling, completion, and workover fluids. Those materials may include barite, hematite, calcium carbonate, magnesium carbonate, various dry salts (e.g., sodium, calcium, zinc chloride, and/or sodium formate), and blends. It is important to know the average specific gravity (ASG) of the material being used. For example, the current API ASG specification for barite is 4.2. The ASG of the dry material you are using should be obtained from the company supplying the product.

(A) "Field" formula for mud weight increase:

(1) Weight material required to increase 100 bbl of mud

$$W_M = 5(\rho_n)(\rho_n - \rho_o) \tag{4.2}$$

where

W_M = number of 100 lb sacks of weight material required for 100 bbl of mud
ρ_o = original mud weight in ppg
ρ_n = new mud weight in ppg

(2) Volume increase based on the amount of weight material added:

$$V_i = \frac{W_M}{3.5(SG_{wm})} \tag{4.3}$$

where

V_i = volume increase in bbl
SG_{wm} = specific gravity of weight material in g/cm^3

Example: Calculate the number of sacks of 4.2 ASG of barite required to increase the density of 100 bbl of 12.0 ppg (ρ_o) mud to 14.0 ppg (ρ_n) and the resultant volume increase.

(1) $W_M = 5(14.0)(14.0 - 12.0) = 140$ sacks of barite required for 100 bbl of 12.0 ppg mud.

(2) $V_i = \dfrac{140}{3.5(4.2)} = 9.5$ bbl of volume increase

(B) Long formula to increase mud weight:

(1)
$$W_M = \frac{(350\,SG_{wm})(\rho_n - \rho_o)}{(8.34\,SG_{wm}) - \rho_n} \tag{4.4}$$

where

350 = weight of 1 barrel of fresh water in lb
8.34 = mud weight of fresh water in lb/gal (ppg)
SG_{wm} = specific gravity of weight material, g/cm^3
ρ_n = new mud weight in ppg
ρ_o = original mud weight in ppg

(2) Volume increase with 100 lb sacks of weight material based on the change in mud weight:

$$V_i = \frac{100(\rho_n - \rho_o)}{(8.34\,SG_{wm}) - \rho_n} \tag{4.5}$$

where

V_i = volume increase in bbl

Example: Calculate the number of sacks of 4.2 ASG barite required to increase the density of 100 barrels of 12.0 ppg (ρ_o) mud to 14.0 ppg (ρ_n):

(1) $W_M = \dfrac{((350)(4.2))(14.0 - 12.0)}{((8.34)(4.2)) - 14.0} = 139.8 \cong 140$ sx of barite per 100 bbl of 12.0 ppg mud

(2) $V_i = \dfrac{100(14.0 - 12.0)}{((8.34)(4.2)) - 14.0} = 9.5$ bbl of volume increase

Spreadsheet 4.1.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Increase MW—*No* base liquid added and *No* volume limit.

4.1.3 Increase mud weight—*No* base liquid added *but* limit final volume

Step 1: Calculate the starting volume required:

$$V_s = \frac{((8.34)(SG_{wm})) - \rho_n}{((8.34)(SG_{wm})) - \rho_o}(V_f) \tag{4.6}$$

where

V_s = starting volume of mud in bbl
ρ_n = new mud weight in ppg
ρ_o = original mud weight in ppg
V_f = final volume of mud in bbl

Step 2: Calculate the amount of weight material to be added:

$$W_{Ma} = (V_f - V_s)((SG_{wm})(3.5)) \tag{4.7}$$

where

W_{Ma} = weight material in 100 lb sacks adjusted for the desired final volume

Example: Calculate the number of sacks of 4.2 ASG barite required to increase the density of 12.0 ppg (ρ_o) mud to end up with 100 bbl of 14.0 ppg (ρ_n) mud:

Step 1: Calculate the starting volume:

$$V_s = \frac{((8.34)(4.2)) - 14.0}{((8.34)(4.2)) - 12.0}(100) = 91 \text{ bbl}$$

Step 2: Calculate the amount of weight material to be added:

$$W_{Ma} = (100 - 91)((4.2)(3.5))$$
$$= 132 \text{ sxs of barite added to 91 bbl of 12.0 mud to result in 100 bbl of 14.0 ppg mud.}$$

Spreadsheet 4.1.3 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Increase MW—*No* base liquid added *but* limit final volume.

4.1.4 Increase the mud weight—*With* base liquid added and *no* volume limit

Adding dry powder to a mud will cause an increase in the viscosity of that mud. It is a general rule that a volume of base liquid must be added to wet the surface of any dry weight material added to an existing mud. From experience, at least 1½ gal of base liquid must be added per 100 lb of weight material added. This base liquid volume has to be "weighed up" to the final mud weight required for the total mud system. Therefore, the amount of weight material calculated with the previous equations is the minimum amount that would be required before adding the additional base liquid. The following equations can be used to calculate the amount of base liquid to be added to the amount of weighting material to reach the desired mud weight.

Step 1: Calculate the mud weight of the weight material-base liquid mixture:

$$MW_X = \frac{((L_b)(SG_{wm})) + ((0.1251)(SG_{wm})(L_b))}{(1 + ((0.1251)(SG_{wm})))} \tag{4.8}$$

where

MW_X = mud weight of weight material-base liquid mixture in ppg
L_b = base liquid mud weight in ppg
SG_{wm} = specific gravity of weight material in g/cm^3

Step 2: Calculate the amount of weight material required:

$$W_{MX} = \frac{((0.42)(MW_X)(\rho_n - \rho_o))}{(MW_X - \rho_n)(1 + ((0.015)(L_b)))}(V_s) \tag{4.9}$$

where

W_{MX} = number of 100 lb sacks of weight material required for the mixture
ρ_n = new mud weight in ppg
ρ_o = original mud weight in ppg
V_s = starting volume in bbl

Step 3: Calculate the volume of wetting (base) liquid to be added:

$$L_W = \frac{(W_{MX})(1.5)}{(42)} \tag{4.10}$$

where

L_W = amount of wetting (base) liquid to be added with weight material in bbl

Step 4: Calculate the final volume with the weight material-base liquid mixture added:

$$V_f = (V_s)\left[1 + \left(\frac{W_{MX}}{(350)(SG_{wm})}\right) + \left(\frac{((1.5)(W_{MX}))}{(4200)}\right)\right] \tag{4.11}$$

where

V_f = final volume of mud in bbl
V_s = starting volume of mud in bbl

Example: Calculate the number of sacks of 4.2 ASG barite required to increase the density of 100 bbl of 12.0 ppg (ρ_o) mud to 14.0 ppg (ρ_n) using fresh water as the wetting agent:

Step 1: Calculate the mud weight of the weight material-base liquid mixture:

$$\mathrm{WM_X} = \frac{((8.34)(4.2)) + ((0.1251)(4.2)(8.34))}{(1 + ((0.1251)(4.2)))}$$
$$= 25.8 \text{ ppg the mud weight of the weight material} - \text{fresh water mixture.}$$

Step 2: Calculate the amount of weight material required:

$$W_{MX} = \frac{((0.42)(25.8))(14.0 - 12.0)}{(25.8 - 14.0)(1 + ((0.015)(8.34)))}(100)$$
$$= 163 \text{ sxs of weight material required for 100 bbl of 12.0 ppg mud.}$$

Step 3: Calculate the volume of wetting (base) liquid to be added:

$$L_W = \frac{(163)(1.5)}{(42)} = 5.8 \text{ bbl}$$

Step 4: Calculate the final volume with the weight material-base liquid mixture added:

$$V_f = (100)\left[1 + \left(\frac{163}{(350)(4.2)}\right) + \left(\frac{((1.5)(163))}{(4200)}\right)\right]$$
$$= 116.9 \text{ bbl final volume of 14.0 ppg mud.}$$

Spreadsheet 4.1.4a in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Increase MW—*With* base liquid added and *No* volume limit.

Spreadsheet 4.1.4b in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 EXTRA This is a spreadsheet that covers all density changes from 9.0 to 19.0 ppg with either Water Base or Oil Base fluids.

4.1.5 Increase mud weight—*With* base liquid added *but* limit final volume

Step 1: Calculate the amount of weight material required per 100 bbl of mud using Eq. (4.9):

Step 2: Calculate the starting volume of mud required to obtain the desired final mud volume:

$$V_s = \frac{(V_f)}{\left(1 + \left(\left(\frac{W_{MX}}{((350)(SG_{wm}))}\right) + \left(\frac{(1.5)(W_{MX})}{4200}\right)\right)\right)} \tag{4.12}$$

Step 3: Calculate the amount of mud that must be dumped to have available pit space for the final volume in the system:

$$V_d = V_f - V_s \tag{4.13}$$

where

V_d = volume of mud in bbl to be dumped out of the system to make room for the mud weight increase.

Step 4: Use Eq. (4.9) to calculate the amount of weight material (W_{MX}) required to increase the mud weight of the reduced starting volume.

Step 5: Calculate the volume increase due to the amount of weight material required to increase the mud weight:

$$V_{WM} = \frac{(W_{MX})}{((3.5)(SG_{wm}))} \tag{4.14}$$

where

V_{WM} = volume of weight material to be added in bbl

Step 6: Calculate the amount of base liquid added to wet the weight material:

$$L_W = (V_f) - ((V_s) + (V_{WM})) \tag{4.15}$$

where

L_W = volume of base liquid in bbl required to wet weight material added to increase the mud weight.

Example: Calculate the amount of mud to be dumped when increasing the mud weight from 12.0 to 14.0 ppg with a final volume of 1000 bbl.

Step 1: Calculate the amount of weight material required per 100 bbl of mud using Eq. (4.9):

$$W_{MX} = \frac{((0.42)(25.8))(14.0 - 12.0)}{(25.8 - 14.0)(1 + ((0.015)(8.34)))}(100) = 163 \text{ sxs}$$

Step 2: Calculate the starting volume of mud required to obtain the desired final mud volume:

$$V_s = \frac{(1000)}{\left(1 + \left(\left(\frac{163}{((350)(4.2))}\right) + \left(\frac{(1.5)(163)}{4200}\right)\right)\right)} = 855 \text{ bbl}$$

Step 3: Calculate the amount of mud that must be dumped to have available pit space for the final volume in the system:

$V_d = 1000 - 855$

= 145 bbl of 12.0 ppg mud must be dumped before increasing the mud weight to 14.0 ppg with a system volume limited to 1000 bbl.

Step 4: Calculate the amount of weight material (W_{MX}) required to increase the mud weight of the reduced starting volume.

$$W_{MX} = \frac{((0.42)(25.8))(14.0 - 12.0)}{(25.8 - 14.0)(1 + ((0.015)(8.34)))}(855) = 1395 \text{ sxs}$$

Step 5: Calculate the volume increase due to the amount of weight material required to increase the mud weight:

$$V_{WM} = \frac{(1395)}{((3.5)(4.2))} = 95 \text{ bbl of barite required for mud weight increase.}$$

Step 6: Calculate the amount of base liquid added to wet the weight material:

$$L_W = (1000) - ((855) + (95))$$
$$= 50 \text{ bbl of base liquid required for wetting the barite weight material.}$$

Spreadsheet 4.1.5 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Increase MW—*With* base liquid added *but* limit final volume.

4.1.6 Increase mud weight—*With* base liquid added *but* limit final volume and *limited* weight material inventory

Step 1: Calculate the amount of weight material-base liquid (W_{MX}) required to increase the mud weight of the total mud system volume:

$$W_{MX} = \frac{(0.42\,MW_X)(\rho_n - \rho_o)}{(MW_X - \rho_n)(1 + (0.015\,L_b))}(V_C) \tag{4.16}$$

where

V_C = current total volume of mud system in bbl

Note: If this amount of weight material required is more than the current inventory, then a new starting volume must be calculated.

Step 2: Calculate the new starting volume for the weight material inventory on the rig:

$$V_{ns} = \left(\frac{W_{MI}}{W_{MX}}\right)(V_C) \tag{4.17}$$

where

V_{ns} = new starting volume for limited weight material inventory in bbl
W_{MI} = weight material inventory on rig in 100 lb sacks

Step 3: Use Eq. (4.13) to calculate the number of barrels that need to be dumped before adding weight material.

Step 4: Recalculate the amount of weight material-base liquid (W_{MX}) required to increase the mud weight of the new starting mud system volume:

$$W_{MX} = \frac{(0.42\,MW_X)(\rho_n - \rho_o)}{(MW_X - \rho_n)(1 + (0.015\,L_b))}(V_{ns}) \tag{4.18}$$

Step 5: Use Eq. (4.14) to calculate the volume increase due to the weight material added to the system.

Step 6: Use Eq. (4.15) to calculate the volume of wetting agent (base liquid) to be added with the weight material.

Example: Calculate the amount of weight material required to increase the mud weight from 12.0 to 14.0 ppg with a limited mud volume and the following data:

Current mud volume:	1000 bbl
Maximum mud volume:	1000 bbl
Weight material inventory:	1400 sacks

Mud weight of the weight material-base liquid mixture = 25.8 ppg

Step 1: Calculate the amount of weight material required to increase the density of 1000 bbl of mud:

$$W_{MX} = \frac{((0.42)(25.8))(14.0 - 12.0)}{(25.8 - 14.0)(1 + ((0.015)(8.34)))}(1000) = 1632 \text{ sxs}$$

Step 2: Calculate the new starting volume for the weight material inventory on the rig:

$$V_{ns} = \left(\frac{1400}{1632}\right)(1000) = 858 \text{ bbl of new starting mud volume.}$$

Step 3: Calculate the number of barrels that need to be dumped before adding weight material:

$$V_d = 1000 - 858 = 142 \text{ bbl}$$

Step 4: Recalculate the amount of weight material-base liquid (W_{MX}) required to increase the mud weight of the new starting mud system volume:

$$W_{MX} = \frac{((0.42)(25.8))(14.0 - 12.0)}{(25.8 - 14.0)(1 + ((0.015)(8.34)))}(858)$$
$$= 1400 \text{ sxs required to increase the mud weight of the new starting volume with limited weight material inventory.}$$

Step 5: Calculate the volume increase due to the weight material added to the system:

$$V_{WM} = \frac{(1400)}{((3.5)(4.2))} = 95 \text{ bbl}$$

Step 6: Calculate the volume of wetting agent (base liquid) to be added with the weight material.

$$L_W = (1000) - ((858) + (95)) = 47 \text{ bbl}$$

Spreadsheet 4.1.6 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Increase MW—*with* base liquid added *but* limit final volume and *Limited* weight material inventory.

4.1.7 Increase mud weight to a maximum mud weight *with* base liquid added *but* with *limited* weight material inventory

$$MW_{MX} = \frac{((100\,W_{MI})(MW_X)(1 + (0.015\,L_b))) + (42(MW_X)(V_s)(\rho_o))}{((100\,W_{MI})(1 + (0.015\,L_b))) + (42(MW_X)(V_s))} \tag{4.19}$$

where

MW_{MX} = maximum mud weight possible in ppg with weight material inventory

Example: Calculate the maximum mud weight with the following data:

Mud system volume:	2000 bbl
Current mud weight:	12.0 ppg
Weight material-base liquid mixture weight:	25.8 ppg
Base liquid mud weight:	8.34 ppg
Weight material inventory:	1000 sks

$$MW_{MX} = \frac{((100(1000))(25.8)(1 + (0.015(8.34)))) + (42(25.8)(2000)(12.0))}{((100(1000))(1 + (0.015(8.34)))) + (42(25.8(2000)))}$$

= 12.7 ppg is the maximum mud weight that can be mixed with only 1000 sacks of weight material of rig inventory with a 2000 bbl system of 12.0 ppg mud.

Spreadsheet 4.1.7 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Increase MW to a max *with* base liquid added *but* with *Limited* weight material inventory.

4.2 Mud weight reduction with base liquid dilution

4.2.1 Mud weight reduction with base liquid

(A) Calculate the volume of base liquid in bbl added to reduce the mud weight:

$$V_{\mathrm{bl}} = \frac{(V_{\mathrm{m}}(\rho_0 - \rho_{\mathrm{n}}))}{(\rho_{\mathrm{n}} - L_{\mathrm{b}})} \tag{4.20}$$

where

V_{bl} = volume of base liquid mixture in bbl added to reduce the mud weight
ρ_{o} = original mud weight in ppg
ρ_{n} = new mud weight in ppg
L_{b} = base liquid mud weight in ppg

Example 1: Determine the number of barrels of fresh water weighing 8.34 ppg required to reduce the mud weight of 100 bbl of water-base mud (WBM) from 14.0 to 12.0 ppg:

$$V_{\mathrm{bl}} = \frac{(100(14.0 - 12.0))}{(12.0 - 8.34)} = 54.6 \text{ bbl of fresh water required.}$$

Example 2: Determine the number of barrels of base oil weighing 6.7 ppg required to reduce the mud weight of 100 bbl of synthetic-base mud (SBM) from 14.0 to 12.0 ppg:

$$V_{\mathrm{bl}} = \frac{(100(14.0 - 12.0))}{(12.0 - 6.7)} = 37.7 \text{ bbl of base oil required.}$$

Note: Adding this much base oil to 100 bbl of SBM may increase the oil/water ratio (OWR) **too** much, so the dilution volume may need to be added as a mixture of base oil and water with the same OWR as the active mud system. To calculate the volume of mixture required, calculate the mud weight of the mixture, then calculate a new V_{bl} value.

$$\mathrm{MW_{OB}} = \left((f_{\mathrm{o}})(\rho_{\mathrm{o}})\right) + \left((f_{\mathrm{w}})(\rho_{\mathrm{b}})\right) \tag{4.21}$$

where

$\mathrm{MW_{OB}}$ = mud weight of oil/brine mixture in ppg
f_{o} = fraction of base oil in mixture
ρ_{o} = mud weight of base oil in ppg
f_{w} = fraction of water in mixture
ρ_{b} = mud weight of brine in ppg

Example: Determine the mud weight of the base oil/brine mixture required to reduce the mud weight of 100 bbl of synthetic-base mud (SBM) from 14.0 to 12.0 ppg and maintain a 75/25 OWR.

Note: The SBM will normally contain calcium chloride brine in the emulsion; therefore, this example uses a 10.0 ppg brine as the water phase of this OWR mixture.

$$\text{Base oil mud weight} = 6.7 \text{ ppg (SG 0.80)}$$

$$\text{Brine water weight} = 10.0 \text{ ppg}$$

$$\text{MW}_{\text{OB}} = ((0.75)(6.7)) + ((0.25)(10.0)) = 7.5 \text{ ppg}$$

$$V_{\text{bl}} = \frac{(100(14.0 - 12.0))}{(12.0 - 7.5)} = 44.4 \text{ bbl}$$

Therefore (44.4×0.75) or 33.3 bbl of base oil will be required to mix with (44.4×0.25) or 11.1 bbl of brine water to reduce the mud weight and keep the OWR the same.

Spreadsheet 4.2.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 MW reduction with base liquid.

4.3 Mixing fluids of different densities

4.3.1 Mixing fluids of different densities formula

(A) Mixing two fluids with *No* volume limit:

$$\rho_{\text{f}} = \frac{((V_1)(\rho_1)) + ((V_2)(\rho_2))}{(V_1 + V_2)} \tag{4.22}$$

where

ρ_{f} = final mud weight in ppg, lb/ft^3, etc.
V_1 = volume of fluid #1 in bbl, gal, etc.
ρ_1 = mud weight of fluid #1 in ppg, lb/ft^3, etc.
V_2 = volume of fluid #2 in bbl, gal, etc.
ρ_2 = mud weight of fluid #2 in ppg, lb/ft^3, etc.

Example: Use the following data:

Volume of fluid #1 = 300 bbl
Mud weight of fluid #1 = 11.0 ppg
Volume of fluid #2 = 100 bbl
Mud weight of fluid #2 = 14.0 ppg

(A) Solve for final mud weight with no volume limit:

$$\rho_{\text{f}} = \frac{((300)(11.0)) + ((100)(14.0))}{(300 + 100)} = 400 \text{ bbl of } 11.75 \cong 11.8 \text{ ppg fluid}$$

(B) Solve for a limited final volume with a known mud weight:

(1)
$$V_2 = \frac{(((V_f)(\rho_f)) - ((V_f)(\rho_1)))}{(\rho_2 - \rho_1)} \tag{4.23}$$

where

V_2 = volume of fluid #2 in bbl, gal, etc.
V_f = final volume in bbl, gal, etc.
ρ_f = final mud weight in ppg, lb/ft^3, etc.
ρ_1 = mud weight of fluid #1 in ppg, lb/ft^3, etc.
ρ_2 = mud weight of fluid #2 in ppg, lb/ft^3, etc.

(2) Determine the volume of fluid #1 in bbl:

$$V_1 = V_f - V_2 \tag{4.24}$$

where

V_1 = volume of fluid #1 in bbl

Example: Use the following data:

Final volume	= 300 bbl
Final mud weight	= 11.5 ppg
Mud weight of fluid #1	= 11.0 ppg
Mud weight of fluid #2	= 14.0 ppg

(1) $V_2 = \frac{(((300)(11.5)) - ((300)(11.0)))}{(14.0 - 11.0)} = 50$ bbl of fluid#2 mud weight

(2) $V_1 = 300 - 50 = 250$ bbl of fluid #1 mud weight

Spreadsheet 4.3.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Mixing fluids of different densities.

4.4 Oil-based mud calculations

4.4.1 Calculate the starting volume of liquid (base oil plus water) required to prepare a desired final volume of nonaqueous fluid NAF (OBM)

Step 1: Calculate the base oil-water mixture mud weight:

$$MW_{owm} = \left(\left(\frac{O_R}{100}\right)((O_{SG})(8.34))\right) + \left(\left(\frac{W_R}{100}\right)(8.34)\right) \tag{4.25}$$

where

MW_{owm} = mud weight for the oil-water mixture in ppg
O_R = oil ratio

O_{SG} = specific gravity of the oil in g/cm^3
W_R = water ratio

Step 2: Calculate the starting volume:

$$V_s = \frac{(((8.34)(WM_{SG})) - (MW_f))}{(((8.34)(WM_{SG})) - (MW_{owm}))} V_f \tag{4.26}$$

where

V_s = starting volume for mud mixture in bbl
WM_{SG} = weight material specific gravity in g/cm^3
MW_f = final mud weight in ppg
V_f = final mud volume in bbl

Step 3: Calculate the volume of the liquid components in bbl:

(a) Calculate the oil content of the liquid phase in bbl:

$$O_C = (V_s)\left(\frac{O_R}{100}\right) \tag{4.27}$$

where

O_C = oil content of the liquid mixture in bbl

(b) Calculate the water content of the liquid phase in bbl:

$$W_C = 100 - O_C \tag{4.28}$$

where

W_C = water content of the liquid mixture in bbl

Step 4: Calculate the volume of weight material in bbl:

$$V_{WM} = V_f - V_s \tag{4.29}$$

where

V_{WM} = volume of weight material to be added in bbl

Step 5: Calculate the sacks of weight material required for the 100 bbl of mud:

$$W_M = (V_{WM})(3.5(WM_{SG})) \tag{4.30}$$

where

W_M = weight material required in sacks
WM_{SG} = specific gravity of the weight material in g/cm^3

Step 6: Recalculate the mud weight to verify the amounts are correct:

$$\mathrm{MW} = \frac{((V_{\mathrm{WM}})(8.34(\mathrm{WM_{SG}}))) + ((\mathrm{MW_{owm}})(V_{\mathrm{s}}))}{(V_{\mathrm{f}})} \tag{4.31}$$

where

MW = final mud weight of new volume in ppg
V_{f} = final volume of new mud in bbl

Example: Prepare 100 bbl of 16.0 ppg mud with a 75/25 OWR using a 0.80 SG base oil and fresh water (no salt added) and weight material specific gravity of 4.2 g/cm^3:

Step 1: Calculate the base oil-water mixture mud weight:

$$\mathrm{MW_{owm}} = \left(\left(\frac{75}{100}\right)((0.80)(8.34))\right) + \left(\left(\frac{25}{100}\right)(8.34)\right) = 7.1 \text{ ppg}$$

Step 2: Calculate the starting volume:

$$V_{\mathrm{s}} = \frac{(((8.34)(4.2)) - (16.0))}{(((8.34)(4.2)) - (7.1))}(100) = 68.1 \text{ bbl}$$

Step 3: Calculate the volume of the liquid components in bbl:

(a) Calculate the oil content of the liquid phase in bbl:

$$O_{\mathrm{C}} = (68.1)\left(\frac{75}{100}\right) = 51.075 \cong 51.1 \text{ bbl of oil}$$

(b) Calculate the water content of the liquid phase in bbl:

$$W_{\mathrm{C}} = 68.1 - 51.1 = 17 \text{ bbl of water}$$

Step 4: Calculate the volume of weight material in bbl:

$$V_{\mathrm{WM}} = 100 - 68.1 = 31.9 \text{ bbl}$$

Step 5: Calculate the sacks of weight material required for the 100 bbl of mud:

$$W_{\mathrm{M}} = (31.9)(3.5(4.2)) = 469 \text{ sx}$$

Step 6: Recalculate the mud weight to verify the amounts are correct:

$$\mathrm{MW} = \frac{((31.9)(8.34(4.2))) + ((7.1)(68.1))}{100} = 16.0 \text{ ppg}$$

Spreadsheet 4.4.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Calculate starting volume of liquid required to prepare a final volume of NAF mud.

4.4.2 Oil/water ratio from retort data

Obtain the percent-by-volume oil and percent-by-volume water from retort analysis or mud still analysis. Using the data obtained, the OWR is calculated as follows:

(a) Calculate the % of base oil in the OWR mixture:

$$O_{\mathrm{OWR}} = \frac{O_{\%}}{O_{\%} + W_{\%}} 100 \tag{4.32}$$

where

O_{OWR} = oil content in the OWR in %
$O_{\%}$ = oil content in the mud in %
$W_{\%}$ = water content in the mud in %

(b) Calculate the % of water in the OWR mixture:

$$W_{\mathrm{OWR}} = \frac{W_{\%}}{(O_{\%} + W_{\%})} 100 \tag{4.33}$$

Example: Calculate the OWR of a mud that has the following data:

Oil content by volume: 51%
Water content by volume: 17%
Solids content by volume: 32%

$$O_{\mathrm{OWR}} = \frac{51}{(51 + 17)} 100 = 75$$

$$W_{\mathrm{OWR}} = \frac{17}{(51 + 17)} 100 = 25$$

The OWR is 75/25.

Spreadsheet 4.4.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 OWR.

4.4.3 Change the OWR

Note: If the OWR is to be ***increased,*** add **OIL**; if it is to be ***decreased,*** add **WATER**.

(a) To increase the oil content in the OWR, the current water content will be changed to a new volume percent in the OWR:

$$V_{\mathrm{nw}} = \frac{W_{\mathrm{o}}}{W_{\mathrm{n}}} \tag{4.34}$$

where

V_{nw} = new volume of base oil-water mixture in bbl when holding the water content constant
W_o = old water content, bbl in 100 bbl of mud
W_n = new water content in % (decimal)

(b) The amount of oil to add is calculated by the following:

$$O_a = V_{nw} - V_o \tag{4.35}$$

where

O_a = volume of oil to be added in bbl
V_o = old volume of base oil-water mixture in bbl

(c) To increase the water content in the OWR, the current oil content will be changed to a new volume:

$$V_{no} = \frac{O_o}{O_n} \tag{4.36}$$

where

V_{no} = new volume of base oil-water mixture in bbl when holding the base oil content constant
O_o = old oil content, bbl in 100 bbl of mud
O_n = new oil content in % (decimal)

(d) The amount of water to add is calculated by the following:

$$W_a = V_{no} - V_o \tag{4.37}$$

where

W_a = volume of water to be added in bbl

Example 1: Increase the OWR from 75/25 to 80/20:

Given: Oil content by volume: 51%
Water content by volume: 17%
Solids content by volume: 32%
OWR: 75/25

In 100 bbl of this mud, there are 68 bbl of liquid (oil plus water). To increase the OWR, add oil. The total liquid volume will be increased by the volume of the oil added, but the water volume will not change. The 17 bbl of water now in the mud represents 25% of the old liquid volume, but it will represent *only* 20% of the new liquid volume.

$$V_{nw} = \frac{17}{0.20}$$

$V_{nw} = 85$ bbl of new liquid volume after adding base oil to 100 bbl of mud volume.

$O_a = 85 - 68 = 17$ bbl of base oil to be added per 100 bbl of mud

Check the calculations.

$$O_{OWR} = \frac{51 + 17}{(68 + 17)}100 = 80$$

The new OWR is 80/20.

Example 2: Change the OWR from 75/25 to 70/30:

As in Example 1, there are 68 bbl of liquid in 100 bbl of this mud. In this case, however, water will be added and the volume of oil will remain constant. The 51 bbl of oil represents 75% of the original liquid volume and 70% of the final volume:

$$V_{no} = \frac{51}{0.70} = 72.8 \approx 73 \text{ bbls}$$

$$W_a = 73 - 68 = 5 \text{ bbl of water added per 100 bbl of mud.}$$

Check the calculations.

$$W_{OWR} = \frac{17 + 5}{(68 + 5)}100 = 30$$

The new OWR is 70/30.

Spreadsheet 4.4.3 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Change the OWR.

4.5 Solids analysis

Note: Steps 1–4 are performed on high salt content muds. For low chloride muds begin with Step 5.

Step 1: Calculate the volume of saltwater in %:

$$W_s = \left[\left(5.88 \times 10^{-8}\right)\left(Cl^{1.2}\right) + 1\right](W_{\%}) \tag{4.38}$$

where

W_s = volume of saltwater in %
Cl = chloride content measured from the filtrate in ppm
$W_{\%}$ = volume of water in mud from the retort in %

Step 2: Calculate the volume of the suspended solids in %:

$$S_s = 100 - O_{\%} - W_s \tag{4.39}$$

where

S_s = volume of suspended solids in %
$O_{\%}$ = oil content in %

Step 3: Calculate the ASG of the saltwater:

$$W_{\mathrm{ASG}}\left[\left(1.94\times10^{-6}\right)\left(\mathrm{Cl}^{0.95}\right)\right]+1 \tag{4.40}$$

where

W_{ASG} = average specific gravity of the saltwater

Step 4: Calculate the ASG of the solids suspended in the mud in g/cm^3:

$$S_{\mathrm{ASG}}=\frac{((12)(\mathrm{MW}))-((W_{\%})(W_{\mathrm{ASG}}))-((O_{\%})(O_{\mathrm{ASG}}))}{S_{\mathrm{s}}} \tag{4.41}$$

where

S_{ASG} = average specific gravity of suspended solids in the mud in g/cm^3
$W_{\%}$ = volume of water in the mud in %
W_{ASG} = specific gravity of water in g/cm^3
$O_{\%}$ = volume of oil in the mud in %
O_{ASG} = specific gravity of base oil being used in mud in g/cm^3 (0.84 for diesel; 0.80 for IO)

Step 5: Calculate the ASG of solids ***without*** salt in the water phase in g/cm^3:

$$S_{\mathrm{fASG}}=\frac{((12)(\mathrm{MW}))-((1.0)(W_{\%}))-((O_{\%})(O_{\mathrm{ASG}}))}{S_{\mathrm{s}}} \tag{4.42}$$

where

S_{fASG} = average specific gravity of solids ***without*** salt in the water phase in g/cm^3

Step 6: Calculate the volume of low gravity solids (LGS) in %:

$$\mathrm{LGS}_{\%}=\frac{(S_{\mathrm{s}})(\mathrm{WM}_{\mathrm{SG}}-S_{\mathrm{ASG}})}{((\mathrm{HGS}_{\mathrm{SG}})-(\mathrm{LGS}_{\mathrm{SG}}))} \tag{4.43}$$

where

$\mathrm{LGS}_{\%}$ = volume of LGS in %

Step 7: Calculate the amount of LGS in lb/bbl:

$$\mathrm{LGS}_{\mathrm{ppb}}=(910)\left(\frac{\mathrm{LGS}_{\%}}{100}\right) \tag{4.44}$$

where

$\mathrm{LGS}_{\mathrm{ppb}}$ = amount of LGS in lb/bbl
910 = weight of 1 barrel of low gravity solids with a 2.6 g/cm^3 AGS in lb

Step 8: Calculate the volume of the weight material in %:

$$\mathrm{HGS}_{\%}=S_{\mathrm{s}}-\mathrm{LGS}_{\%} \tag{4.45}$$

where

$\mathrm{HGS}_{\%}$ = volume of high specific gravity weight material in %

Step 9: Calculate the amount of high specific gravity weight material in pounds (lb):

$$\mathrm{HGS_{ppb}} = (\mathrm{HGS_{\%}})(3.5(\mathrm{SG_{WM}})) \tag{4.46}$$

where

$\mathrm{HGS_{ppb}}$ = amount of high specific gravity weight material in lb/bbl

Step 10: Calculate the amount of bentonite (high-quality LGS) in the mud:

If the cation exchange capacity (CEC) of the formation clays and the methylene blue test (MBT) of the mud are known:

(a) Calculate the amount of bentonite in the mud in lb/bbl:

$$B_{\mathrm{ppb}} = \left(\frac{1}{\left(1 - \left(\frac{F_{\mathrm{CEC}}}{65}\right)\right)}\right)\left(((M_{\mathrm{MBT}}) - 9)\left(\frac{F_{\mathrm{CEC}}}{65}\right)\right)(\mathrm{LGS_{\%}}) \tag{4.47}$$

where

B_{PPB} = amount of bentonite in the mud in lb/bbl
F_{CEC} = CEC of the formation solids
M_{MBT} = MBT of the mud

(b) Calculate the volume of bentonite in the mud in %:

$$B_{\%} = \frac{B_{\mathrm{ppb}}}{9.1} \tag{4.48}$$

where

$B_{\%}$ = amount of bentonite in the mud in %

Alternate Step 10: If the CEC of the formation clay is ***not known***:

(a) Estimate the volume of bentonite in %:

$$B_{\mathrm{E\%}} = \frac{(M_{\mathrm{MBT}} - \mathrm{LGS_{\%}})}{8} \tag{4.49}$$

where

$B_{\mathrm{E\%}}$ = estimated volume of bentonite in %

(b) Estimate the amount of bentonite in the mud in lb/bbl:

$$B_{\mathrm{Eppb}} = (9.1)(B_{\mathrm{E\%}}) \tag{4.50}$$

Step 11: Calculate the volume of drill solids in %:

$$\mathrm{DS_{\%}} = \mathrm{LGS_{\%}} - B_{\%} \tag{4.51}$$

where

$\mathrm{DS_{\%}}$ = volume of drill solids in %

Step 12: Calculate the amount of drill solids in the mud in lb/bbl:

$$\mathrm{DS_{ppb}} = 9.1(\mathrm{DS_{\%}}) \tag{4.52}$$

where

$\mathrm{DS_{ppb}}$ = amount of drill solids in lb/bbl

Step 13: Calculate the drill solids/bentonite ratio:

$$\mathrm{DS}/B_{\mathrm{R}} = \frac{(\mathrm{DS_{ppb}})}{(B_{\mathrm{ppb}})} \tag{4.53}$$

where

$\mathrm{DS}/B_{\mathrm{R}}$ = drill solids to bentonite ratio

Note: Field experience has shown that a ratio of 2.0 or less is optimum.

Example:

Mud weight = 16.0 ppg
Chlorides = 73,000 ppm
MBT of mud = 30 lb/bbl
CEC of shale = 7 lb/bbl
ASG of shale = 2.6 g/cm^3

Retort analysis:

Water = 57.0% by volume
Oil = 7.5% by volume (0.84 ASG, diesel oil)
Solids = 35.5% by volume (4.2 ASG barite)

Step 1: Calculate the volume of the saltwater in %:

$$W_{\mathrm{s}} = \left(\left(\left(5.88 \times 10^{-8}\right)\left(73,000^{1.2}\right)\right) + 1\right)(57) = 59.3\%.$$

Step 2: Calculate the volume of suspended solids in %:

$$S_{\mathrm{s}} = 100 - 7.5 - 59.3 = 33.2\%$$

Step 3: Calculate the ASG of the saltwater:

$$W_{\mathrm{ASG}} = \left(\left(1.94 \times 10^{-6}\right)\left(73,000^{0.95}\right)\right) + 1 = 1.081$$

Step 4: Calculate the **ASG** of the solids suspended in the mud:

$$S_{\mathrm{ASG}} = \frac{((12)(16.0)) - ((59.3)(1.081)) - ((7.5)(0.84))}{(33.2)} = 3.66 \text{ g/cm}^3$$

Step 5: Because a high chloride example is being used, Step 5 is omitted.
Step 6: Calculate the volume of LGS in the mud in %:

$$\mathrm{LGS}_{\%} = \frac{((33.2)(4.2 - 3.66))}{(4.2 - 2.6)} = 11.2\%$$

Step 7: Calculate the amount of LGS in the mud in lb/bbl:

$$\mathrm{LGS}_{\mathrm{ppb}} = (910)\left(\frac{11.2}{100}\right) = 101.9\ \mathrm{lb/bbl}$$

Step 8: Calculate the volume of the weight material in %:

$$\mathrm{HGS} = 33.2 - 11.2 = 22.0\%\text{volume of barite weight material}$$

Step 9: Calculate the amount of high specific gravity weight material in pounds (lb/bbl):

$$\mathrm{HGS}_{\mathrm{ppb}} = (22.0)((3.5)(4.2)) = 323.4\ \mathrm{lb/bbl}$$

Step 10: Calculate the amount of bentonite (high quality LGS) in the mud:

(a) Calculate the amount of bentonite in the mud in lb/bbl:

$$B_{\mathrm{ppb}} = \left(\frac{1}{(1 - (\frac{7.0}{65}))}\right)\left(((30.0) - 9)\left(\frac{7.0}{65}\right)\right)(11.2) = 28.4\ \mathrm{lb/bbl}$$

(b) Calculate the volume of bentonite in the mud in %:

$$B_{\%} = \frac{(28.4)}{9.1} = 3.1\%$$

Step 11: Calculate the volume of drill solids in %:

$$\mathrm{DS}_{\%} = 11.2 - 3.1 = 8.1\%$$

Step 12: Calculate the amount of drill solids in the mud in lb/bbl:

$$\mathrm{DS}_{\mathrm{ppb}} = 9.1\ (8.1) = 73.7\,\mathrm{lb/bbl}$$

Step 13: Calculate the drill solids/bentonite ratio:

$$\mathrm{DS}/B_{\mathrm{R}} = \frac{(73.7)}{(28.4)} = 2.6$$

Spreadsheet 4.5 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Solids analysis.

4.6 Solids fractions (barite treated muds)

4.6.1 Calculate the maximum recommended solids fraction in % based on the mud weight

$$S_{\mathrm{RM}} = ((2.917)(\mathrm{MW})) - (14.17) \tag{4.54}$$

where

S_{RM} = maximum recommended solids content in % by volume
MW = mud weight in ppg

Example: Calculate the maximum recommended solids content in % by volume with the following data:

$$\text{Mud weight} = 14.0\,\text{ppg}$$

$$S_{RM} = ((2.917)(14.0)) - (14.17) = 26.7\%$$

Spreadsheet 4.6.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Max recommended solids fraction.

4.6.2 Calculate the maximum recommended LGS fraction in % based on the mud weight

$$\text{LGS}_{RM} = \left(\left(\frac{S_{RM}}{100}\right) - \left((0.3125)\left(\left(\frac{MW}{8.34}\right) - 1\right)\right)\right)(200) \tag{4.55}$$

where

LGS_{RM} = maximum recommended low gravity solids fractions, % by volume

Example: Calculate the maximum recommended solids content and LGS content in % with a 14.0 ppg water-based mud:

$$S_{RM} = ((2.917)(14.0)) - (14.17)$$
$$= 26.7\%\text{maximum recommended solids content}$$

$$\text{LGS}_{RM} = \left(\left(\frac{26.7}{100}\right) - \left((0.3125)\left(\left(\frac{14.0}{8.34}\right) - 1\right)\right)\right)(200) = 11.0\%$$

Spreadsheet 4.6.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Max recommended LGS fraction.

4.7 Dilution of mud system

4.7.1 Calculate the volume of dilution in bbl required to reduce the solids content in the mud system

(A) No limit on final volume:

$$V_d = \frac{(V_s)(\text{LGS}_o - \text{LGS}_d)}{(\text{LGS}_d - \text{LGS}_a)} \tag{4.56}$$

where

V_d = volume of dilution with base liquid or mud in bbl
LGS_o = original LGS content of mud to be diluted in %
LGS_d = desired LGS content of mud in %
LGS_a = low gravity solids from bentonite or chemicals added in the dilution volume to the mud in %

Example: Calculate the volume of dilution required to change the LGS content from 6% to 4% in 1000 bbl of mud with fresh water:

$$V_d = \frac{(1000)(6.0 - 4.0)}{(4.0 - 0.0)} = 500 \text{ bbl of dilution volume}$$

Verify the final LGS content in the new mud volume:

$$\left((1000)\left(\frac{6}{100}\right)\right) = 60 \text{ bbl of LGS content in mud to be diluted}$$

$$\frac{(60)}{(1000 + 500)} = (0.04)(100) = 4\%\text{LGS in final volume}$$

Example: Calculate the volume of dilution with a 2% bentonite slurry required to change the LGS content from 6% to 4% in 1000 bbl of mud:

$$V_d = \frac{(1000)(6.0 - 4.0)}{(4.0 - 2.0)} = 1000 \text{ bbl of dilution volume}$$

Verify the final LGS content in the new mud volume:

$$\left((1000)\left(\frac{6.0}{100}\right)\right) + \left((1000)\left(\frac{2.0}{100}\right)\right) = 80 \text{ bbl of LGS content in mud mixture}$$

$$\frac{(80)}{(1000 + 1000)} = (0.04)(100) = 4\%\text{LGS in final volume}$$

Spreadsheet 4.7.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Volume of dilution required to reduce the solids content.

4.7.2 Displacement—Barrels of water/slurry required

(A) Final volume limited:

$$V_j = \frac{(V_s)(LGS_o - LGS_d)}{(LGS_o - LGS_a)} \tag{4.57}$$

where

V_j = volume of mud in bbl to be jetted and base liquid or slurry to be added to maintain constant circulating volume.

Example: Calculate the volume of mud to be jetted or dumped from 1000 barrels to change the LGS content from 6% to 4% and maintain the mud system volume at 1000 barrels with untreated dilution fluid:

$$V_j = \frac{(1000)(6.0 - 4.0)}{(6.0 - 0.0)} = 333.3 \text{ bbl}$$

Verify the final LGS content in the new mud volume:

$$\left((1000 - 333.3)\left(\frac{6.0}{100}\right)\right) = 40.0 \text{ bbl of LGS content in mud to be diluted}$$

$$\frac{(40)}{(666.7 + 333.3)} = (0.04)(100) = 4\%\text{LGS in final volume}$$

Example: Calculate the volume of mud to be jetted or dumped to change the LGS content from 6% to 4% with a 2% bentonite slurry and maintain the mud system volume at 1000 bbl:

$$V_j = \frac{(1000)(6.0 - 4.0)}{(6.0 - 2.0)} = 500 \text{ bbl}$$

Verify the final LGS content in the new mud volume:

$$\left((1000 - 500)\left(\frac{6.0}{100}\right)\right) + \left((500)\left(\frac{2.0}{100}\right)\right) = 40 \text{ bbl of LGS content in mud mixture}$$

$$\frac{(40)}{(500 + 500)} = (0.04)(100) = 4\%\text{LGS in final volume}$$

Spreadsheet 4.7.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Displacement—Barrels of water or slurry required.

4.8 Evaluation of hydrocyclones

4.8.1 Calculate the mass of solids (for an unweighted mud) and the volume of water discarded by one cone of a hydrocyclone (desander or desilter) with a water-based mud

$$H_d = \left(\frac{(MW_d - 8.34)}{13.37}\right)(100) \tag{4.58}$$

where

H_d = volume fraction of solids discarded by the hydrocyclone in %
MW_d = mud weight of solids discharged in ppg

Example: Calculate the percent of solids discharged by one hydrocyclone with a 16.0 ppg underflow:

$$H_d = \left(\frac{(16.0 - 8.34)}{13.37}\right)(100) = 57.3\%$$

Spreadsheet 4.8.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Solids volume discarded by one hydrocyclone.

4.8.2 Calculate the mass rate of solids in gal/h

$$S_{MR} = \left(19,530\left(\frac{H_d}{100}\right)\right)\left(\frac{V_Q}{t}\right) \tag{4.59}$$

where

S_{MR} = mass rate of solids discharged by one cone of a hydrocyclone in lb/h
V_Q = volume of slurry collected in quarts
t = time required to collect sample slurry in seconds

Example: Calculate the mass rate of solids discharged by one hydrocyclone with the following data in lb/h:

Cone underflow solids = 57.3%
Volume of slurry = 2 quarts
Time to sample = 45 s

$$S_{MR} = \left(19,530\left(\frac{57.3}{100}\right)\right)\left(\frac{2.0}{45}\right) = 497.4 \text{ lb/h}$$

Spreadsheet 4.8.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Mass rate of solids discarded.

4.8.3 Calculate the volume of liquid ejected by one cone of a hydrocyclone in gal/h

Step 1: Calculate the volume fraction of solids discharged using Eq. (4.58).
Step 2: Calculate the mass rate of solids discharged using Eq. (4.59).
Step 3: Calculate the volume rate of liquid ejected by one cone:

$$V_C = (900)\left(1 - \left(\frac{H_d}{100}\right)\right)\left(\frac{V_Q}{t}\right) \tag{4.60}$$

where

V_C = volume of liquid ejected by one cone of a hydrocyclone in gal/h

Example: Calculate the evaluation of a single hydrocyclone cone with the following data:

Average mud weight of slurry sample collected = 16.0 ppg
Sample collection time = 45 s
Volume of slurry sample collected = 2 quarts

Step 1: Calculate the volume fraction of solids discharged using Eq. (4.58).

$$H_d = \left(\frac{(16.0 - 8.34)}{13.37}\right)(100) = 57.3\%$$

Step 2: Calculate the mass rate of solids discharged using Eq. (4.59).

$$S_{MR} = \left(19,530\left(\frac{57.3}{100}\right)\right)\left(\frac{2.0}{45}\right) = 497.4 \text{ lb/h}$$

Step 3: Calculate the volume rate of liquid ejected by one cone in gal/h:

$$V_C = (900)\left(1 - \left(\frac{57.3}{100}\right)\right)\left(\frac{2.0}{45}\right) = 17.1 \text{ gal/h}$$

Spreadsheet 4.8.3 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Volume of liquid ejected by one hydrocyclone.

4.8.4 Calculate the theoretical cut point of hydrocyclone

$$H_{CP} = (144.7)\left(\left(\frac{PV}{(162.5 - ((MW)(7.48)))}\right)^{0.5}\right) \tag{4.61}$$

where

H_{CP} = hydrocyclone cut point in microns (μ)
PV = plastic viscosity in cP
MW = mud weight of cone feed in ppg

Example: Calculate the theoretical cut point of a hydrocyclone discharge with the following data:

Mud weigh = 9.0 ppg
Plastic viscosity = 10 cP

$$H_{CP} = (144.7)\left(\left(\frac{10}{(162.5 - ((9.0)(7.48)))}\right)^{0.5}\right) = 46.9 \text{ microns}$$

Spreadsheet 4.8.4 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Theoretical cut point of hydrocyclone.

4.9 Evaluation of centrifuge

4.9.1 Calculate the centrifugal force of a centrifuge bowl

$$F_{CB} = \left(1.42 \times 10^{-5}\right)\left(\mathrm{RPM}^2\right)\left(B_D\right) \tag{4.62}$$

where

F_{CB} = centrifugal force of a centrifuge bowl
RPM = RPM of bowl
B_D = bowl diameter in inches

Example: Calculate the centrifugal force of a centrifuge bowl with the following data:

$$\begin{aligned}\text{Bowl RPM} &= 1950\,\text{rpm}\\ \text{Bowl diameter} &= 18\,\text{in.}\end{aligned}$$

$$F_{CB} = \left(1.42 \times 10^{-5}\right)\left(1950^2\right)(18.0) = 972\ \text{G's}$$

Spreadsheet 4.9.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Centrifugal force of a centrifuge bowl.

4.9.2 Evaluate the centrifuge underflow

(a) Calculate the centrifuge underflow mud volume in gal/min:

$$V_U = \frac{\left(V_F(\mathrm{MW_F} - \mathrm{MW_O})\right) - \left(V_D(\mathrm{MW_O} - \mathrm{MW_D})\right)}{(\mathrm{MW_U} - \mathrm{MW_O})} \tag{4.63}$$

where

V_U = volume of centrifuge underflow in gal/min
V_F = volume of mud feed into centrifuge in gal/min
$\mathrm{MW_F}$ = mud weight of feed to centrifuge in ppg
$\mathrm{MW_O}$ = mud weight of centrifuge overflow in ppg
V_D = volume of dilution liquid in gal/min
$\mathrm{MW_D}$ = mud weight of dilution liquid in pp.
$\mathrm{MW_U}$ = mud weight of underflow in ppg

(b) Calculate the fraction of old mud in underflow in %:

$$V_{OM} = \frac{((WM_{SG}(MW_D)) - MW_U)}{((WM_{SG}(MW_D)) - (MW_F)) + \left(\left(\frac{V_D}{V_F}\right)((WM_{SG}(MW_D)) - (MW_D))\right)}(100) \tag{4.64}$$

where

V_{OM} = volume fraction of old mud in underflow in %
WM_{SG} = weight material specific gravity in g/cm^3

(c) Calculate the mass rate of clay (LGS) in the underflow in lb/min:

$$W_{UC} = \frac{\left(\left(LGS_{ppb}\right)\left((V_F) - \left((V_U)\left(\frac{V_{OM}}{100}\right)\right)\right)\right)}{42} \tag{4.65}$$

where

W_{UC} = weight of clay in the centrifuge underflow in lb/min
LGS_{ppb} = LGS clay content in lb/bbl

(d) Calculate the mass rate of additives in the underflow going into the mixing pit in lb/min:

$$W_{UA} = \frac{\left((A)\left((V_F) - \left((V_U)\left(\frac{V_{OM}}{100}\right)\right)\right)\right)}{42} \tag{4.66}$$

where

W_{UA} = weight of mud additives in the underflow going into the mixing pit in lb/min
A = additive content in lb/bbl

(e) Calculate the base liquid flow rate going into mixing pit in gal/min:

$$V_{UL} = \frac{(V_F)(((MW_D)(WM_{SG}) - MW_F)) - (V_U((MW_D)(WM_{SG}) - MW_U)) - (0.613(W_{UC}) - (0.613(W_{UA})))}{(((MW_D)(WM_{SG})) - MW_D)} \tag{4.67}$$

where

V_{UL} = volume of base liquid going into the mixing pit in gal/min

(f) Calculate the percent of barite in the underflow:

(1) Determine the amount of water wetting the barite in the underflow in gal/100 lb:

$$V_{WU} = \frac{(100) - ((2.86)(MW_U))}{(MW_U - 8.34)} \tag{4.68}$$

where

V_{WU} = volume of water wetting the barite in the underflow in gal/100 lb

(2) Determine the percent barite in the underflow in %:

$$B_{\%U} = \left(\frac{(100)}{((V_{WU})(8.34)) + (100)}\right)(100) \tag{4.69}$$

where

$B_{\%U}$ = percent of barite in the underflow in %

(g) Calculate the total volume of barite and base liquid mixture in the underflow in gal/min:

$$V_{BM} = (V_{fd} + V_F) - V_U - V_{UL} - \left(\frac{W_{UC}}{21.7}\right) - \left(\frac{W_{UA}}{21.7}\right) \tag{4.70}$$

where

W_{BM} = weight of barite and base liquid mixture in the underflow in gal/min

(h) Calculate the weight of the barite in the underflow in lb/min:

$$W_{BU} = \left(\frac{B_{\%U}}{100}\right)(MW_U)(V_{BM}) \tag{4.71}$$

Example: Calculate the amount of barite in the underflow with the following data:

Feed mud density into centrifuge	= 16.2 ppg
Feed mud volume into centrifuge	= 16.5 ppg
Dilution water density	= 8.34 ppg
Dilution water volume	= 10.5 gal/min
Underflow mud density	= 23.4 ppg
Overflow mud density	= 9.3 ppg
Clay content of mud	= 22.5 lb/bbl
Additive content of mud	= 6 lb/bbl

(a) Calculate the centrifuge underflow mud volume in gal/min:

$$V_U = \frac{(16.5(16.2 - 9.3)) - (10.5(9.3 - 8.34))}{(23.4 - 9.3)} = 7.4 \text{ gal/min}$$

(b) Calculate the fraction of old mud in underflow in %:

$$V_{OM} = \frac{((4.2(8.34)) - 23.4)}{((4.2(8.34)) - (16.2)) + \left(\left(\frac{10.5}{16.5}\right)((4.2(8.34)) - (8.34))\right)}(100) = 32.5\%$$

(c) Calculate the mass rate of clay (LGS) in the underflow in lb/min:

$$W_{UC} = \frac{\left((22.5)\left(16.5 - \left((7.4)\left(\frac{32.5}{100}\right)\right)\right)\right)}{42} = 7.55 \text{ lb/min}$$

(d) Calculate the mass rate of additives in the underflow going into the mixing pit in lb/min:

$$W_{UA} = \frac{\left((6.0)\left((16.5) - \left((7.4)\left(\frac{32.3}{100}\right)\right)\right)\right)}{42} = 2.0 \text{ lb/min}$$

(e) Calculate the base liquid flow rate going into mixing pit in gal/min:

$$V_{UL} = \frac{(16.5)(((8.34)(4.2) - 16.2)) - (7.4((8.34)(4.2) - 23.4)) - (0.613(7.56) - (0.613(2.0)))}{(((8.34)(4.2)) - 8.34)}$$

$$= 8.2 \text{ gal/min}$$

(f) Calculate the percent of barite in the underflow:

(1) Determine the amount of water wetting the barite in the underflow in gal/100 lb:

$$V_{WU} = \frac{(100) - ((2.86)(23.4))}{(23.4 - 8.34)} = 2.2 \text{ gal/100 lb barite}$$

(2) Determine the percent barite in the underflow in %:

$$B_{\%U} = \left(\frac{(100)}{((2.2)(8.34)) + (100)}\right)(100) = 84.5\%$$

(g) Calculate the total volume of barite and base liquid mixture in the underflow in gal/min:

$$V_{BM} = (10.5 + 16.5) - 7.4 - 8.2 - \left(\frac{7.56}{21.7}\right) - \left(\frac{2.0}{21.7}\right) = 10.96 \text{ gal/min}$$

(h) Calculate the weight of the barite in the underflow in lb/min:

$$W_{BU} = \left(\frac{84.5}{100}\right)(23.4)(10.96) = 216.7 \text{ lb/min}$$

Spreadsheet 4.9.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Evaluate the centrifuge underflow.

4.10 Mud volume required to drill 1000 ft of hole

Mud volume is required to fill the hole that is drilled with the bit. That volume can be calculated based on the bit size and length of hole drilled. The cuttings that are removed from the hole are coated with mud used to drill that section. Field test have indicated that the amount of mud that accompanies the cuttings is related to the size of the bit teeth cutting the formation. The "MTCR" (mud to cutting ratio) can be utilized to estimate the amount of mud discharged with the cuttings that are separated by the shale shakers. Washout or hole enlargement should be included in the calculation.

Calculate the amount of mud required to drill 1000 ft of 10 in. (9⅞″ bit) hole:

(a) Determine the volume of hole drilled:

$$V_{HD} = \left((D_H)\left(1 + \frac{(W_O)}{100}\right)\right)^2 \tag{4.72}$$

where

V_{HD} = volume of hole drilled in bbl
W_O = hole washout in %

(b) Determine the volume of mud separated with the cuttings in bbl:

$$V_{MS} = (\text{MTCR})(V_{HD}) \tag{4.73}$$

where

V_{MS} = volume of mud separated with cuttings in bbl

(c) Determine the total volume of mud required to drill 1000 ft of 10.0 in. hole with 10% washout:

$$V_T = (V_{HD}) + (V_{MS}) \tag{4.74}$$

Example: Calculate the volume of mud required to drill 1000 of 10 in. hole with 10% washout:

(a) Determine the volume of hole drilled:

$$V_{HD} = \left((10.0)\left(1 + \frac{(10.0)}{100}\right)\right)^2 = 121.0 \text{ bbl}$$

(b) Determine the volume of mud separated with the cuttings with a 1.0 MTCR in bbl:

$$V_{MS} = (1.0)(121.0) = 121.0 \text{ bbl}$$

(c) Determine the total volume of mud required to drill 1000 ft of 10.0 in. hole with 10% washout:

$$V_T = (121.0) + (121.0) = 242.0 \text{ bbl}$$

Spreadsheet 4.10 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Mud volume required to drill 1000 ft of hole.

4.11 Determine the downhole density of the base oil or brine in the mud at depth of interest in ppg

This calculation is made for a specific depth location in the well at the conditions that occur at that depth. To determine the ESD for the entire well, values must be calculated for each depth interval and integrated in a stepwise procedure down to the total well depth.

(a) Determine the volume of salt in a 19.3% (v/v) brine phase of the mud in %:

$$V_{\text{salt}} = \left(\left(13.091 \times 10^{-4}\right)(\text{CaCl}_2)\right) + \left(\left(8.44 \times 10^{-5}\right)\left(\text{CaCl}_2{}^2\right)\right) \tag{4.75}$$

where

V_{salt} = volume of $CaCl_2$ salt in the brine in %
$CaCl_2$ = weight percent of calcium chloride in %

(b) Determine the volume of brine in the mud in %:

$$V_{\text{B}} = V_{\text{salt}} + V_{\text{W}} \tag{4.76}$$

where

V_{B} = volume of brine in the mud in %
V_{W} = volume of water in the mud from the retort test in %

(c) Determine the corrected volume of solids in the mud in %:

$$V_{\text{CS}} = (100.0 - V_{\text{O}} - V_{\text{B}}) \tag{4.77}$$

where

V_{CS} = volume of corrected solids in the mud in %
V_{O} = volume of oil determined in the retort test of the mud in %
V_{B} = volume of brine in %

(d) Determine the density of the base oil or brine in the mud in ppg:

$$\text{MW}_f = \left[(a_1) + ((b_1)(P)) + \left((c_1)\left(P^2\right)\right) + ((a_2) + ((b_2)(P)) + \left(\left((c_2)\left(P^2\right)\right)(T)\right))\right] \tag{4.78}$$

where

MW_{f} = density of the base oil (MW_{O})or brine fluid (MW_{B}) at depth of interest in ppg
a_1 = density correction coefficient from Table 4.1 for pressure in ppg
b_1 = density correction coefficient from Table 4.1 for pressure in ppg/psi
c_1 = density correction coefficient from Table 4.1 for pressure in ppg/psi^2
P = pressure at depth of interest in psi
a_2 = density correction coefficient from Table 4.1 for temperature in ppg/°F

TABLE 4.1 Temperature and pressure coefficients for determining fluid density.

	$CaCl_2$ (19.3%, w/w)	Diesel	Mineral oil	Internal olefin	Paraffin
Pressure coefficients					
a_1 (ppg)	9.9952	7.3183	6.9912	6.8538	6.9692
b_1 (ppg/psi)	1.77×10^{-5}	5.27×10^{-5}	2.25×10^{-5}	2.23×10^{-5}	3.35×10^{-5}
c_1 (ppg/psi^2)	6.0×10^{-11}	-8.0×10^{-10}	-1.0×10^{-10}	-2.0×10^{-10}	-5.0×10^{-10}
Temperature coefficients					
a_2 (ppg/°F)	-2.75×10^{-3}	-3.15×10^{-3}	3.28×10^{-3}	-3.39×10^{-3}	3.46×10^{-3}
b_2 (ppg/psi/°F)	3.49×10^{-8}	7.46×10^{-8}	1.17×10^{-7}	1.12×10^{-7}	-1.64×10^{-8}
c_2 (ppg/psi^2/°F)	-9.0×10^{-13}	-1.0×10^{-12}	-3.0×10^{-12}	-2.0×10^{-12}	2.0×10^{-13}

Ref: Table 3 of API RP 13D, June 2006, p. 26.

b_2 = density correction coefficient from Table 4.1 for temperature in ppg/psi/°F
c_2 = density correction coefficient from Table 4.1 for temperature in ppg/psi^2/°F
T = temperature at depth of interest in °F

(e) Determine the static density of the mud at a specific depth in the well in ppg:

$$\mathrm{ESD_{DI}} = \frac{\left(\left(\frac{V_O}{100}\right)(\mathrm{MW_O})\right) + \left(\left(\frac{V_B}{100}\right)(\mathrm{MW_B})\right) + \left(\left(\frac{V_{CS}}{100}\right)((S_{ASG})(8.34))\right)}{\left(\frac{V_T}{100}\right)} \quad (4.79)$$

where

$\mathrm{ESD_{DI}}$ = equivalent static density at depth of interest in ppg
V_O = volume of oil in the mud from the retort test in %
$\mathrm{MW_O}$ = weight of oil in ppg
V_B = volume of brine in %
$\mathrm{MW_B}$ = weight of bring in ppg
V_S = volume of solids in the mud from the retort test in %
S_{ASG} = average specific gravity of the solids in g/cm^3
V_T = total of volume of mud tested in the retort in %

Example: Determine the downhole density of a 12.0 ppg OBM at a depth of interest of 6000 ft in ppg with the following data:

Depth of interest	= 6000 ft
Mud weight	= 12.0 ppg
Oil content	= 64% by retorted volume
Water content	= 16% by retorted volume
$CaCl_2$ content	= 19.3% by volume

(a) Determine the volume of salt in a 19.3% (v/v) brine phase of the mud in %:

$$V_{salt} = \left(\left(13.091 \times 10^{-4}\right)(19.3)\right) + \left(\left(8.44 \times 10^{-5}\right)\left(19.3^2\right)\right) = 0.0567\%$$

(b) Determine the volume of brine in the mud in %:

$$V_B = (0.0567) + (16.0) = 16.0567\%$$

(c) Determine the corrected volume of solids in the mud in %:

$$V_{CS} = (100.0 - 64.0 - 16.0567) = 19.94\%$$

(d) Determine the density of the internal olefin in the mud in ppg:

Use a pressure of $(6000)(0.052)(12.0) = 3744$ psi

and a temperature of $(1.0)(60) + (80) = 140^{\circ}$F.

$$\begin{aligned}\mathrm{MW_O} &= \left[\left((6.8538) + \left(\left(2.23\times10^{-5}\right)(3744)\right) + \left(\left(-2\times10^{-10}\right)\left(3744^2\right)\right)\right)\right] \\ &+ \left[\left(\left(-3.39\times10^{-3}\right) + \left(\left(1.12\times10^{-12}\right)(3744)\right) + \left(\left(-2.0\times10^{-12}\right)\left(3744^2\right)\right)\right)(140)\right] \\ &= 6.51\,\mathrm{ppg}\end{aligned}$$

(e) Determine the density of the brine in the mud in ppg:

$$\begin{aligned}\mathrm{MW_B} &= \left[\left((9.992) + \left(\left(1.77\times10^{-5}\right)(3744)\right) + \left(\left(6.0\times10^{-11}\right)\left(3744^2\right)\right)\right)\right] \\ &+ \left[\left(\left(-2.75\times10^{-3}\right) + \left(\left(3.49\times10^{-8}\right)(3744)\right)\right.\right. \\ &+ \left.\left.\left(\left(-9.0\times10^{-13}\right)\left(3744^2\right)\right)\right)(140)\right] = 9.69\,\mathrm{ppg}\end{aligned}$$

(f) Determine the static density of the mud at a specific depth of 6000 ft in the well in ppg:

$$\begin{aligned}\mathrm{ESD_{DI}} &= \frac{\left(\left(\frac{64.0}{100}\right)(6.51)\right) + \left(\left(\frac{16.0567}{100}\right)(9.69)\right) + \left(\left(\frac{19.94}{100}\right)((3.82)(8.34))\right)}{\left(\frac{100}{100}\right)} \\ &= 12.1\ \mathrm{ppg}\end{aligned}$$

Spreadsheet 4.11 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Downhole MW of OBM at depth of interest.

Appendix: Supplementary material

Supplementary material related to this chapter can be found online at https://www.elsevier.com/books-and-journals/book-companion/9780323905497.

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CHAPTER

5

Cementing calculations

5.1 Cement additive calculations

Step 1: Calculate the weight of additive per sack of cement in lb:

$$W_{ca} = (V_a)(94.0) \tag{5.1}$$

where

W_{ca} = weight of cement additive in lb
V_a = volume percent (%) of additive for cement mixture
94.0 = pounds of cement in one sack (sx)

Step 2: Total water requirement for each sack of cement in gal/sx:

$$V_{cwt} = V_{cw} + V_{aw} \tag{5.2}$$

where

V_{cwt} = total volume of water required for each sack of cement in gal/sx
V_{cw} = volume of water required for cement portion in gal/sk (Table 5.1)
V_{aw} = volume of water required for additive portion in gal/sk (Table 5.1)

Step 3: Calculate the total volume of the cement slurry in gal/sx:

$$V_{cs} = \left(\frac{94.0}{(SG_{cmt})(8.33)}\right) + \left(\frac{W_{ca}}{(SG_a)(8.33)}\right) + V_{cwt} \tag{5.3}$$

where

V_{cs} = total volume of cement slurry in gal/sx
SG_{cmt} = specific gravity of dry cement in gm/cm^3
SG_a = specific gravity of dry additive in gm/cm^3
8.33 = density of fresh water in lb/gal

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TABLE 5.1 Water requirements and specific gravity of common cement additives.

Material	Water requirement (gal/94 lb/sk)	Specific gravity
API class cement		
Class A & B	5.2	3.14
Class C	6.3	3.14
Class D & E	4.3	3.14
Class G	5.0	3.14
Class H	4.3–5.2	3.14
Chem Comp cement	6.3	3.14
Common cement additives		
Attapulgite	1.3 for 2% gel in cement	2.89
Cement Fondu	4.5	3.23
Lumnite cement	4.5	3.20
Trinity Lite-weight cement	9.7	2.80
Bentonite	1.3 for 2% gel in cement	2.65
Calcium carbonate powder	0	1.96
Calcium chloride	0	1.96
Cal-Seal (gypsum cement)	4.5	2.70
CFR-1	0	1.63
CFR-2	0	1.30
D-Air-1	0	1.35
D-Air-2	0	1.005
Diacel A	0	2.62
Diacel D	3.3–7.4 for 10% in cement	2.10
Diacel LWL	0 (up to 0.7%)	1.36
	0.8:1/1% in cement	
Gilsonite	2 for 50 lb/ft^3	1.07
Halad-9	0 (up to 5%)/0.4–0.5 over 5%	1.22
Halad-14	0	1.31
HR-4	0	1.56
HR-5	0	1.41

TABLE 5.1 Water requirements and specific gravity of common cement additives—cont'd

Material	Water requirement (gal/94lb/sk)	Specific gravity
HR-7	0	1.30
HR-12	0	1.22
HR-15	0	1.57
Hydrated lime	14.4	2.20
Hydromite	2.82	2.15
Iron carbonate	0	3.70
LA-2 Latex	0.8	1.10
NF-D	0	1.30
Perlite regular	4/8lb/ft^3	2.20
Perlite 6	6/38lb/ft^3	–
Pozmix A	4.6–5.0	2.46
Salt (NaCl)	0	2.17
Sand Ottawa	0	2.63
Silica flour	1.6 for 35% flour in cement	2.63
Coarse silica	0	2.63
Spacer sperse	0	1.32
Spacer mix (liquid)	0	0.932
Tuf Additive No. 1	0	1.23
Tuf Additive No. 2	0	0.88
Tuf Plug	0	1.28

Step 4: Calculate the yield of the cement slurry in ft^3/sx:

$$Y_{cs} = \left(\frac{V_{cs}}{7.48}\right) \tag{5.4}$$

where

Y_{cs} = yield of cement slurry mixture in ft^3/sx
7.48 = volume of fresh water in gal/ft^3

Step 5: Calculate the cement slurry density in lb/gal:

$$W_{cs} = \left(\frac{94.0 + W_{ca} + ((8.33)(V_{cwt}))}{V_{cs}}\right) \tag{5.5}$$

where

W_{cs} = weight of cement slurry in lb/gal

Example: Using a Class A cement plus 4% bentonite with normal mixing water, calculate the following:

1. Amount of bentonite to add in lb/sx
2. Total water volume required for the slurry in gal/sk determined from Table 5.1. (In this case—5.2 gal/94 lb sx of cement. Water volume for the bentonite additive from Table 5.1 is $1.3 \times 2 = 2.6$ gal.)
3. Slurry yield in ft^3/sx
4. Slurry weight in lb/gal

Step 1: Calculate the weight of the bentonite additive:

$$W_{ca} = (0.04)(94.0) = 3.76\,\text{lb/sx}$$

Step 2: Calculate the total water requirement per sack of cement used from Table 5.1:

$$V_{cwt} = 5.2 + 2.6 = 7.8\,\text{gal/sx}$$

Step 3: Calculate the total volume of the cement slurry:

$$V_{cs} = \left(\frac{94.0}{(3.14)(8.33)}\right) + \left(\frac{3.76}{(2.65)(8.33)}\right) + 7.8 = 11.56\ \text{gal/sx.}$$

Step 4: Calculate the yield of the cement slurry:

$$Y_{cs} = \left(\frac{11.56}{7.48}\right) = 1.55\,ft^3/\text{sx}$$

Step 5: Calculate the cement slurry density:

$$W_{cs} = \left(\frac{94.0 + 3.76 + ((8.33)(7.8))}{11.56}\right) = 14.08\ \text{lb/gal}$$

Spreadsheet 5.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Cement additive calculations.

5.2 Water requirements

Step 1: Calculate the weight of cement additive materials in lb/sx:

$$W_{ca} = (94.0)\left(\frac{V_a}{100}\right) \tag{5.6}$$

where

W_{ca} = weight of cement additive materials in lb/sx
V_a = volume percent (%) of additive to be mixed

Step 2: Calculate the volume of cement and additives:

$$V_{ca} = \left(\frac{94.0}{(SG_{cmt})(8.33)} + \left(\frac{W_{ca}}{(SG_a)(8.33)}\right)\right) \tag{5.7}$$

where

V_{ca} = volume of cement and additives in gal/sx
SG_{cmt} = specific gravity of cement in gm/cm^3
SG_a = specific gravity of additive in gm/cm^3

Step 3: Calculate the water requirement for the slurry:

$$V_{cwt} = \frac{(94.0 + W_{ca}) - ((V_{ca})(W_{cs}))}{(W_{cs} - 8.33)} \tag{5.8}$$

where

W_{cwt} = water requirement per sack of cement in gal/sx
W_{cs} = weight of cement slurry in lb/gal

Step 4: Calculate the yield of the cement slurry in ft^3/sx:

$$Y_{cs} = \left(\frac{V_{cs} + V_{cwt}}{7.48}\right) \tag{5.9}$$

where

Y_{cs} = yield of cement slurry mixture in ft^3/sx
7.48 = volume of fresh water in gal/ft^3

Step 5: Calculate the cement slurry density in lb/gal:

$$W_{cs} = \left(\frac{94.0 + W_{ca} + ((8.33)(V_{cwt}))}{(V_{cs} + V_{cwt})}\right) \tag{5.10}$$

where

W_{cs} = weight of cement slurry in lb/gal

Example: Using a Class H cement plus 6% bentonite to be mixed at 14.0 lb/gal.

Calculate the following:

1. Bentonite requirement in lb/sx
2. Water requirement in gal/sx
3. Slurry yield in ft^3/sx
4. Check the slurry weight in lb/gal

Step 1: Calculate the weight of cement additive materials:

$$W_{ca} = (94.0)\left(\frac{6}{100}\right) = 5.64 \text{ lb per sx of bentonite}$$

Step 2: Calculate the volume of cement and additives:

$$V_{ca} = \left(\frac{94.0}{(3.14)(8.33)} + \left(\frac{5.64}{(2.65)(8.33)}\right)\right) = 3.85 \text{ gal per sx of cement}$$

Step 3: Calculate the water requirement for the slurry:

$$V_{cwt} = \frac{(94.0 + 5.64) - ((3.85)(14.0))}{(14.0 - 8.33)} = 8.07 \text{ gal per sx of cement}$$

Step 4: Calculate the yield of the cement slurry:

$$Y_{cs} = \left(\frac{3.85 + 8.06}{7.48}\right) = 1.59 \text{ ft}^3 \text{ per sx}$$

Step 5: Recheck the cement slurry density:

$$W_{cs} = \left(\frac{94.0 + 5.64 + ((8.33)(8.06))}{(3.85 + 8.06)}\right) = 14.0 \text{ lb per gal}$$

Spreadsheet 5.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Water requirements.

5.3 Field cement additive calculations

When bentonite is to be prehydrated, the amount of bentonite added is calculated based on the total amount of mixing water used.

Step 1: Calculate the volume of mixing water required in gal:

$$V_{mwr} = (V_{cmt})(V_{mwx}) \tag{5.11}$$

where

V_{mwr} = volume of mixing water required in gallons
V_{cmt} = amount of cement used in sacks
V_{mwx} = amount of mixing water per sack of cement in gal/sx

Step 2: Calculate the total weight of mixing water in lb:

$$W_{mwt} = (V_{mwr})(8.33) \tag{5.12}$$

where

W_{mwt} = total weight of mixing water in lb

Step 3: Calculate the amount of bentonite required in lb:

$$W_b = (W_{mwt})\left(\frac{V_b}{100}\right) \tag{5.13}$$

where

W_b = weight of bentonite required in lb
V_b = volume of bentonite in %

Note: Other additives are calculated based on the weight of the cement.

Step 4: Calculate the weight of the cement:

$$W_{cmt} = (V_{cmt})(94.0) \tag{5.14}$$

Step 5: Calculate the weight of the other additives:

$$W_a = (W_{cmt})\left(\frac{V_a}{100}\right) \tag{5.15}$$

where

W_a = weight of additive in lb
V_a = volume of additive in %

Cement program:	Cement	= 240 sk
	Slurry density	= 13.8 lb/gal
	Mixing water	= 8.6 gal/sk
	Bentonite to be *prehydrated*	= 1.5%
	Halad (fluid loss from Table 5.1)	= 0.50%
	CFR-2 (dispersant from Table 5.1)	= 0.40%

Step 1: Calculate the volume of mixing water required in gal:

$$V_{mwr} = (240)(8.6) = 2064 \text{ gal}$$

Step 2: Calculate the total weight of mixing water in lb:

$$W_{\text{mwt}} = (2064)(8.33) = 17,193.1 \text{ lb}$$

Step 3: Calculate the amount of bentonite required in lb:

$$W_{\text{b}} = (17,193.1)\left(\frac{1.5}{100}\right) = 257.9 \cong 258 \text{ lb}$$

Note: Other additives are calculated based on the weight of the cement.

Step 4: Calculate the weight of the cement in lb:

$$W_{\text{cmt}} = (240)(94.0) = 22,560 \text{ lb}$$

Step 5: Calculate the weight of the other additives:

$$W_{\text{a}} = (22,560)\left(\frac{0.50}{100}\right) = 112.8 \text{ lb of Halad}$$

$$W_{\text{a}} = (22,560)\left(\frac{0.40}{100}\right) = 90.2 \text{ lb of CFR} - 2$$

Spreadsheet 5.3 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Field cement additive calculations.

5.4 Weighted cement calculations

Step 1: Calculate the amount of high density additive required per sack of cement to achieve a required cement slurry density in lb:

$$W_{\text{ca}} = \frac{(((W_{\text{cs}})(11.207983))/\text{SG}_{\text{cmt}}) + ((W_{\text{cs}})(V_{\text{cwt}})) - 94.0 - ((8.33)(W_{\text{cs}}))}{\left(1 + \left(\frac{V_{\text{wa}}}{100}\right)\right) - \left(\frac{W_{\text{cs}}}{(\text{SG}_{\text{a}})(8.33)}\right) - \left((W_{\text{cs}})\left(\frac{V_{\text{wa}}}{100}\right)\right)} \tag{5.16}$$

where

W_{ca} = additive required per sack of cement in lb
W_{cs} = required cement slurry density in lb/gal
SG_{cmt} = specific gravity of cement
V_{cwt} = water requirement of cement in gal/sk
V_{wa} = water requirement of additive in gal/sk
SGa = specific gravity of additive (Table 5.2)

TABLE 5.2 Weighting agents for cement.

Additive	Water requirement (gal/94 lb/sk)	Specific gravity
Hematite	0.34	5.02
Ilmenite	0	4.67
Barite	2.5	4.23
Sand	0	2.63

Example: Calculate how much hematite (in lb/sk) is required to increase the density of Class H cement to 17.5 lb/gal:

Water requirement of cement = 4.3 gal/sk
Water requirement of additive (hematite) = 0.34 gal/sk
Specific gravity of cement = 3.14
Specific gravity of additive (hematite) = 5.02

$$W_a = \frac{(((17.5)(11.207983))/3.14) + ((17.5)(4.3)) - 94.0 - ((8.33)(4.3))}{\left(1 + \left(\frac{0.34}{100}\right)\right) - \left(\frac{17.5}{(5.02)(8.33)}\right) - \left((17.5)\left(\frac{0.34}{100}\right)\right)}$$
$$= 15.0 \text{ lb per sack of cement}$$

Spreadsheet 5.4 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Weighted cement calculations.

5.5 Calculate the number of sacks required for cement job

If the number of feet to be cemented is known, use the following:
Step 1: Calculate the following capacities:

(a) Annular capacity in ft^3/ft:

$$V_{acf} = \left(\frac{D_h^2 - D_p^2}{183.35}\right) \tag{5.17}$$

where

V_{acf} = annular capacity in ft^3/ft

(b) Casing capacity in ft^3/ft:

$$V_{cc} = \left(\frac{D_i^2}{183.35}\right) \tag{5.18}$$

where

V_{cc} = casing capacity in ft^3/ft

(c) Casing capacity in bbl/ft:

$$V_{pc} = \left(\frac{D_i^2}{1029.4}\right) \tag{5.19}$$

where

V_{pc} = casing capacity in bbl/ft

Step 2: Calculate the number of sacks of **LEAD** or **FILLER** cement required:

$$SR_L = \frac{(L_{cmt})(V_{ac})(1 + (V_{\%e}/100))}{Y_L} \tag{5.20}$$

where

SR_L = cement required for **LEAD** job in sk
L_{cmt} = length of section to be cemented in ft
$V_{\%e}$ = excess volume for job in %
Y_L = yield of **LEAD** cement in ft^3/sk

Step 3: Calculate the number of sacks of **TAIL** or **NEAT** cement required:

$$SR_{Ta} = \frac{(L_{cmt})(V_{ac})(1 + (V_{\%e}/100))}{Y_T} \tag{5.21}$$

where

SR_{Ta} = cement required in annulus for the **TAIL** job in sk
Y_T = yield of **TAIL** cement in ft^3/sk

Step 4: Calculate the amount of cement required inside the casing for the **TAIL** job in sks:

$$SR_{Tc} = \frac{(L_{cmt})(V_{cc})}{Y_T} \tag{5.22}$$

where

SR_{Tc} = cement required inside the casing for the **TAIL** job in sks

Step 5: Calculate the total sacks of **TAIL** cement required:

$$SR_{Tt} = SR_{Ta} + SR_{Tc} \tag{5.23}$$

where

SR_{Tt} = total cement required for the **TAIL** job in sks

Step 6: Calculate the barrels of mud required to bump the top plug:

$$V_{fc} = (V_{pc})(L_{fc}) \tag{5.24}$$

where

V_{fc} = casing capacity down to the float collar in bbl
L_{fc} = length of casing from the surface to the float collar in ft

Step 7: Calculate the number of strokes required to bump the plug:

$$S_{bp} = \left(\frac{V_{pc}}{O_p}\right) \tag{5.25}$$

where

S_{bp} = strokes to bump the plug

Example: Calculate the following based on the data listed below:

1. How many sacks of LEAD cement will be required?
2. How many sacks of TAIL cement will be required?
3. How many barrels of mud will be required to bump the plug?
4. How many strokes will be required to bump the top plug?

Data:	Casing setting depth	= 3000 ft
	Hole size	= 17½ in.
	Casing—54.5 lb/ft	= 13⅜ in.
	Casing ID	= 12.615 in.
	Float collar (number of feet above shoe)	= 44 ft
	Pump (5½ in. by 14 in. duplex @ 90% eff)	= 0.112 bbl/stk

Cement program:	LEAD cement (13.8 lb/gal)	= 2000 ft
	Slurry yield	= 1.59 ft^3/sk
	TAIL cement (15.8 lb/gal)	= 1000 ft
	Slurry yield	= 1.15 ft^3/sk
	Excess volume	= 50%

Step 1: Calculate the following capacities:

(a) Annular capacity in ft^3/ft:

$$V_{ac} = \frac{(17.5^2 - 13.375^2)}{183.35} = 0.6946 \text{ ft}^3/\text{ft}$$

(b) Casing capacity in ft^3/ft:

$$V_{cc} = \left(\frac{12.615^2}{183.35}\right) = 0.8679 \text{ ft}^3/\text{ft}$$

(c) Casing capacity in bbl/ft:

$$V_{pc} = \left(\frac{12.615^2}{1029.4}\right) = 0.1546 \text{ bbl/ft}$$

Step 2: Calculate the number of sacks of LEAD or FILLER cement required:

$$SR_L = \frac{(2000)(0.6946)\left(1 + \left(\frac{50}{100}\right)\right)}{1.59} = 1311 \text{ sx}$$

Step 3: Calculate the number of sacks of TAIL or NEAT cement required:

$$SR_{Ta} = \frac{(1000)(0.6946)\left(1 + \left(\frac{50}{100}\right)\right)}{1.15} = 906 \text{ sx}$$

Step 4: Calculate the amount of cement required inside the casing for the TAIL job in sks:

$$SR_{Tc} = \frac{(44)(0.8679)}{1.15} = 33 \text{ sx}$$

Step 5: Calculate the total sacks of TAIL cement required:

$$SR_{Tt} = 906 + 33 = 939 \text{ sx}$$

Step 6: Calculate the barrels of mud required to bump the top plug:

$$V_{fc} = (0.1546)(3000 - 44) = 457 \text{ bbl}$$

Step 7: Calculate the number of strokes required to bump the top plug:

$$S_{bp} = \left(\frac{457}{0.112}\right) = 4080 \text{ stks}$$

Spreadsheet 5.5 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Number of sacks required for cement job.

5.6 Calculations for the number of feet to be cemented

If the number of sacks of cement is known, use the following:

Step 1: Calculate the following capacities:

(a) Calculate annular capacity with Eq. (5.17) in ft^3/ft:
(b) Calculate casing capacity with Eq. (5.18) in ft^3/ft:
(c) Calculating casing capacity with Eq. (5.18) in bbl/ft:

Step 2: Calculate the slurry volume in ft^3:

$$V_{cs} = (ST_t)(Y_s) \tag{5.26}$$

where

V_{cs} = volume of cement slurry in ft^3

Step 3: Calculate the amount of cement to be left in casing in ft^3:

$$V_{cc} = (L_{csg} - D_{st})(V_{cc}) \tag{5.27}$$

where

V_{cc} = volume of cement left in casing in ft^3

Step 4: Calculate the height of cement in the annulus in ft:

$$H_{cmt} = \frac{\left(\frac{(V_{cs} - V_{cc})}{V_{ac}}\right)}{\left(1 + \left(\frac{V_{\%c}}{100}\right)\right)} \tag{5.28}$$

where

H_{cmt} = height of cement slurry in the annulus in ft

Step 5: Calculate the depth of the top of the cement slurry in the annulus in ft:

$$D_{tcmt} = (L_c - L_{cmt}) \tag{5.29}$$

where

D_{tcmt} = depth of the top of the cement slurry in the annulus in ft

Step 6: Calculate the volume of mud required to displace the cement in bbl:

$$V_{dcmt} = (L_p - L_{as})(V_p) \tag{5.30}$$

where

V_{dcmt} = volume of mud required to displace cement slurry in bbl
L_{as} = length of distance between cementing tool and casing shoe in ft

Step 7: Calculate the number of strokes required to displace the cement slurry:

$$S_{\text{dcmt}} = \left(\frac{V_{\text{dcmt}}}{O_{\text{p}}}\right) \tag{5.31}$$

where

S_{dcmt} = the number of strokes required to displace the cement

Example: Calculate the following from the data listed below:

1. Height of the cement in the annulus in ft
2. Amount of the cement in the casing in ft^3
3. Depth of the top of the cement in the annulus in ft
4. Volume of mud required to displace the cement in bbl
5. Number of strokes required to displace the cement

Data: Casing setting depth = 3000 ft
Hole size = 17½ in.
Casing (54.5 lb/ft) = 13⅜ in.
Casing ID = 12.615 in.
Drill pipe (5.0 in., 19.5 lb/ft) = 0.01776 bbl/ft
Pump (7 × 12 in. triplex @ 95% eff.) = 0.136 bbl/stk
Cementing tool (number of feet above shoe) = 100 ft

Cementing program: **NEAT** cement = 500 sk
Slurry yield = 1.15 ft^3/sk
Excess volume = 50%=

Step 1: Calculate the following capacities:

(a) Annular capacity between casing and hole in ft^3/ft:

$$V_{\text{ac}} = \left(\frac{(17.5_2 - 13.375_2)}{183.35}\right) = 0.6946 \text{ ft}^3/\text{ft}$$

(b) Casing capacity in ft^3/ft:

$$V_{\text{cc}} = \left(\frac{12.615^2}{183.35}\right) = 0.8679 \text{ ft}^3/\text{ft}$$

Step 2: Calculate the cement slurry volume in ft^3:

$$V_{\text{cs}} = (500)(1.15) = 575 \text{ ft}^3$$

Step 3: Calculate the amount of cement, ft^3, to be left in the casing:

$$V_{\text{cc}} = (3000 - 2900)(0.8679) = 86.8 \text{ ft}^3$$

Step 4: Calculate the height of the cement in the annulus in ft:

$$H_{\text{cmt}} = \frac{\left(\frac{575-86.8}{0.6946}\right)}{\left(1 + \left(\frac{50}{100}\right)\right)} = 469 \text{ ft}$$

Step 5: Calculate the depth of the top of the cement in the annulus:

$$D_{\text{tcmt}} = (3000 - 469) = 2531 \text{ ft}$$

Step 6: Calculate the number of barrels of mud required to displace the cement:

$$V_{\text{dcmt}} = (3000 - 100)(0.01776) = 51.5 \text{ bbl}$$

Step 7: Calculate the number of strokes required to displace the cement:

$$S_{\text{dcmt}} = \left(\frac{51.5}{0.136}\right) = 379 \text{ stks}$$

Spreadsheet 5.6 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Number of feet to be cemented.

5.7 Setting a balanced cement plug

Step 1: Calculate the following capacities:

(a) Calculate the annular capacity between pipe or tubing and hole or casing in ft^3/ft:

$$V_{\text{ac}} = \left(\frac{D_{\text{h}}^2 - D_{\text{p}}^2}{183.35}\right) \tag{5.32}$$

(b) Calculate the annular capacity between pipe or tubing and hole or casing in ft/bbl:

$$V_{\text{acf}} = \left(\frac{1029.4}{D_{\text{h}}^2 - D_{\text{p}}^2}\right) \tag{5.33}$$

(c) Hole or casing capacity in ft^3/ft:

$$V_{cc} = \left(\frac{D_{\text{i}}^2}{183.35}\right) \tag{5.34}$$

(d) Drill pipe or tubing capacity in ft^3/ft:

$$V_{dpc} = \left(\frac{D_i^2}{183.35}\right) \tag{5.35}$$

(e) Drill pipe or tubing capacity in bbl/ft:

$$V_{pcb} = \left(\frac{D_i^2}{1029.4}\right) \tag{5.36}$$

Step 2: Calculate the number of SACKS of cement required for a given length of plug, OR determine the FEET of plug for a given number of sacks of cement:

(a) Determine the number of SACKS of cement required for a given length of plug:

$$SR_{plug} = \frac{(L_{plug})(V_{acf})\left(1 + \left(\frac{V_{\%e}}{100}\right)\right)}{Y_{cs}} \tag{5.37}$$

where

SR_{plug} = cement required for a given length of plug in sk
L_{plug} = length of plug in ft
V_{acf} = hole or casing capacity in ft^3/ft
$V_{\%e}$ = excess volume for job in %
Y_{cs} = yield of cement slurry in ft^3/sk

Note: If no excess is to be used, enter ZERO for excess volume.

(b) Determine the number of FEET of plug for a given number of sacks of cement:

$$L_{plug} = \left(\frac{\left((SR_{plug})(Y_{cs})/V_{ac}\right)}{\left(1 + \left(\frac{V_{\%e}}{100}\right)\right)}\right) \tag{5.38}$$

Note: If no excess is to be used, enter ZERO for excess volume.

Step 3: Calculate the spacer volume (usually water) to be pumped behind the cement slurry to balance the plug in bbl:

$$V_{\text{spacer behind}} = \left(\frac{V_{ac}}{\left(1 + \left(\frac{V_{\%e}}{100}\right)\right)}\right)(V_{\text{spacer ahead}})(V_{pc}) \tag{5.39}$$

where

$V_{\text{spacer behind}}$ = volume of spacer pumped behind cement plug in bbl
$V_{\text{spacer ahead}}$ = volume of spacer pumped ahead of cement plug in bbl

Note: If no excess is to be used, enter ZERO for excess volume.

Step 4: Calculate the plug length before the pipe is withdrawn in ft:

$$L_{\text{plug}} = \frac{(\text{SR}_{\text{plug}})(Y_{\text{cs}})}{\left(\left((V_{\text{ac}})\left(1 + \left(\frac{V_{\%\text{e}}}{100}\right)\right)\right) + (V_{\text{pf}})\right)} \tag{5.40}$$

where

SR_{cmt} = cement required for plug in sk

Note: If no excess is to be used, enter ZERO for excess volume.

Step 5: Calculate the fluid volume required to spot the plug in bbl:

$$V_{\text{displace}} = (L_{\text{p}} - L_{\text{plug}})(V_{\text{pc}}) - V_{\text{spacer behind}} \tag{5.41}$$

where

V_{displace} = volume of fluid required to spot the plug in bbl
L_{p} = length of pipe or tubing in ft

Example 1: A 300 ft plug is to be placed at a depth of 5000 ft. The open-hole size is 8½ in. and the drill pipe is 3½ in.—13.3 lb/ft; ID—2.764 in. Ten barrels of water will be pumped ahead of the slurry. Use a slurry yield of 1.15 ft^3/sk. Use 25% as the excess for the slurry volume:

Determine the following:

1. Number of sacks of cement required
2. Volume of water to be pumped behind the slurry to balance the plug
3. Plug length before the pipe is withdrawn
4. Amount of mud required to spot the plug plus the spacer behind the plug

Step 1: Calculate the following capacities:

(a) Annular capacity between drill pipe and hole in ft^3/ft:

$$V_{\text{ac}} = \left(\frac{8.5^2 - 3.5^2}{183.35}\right) = 0.3272 \text{ ft}^3/\text{ft}$$

(b) Annular capacity between drill pipe and hole in ft/bbl:

$$V_{\text{acf}} = \left(\frac{1,029.4}{8.5^2 - 3.5^2}\right) = 17.1569 \text{ ft/bbl}$$

(c) Hole capacity in ft^3/ft:

$$V_{\text{cc}} = \left(\frac{8.5^2}{183.35}\right) = 0.3941 \text{ ft}^3/\text{ft}$$

(d) Drill pipe capacity in bbl/ft:

$$V_{\text{pb}} = \left(\frac{2.764^2}{1029.4}\right) = 0.00742 \text{ bbl/ft}$$

(e) Drill pipe capacity in ft^3/ft:

$$V_{\text{pf}} = \left(\frac{2.764^2}{183.35}\right) = 0.0417 \text{ ft}^3/\text{ft}$$

Step 2: Calculate the number of sacks of dry cement required:

$$\text{SR}_{\text{L}} = \frac{(300)(0.3941)\left(1 + \left(\frac{25}{100}\right)\right)}{1.15} = 128.5 \cong 129 \text{ sk}$$

Step 3: Calculate the spacer volume (water) to be pumped behind the cement slurry to balance the plug in bbl:

$$V_{\text{spacer behind}} = \left(\frac{17.1568}{\left(1 + \left(\frac{25}{100}\right)\right)}\right)(10)(0.00742) = 1.018 \text{ bbl}$$

Step 4: Calculate the plug length before the pipe is withdrawn in ft:

$$L_{\text{plug}} = \frac{(129)(1.15)}{\left(\left((0.3272)\left(1 + \left(\frac{25}{100}\right)\right)\right) + (0.0417)\right)} = 329 \text{ ft}$$

Step 5: Calculate the volume of the displacing fluid required to spot the plug in bbl:

$$V_{\text{displace}} = \lfloor (5000 - 329)(0.00742) \rfloor - 1.0 = 33.6 \text{ bbl}$$

Example 2: Determine the number of **FEET** of plug for a given number of **SACKS** of cement:

A cement plug with 100 sk of cement is to be used in an 8½ in. hole. Use $1.15 \text{ ft}^3/\text{sk}$ for the cement slurry yield. The capacity of 8½ in. hole $= 0.3941 \text{ ft}^3/\text{ft}$. Use 50% as excess slurry volume:

$$L_{\text{plug}} = \left(\frac{\left(\frac{(100)(1.15)}{0.3941}\right)}{1 + \left(\frac{50}{100}\right)}\right) = 194.5 \text{ ft}$$

Spreadsheet 5.7 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Setting a balanced cement plug.

5.8 Differential hydrostatic pressure between cement in the annulus and mud inside the casing

1. Determine the hydrostatic pressure exerted by the cement and any mud remaining in the annulus.
2. Determine the hydrostatic pressure exerted by the mud and cement remaining in the casing.
3. Determine the differential pressure.

Example: 9⅝ in. casing—43.5 lb/ft in 12¼ in. hole:

Well depth	= 8000 ft
Cementing program:	
LEAD slurry 2000 ft	= 13.8 lb/gal
TAIL slurry 1000 ft	= 15.8 lb/gal
Mud weight	= 10.0 lb/gal
Float collar (no. of feet above shoe)	= 44 ft

Step 1: Calculate the total hydrostatic pressure of cement and mud in the annulus:

(a) Hydrostatic pressure of mud in annulus in psi:

$$HP_{ma} = (10.0)(0.052)(5000) = 2600\,\text{psi}$$

(b) Hydrostatic pressure of **LEAD** cement in psi:

$$HP_{L} = (13.8)(0.052)(2000) = 1435\,\text{psi}$$

(c) Hydrostatic pressure of **TAIL** cement in psi:

$$HP_{T} = (15.8)(0.052)(1000) = 822\,\text{psi}$$

(d) Total hydrostatic pressure in annulus in psi:

$$HP_{ta} = (2600 + 1435 + 822) = 4857\,\text{psi}$$

Step 2: Calculate the total pressure inside the casing in psi:

(a) Pressure exerted by the mud in psi:

$$HP_{m} = (10.0)(0.052)(8000 - 44) = 4137\,\text{psi}$$

(b) Pressure exerted by the cement in psi:

$$HP_{cmt} = (15.8)(0.052)\,(44) = 36\,\text{psi}$$

(c) Total pressure inside the casing in psi:

$$HP_{csg} = (4137 + 36) = 4173\,\text{psi}$$

Step 3: Calculate the differential pressure in psi:

$$P_d = (4857 - 4173) = 684 \text{ psi}$$

Spreadsheet 5.8 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Differential HP between annular cement and mud inside the casing.

5.9 Hydraulicing casing

These calculations will determine if the casing will hydraulic out (move upward) when cementing.

Step1: Calculate the buoyancy factor:

$$\text{BF} = \frac{65.5 - W_m}{65.5} \tag{5.42}$$

where

BF = buoyancy factor
W_m = weight of mud in ppg

Step 2: Calculate the difference in the pressure gradient between the cement and the mud in psi/ft:

$$\text{PG}_d = (W_{cmt} - W_m)(0.052) \tag{5.43}$$

where

PG_d = difference in the pressure gradient between the cement and the mud in psi/ft
W_{cmt} = weight of the cement in ppg

Step 3: Calculate the differential pressure (P_d) between the cement and the mud in psi:

$$P_d = (\text{PG}_d)(L_{csg}) \tag{5.44}$$

where

P_d = differential pressure between the cement and the mud in psi
L_{csg} = length of the casing in ft

Step 4: Calculate the area of the casing below the shoe in sq. in.:

$$A_{bcs} = (D_c^2)(0.7854) \tag{5.45}$$

where

A_{bcs} = area below the casing shoe in sq. in.
D_c = outside diameter of casing in in.

Step 5: Calculate the upward force (F) in lb. This is the weight or total force acting at the bottom of the shoe:

$$F_{\text{up}} = (A_{\text{bcs}})(P_{\text{d}}) \tag{5.46}$$

where

F_{up} = upward force on casing in lb

Step 6: Calculate the downward force (W) in lb. This is the weight of the casing:

$$F_{\text{down}} = \left(W_{\text{csg}}\right)\left(L_{\text{csg}}\right)(\text{BF}) \tag{5.47}$$

where

F_{down} = downward force or weight of the casing in lb
W_{csg} = weight of the casing in lb/ft

Step 7: Calculate the difference in these forces in lb:

$$F_{\text{d}} = \left(F_{\text{down}} - F_{\text{up}}\right) \tag{5.48}$$

where

F_{d} = differential force in lb

Step 8: Calculate the pressure required to balance these forces so that the casing will not hydraulic out of the hole (move upward) in psi:

$$P_{\text{b}} = \left(\frac{F_{\text{d}}}{A_{\text{bcs}}}\right) \tag{5.49}$$

where

P_{b} = pressure to balance forces in psi

Step 9: Calculate the mud weight increase required to balance the pressure in lb/gal:

$$W_{\text{mb}} = \left(\frac{(P_{\text{b}}/0.052)}{L_{\text{csg}}}\right) \tag{5.50}$$

where

W_{mb} = mud weight increase required to balance pressure in lb/gal
L_{csg} = length of casing in ft

Step 10: Calculate the new mud weight in lb/gal:

$$W_{\text{mn}} = (W_{\text{m}} + W_{\text{mb}}) \tag{5.51}$$

where

W_{mn} = new mud weight to balance pressure in lb/gal

Check the forces with the *new* mud weight with the following equations:

(a) $BF_N = \frac{(65.5 - W_{mn})}{65.5}$
(b) $PG_d = (W_{cmt} - W_{mn})(0.052)$
(c) $P_d = (PG)\ (L_{csg})$
(d) $F_{up} = (A_{bcs})(P_d)$
(e) $F_{dowm} = (W_{csg})(L_{csg})(BF_N)$
(f) $F_d = (F_{down} - F_{up})$

Example: Calculate the forces and new mud weight with the following data:

Casing size = 13⅜ in.
Casing weight = 54 lb/ft
Mud weight = 8.8 lb/gal
Cement weight = 15.8 lb/gal
Casing depth = 1000 ft

Step 1: Calculate the Buoyancy Factor

$$BF = \frac{(65.5 - 8.8\ )}{65.5} = 0.8656$$

Step 2: Calculate the difference in pressure gradient between the cement and the mud in psi/ft:

$$PG_d = (15.8 - 8.8)(1000) = 0.364 \text{ psi}$$

Step 3: Calculate the differential pressure between the cement and the mud in psi:

$$P_d = (0.364)(1000) = 364 \text{ psi}$$

Step 4: Calculate the area of the casing below the shoe in sq. in.:

$$A_{bcs} = \left(13.375^2\right)(0.7854) = 140.5 \text{ in.}^2$$

Step 5: Calculate the upward force (F) in lb. This is the weight or total force acting at the bottom of the shoe:

$$F_{up} = (140.5)(364) = 51,142 \text{ lb}$$

Step 6: Calculate the downward force (W) in lb. This is the weight of the casing:

$$F_{down} = (54.0)(1000)(0.8656) = 46,742.4 \text{ lb}$$

Step 7: Calculate the difference in these forces in lb:

$$F_d = (46,742 - 51,142) = -4400 \text{ lb}$$

The resultant force is **NEGATIVE!**

Therefore: Unless the casing is tied down or stuck, it will "hydraulic" out of the hole (move upward).

Step 8: Calculate the pressure required to balance these forces so that the casing will not hydraulic out of the hole (move upward) in psi:

$$P_b = \frac{4400}{140.5} = 31.3 \text{ psi}$$

Step 9: Calculate the mud weight increase required to balance the pressure in lb/gal:

$$W_{mb} = \left(\frac{\left(\frac{31.3}{0.052}\right)}{1000}\right) = 0.60 \text{ ppg}$$

Note: Add 0.1 ppg to the mud weight to balance the pressure for a safety factor. (This value may be increased if a larger safety factor is desired).

Step 10: Calculate the new mud weight in lb/gal:

$$W_{mn} = (8.8 + 0.6 + 0.1) = 9.5 \text{ ppg}$$

This new mud weight should be used to displace the cement for this casing job. Another option would be to put pressure on the cementing head when the plug is bumped. In this example, the pressure option would amount to the following:

$$P_b = (9.5 - 8.8)(0.052)(1000) = 36.4 \text{ psi}$$

Check the forces with the new mud weight:

(a) $BF_N = \dfrac{(65.5 - 9.5)}{65.5} = 0.8550$
(b) $PG_d = (15.8–9.5)(0.052) = 0.328 \text{ psi/ft}$
(c) $P_d = (0.328)(1000) = 328 \text{ psi}$
(d) $F_{up} = (140.5)(328) = 46{,}084 \text{ lb}$
(e) $F_{down} = (54.0)(1000)(0.8550) = 46{,}170 \text{ lb}$
(f) $F_d = (46{,}170–46{,}068) = +\ 86.0 \text{ lb}$

The casing will not move upward with this resulting force.

Spreadsheet 5.9 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Hydraulicing casing.

5.10 Pump strokes to bump the plug

The strokes required to bump the plug on deep casing jobs will vary according to the type of mud system used to displace the cement.

Step 1: Calculate the volume of the casing required to bump the plug:

$$V_{csg} = \left(\frac{(C_{ID}^2)}{1029.4}\right)(D_{plug}) \tag{5.52}$$

where

V_{csg} = volume of casing in bbl
C_{ID} = casing ID in inches
D_{plug} = depth to plug in ft

Step 2: Calculate the compressibility of the mud:

$$C_m = \left(3\left(\frac{W_m}{100}\right)\right) + \left(5\left(\frac{O_m}{100}\right)\right) + \left(0.2\left(\frac{S_m}{100}\right)\right) \tag{5.53}$$

where

C_m = compressibility of the mud
W_m = water content of the mud in %
O_m = oil content of the mud in %
S_m = solids content of the mud in %

Step 3: Calculate the psi change per barrel pumped:

$$P_c = \frac{(1,000,000)}{(C_m)(V_{csg})} \tag{5.54}$$

where

P_c = pressure change per barrel pumped in psi

Step 4: Calculate the volume of mud that will be compressed to achieve a certain pressure:

$$V_c = \frac{P_f}{P_c} \tag{5.55}$$

where

V_c = volume of compressed mud in bbl
P_f = final pressure applied to mud in psi

Step 5: Calculate the total volume of mud to be pumped in bbl

$$V_{mT} = (V_{csg}) + (V_c) \tag{5.56}$$

where

V_{mT} = total volume of mud pumped to bump plug in bbl

Step 6: Calculate the number of strokes required to bump the plug:

$$S_{bp} = \frac{V_{mT}}{Q} \tag{5.57}$$

where

S_{bp} = strokes to bump the plug

Example: Determine the number of strokes required to bump the plug with the following data:

Casing ID = 10.136 in
MD to plug = 14,960 ft
Pump output = 0.128 bbl/stk

Mud content = %	*Water*	*Oil*	*Solids*
WBM	72.0	8.0	20.0
OBM	8.0	72.0	20.0

Step 1: Calculate the volume of the casing required to bump the plug:

$$V_{csg} = \left(\frac{(10.136^2)}{1029.4}\right)(14, 960) = 1493.1 \text{ bbl}$$

Step 2: Calculate the compressibility of the WBM:

$$C_m = \left(3\left(\frac{72.0}{100}\right)\right) + \left(5\left(\frac{8.0}{100}\right)\right) + \left(0.2\left(\frac{20.0}{100}\right)\right) = 2.6$$

Step 3: Calculate the psi change per compressed barrel pumped:

$$P_c = \frac{(1, 000, 000)}{(2.6)(1439.1)} = 257.6 \text{ psi}$$

Step 4: Calculate the volume of mud that will be compressed to achieve a certain pressure:

$$V_c = \frac{2000}{257.6} = 7.8 \text{ bbl}$$

Step 5: Calculate the total volume of mud to be pumped in bbl

$$V_{mT} = (1493.1) + (7.8) = 1500.9 \text{ bbl}$$

Step 6: Calculate the number of strokes required to bump the plug:

$$S_{bp} = \frac{1500.9}{Q0.128} = 11, 726 \text{ stks}$$

Repeat Steps 2–6 for oil base mud:

Step 2: Calculate the compressibility of the OBM:

$$C_m = \left(3\left(\frac{8.0}{100}\right)\right) + \left(5\left(\frac{72.0}{100}\right)\right) + \left(0.2\left(\frac{20.0}{100}\right)\right) = 3.88$$

Step 3: Calculate the psi change per compressed barrel pumped:

$$P_c = \frac{(1,000,000)}{(3.88)(1493.1)} = 172.6 \text{ psi}$$

Step 4: Calculate the volume of mud that will be compressed to achieve a certain pressure:

$$V_c = \frac{2000}{172.6} = 11.6 \text{ bbl}$$

Step 5: Calculate the total volume of mud to be pumped in bbl

$$V_{mT} = (1493.1) + (11.6) = 1504.7 \text{ bbl}$$

Step 6: Calculate the number of strokes required to bump the plug:

$$S_{bp} = \frac{1504.7}{Q0.128} = 11,756 \text{ stks}$$

Spreadsheet 5.10 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Strokes to bump the plug.

Appendix: Supplementary material

Supplementary material related to this chapter can be found online at https://www.elsevier.com/books-and-journals/book-companion/9780323905497.

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CHAPTER

6

Well hydraulics

6.1 System pressure losses

6.1.1 Determine the pressure loss in the surface system in psi

$$\Delta P_{SS} = (C_{SS})(MW)\left(\frac{Q}{100}\right)^{1.86} \tag{6.1}$$

where

ΔP_{SS} = pressure loss in surface system in psi
C_{SS} = constant for surface system from Table 6.1
Q = pump output in gpm

Example:

Mud weight = 12.0 ppg
Surface case = 4
Constant for surface case = 0.15
Pump output = 400 gpm

$$\Delta P_{SS} = (0.15)(12.0)\left(\frac{400}{100}\right)^{1.86}$$

$$\Delta P_{SS} = 23.7\,\text{psi}$$

Spreadsheet 6.1.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Pressure loss in the surface system.

Formulas and Calculations for Drilling, Production, and Workover
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TABLE 6.1 Surface system cases.

Case	Standpipe (ft × ID in.)	Hose (ft × ID in.)	Swivel (ft×ID in.)	Kelly (ft×ID in.)	C_{SS}
1	40.0 × 3.0	45.0 × 2.0	4.0 × 2.0	40.0 × 2.25	1.00
2	40.0 × 3.5	55.0 × 2.5	5.0 × 2.5	40.0 × 3.25	0.36
3	45.0 × 4.0	55.0 × 3.0	5.0 × 2.5	40.0 × 3.25	0.22
4	45.0 × 4.0	55.0 × 3.0	6.0 × 3.0	40.0 × 4.00	0.15
5	100.0 × 5.0	85.0 × 3.5	22.0 × 3.5		0.15

Ref: Table 4 of API RP 13D, June 2006, p. 28.

6.1.2 Determine the pressure loss in the drill string (drill pipe—DP, heavy weight drill pipe—HWDP, or drill collars—DC) in psi

(1) Determine the fluid velocity down the pipe in ft/s:

$$v_P = \frac{(0.408)(Q)}{D_{pID}^2} \tag{6.2}$$

where

V_p = fluid velocity down the drill string in ft/s
D_{pID} = internal diameter of the pipe in inches

(2) Determine the n value for the pipe (flow behavior index) in the pipe:

$$n_p = 3.32 \log\left(\frac{\theta 600}{\theta 300}\right) \tag{6.3}$$

where

n_p = flow behavior index for the pipe (dimensionless)
$\theta 600$ = viscometer reading at 600 rpm
$\theta 300$ = viscometer reading at 300 rpm

(3) Determine the K value (consistency factor) in the pipe in Poise:

$$K_p = \frac{5.11(\theta 600)}{(1022^{n_p})} \tag{6.4}$$

where

K_p = consistency factor in Poise

(4) Determine the effective viscosity in the pipe in cP:

$$\mu_{ep} = 100(K_p)\left[\frac{96(V_p)}{D_{pID}}\right]^{(n_p-1)} \tag{6.5}$$

where

μ_{ep} = effective viscosity in the pipe in cP

(5) Determine the Reynolds number for the pipe:

$$Re_p = \frac{928(V_p)(D_{pID})(MW)}{(\mu_{ep})\left[\frac{(3n_p+1)}{(4n_p)}\right]^{n_p}} \tag{6.6}$$

where

R_{ep} = Reynolds number for the pipe (dimensionless)

(6) Determine the Reynolds number for the change from laminar to transitional flow for the pipe:

$$Re_L = 3470 - 1370(n_p) \tag{6.7}$$

where

Re_L = Reynolds number for the change from laminar to transitional for the pipe (dimensionless)

(7) Determine the Reynolds number for the change from transitional to turbulent flow for the pipe:

$$Re_T = 4270 - 1370(n_p) \tag{6.8}$$

where

Re_T = Reynolds number for the change from transitional to turbulent flow for the pipe (dimensionless)

(8) Determine the type of flow, then determine the friction factor:

a. If the $Re_p < Re_L$, select the laminar flow equation to determine the friction factor:

$$f_p = \frac{16}{Re_p} \tag{6.9}$$

b. If the $Re_L < Re_p < Re_T$, use the transitional flow equation to determine the friction factor:

$$f_p = \left[\frac{Re_p - Re_L}{800}\right]\left[\frac{(\log(n_p) + 3.93)/50}{Re_T\frac{([1.75 - \log(n_p)])}{7}} - \frac{16}{Re_L}\right] + \frac{16}{Re_L} \tag{6.10}$$

c. If the $Re_p > Re_T$, use the turbulent flow equation to determine the friction factor:

$$f_p = \frac{\left(\left(\log\left(n_p\right) + 3.93\right)/50\right)}{Re_p^{\frac{\left(\left[1.75 - \log\left(n_p\right)\right]\right)}{7}}} \tag{6.11}$$

d. Determine the pressure loss for the interval:

$$\Delta P_{pI} = \frac{\left(f_p\right)\left(V_p^2\right)(MW)}{25.8\left(D_{pID}\right)}\left(L_I\right) \tag{6.12}$$

where

ΔP_{pI} = pressure loss in the pipe for the internal in psi
L_I = length of the interval in ft

Example: Determine the pressure loss for the following drill string:

Data:	Mud weight	= 12.0 ppg
	Drill pipe size	= 5.0 × 4.276 in.
	Drill pipe length	= 13,420 ft
	Heavy weight drill pipe size	= 5.0 × 3.0 in.
	Heavy weight drill pipe length	= 930 ft
	Drill collar size	= 6¾ × 3.0 in. (98.0 lb/ft)
	Drill collar length	= 650 ft
	Pump output	= 400 gpm
	θ600 viscometer value	= 66
	θ300 viscometer value	= 40

a. Determine the pressure loss for the Drill Pipe:

(a) Determine the fluid velocity down the pipe in ft/s:

$$v_P = \frac{(0.408)(400)}{\left(4..276^2\right)} = 8.925 \text{ ft/s}$$

(b) Determine the n value for the pipe (flow behavior index) in the pipe:

$$n_p = 3.32\ log\left(\frac{66}{40}\right) = 0.7220$$

(c) Determine the K value (consistency factor) in the pipe in Poise:

$$K_p = \frac{5.11(66)}{\left(1022^{0.7220}\right)} = 2.265 \text{ Poise}$$

(d) Determine the effective viscosity in the pipe in cP:

$$\mu_{ep} = 100(2.265)\left(\frac{96(8.925)}{4.276}\right)^{0.7220-1} = 51.9 \text{ cP}$$

(e) Determine the Reynolds number for the pipe:

$$Re_p = \frac{928(8.925)(4.276)(12.0)}{(51.9)\left(\frac{3(0.7220)+1}{4(0.7220)}\right)^{0.7220}} = 7662.8$$

(f) Determine the Reynolds number for the change from laminar to transitional flow for the pipe:

$$Re_L = 3470 - 1370(0.7220) = 2480.9$$

(g) Determine the Reynolds number for the change from transitional to turbulent flow for the pipe:

$$Re_T = 4270 - 1370(0.7220) = 3280.9$$

(h) The Re_p value of 7662.8 is larger than the Re_T value of 3280.9; therefore, the fluid is in turbulent flow. Use Eq. (6.11) to determine the friction factor:

$$f_p = \frac{((log\,(0.7220) + 3.93)/50)}{7662.8\left(\frac{(1.75 - log\,(0.7220))}{7}\right)} = 0.006760.$$

(i) Determine the pressure loss for the drill pipe:

$$\Delta P_{dp} = \frac{(0.006760)\left(8.925^2\right)(12.0)}{25.8(4.276)}(13,420) = 786.0 \text{ psi}$$

b. Determine the pressure loss for the Heavy Weight Drill Pipe.

(a) Determine the fluid velocity down the HWDP in ft/s:

$$v_P = \frac{(0.408)(400)}{\left(3.0^2\right)} = 18.13 \text{ ft/s}$$

(b) Determine the n value for the pipe (flow behavior index) in the HWDP:

$$n_p = 3.32\ log\left(\frac{66}{40}\right) = 0.7220$$

(c) Determine the K value (consistency factor) in the HWDP in Poise:

$$K_p = \frac{5.11(66)}{\left(1022^{0.7220}\right)} = 2.265 \text{ Poise}$$

(d) Determine the effective viscosity in the HWDP in cP:

$$\mu_{ep} = 100(2.265)\left(\frac{96(18.13)}{(3.0)}\right)^{(0.7220-1)} = 38.6 \text{ cP}$$

(e) Determine the Reynolds number for the HWDP:

$$Re_p = \frac{928(18.13)(3.0)(12.0)}{(38.6)\left(\frac{3(0.7220)+1}{4(0.7220)}\right)^{0.7220}} = 14,684$$

(f) Determine the Reynolds number for the change from laminar to transitional flow for the HWDP:

$$Re_L = 3470 - 1370(0.7220) = 2480.9$$

(g) Determine the Reynolds number for the change from transitional to turbulent flow for the HWDP:

$$Re_T = 4270 - 1370(0.7220) = 3280.9$$

(h) The Re_p value of 14,684 is larger than the Re_T value of 3280.9; therefore, the fluid is in turbulent flow. Use Eq. (6.11) to determine the friction factor:

$$f_p = \frac{((log\,(0.7220) + 3.93)/50)}{14,684\left(\frac{(1.75 - log\,(0.7220))}{7}\right)} = 0.005671$$

(i) Determine the pressure loss for the HWDP:

$$\Delta P_{dp} = \frac{(0.005671)(18.13^2)(12.0)}{25.8(3.0)}(930) = 268.7 \text{ psi}$$

c. Determine the pressure loss for the Drill Collars.

(1) Determine the fluid velocity down the DC in ft/s:

$$v_P = \frac{(0.408)(400)}{(3.0^2)} = 18.13 \text{ ft/s}$$

(2) Determine the n value for the pipe (flow behavior index) in the DC:

$$n_p = 3.32\ log\left(\frac{66}{40}\right) = 0.7220$$

(3) Determine the K value (consistency factor) in the DC in Poise:

$$K_{\mathrm{p}} = \frac{5.11(66)}{\left(1022^{0.7220}\right)} = 2.265 \text{ Poise}$$

(4) Determine the effective viscosity in the DC in cP:

$$\mu_{\mathrm{ep}} = 100(2.265)\left(\frac{96(18.13)}{(3.0)}\right)^{(0.7220-1)} = 38.6 \text{ cP}$$

(5) Determine the Reynolds number for the DC:

$$Re_{\mathrm{p}} = \frac{928(18.13)(3.0)(12.0)}{(38.6)\left(\frac{3(0.7220)+1}{4(0.7220)}\right)^{0.7220}} = 14,684$$

(6) Determine the Reynolds number for the change from laminar to transitional flow for the DC:

$$Re_{\mathrm{L}} = 3470 - 1370(0.7220) = 2480.9$$

(7) Determine the Reynolds number for the change from transitional to turbulent flow for the DC:

$$Re_{\mathrm{T}} = 4270 - 1370(0.7220) = 3280.9$$

(8) The Re_{p} value of 14,684 is larger than the Re_{T} value of 3280.9; therefore, the fluid is in turbulent flow. Use Eq. (6.11) to determine the friction factor:

$$f_{\mathrm{p}} = \frac{((log\,(0.7220) + 3.93)/50)}{14,684\left(\frac{(1.75 - log\,(0.7220))}{7}\right)} = 0.005671$$

(9) Determine the pressure loss for the DC:

$$\Delta P_{\mathrm{dp}} = \frac{(0.005671)\left(18.13^2\right)(12.0)}{25.8(3.0)}(650) = 187.8 \text{ psi}$$

d. The total pressure loss for the drill string is:

$$(23.7) + (786.0) + (268.7) + (187.8) = 1266.2 \text{ psi}$$

Spreadsheet 6.1.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Pressure loss in the drill string.

6.1.3 Determine the pressure loss at the bit in psi

$$\Delta P_b = \frac{156.5(Q^2)(MW)}{\left[\left(D_{J1}{}^2\right) + \left(D_{J2}{}^2\right) + \left(D_{J3}{}^2\right) + \ldots\left(D_{Jn}{}^2\right)\right]^2} \tag{6.13}$$

where

ΔP_b = pressure loss at the bit in psi
Q = pump output in gpm
MW = mud weight in ppg
D_J = diameter of the bit nozzles #1, #2, and #3 in 32nds of an inch (rounded to the nearest whole number).

Example: Determine the pressure loss at the bit:

Data: Mud weight = 12.0 ppg
Pump output = 400 gpm
Nozzle size = 3 × 12/32nds jets

$$\Delta P_b = \frac{156.5(400^2)(12.0)}{\left[\left(12^2\right) + \left(12^2\right) + \left(12^2\right)\right]^2} = 1610 \text{ psi}$$

Spreadsheet 6.1.3 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Pressure loss at the bit.

6.1.4 Determine the pressure loss in the annulus in psi

(1) Determine the fluid velocity in the annulus in ft/s:

$$V_a = \frac{(0.408)(Q)}{\left(\left(D_h^2\right) - \left(D_p^2\right)\right)} \tag{6.14}$$

where

V_a = fluid velocity in the annulus in ft/s
D_h = diameter of the hole or the internal diameter of the casing in inches
D_p = outside diameter of the pipe in inches.

(2) Determine the n value for the pipe (flow behavior index) in the annulus:

$$n_a = 0.5 \log\left(\frac{\theta 300}{\theta 3}\right) \tag{6.15}$$

where

n_a = flow behavior index for the annulus (dimensionless)
$\theta 300$ = viscometer reading at 300 rpm
$\theta 3$ = viscometer reading at 3 rpm

(3) Determine the K value (consistency factor) the annulus in Poise:

$$K_a = \frac{5.11(\theta 300)}{(511^{n_a})} \tag{6.16}$$

where

K_a = consistency factor in Poise

(4) Determine the effective viscosity in the annulus in cP:

$$\mu_{ea} = 100(K_a)\left[\frac{144(V_a)}{(D_h - D_p)}\right]^{(n_p - 1)} \tag{6.17}$$

where

μ_{ea} = effective viscosity in the annulus in cP

(5) Determine the Reynolds number for the annulus:

$$Re_a = \frac{928(V_a)(D_h - D_p)(\mathrm{MW})}{(\mu_{ea})\left[\frac{2n_a + 1}{3n_a}\right]^{n_a}} \tag{6.18}$$

where

Re_a = Reynolds number for the annulus (dimensionless)

(6) Determine the Reynolds number for the change from laminar to transitional flow for the annulus:

$$Re_L = 3470 - 1370(n_a) \tag{6.19}$$

where

Re_L = Reynolds number for the change from laminar to transitional for the annulus (dimensionless)

(7) Determine the Reynolds number for the change from transitional to turbulent flow for the annulus:

$$Re_T = 4270 - 1370(n_a) \tag{6.20}$$

where

Re_T = Reynolds number for the change from transitional to turbulent flow for the annulus (dimensionless)

(8) Determine the type of flow, then determine the friction factor:

a. If the $Re_a < Re_L$, select the laminar flow equation to determine the friction factor:

$$f_a = \frac{24}{Re_a} \tag{6.21}$$

b. If the $Re_L < Re_a < Re_T$, use the transitional flow equation to determine the friction factor:

$$f_a = \left[\frac{Re_a - Re_L}{800}\right]\left[\frac{\left(\left(\log(n_a) + 3.93\right)/50\right)}{Re_T\left(\left[1.75 - \log(n_a)\right]/7\right)} - \frac{24}{Re_L}\right] + \frac{24}{Re_L} \tag{6.22}$$

c. If the $Re_a > Re_T$, use the turbulent flow equation to determine the friction factor:

$$f_a = \frac{\left(\left(\log(n_a) + 3.93\right)/50\right)}{Re_a\left(\left[1.75 - \log(n_a)\right]/7\right)} \tag{6.23}$$

d. Determine the pressure loss in the annulus for the interval:

$$\Delta P_{aI} = \frac{\left(f_p\right)\left(V_P^2\right)(MW)}{25.8\left(D_h - D_p\right)}\left(L_I\right) \tag{6.24}$$

where

ΔP_{aI} = pressure loss in the annulus in psi
L_I = length of the annulus interval in ft

Example: Determine the pressure loss in the annulus:

Data:	Mud weight	= 12.0 ppg
	Well depth	= 15,000 ft
	Bit size	= 8½ in.
	Casing size	= 10¾ × 9.850 in. (N-80, 51 lb/ft)
	Casing depth	= 12,000 ft
	Drill pipe size	= 5.0 × 4.276 in.
	Drill pipe length	= 13,420 ft
	Heavy weight drill pipe size	= 5.0 × 3.0 in.
	Heavy weight drill pipe length	= 930 ft
	Drill collar size	= 6¾ × 3.0 (98.0 lb/ft)
	Drill collar length	= 650 ft
	θ300 viscometer value	= 40
	θ3 viscometer value (gel strength)	= 8 lb/100 ft^2
	Pump output	= 400 gpm

A. Determine the pressure loss in the DP/casing annulus:

(1) Determine the fluid velocity in the DP/Casing annulus in ft/s:

$$V_a = \frac{(0.408)(400)}{\left((9.850^2) - (5.0^2)\right)} = 2.266 \text{ ft/s}$$

(2) Determine the n value for the pipe (flow behavior index) in the DP/Casing annulus:

$$n_a = 0.5 \; log\left(\frac{40}{8}\right) = 0.3495$$

(3) Determine the K value (consistency factor) the DP/Casing annulus in Poise:

$$K_p = \frac{5.11(40)}{\left(511^{0.3495}\right)} = 23.1 \text{ Poise}$$

(4) Determine the effective viscosity in the DP/Casing annulus in cP:

$$\mu_{ea} = 100(23.1)\left[\frac{144(2.266)}{9.850 - 5.0}\right]^{(0.3495-1)} = 149.5 \text{ cP}$$

(5) Determine the Reynolds number for the DP/Casing annulus:

$$Re_a = \frac{928(2.266)(9.850 - 5.0)(12.0)}{(149.5)\left(\left[\frac{2(0.3495) + 1}{3(0.3495)}\right]^{0.3495}\right)} = 691.6$$

(6) Determine the Reynolds number for the change from laminar to transitional flow for the DP/annulus:

$$Re_L = 3470 - 1370(0.3495) = 2991.2$$

(7) Determine the Reynolds number for the change from transitional to turbulent flow for the DP/annulus:

$$Re_T = 4270 - 1370(0.3495) = 3791.2$$

(8) The type of flow is laminar; therefore, use Eq. (6.21) to determine the friction factor:

$$f_a = \frac{24}{691.6} = 0.0347$$

(9) Determine the pressure loss for the DP/Casing annulus interval:

$$\Delta P_a = \frac{(0.0347)\left(2.266^2\right)(12.0)}{25.8(9.850 - 5.0)}(12,\,000) = 205.0 \text{ psi}$$

B. Determine the pressure loss in the DP/open hole annulus:

(1) Determine the fluid velocity in the DP/open hole annulus in ft/s:

$$V_a = \frac{(0.408)(400)}{\left((8.5^2) - (5.0^2)\right)} = 3.454 \text{ ft/s}$$

(2) Determine the n value for the pipe (flow behavior index) in the DP/open hole annulus:

$$n_a = 0.5 \; log\left(\frac{40}{8}\right) = 0.3495$$

(3) Determine the K value (consistency factor) the DP/open hole annulus in Poise:

$$K_p = \frac{5.11(40)}{511^{0.3495}} = 23.1 \text{ Poise}$$

(4) Determine the effective viscosity in the DP/open hole annulus in cP:

$$\mu_{ea} = 100(23.1)\left[\frac{144(3.454)}{8.5 - 5.0}\right]^{(0.3495-1)} = 91.9 \text{ cP}$$

(5) Determine the Reynolds number for the DP/open hole annulus:

$$Re_a = \frac{928(3.454)(8.5 - 5.0)(12.0)}{(91.9)\left(\left[\frac{2(0.3495) + 1}{3(0.3495)}\right]^{0.3495}\right)} = 1237.5$$

(6) Determine the Reynolds number for the change from laminar to transitional flow for the DP/open hole annulus:

$$Re_L = 3470 - 1370(0.3495) = 2991.2$$

(7) Determine the Reynolds number for the change from transitional to turbulent flow for the DP/open hole annulus:

$$Re_T = 4270 - 1370(0.3495) = 3791.2$$

(8) The type of flow is laminar; therefore, use Eq. (6.21) to determine the friction factor:

$$f_a = \frac{24}{1237.5} = 0.01939$$

(9) Determine the pressure loss for the DP/open hole annulus interval:

$$\Delta P_a = \frac{(0.01939)\left(3.454^2\right)(12.0)}{25.8(8.5 - 5.0)}(13,\, 420 - 12,\, 000) = 43.7 \text{ psi}$$

C. Determine the pressure loss for the HWDP/open hole annulus:

(1) Determine the fluid velocity in the HWDP/open hole annulus in ft/s:

$$V_a = \frac{(0.408)(400)}{\left((8.5^2) - (5.0^2)\right)} = 3.454 \text{ ft/s}$$

(2) Determine the n value for the pipe (flow behavior index) in the HWDP/open hole annulus:

$$n_a = 0.5 \; log\left(\frac{40}{8}\right) = 0.3495$$

(3) Determine the K value (consistency factor) the HWDP/open hole annulus in Poise:

$$K_p = \frac{5.11(40)}{\left(511^{0.3495}\right)} = 23.1 \text{ Poise}$$

(4) Determine the effective viscosity in the HWDP/open hole annulus in cP:

$$\mu_{ea} = 100(23.1)\left[\frac{144(3.454)}{8.5 - 5.0}\right]^{(0.3495-1)} = 91.9 \text{ cP}$$

(5) Determine the Reynolds number for the HWDP/open hole annulus:

$$Re_a = \frac{928(3.454)(8.5 - 5.0)(12.0)}{(91.9)\left(\left[\frac{2(0.3495) + 1}{3(0.3495)}\right]^{0.3495}\right)} = 1237.5$$

(6) Determine the Reynolds number for the change from laminar to transitional flow for the HWDP/open hole annulus:

$$Re_L = 3470 - 1370(0.3495) = 2991.2$$

(7) Determine the Reynolds number for the change from transitional to turbulent flow for the HWDP/open hole annulus:

$$Re_T = 4270 - 1370(0.3495) = 3791.2$$

(8) The type of flow is laminar; therefore, use Eq. (6.21) to determine the friction factor:

$$f_a = \frac{24}{1237.5} = 0.01939$$

(9) Determine the pressure loss for the HWDP/open hole annulus interval:

$$\Delta P_a = \frac{(0.01939)(3.454^2)(12.0)}{25.8(8.5-5.0)}(930) = 28.6 \text{ psi}$$

D. Determine the pressure loss for the DC/open hole annulus:

(1) Determine the fluid velocity in the DC/open hole annulus in ft/s:

$$V_a = \frac{(0.408)(400)}{((8.5^2)-(6.75^2))} = 6.12 \text{ ft/s}$$

(2) Determine the n value for the pipe (flow behavior index) in the DC/open hole annulus:

$$n_a = 0.5 \ log\left(\frac{40}{8}\right) = 0.3495$$

(3) Determine the K value (consistency factor) the DC/open hole annulus in Poise:

$$K_p = \frac{5.11(40)}{(511^{0.3495})} = 23.1 \text{ Poise}$$

(4) Determine the effective viscosity in the DC/open hole annulus in cP:

$$\mu_{ea} = 100(23.1)\left[\frac{144(6.12)}{8.5-6.75}\right]^{(0.3495-1)} = 40.4 \text{ cP}$$

(5) Determine the Reynolds number for the DC/open hole annulus:

$$Re_a = \frac{928(6.12)(8.5-6.75)(12.0)}{(40.4)\left(\left[\frac{2(0.3495)+1}{3(0.3495)}\right]^{0.3495}\right)} = 2493.9$$

(6) Determine the Reynolds number for the change from laminar to transitional flow for the DC/open hole annulus:

$$Re_L = 3470 - 1370(0.3495) = 2991.2$$

(7) Determine the Reynolds number for the change from transitional to turbulent flow for the DC/open hole annulus:

$$Re_T = 4270 - 1370(0.3495) = 3791.2$$

(8) The type of flow is laminar; therefore, use Eq. (6.21) to determine the friction factor:

$$f_a = \frac{24}{2493.9} = 0.009624$$

(9) Determine the pressure loss for the DC/open hole annulus interval:

$$\Delta P_a = \frac{(0.009624)\left(6.12^2\right)(12.0)}{25.8(8.5-6.75)}(650) = 62.3 \text{ psi}$$

E. The total pressure loss for the annulus is:

$$(205.0) + (43.7) + (28.6) + (62.3) = 339.6 \text{ psi}$$

F. The pressure loss for total system of drill string, bit and annulus is:

$$(1266.2) + (1610) + (339.6) = 3215.8 \text{ psi}$$

Spreadsheet 6.1.4 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Pressure loss in the annulus.

6.1.5 Critical velocity and pump rate

Determine the velocity and flow rate where the fluid enters turbulent flow.

A. Inside the **PIPE**.

(1) Determine n_p (dimensionless):

$$n_p = 3.32 \log\left(\frac{\theta 600}{\theta 300}\right) \tag{6.25}$$

where

n_p = flow index in the pipe
$\theta 600$ = viscometer reading at 600 rpm
$\theta 300$ = viscometer reading at 300 rpm

(2) Determine K_p in Poise:

$$K_p = \frac{\theta 600}{(1022^{n_p})} \tag{6.26}$$

where

K_p = consistency in the pipe.

(3) Determine the critical velocity in the pipe:

$$V_{pc} = \left(\frac{\left(81,600\left(K_p\right)\left(n_p^{0.387}\right)\right)}{\left(D_p^{n_p}\right)(MW)}\right)^{\left(\frac{1}{(2-n_p)}\right)} \tag{6.27}$$

where

V_{pc} = critical velocity in ft/min
D_p = internal diameter of pipe in inches
MW = mud weight in ppg

(4) Determine critical flow rate in the pipe in gpm:

$$Q_{pc} = \frac{V_{pc}\left(D_p^2\right)}{24.5} \tag{6.28}$$

where

Q_{pc} = critical flow rate in the pipe in gpm

Example:

Mud weight = 12.0 ppg
$\theta 600$ value = 66
$\theta 300$ value = 40
Pipe ID = 4.276 in.

(1) Determine n_p:

$$n_p = 3.32 \log\left(\frac{66}{40}\right) = 0.7220$$

(2) Determine K_p:

$$K_p = \frac{66}{\left(1022^{0.7220}\right)} = 0.4433$$

(3) Determine the critical velocity in the pipe:

$$V_{pc} = \left(\frac{\left(81,600(0.4433)\left(0.7220^{0.387}\right)\right)}{\left(4.276^{0.7220}\right)(12.0)}\right)^{\left(\frac{1}{(2-0.7220)}\right)} = 210.4 \text{ ft/min}$$

(4) Determine the critical flow rate in the pipe:

$$Q_{pc} = \frac{210.4\left(4.276^2\right)}{24.5} = 157.0 \text{ gpm}$$

B. In the **ANNULUS**.

(1) Determine n_a (dimensionless):

$$n_a = 3.32 \log\left(\frac{\theta 600}{\theta 300}\right) \tag{6.29}$$

where

n_a = flow index in the annulus
$\theta 600$ = viscometer reading at 600 rpm
$\theta 300$ = viscometer reading at 300 rpm

(2) Determine K_a:

$$K_a = \frac{\theta 600}{(1022^{n_a})} \tag{6.30}$$

where

K_a = consistency in the annulus.

(3) Determine the critical velocity in the annulus:

$$V_{ac} = \left(\frac{(81,600(K_a)(n_a^{0.387}))}{((D_h - D_p)^{n_a})(MW)}\right)^{\left(\frac{1}{(2-n_a)}\right)} \tag{6.31}$$

where

V_{ac} = critical velocity in ft/min
D_h = hole or casing size in inches
D_p = outside diameter of pipe in inches
MW = mud weight in ppg

(4) Determine critical flow rate in the annulus in gpm:

$$Q_{ac} = \frac{V_{ac}\left(D_h^2 - D_p^2\right)}{24.5} \tag{6.32}$$

where

Q_{ac} = critical flow rate in the annulus in gpm

Example:

Mud weight = 12.0 ppg
$\theta 600$ value = 66
$\theta 300$ value = 40
Hole diameter = 8½ in.
Pipe OD = 5.0 in.

(1) Determine n_a:

$$n_a = 3.32 \log\left(\frac{66}{40}\right) 0.7220$$

(2) Determine K_a:

$$K_a = \frac{66}{(1022^{0.7220})} = 0.4433$$

(3) Determine the critical velocity in the annulus:

$$V_{ac} = \left(\frac{(81,600(0.4433)(0.7220^{0.387}))}{((8.5-5.0)^{0.7220})(12.0)}\right)^{\left(\frac{1}{(2-0.7720)}\right)} = 235.6 \text{ ft/min}$$

(4) Determine the critical flow rate in the annulus:

$$Q_{ac} = \frac{235.6(8.5^2 - 5.0^2)}{24.5} = 454.4 \text{ gpm}$$

Spreadsheet 6.1.5 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Critical velocity and pump rate.

6.2 Equivalent circulating "density" ECD (ppg)

Definition: ECD takes into account the friction loss in the annulus due to circulation of the drilling mud with the pumps on.

A. ECD without cuttings:

$$\text{ECD} = \left(\frac{\Delta P_a}{0.052(D_{TVD})}\right) + MW \tag{6.33}$$

where

ECD = equivalent circulating density in ppg
ΔP_a = annulus friction pressure loss in psi
D_{TVD} = total vertical depth in ft
MW = current mud weight in ppg

Example: Determine the ECD at the casing shoe with the previous pressure loss data.

Data: Mud weight = 12.0 ppg
Pressure loss in the DP/Casing shoe annulus = 205.0 psi
Casing Shoe TVD = 12,000 ft

$$\text{ECD} = \left(\frac{205.0}{0.052(12,000)}\right) + 12.0 = 12.3 \text{ ppg}$$

B. ECD with Cuttings

$$\mathrm{ECD_c} = \left(\left(1 - \left(\frac{V_{ca}}{100}\right)\right)(\mathrm{ESD_a})\right) + \left(\left(\frac{V_{ca}}{100}\right)((G_c)(8.34))\right) + \left(\frac{(\Delta P_{aT})}{(0.052)(D_{TVD})}\right) \tag{6.34}$$

where

$\mathrm{ECD_C}$ = equivalent circulating density with cuttings in the annulus in ppg
V_{Ca} = volume of cuttings in the annulus in %
$\mathrm{ESD_a}$ = equivalent static density in the annulus in ppg
G_C = specific gravity of the cutting in g/cm^3
ΔP_{aT} = total pressure loss in the annulus in psi

Example: Determine the ECD at the casing shoe with the following data:

Data: ESD at the casing shoe = 12.0 ppg
Volume of cuttings in the annulus = 4.5%
Specific gravity of the cutting = $2.5\,g/cm^3$
Pressure loss in the annulus = 205.0 psi
Casing shoe TVD = 12,000 ft

$$\mathrm{ECD_c} = \left(\left(1 - \left(\frac{4.5}{100}\right)\right)(12.0)\right) + \left(\left(\frac{4.5}{100}\right)((2.5)(8.34))\right) + \left(\frac{(205.0)}{(0.052)(12,000)}\right) = 12.7 \text{ ppg}$$

Spreadsheet 6.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 ECD.

6.3 Surge and swab pressure loss

This technique is based on converting the drill pipe speed into equivalent mud velocity in the annulus. The time to run or pull the middle joint in a stand through the rotary table is measured in seconds from the box to the pin. This mud velocity caused by the displacement of the pipe is then used in the annular pressure loss equations to determine an equivalent mud weight in ppg.

6.3.1 Determine the pressure surge for the plugged pipe case

This case applies to a drill string that has a bit with nozzles. It assumes laminar flow around the drill pipe and the heavy weight drill pipe but turbulent flow around the drill collars.

A. Determine the surge pressure around the drill pipe.

(1) Determine the velocity of the pipe movement with an average joint length of 31 ft.

$$V_{dp} = \frac{(31)(60)}{t} \tag{6.35}$$

where

V_{dp} = velocity of the drill pipe movement in ft/min
t = time of the movement of the middle joint in the stand from box to pin through the rotary Table.

(2) Determine the n number:

$$n = 3.32 \log \left(\frac{\theta 600}{\theta 300} \right) \tag{6.36}$$

where

n = n number (dimensionless)
$\theta 600$ = viscometer reading at 600 rpm
$\theta 300$ = viscometer reading at 300 rpm

(3) Determine the K factor:

$$K = \frac{\theta 300}{511^n} \tag{6.37}$$

K = consistency factor

(4) Determine the fluid velocity around the drill pipe in ft/min

$$v_{fdp} = \left(0.45 + \left(\frac{D_p^2}{D_h^2 - D_p^2} \right) \right) V_{dp} \tag{6.38}$$

where

V_{fdp} = fluid velocity around the drill pipe in ft/min
D_p = drill pipe or heavy weight drill pipe OD in inches
D_h = hole diameter or casing ID in inches

(5) Determine the maximum fluid velocity:

$$V_{dpm} = (V_{fdp})(1.5) \tag{6.39}$$

where

V_{dpm} = maximum fluid velocity around the drill pipe in ft/min

(6) Determine the shear rate of the mud moving around the drill pipe.

$$\gamma_m = \frac{2.4(V_{dpm})}{(D_h - D_p)} \tag{6.40}$$

where

γ_m = shear rate of the mud in s^{-1}

(7) Calculate the shear stress of the mud moving around the drill pipe:

$$\tau = K(\gamma_m^n) \tag{6.41}$$

where

τ = shear stress of the mud in $lb/100\,ft^2$

(8) Determine the pressure loss for the drill pipe interval:

$$\Delta P_{dp} = \left(\frac{(3.33(\tau))}{(D_h - D_p)}\right)\left(\frac{L_p}{1000}\right) \tag{6.42}$$

where

ΔP_{dp} = pressure loss due to drill pipe movement in psi
L_p = length of pipe in ft

Note: Use the preceding Eqs. (6.35)–(6.42) to calculate the surge pressure for the heavy weight drill pipe when it has the same OD size as the drill pipe.

B. Determine the surge pressure around the drill collars (fluid in turbulent flow).

(1) Determine the velocity of the pipe movement with an average joint length of 31 ft.

$$V_{dc} = \frac{(31)(60)}{t} \tag{6.43}$$

where

V_{dc} = velocity of the drill collar movement in ft/min
t = time of the movement of the middle joint in the stand from box to pin through the rotary table.

(2) Determine the fluid velocity around the drill pipe in ft/min

$$v_{\text{fdc}} = \left(0.45 + \left(\frac{D_{\text{p}}^2}{D_{\text{h}}^2 - D_{\text{p}}^2}\right)\right) V_{\text{dp}} \tag{6.44}$$

where

V_{fdc} = fluid velocity around the drill collar in ft/min
D_{p} = drill collar OD in inches
D_{h} = hole diameter or casing ID in inches

(3) Determine the maximum fluid velocity:

$$V_{\text{dcm}} = \left(V_{\text{fdp}}\right)(1.5) \tag{6.45}$$

where

V_{dcm} = maximum fluid velocity around the drill collar in ft/min

(4) Convert the equivalent velocity of the mud due to pipe movement to equivalent flowrate:

$$Q_{\text{dc}} = \frac{(V_{\text{dcm}})\left(D_{\text{h}}^2 - D_{\text{p}}^2\right)}{24.5} \tag{6.46}$$

where

Q_{dc} = equivalent flowrate around the drill collars in gpm

(5) Determine the Plastic Viscosity of the mud:

$$PV = (\theta 600) - (\theta 300) \tag{6.47}$$

where

PV = plastic viscosity in cP

(6) Determine the pressure loss for the drill collar interval:

$$\Delta P_{\text{dc}} = \frac{(0.000077)\left(MW^{0.8}\right)\left(Q_{\text{dc}}^{1.8}\right)\left(PV^{0.2}\right)(L_{\text{dc}})}{\left(\left(D_{\text{h}} - D_{\text{p}}\right)^3\right)\left(\left(D_{\text{h}} + D_{\text{p}}\right)^{1.8}\right)} \tag{6.48}$$

where

ΔP_{dc} = pressure loss due to drill collar movement in psi
L_{dc} = length of drill collars in ft

(7) Add the pressure loss from the all of the intervals to obtain the total pressure loss for the total string:

$$\Delta P_{\text{T}} = \left(\Delta P_{\text{dp}}\right) + \left(\Delta P_{\text{dc}}\right) \tag{6.49}$$

where

ΔP_{T} = total pressure loss for the drill string in psi

(8) If the bit depth is less than the casing depth or total measured depth (or out of the casing and in the open hole), the pressure loss should be added to the hydrostatic head of the mud at the casing shoe or TVD to determine the total surge pressure in ppg.

$$\Delta P_{su} = \frac{(\Delta P_T)}{\big((0.052)(D_{csg})\big)} + MW \tag{6.50}$$

where

ΔP_{su} = SURGE pressure on the formation at the casing shoe in psi
MW = mud weight in ppg
D_{csg} = total vertical depth of casing shoe in ft

(9) Calculate the SWAB pressure at the casing shoe:

$$\Delta P_{sw} = MW - \frac{(\Delta P_T)}{\big((0.052)(D_{csg})\big)} \tag{6.51}$$

where

ΔP_{sw} = SWAB pressure on the formation at the casing shoe in ppg

Example: Calculate the surge and swab pressure casing shoe:

Data:	Well depth	= 15,000 ft
	Mud weight	= 15.0 ppg
	Bit depth	= 10,000 ft
	Casing ID	= 8.681 in.
	Casing length	= 12,000 ft (also TVD)
	Drill pipe OD	= 4.5 in.
	Drill pipe length	= 8370 ft
	Heavy weight drill pipe OD	= 4.5 in.
	Heavy weight drill pipe length	= 930 ft
	Drill collar OD	= 6¼ in.
	Drill collar length	= 700 ft
	Pipe running speed (middle joint)	= 10 s
	θ600 value	= 140
	θ300 value	= 80
	Plastic viscosity	= 60 cP
	Fracture gradient at the casing shoe	= 17.7 ppg
	Pore pressure at the casing shoe	= 12.0 ppg

A. Determine the surge pressure for the drill pipe and heavy weight drill pipe.

(1) Determine the velocity of the pipe movement with an average joint length of 31 ft.

$$V_{dp} = \frac{(31)(60)}{10} = 186 \text{ ft/min}$$

(2) Determine the n number:

$$n = 3.32 \log\left(\frac{140}{80}\right) = 0.8069$$

(3) Determine the K factor:

$$K = \frac{80}{(511^{n0.8069})} = 0.5220$$

(4) Determine the fluid velocity around the drill pipe in ft/min

$$v_{\text{fdp}} = \left(0.45 + \left(\frac{4.5^2}{8.681^2 - 4.5^2}\right)\right)(186) = 152.0 \text{ ft/min}$$

(5) Determine the maximum fluid velocity:

$$V_{\text{dpm}} = (152.0)(1.5) = 228.0 \text{ ft/min}$$

(6) Determine the shear rate of the mud moving around the drill pipe.

$$\gamma_{\text{m}} = \frac{2.4(228.0)}{(8.681 - 4.5)} = 130.9 \text{ s}^{-1}$$

(7) Calculate the shear stress of the mud moving around the drill pipe:

$$\tau = (0.5220)(130.9)^{0.8069} = 26.7 \text{ lb}/100\text{ft}^2$$

(8) Determine the pressure loss for the drill pipe and heavy weight drill pipe interval:

$$\Delta P_{\text{dp}} = \left(\frac{(3.33(26.7))}{(8.681 - 4.5)}\right)\left(\frac{8,\,370 + 930}{1000}\right) = 197.8 \text{ psi}$$

B. Determine the surge pressure around the drill collars (assume the fluid is in turbulent flow).

a. Determine the velocity of the pipe movement with an average joint length of 31 ft.

$$V_{\text{dp}} = \frac{(31)(60)}{10} = 186.0 \text{ ft/min}$$

b. Determine the fluid velocity around the drill collars:

$$v_{\text{fdc}} = \left(0.45 + \left(\frac{6.25^2}{8.681^2 - 6.25^2}\right)\right)(186.0) = 283.9 \text{ ft/min}$$

c. Determine the maximum fluid velocity:

$$V_{\text{dcm}} = (283.9)(1.5) = 425.9 \text{ ft/min}$$

d. Convert the equivalent velocity of the mud due to pipe movement to equivalent flowrate:

$$Q_{dc} = \frac{(425.9)\left(8.681^2 - 6.25^2\right)}{24.5} = 631.0 \text{ gpm}$$

e. Determine the Plastic Viscosity of the mud:

$$PV = (140) - (80) = 60 \text{ cP}$$

f. Determine the pressure loss for the drill collar interval:

$$\Delta P_{dc} = \frac{(0.000077)\left(15.0^{0.8}\right)\left(631.0^{1.8}\right)\left(60^{0.2}\right)(700)}{\left((8.681 - 6.25)^3\right)\left((8.681 + 6.25)^{1.8}\right)} = 62.7 \text{ psi}$$

g. Add the pressure loss from the all of the intervals to obtain the total pressure loss for the total string:

$$\Delta P_T = (197.8) + (62.7) = 260.5 \text{ psi}$$

h. Determine the SURGE pressure at the casing shoe in ppg.

$$\Delta P_{su} = \frac{(260.5)}{((0.052)(12,\,000))} + 15.0 = 15.4 \text{ ppg}$$

i. Calculate the SWAB pressure at the casing shoe:

$$\Delta P_{sw} = 15.0 - \frac{(260.5)}{((0.052)(12,\,000))} = 14.6 \text{ ppg}$$

Spreadsheet 6.3.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Pressure surge for the plugged pipe.

6.3.2 Determine the pressure surge for the open pipe case

All of the same equations except **one** should be utilized to calculate the surge and swab pressure when moving the drill string in the wellbore. The only change required is to use the following equation to calculate the fluid velocity with the open pipe.

(1) Determine the fluid velocity around and into open pipe in ft/min

$$v_{op} = \left(0.45 + \left(\frac{D_p^2}{D_h^2 - D_p^2 + D_{ID}^2}\right)\right) \tag{6.52}$$

where

v_{op} = fluid velocity around the open pipe in ft/min
D_p = OD of the pipe in inches

D_h = diameter of the hole or ID of the casing in inches
D_{ID} = ID of the drill pipe, heavy weight drill pipe or drill collars in inches

All other equations are the same.
Example: Calculate the surge and swab pressure at the casing shoe:

Data:	Well depth	= 15,000 ft
	Mud weight	= 15.0 ppg
	Bit depth	= 10,000 ft
	Casing ID	= 8.681 in.
	Casing length	= 12,000 ft (also TVD)
	Drill pipe size	= 4.5 × 3.826 in.
	Drill pipe length	= 8370 ft
	Heavy weight drill pipe size	= 4.5 × 2.750 in.
	Heavy weight drill pipe length	= 930 ft
	Drill collar size	= 6¼ × 2.750 in.
	Drill collar length	= 700 ft
	Pipe running speed (middle joint)	= 10 s
	θ600 value	= 140
	θ300 value	= 80
	Plastic viscosity	= 60 cP
	Fracture gradient at the casing shoe	= 17.7 ppg
	Pore pressure at the casing shoe	= 12.0 ppg

A. Determine the surge pressure for the drill pipe and heavy weight drill pipe.

(1) Determine the velocity of the pipe movement with an average joint length of 31 ft.

$$V_{dp} = \frac{(31)(60)}{10} = 186 \text{ ft/min}$$

(2) Determine the n number:

$$n = 3.32 \log\left(\frac{140}{80}\right) = 0.8069$$

(3) Determine the K factor:

$$K = \frac{80}{(511^{n0.8069})} = 0.5220$$

(4) Determine the fluid velocity around the drill pipe in ft/min

$$v_{fdp} = \left(0.45 + \left(\frac{4.5^2}{8.681^2 - 4.5^2 + 3.826^2}\right)\right)(186) = 137.7 \text{ ft/min}$$

(5) Determine the maximum fluid velocity:

$$V_{\text{dpm}} = (137.7)(1.5) = 206.5 \text{ ft/min}$$

(6) Determine the shear rate of the mud moving around the drill pipe.

$$\gamma_{\text{m}} = \frac{2.4(206.5)}{(8.681 - 4.5)} = 118.5 \text{ s}^{-1}$$

(7) Calculate the shear stress of the mud moving around the drill pipe:

$$\tau = (0.5220)(118.5)^{0.8069} = 24.6 \text{ lb/100ft}^2$$

(8) Determine the pressure loss for the drill pipe interval:

$$\Delta P_{\text{dp}} = \left(\frac{(3.33(24.6))}{(8.681 - 4.5)}\right)\left(\frac{8,370 + 930}{1000}\right) = 182.2 \text{ psi}$$

B. Determine the surge pressure around the drill collars (fluid in turbulent flow).

(1) Determine the velocity of the pipe movement with an average joint length of 31 ft.

$$V_{\text{dp}} = \frac{(31)(60)}{10} = 186.0 \text{ ft/min}$$

(2) Determine the fluid velocity around the drill collars:

$$v_{\text{fdp}} = \left(0.45 + \left(\frac{6.25^2}{8.681^2 - 6.25^2 + 2.750^2}\right)\right)(186) = 249.4 \text{ ft/min}$$

(3) Determine the maximum fluid velocity:

$$V_{\text{dcm}} = (249.4)(1.5) = 374.1 \text{ ft/min}$$

(4) Convert the equivalent velocity of the mud due to pipe movement to equivalent flowrate:

$$Q_{\text{dc}} = \frac{(374.1)\left(8.681^2 - 6.25^2\right)}{24.5} = 554.2 \text{ gpm}$$

(5) Determine the Plastic Viscosity of the mud:

$$PV = (140) - (80) = 60 \text{ cP}$$

(6) Determine the pressure loss for the drill collar interval:

$$\Delta P_{\text{dc}} = \frac{(0.000077)\left(15.0^{0.8}\right)\left(554.2^{1.8}\right)\left(60^{0.2}\right)(700)}{\left((8.681 - 6.25)^3\right)\left((8.681 + 6.25)^{1.8}\right)} = 49.7 \text{ psi}$$

(7) Add the pressure loss from the all of the intervals to obtain the total pressure loss for the total string:

$$\Delta P_T = (182.2) + (49.7) = 231.9 \text{ psi}$$

(8) Determine the SURGE pressure at the casing shoe in ppg.

$$\Delta P_{su} = \frac{(231.9)}{((0.052)(12,000))} + 15.0 = 15.4 \text{ ppg}$$

(9) Calculate the SWAB pressure at the casing shoe:

$$\Delta P_{sw} = 15.0 - \frac{(231.9)}{((0.052)(12,000))} = 14.6 \text{ ppg}$$

Spreadsheet 6.3.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Pressure surge for the open pipe.

6.4 Equivalent spherical diameter for drilled cuttings size used in slip velocity equations

Step 1: Determine the length, width, and thickness dimension of the cutting in inches.

Step 2: Convert these measurements to eights of-an-inch.

Step 3: Determine the volume of the cutting:

$$V_C = (L)(W)(T) \tag{6.53}$$

where

V_C = volume of the cutting in eights of-an-inch
L = length of the cutting
W = width of the cutting
T = thickness of the cutting

Step 4: Determine the equivalent spherical diameter from Table 6.2 to use in the slip velocity equation.

Example: Determine the equivalent spherical diameter for the following:

Step 1: Determine the length, width, and thickness dimension of the cutting in in.

Data: Length = 1 in.
Width = ½ in.
Thickness = ¼ in.

TABLE 6.2 Equivalent spherical diameter for cuttings.

Volume $\left(\frac{\text{inches}}{8}\right)$	Equivalent diameter (in.)	Equivalent diameter (decimal)
1	1/8	0.125
4	1/4	0.250
14	3/8	0.375
34	1/2	0.500
65	5/8	0.625
110	3/4	0.750
180	7/8	0.875
270	1	1.000
380	9/8	1.125
520	5/4	1.250
700	11/8	1.375
900	3/2	1.500

Step 2: Convert these measurements to eights of-an-inch.

$$\begin{aligned}\text{Length} &= 8\\ \text{Width} &= 4\\ \text{Thickness} &= 2\end{aligned}$$

Step 3: Determine the volume of the cutting:

$$V_C = (8)(4)(2) = 64$$

Step 4: Determine the equivalent spherical diameter:

$$D_p = \frac{\left(\sqrt[3]{\frac{\left(3\left(\frac{V_c}{8}\right)\right)}{4(\pi)}}\right)}{2} \tag{6.54}$$

where

D_p = equivalent spherical diameter of cutting in inches
V_c = volume of the cutting in 8th of an inch
π = pi constant 3.14

Example: Determine the equivalent spherical diameter of a cutting with a volume of 64:

$$D_p = \frac{\left(\sqrt[3]{\frac{\left(3\left(\frac{64}{8}\right)\right)}{4(3.14)}}\right)}{2} = 0.620 \text{ in.}$$

The equivalent spherical diameter for 64 is (approximately) ⅝ in. or **0.625** and can be utilized as the particle size in the slip velocity equation.

Spreadsheet 6.4 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Equivalent spherical diameter for drilled cuttings size.

6.5 Slip velocity of cuttings in the annulus

These calculations provide the slip velocity of a cutting of a specific size and weight in a given fluid. The annular velocity and the cutting net rise velocity are also calculated.

(1) Annular velocity, ft/min:

$$AV = \frac{(24.5)(Q)}{\left(D_h^2 - D_p^2\right)} \tag{6.55}$$

(2) Cutting slip velocity, ft/min:

$$V_s = 0.45\left(\frac{PV}{(MW)(C_d)}\right)\left[\left(\sqrt{\frac{36,800}{\left(\frac{PV}{(MW)(C_d)}\right)^2}(C_d)\left(\frac{\rho_c}{MW} - 1\right) + 1}\right) - 1\right] \tag{6.56}$$

where

V_s = slip velocity, ft/min
PV = plastic viscosity, cP
MW = mud weight, ppg
C_d = diameter of cutting, in.
ρ_c = density of cutting, ppg

Example: Using the following data, determine the annular velocity, ft/min; the cuttings slip velocity, ft/min, and the cutting net rise velocity, ft/min:

Data: Mud weight (MW) = 11.0 ppg
Plastic viscosity (PV) = 13 cps
Diameter of cutting = 0.25 in.
Density of cutting = 22.0 ppg (8.33 ppg × specific gravity, 2.64)

Flow rate = 520 gpm
Diameter of hole = 12¼ in.
Drill pipe OD = 5.0 in.

Annular velocity, ft/min:

$$AV = \frac{(24.5)(520)}{\left(12.25^2 - 5.0^2\right)} = 101.9 \text{ ft/min}$$

Cutting slip velocity, ft/min:

$$V_s = 0.45\left(\frac{13}{(11.0)(0.25)}\right)\left[\left(\sqrt{\frac{36,800}{\left(\frac{13}{(11.0)(0.25)}\right)^2}(0.25)\left(\frac{22.0}{11.0}-1\right)+1}\right)-1\right]$$

$$= 41.1 \text{ ft/min}$$

Cutting net rise velocity:

Annular velocity	= 101.9 ft/min
Cutting slip velocity	= −41.1 ft/min
Cutting net rise velocity	= 60.8 ft/min

Method 2:

(1) Determine n:

$$n = 3.32\ log\left(\frac{\theta 600}{\theta 300}\right) \tag{6.57}$$

(2) Determine K:

$$K = \frac{\theta 300}{511^n} \tag{6.58}$$

(3) Determine annular velocity, ft/min:

$$AV = \frac{(24.5)(Q)}{\left(D_h^2 - D_p^2\right)} \tag{6.59}$$

(4) Determine viscosity (μ):

$$\mu = \left(\frac{(2.4)(v)}{\left(D_h - D_p\right)}\left(\frac{2(n)+1}{(3(n))}\right)\right)^n\left(\frac{(200)(K)\left(D_h - D_p\right)}{v}\right) \tag{6.60}$$

(5) Slip velocity (V_s), ft/min:

$$V_s = \frac{\left((\rho_c - MW)^{0.667}\right)(175)(C_d)}{\left(MW^{0.333}\right)\left(\mu^{0.333}\right)} \tag{6.61}$$

Nomenclature:

n = Dimensionless
K = Dimensionless

$\theta 600 =$ 600 viscometer dial reading
$\theta 300 =$ 300 viscometer dial reading
$Q =$ Circulation rate, gpm
$D_h =$ Hole diameter, in.
$D_p =$ Pipe or collar OD, in.
$v =$ Annular velocity, ft/min
$\mu =$ Mud viscosity, cP
$\rho_c =$ Cutting density, ppg
$C_d =$ Cutting diameter, in.

Example: Using the data listed below, determine the annular velocity, cuttings slip velocity, and the cutting net rise velocity:

Data: Mud weight (MW) = 11.0 ppg
Θ600 viscometer reading = 36
Θ300 viscometer reading = 23
Plastic viscosity (PV) = 13 cps
Density of cutting = 22.0 ppg
Hole diameter = 12¼ in.
Drill pipe OD = 5.0 in.
Circulation rate = 520 gpm

(1) Determine n:

$$n = 3.32 \log\left(\frac{36}{23}\right) = 0.6460$$

(2) Determine K:

$$K = \frac{23}{511^{0.64599}} = 0.4094$$

(3) Determine annular velocity, ft/min:

$$v = \frac{(24.5)(520)}{\left(12.25^2 - 5.0^2\right)} = 101.9 \text{ ft/min}$$

(4) Determine mud viscosity, cP:

$$\mu = \left(\left(\frac{(2.4)(101.9)}{(12.25 - 5.0)}\left(\frac{2(0.6460) + 1}{(3(0.6460))}\right)\right)^{0.6460}\right)\left(\frac{(200)(0.4094)(12.25 - 5.0)}{101.9}\right)$$
$$= 62.9 \text{ cP}$$

(5) Determine cutting slip velocity, ft/min:

$$V_s = \frac{\left((22.0 - 11.0)^{0.667}\right)(175)(0.25)}{\left(11.0^{0.333}\right)\left(62.9^{0.333}\right)} = 24.5 \text{ ft/min}$$

(6) Determine cutting net rise velocity, ft/min:

Annular velocity	= 101.9 ft/min
Cutting slip velocity	= −24.5 ft/min
Cutting net rise velocity	= 77.4 ft/min

Spreadsheet 6.5 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Slip velocity of cuttings.

6.6 Carrying capacity index

Determine the carrying capacity index (CCI) developed to indicate hole cleaning efficiency in vertical and low-angle wells.

Step 1: Determine the flow behavior index (n) for the Herschel-Bulkley fluids model:

$$n_{HB} = 3.32 \log_{10} \frac{\left(2\left(\overline{PV}\right) + \left(\overline{YP}\right)\right)}{\left(\overline{PV} + \overline{YP}\right)} \tag{6.62}$$

where

n_{HB} = flow behavior index for the Herschel-Bulkley fluids model
PV = plastic viscosity in cP
YP = yield point in lb/100 ft^2

Step 2: Determine the consistency factor (K) for the Herschel-Bulkley fluids model:

$$K_{HB} = \left(511^{(1-n_{HB})}\right)(PV + YP) \tag{6.63}$$

where

K_{HB} = consistency factor for the Herschel-Bulkley fluids model in lb/100 ft^2

Step 3: Determine the carrying capacity index:

$$CCI = \frac{(MW)(K_{HB})(V_a)}{400,\,00} \tag{6.64}$$

where

CCI = carrying capacity index (dimensionless)
MW = mud weight in ppg
K_{HB} = consistency factor in lb/100 ft^2
V_a = annular velocity in ft/min

Example: Determine the carrying capacity index with the following:

Data: Mud weight = 10.0 ppg
Annular velocity = 125 ft/min
Plastic viscosity = 18 cP
Yield point = 15 lb/100 ft^2

Step 1: Determine the flow behavior index (n) for the Herschel-Bulkley fluids model:

$$n_{HB} = 3.32 \log_{10} \frac{\left(2(\overline{18}) + (15)\right)}{(18 + 15)} = 0.6277$$

Step 2: Determine the consistency factor (K) for the Herschel-Bulkley fluids model:

$$K_{HB} = \left(511^{(1-0.6277)}\right)(18 + 15) = 336.4 \ \frac{\text{lb}}{100 \text{ ft}^2}$$

Step 3: Determine the carrying capacity index:

$$\text{CCI} = \frac{(10.0)(336.4)(125)}{400,\,000} = 1.05$$

Note: A CCI of 1.0 or greater indicates excellent hole cleaning. If the CCI is less than 0.5 then the hole cleaning is poor. If the ROP is above 200 ft/h, a CCI of 2.0 or above will indicate good cleaning.

Spreadsheet 6.6 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Carrying capacity index.

6.7 Pressure required to break circulation

(A) Pressure required to break the mud's gel strength inside the drill string in psi.

Step 1: Calculate the pressure required to break the mud's gel strength inside the drill string:

$$P_{gsds} = \left(\frac{(L_{ds})(\tau_{gs})}{300(D_i)}\right) \tag{6.65}$$

where

P_{gsds} = pressure required to break gel strength inside the drill string in psi
τ_{gs} = 10 min gel strength of drilling fluid in lb/100 ft^2
D_i = inside diameter of drill pipe in inches
L_{ds} = length of drill string in ft

Step 2: Calculate the pressure required to break the mud's gel strength in the annulus:

$$P_{\text{gsa}} = \left(\frac{(L_{\text{ds}})(\tau_{\text{gs}})}{300(D_{\text{h}} - D_{\text{i}})}\right) \tag{6.66}$$

where

P_{gsa} = pressure required to break gel strength in the annulus in psi
D_{h} = diameter of the annulus in inches

Step 3: Calculate the total pressure required to break circulation:

$$P_{\text{gst}} = \left(P_{\text{gsds}} + P_{\text{gsa}}\right) \tag{6.67}$$

where

P_{gst} = pressure required to break circulation in the well in psi

Example: Calculate the pressure required to break circulation in the well with this:

Data: Gel strength (10 or 30 min) = 18 lb/100 ft^2
Drill string = 6⅝ × 5.965 in.
Hole size = 12¼ in.
Depth (MD) = 15,000 ft

Step 1: Calculate the pressure required to break the mud's gel strength inside the drill string:

$$P_{\text{gsds}} = \left(\frac{(15,000)(18)}{300(5.965)}\right) = 150.9 \approx 151.0 \text{ psi}$$

Step 2: Calculate the pressure required to break the mud's gel strength in 8the annulus:

$$P_{\text{gsa}} = \left(\frac{(15,000)(18)}{300(12.25 - 6.625)}\right) = 160 \text{ psi}$$

Step 3: Calculate the total pressure required to break circulation:

$$P_{\text{gsa}} = (151 + 160) = 311 \text{ psi}$$

(B) Calculate the effective gel strength based on the actual pressure required to break the circulation.

$$\tau_{\text{egs}} = \frac{300(P_{\text{bc}})(D_{\text{i}}(D_{\text{h}} - D_{\text{ds}}))}{(L_{\text{ds}})(D_{\text{i}} + D_{\text{h}} - D_{\text{ds}})} \tag{6.68}$$

where

τ_{egs} = effective gel strength based on pressure required to break circulation in lb/100 ft^2
P_{bc} = actual pressure required to break circulation in the well in psi

Example: Calculate the effective gel strength with the following:

Data: Pressure required to break circulation = 475 psi
Drill string length = 15,000 ft
Hole size = 12¼ in.
Drill string size = 6⅝ × 5.965 in.

$$\tau_{egs} = \frac{300(475)(5.965(12.25 - 6.625))}{(15,000)(5.965 + 12.25 - 6.625)} = 27.5\,\text{lb}/100\text{ft}^2$$

Spreadsheet 6.7 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Pressure required to break circulation.

6.8 Initial gel strength guidelines for top hole drilling in high angle wells (after Zamora)

$$\tau_I = 8.75[(D_C)(((C_{SG})(8.34)) - MW)]^{0.5} \tag{6.69}$$

where

τ_I = initial gel strength in lb/100 ft^2
D_C = equivalent spherical diameter of cutting in in.
C_{SG} = SG of cutting in g/cm^3 (Shallow formations may range from 2.5 to 2.0 g/cm^3)
MW = mud weight in ppg

Example 1: Determine the initial gel strength recommended for the following conditions:

Data: D_C = 0.625 in. (Gumbo particle with soft clay)
C_{SG} = 2.0 g/cm^3
MW = 9.5 ppg

$$\tau_I = 8.75[(0.625)(((2.0)(8.34)) - 9.5)]^{0.5} = 18.5 \approx 19\,\text{lb}/100\text{ft}^2$$

Example 2: Determine the initial gel strength recommended for the following conditions:

Data: D_C = 0.125 in. (sand particle)
C_{SG} = 2.5 g/cm^3
MW = 9.5 ppg

$$\tau_I = 8.75[(0.125)(((2.5)(8.34)) - 9.5)]^{0.5} = 10.4 \approx 11\,\text{lb}/100\text{ft}^2$$

Spreadsheet 6.8 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Initial gel strength guidelines.

6.9 Bit nozzle selection—Optimized hydraulics

These series of formulas will determine the correct jet sizes when optimizing for jet impact or hydraulic horsepower and optimum flow rate for two or three nozzles.

(1) Determine Bit Nozzles area in sq. inches:

$$B_{\mathrm{Na}} = \frac{\left(N_1^2\right) + \left(N_2^2\right) + \left(N_3^2\right)}{1303.8} \tag{6.70}$$

where

B_{Na} = bit nozzle area in sq. in.

(2) Determine the bit nozzle pressure loss in psi:

$$\text{(a)}\ \ \Delta P_{\mathrm{b1}} = \frac{\left(Q_1^2\right)(MW)}{(10,858)(B_{\mathrm{Na}})} \tag{6.71}$$

where

ΔP_{b1} = bit pressure loss with pump pressure #1 in psi
Q_1 = pump rate 1
MW = mud weight in ppg

$$\text{(b)}\ \ \Delta P_{\mathrm{b2}} = \frac{\left(Q_2^2\right)(MW)}{(10,858)(B_{\mathrm{Na}})} \tag{6.72}$$

where

ΔP_{b2} = bit pressure loss with pump pressure #2 in psi
Q_2 = pump rate 2
MW = mud weight in ppg

(3) Determine the total pressure loss except bit nozzle pressure loss in psi:

$$\text{(a)}\ \ \Delta P_{\mathrm{c1}} = \left(\Delta P_{\mathrm{p1}}\right) - \left(\Delta P_{\mathrm{b1}}\right) \tag{6.73}$$

$$\text{(b)}\ \ \Delta P_{\mathrm{c2}} = \left(\Delta P_{\mathrm{p2}}\right) - \left(\Delta P_{\mathrm{b2}}\right) \tag{6.74}$$

where

ΔP_{c1} = total pressure loss with pump pressure #1 in psi
ΔP_{c2} = total pressure loss with pump pressure #2 in psi

(4) Determine slope of the line M:

$$M = \frac{log\left(\frac{\Delta P_{\mathrm{c1}}}{\Delta P_{\mathrm{c2}}}\right)}{log\left(\frac{Q_1}{Q_2}\right)} \tag{6.75}$$

(5) Determine the optimum pressure loss:
(a) For impact force:

$$P_{\text{optIF}} = \frac{2}{(M+2)}(P_{\max}) \tag{6.76}$$

where

P_{optIF} = optimum pressure loss for impact force

(b) For hydraulic horsepower:

$$P_{\text{optHH}} = \frac{1}{(M+2)}(P_{\max}) \tag{6.77}$$

where

P_{optHH} = optimum pressure loss for hydraulic horsepower

(6) Determine the optimum flow rate:
(a) For impact force:

$$Q_{\text{optIF}} = \left(\left(\frac{P_{\text{optIF}}}{P_{\max}}\right)^{\left(\frac{1}{M}\right)}\right)(Q_1) \tag{6.78}$$

where

Q_{optIF} = optimum flow rate for impact force

(b) For hydraulic horsepower:

$$Q_{\text{optHH}} = \left(\left(\frac{P_{\text{optHH}}}{P_{\max}}\right)^{\left(\frac{1}{M}\right)}\right)(Q_1) \tag{6.79}$$

where

Q_{optHH} = optimum flow rate for hydraulic horsepower

(7) Determine the pressure at the bit:
(a) For impact force:

$$P_{\text{bIF}} = P_{\max} - P_{\text{optIF}} \tag{6.80}$$

where

P_{bIF} = bit pressure for impact force

(b) For hydraulic horsepower:

$$P_{\text{bHH}} = P_{\max} - P_{\text{optHH}} \tag{6.81}$$

where

P_{bHH} = bit pressure for hydraulic horsepower

(8) Determine the nozzle area:
(a) For impact force:

$$B_{\mathrm{NaIF}} = \sqrt{\frac{{Q_{\mathrm{optIF}}}^2(MW)}{(10,858)(P_{\max})}} \tag{6.82}$$

where

B_{NaIF} = bit nozzle area for impact force

(b) For hydraulic horsepower:

$$B_{\mathrm{NaHH}} = \sqrt{\frac{{Q_{\mathrm{optHH}}}^2(MW)}{(10,858)(P_{\max})}} \tag{6.83}$$

where

B_{NaHH} = bit nozzle for hydraulic horsepower

(9) Determine the nozzles size:
(A) For **THREE** nozzles:
(1) Impact Force:

$$B_{\mathrm{N3IF}} = \left(\sqrt{\frac{(B_{\mathrm{NaIF}})}{(3(0.7854))}}\right)(32) \tag{6.84}$$

where

B_{N3IF} = 3 nozzle size for impact force in 32nds

(2) Hydraulic Horsepower:

$$B_{\mathrm{N3HH}} = \left(\sqrt{\frac{(B_{\mathrm{NaHH}})}{(3(0.7854))}}\right)(32) \tag{6.85}$$

where

B_{N3HH} = 3 nozzle size for hydraulic horsepower in 32nds

(B) For **TWO** nozzles:
(1) Impact Force:

$$B_{\mathrm{N2IF}} = \left(\sqrt{\frac{(B_{\mathrm{NaIF}})}{(2(0.7854))}}\right)(32) \tag{6.86}$$

where

B_{N2IF} = 2 nozzle size for impact force in 32nds

(2) Hydraulic Horsepower:

$$B_{\mathrm{N2HH}} = \left(\sqrt{\frac{(B_{\mathrm{NaHH}})}{(2(0.7854))}}\right)(32) \tag{6.87}$$

where

B_{N2HH}=2 nozzle size for hydraulic horsepower in 32nds

Example: Optimize bit hydraulics on a well with the following: Select the proper jet sizes for impact force and hydraulic horsepower for two jets and three jets:

Data:	Mud weight	= 13.0 ppg
	Jet sizes	= 3 × 17/32nds
	Maximum pump pressure (P_{max})	= 3000 psi
	Pump pressure 1 (P_{p1})	= 3000 psi
	Pump rate 1 (Q_1)	= 420 gpm
	Pump pressure 2 (P_{p2})	= 1300 psi
	Pump rate 2 (Q_2)	= 275 gpm

(1) Determine the nozzle area:

$$B_{\mathrm{Na}} = \frac{\left(17^2\right) + \left(17^2\right) + \left(17^2\right)}{1303.8} = 0.6650$$

(2) Determine the bit nozzle pressure loss in psi:

(a) $\Delta P_{\mathrm{b1}} = \dfrac{\left(420^2\right)(13.0)}{(10,858)\left(0.6650^2\right)} = 477.6 \text{ psi} \cong 478 \text{ psi}$

(b) $\Delta P_{\mathrm{b1}} = \dfrac{\left(275^2\right)(13.0)}{(10,858)\left(0.6650^2\right)} = 204.7 \text{ psi} \cong 205 \text{ psi}$

(3) Determine the total pressure losses except bit nozzle pressure loss in psi:

(a) $\Delta P_{\mathrm{c1}} = (3000) - (478) = 2522$ psi

(b) $\Delta P_{\mathrm{c2}} = (1300) - (205) = 1095$ psi

(4) Determine slope of the line (M):

$$M = \frac{log\left(\frac{2522}{1095}\right)}{log\left(\frac{420}{275}\right)} = 1.97$$

(5) Determine the optimum pressure losses in psi:

(a) For impact force:

$$P_{\mathrm{optIF}} = \frac{2}{(1.97 + 2)}(3000) = 1511 \text{ psi}$$

(b) For hydraulic horsepower:

$$P_{\text{optHH}} = \frac{1}{(1.97 + 2)}(3000) = 1010 \text{ psi}$$

(6) Determine the optimum flow rate:

(a) For impact force:

$$Q_{\text{optIF}} = \left(\left(\frac{1511}{3000}\right)^{\left(\frac{1}{1.97}\right)}\right)(420) = 296.5 \text{ gpm} \cong 297 \text{ gpm}$$

(b) For hydraulic horsepower:

$$Q_{\text{optHH}} = \left(\left(\frac{1010}{3000}\right)^{\left(\frac{1}{1.97}\right)}\right)(420) = 241.7 \text{ gpm} \cong 242 \text{ gpm}$$

(7) Determine pressure losses at the bit:

(a) For impact force:

$$\Delta P_{\text{bIF}} = (3000) - (1511) = 1489 \text{ psi}$$

(b) For hydraulic horsepower:

$$\Delta P_{\text{bHH}} = (3000) - (1010) = 1990 \text{ psi}$$

(8) Determine nozzle area:

(a) For impact force:

$$B_{\text{NaIF}} = \sqrt{\frac{(297^2)(13.0)}{(10,858)(1489)}} = 0.2663 \text{ sq.in.}$$

(b) For hydraulic horsepower:

$$B_{\text{NaHH}} = \sqrt{\frac{(242^2)(13.0)}{(10,858)(1990)}} = 0.1877 \text{ sq.in.}$$

(9) Determine nozzle size in 32nd in.:

(A) For **THREE** nozzles:

(1) Impact Force:

$$B_{\text{N3IF}} = \left(\sqrt{\frac{(0.2663)}{(3(0.7854))}}\right)(32) = 10.76 \text{ sq.in.}$$

(2) Hydraulic Horsepower:

$$B_{\text{N3HH}} = \left(\sqrt{\frac{(0.1877)}{(3(0.7854))}}\right)(32) = 9.03 \text{ sq.in.}$$

(B) For **TWO** nozzles:

(1) Impact Force:

$$B_{\text{N2IF}} = \left(\sqrt{\frac{(0.2663)}{(2(0.7854))}}\right)(32) = 13.18 \text{ sq.in.}$$

(2) Hydraulic Horsepower:

$$B_{\text{N2HH}} = \left(\sqrt{\frac{(0.1877)}{(2(0.7854))}}\right)(32) = 11.06 \text{ sq.in.}$$

Note: Usually the nozzle size will have a decimal fraction. The fraction times 3 will determine how many nozzles should be larger than that calculated.

(3) Determine the final selection of nozzles:

(A) For **THREE** nozzles:

(1) Impact force:

$$(0.76)(3) = 2.28 \text{ rounded to } 2.3$$

So: 1 jet = 10/32nds
2 jets = 11/32nds

(2) Hydraulic horsepower:

$$(0.03)(3) = 0.09 \text{ rounded to } 0.1$$

So: 3 jets = 9/32nds

(B) For **TWO** nozzles:

(1) Impact force:

$$(0.18)(2) = 0.36 \text{ rounded to } 0.4$$

So: 2 jets = 13/32nds

(2) Hydraulic horsepower:

$$(0.06)(2) = 0.12 \text{ rounded to } 0.1$$

So: 2 jets = 11/32nds

Spreadsheet 6.9 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Bit nozzle selection.

6.10 Hydraulic analysis

This sequence of calculations is designed to quickly and accurately analyze various parameters of existing bit hydraulics.

(1) Annular velocity:

$$AV = \frac{(24.5)(Q)}{\left(D_{\mathrm{h}}^2 - D_{\mathrm{p}}^2\right)} \tag{6.88}$$

where

AV = annular velocity in ft/min

(2) Jet nozzle pressure loss:

$$P_{\mathrm{b}} = \frac{(156.5)\left(Q^2\right)(MW)}{\left(\left(N_1^2\right) + \left(N_2^2\right) + \left(N_3^2\right) + \left(N_n^2\right)\right)^2} \tag{6.89}$$

where

P_{b} = jet nozzle pressure loss in psi
N = nozzle size in 32nds

(3) System hydraulic horsepower available:

$$\mathrm{HHP_S} = \frac{\left(P_{\mathrm{p}}\right)(Q)}{1714} \tag{6.90}$$

where

$\mathrm{HHP_S}$ = system hydraulic horsepower available
P_{p} = pump pressure in psi
Q = pump output in gpm

(4) Hydraulic horsepower at bit:

$$\mathrm{HHP_b} = \frac{(P_{\mathrm{b}})(Q)}{1714} \tag{6.91}$$

where

$\mathrm{HHP_b}$ = hydraulic horsepower at the bit

(5) Hydraulic horsepower per square inch of bit diameter:

$$\mathrm{HSI} = \frac{(\mathrm{HHP_b})(1.27)}{\left(D_{\mathrm{b}}^2\right)} \tag{6.92}$$

where

HSI = bit hydraulic horsepower per square inch
D_{b} = diameter of bit in inches

(6) Percent pressure loss at bit:

$$P_{b\%} = \left(\frac{(P_b)}{(P_p)}\right)(100) \tag{6.93}$$

where

$P_b\%$ = pressure loss at the bit in percent

(7) Jet velocity:

$$V_N = \frac{(417.2)(Q)}{\left((N_1^2) + (N_2^2) + (N_3^2) + (N_N^2)\right)} \tag{6.94}$$

where

V_N = jet nozzle velocity in ft/s

(8) Impact force:

$$IF = \frac{(MW)(V_N)(Q)}{1930} \tag{6.95}$$

where

IF = impact force in lb.

(9) Impact force per square inch of bit area:

$$IFSI = \frac{(IF)(1.27)}{(D_b^2)} \tag{6.96}$$

where

IFSI = impact force per square inch of bit area

Example:

Mud weight	= 12.0 ppg
Circulation rate	= 520 gpm
Bit size	= 12¼ in.
Nozzle sizes	= 3 × 12/32nds in.
Drill pipe OD	= 5.0 in.
Surface pressure	= 3000 psi

(1) Annular velocity:

$$AV = \frac{(24.5)(520)}{(12.25^2 - 5.0^2)} = 101.9 \text{ ft/min}$$

(2) Jet nozzle pressure loss:

$$P_b = \frac{(156.5)(520^2)(12.0)}{\left((12^2) + (12^2) + (12^2)\right)^2} = 2721 \text{ psi}$$

(3) System hydraulic horsepower available:

$$\mathrm{HHP_S} = \frac{(3000)(520)}{1714} = 910$$

(4) Hydraulic horsepower at bit:

$$\mathrm{HHP_b} = \frac{(2721)(520)}{1714} = 825.5 \cong 826$$

(5) Hydraulic horsepower per square inch of bit area:

$$\mathrm{HSI} = \frac{(826)(1.27)}{\left(12.25^2\right)} = 6.99$$

(6) Percent pressure loss at bit:

$$P_{\mathrm{b}\%} = \left(\frac{(2721)}{(3000)}\right)(100) = 90.7\%$$

(7) Jet velocity:

$$V_{\mathrm{N}} = \frac{(417.2)(520)}{\left(\left(12^2\right) + \left(12^2\right) + \left(12^2\right)\right)} = 502.2 \text{ ft/s}$$

(8) Impact force:

$$\mathrm{IF} = \frac{(12.0)(502.2)(520)}{1930} = 1623.7 \text{ lb}$$

(9) Impact force per square inch of bit area:

$$\mathrm{IFSI} = \frac{(1623.7)(1.27)}{\left(12.25^2\right)} = 13.7 \text{ sq.in.}$$

Spreadsheet 6.10 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Hydraulic analysis.

6.11 Minimum flowrate for PDC bits

$$Q_{\mathrm{m}} = (12.72)\left(D_{\mathrm{b}}^{1.47}\right) \tag{6.97}$$

where

Q_{m} = minimum flow rate for PDC bits in gpm

Example: Determine the minimum flowrate for a 12¼ in. PDC bit:

$$Q_{\mathrm{m}} = (12.72)\left(12.25^{1.47}\right) = 505.9 \cong 506 \text{ gpm}$$

Spreadsheet 6.11 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Minimum flowrate for PDC bits.

6.12 Critical RPM: RPM to avoid due to excessive vibration (accurate to approximately 15%)

$$\mathrm{RPM_C} = \left(\frac{(33,055)}{(L^2)}\right)\left(\sqrt{\left(\left(D_{\mathrm{pOD}}^2\right) + \left(D_{\mathrm{pID}}^2\right)\right)}\right) \tag{6.98}$$

where

$\mathrm{RPM_C}$ = critical RPM to avoid excessive vibration
L = length of one joint of drill pipe
D_{pOD} = OD of drill pipe in inches
D_{pID} = ID of drill pipe in inches

Example: Determine the critical RPM of a drill string with the following dimensions:

Length of one joint of drill pipe = 31 ft
Drill pipe outside diameter = 5.0 in.
Drill pipe inside diameter = 4.276 in.

$$\mathrm{RPM_C} = \left(\frac{(33,055)}{(31^2)}\right)\left(\sqrt{\left((5.0^2) + (4.276^2)\right)}\right) = 226.3 \text{ rpm}$$

Note: As a rule of thumb, for 5.0 in. drill pipe, do not exceed 200 RPM at any depth.

Spreadsheet 6.12 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Critical RPM.

Appendix: Supplementary material

Supplementary material related to this chapter can be found online at https://www.elsevier.com/books-and-journals/book-companion/9780323905497.

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CHAPTER

7

Drilling and completion calculations

7.1 Control drilling—Maximum drilling rate (MDR) when drilling large diameter holes (14¾ in. and larger) in ft/h

$$\text{MDR} = \frac{67 \times (MW_o - MW_i) \times (R_c)}{D_h{}^2} \tag{7.1}$$

where

MDR = maximum drilling rate in ft/h
MW_o = mud weight out in ppg
MW_i = mud weight in in ppg
R_c = rate of circulation in gpm

Example: Determine the MDR, ft/h, necessary to keep the mud weight coming out at 9.7 ppg at the flow line.

Data: Mud weight in = 9.0 ppg
Circulation rate = 530 gpm
Hole size = 17½ in.

$$\text{MDR} = \frac{67 \times (9.7 - 9.0) \times (530)}{17.5^2}$$

$$\text{MDR} = \frac{24{,}857}{306.25} = 81.16 \frac{\text{ft}}{\text{h}}$$

Spreadsheet 7.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 MDR for large diameter holes.

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7.2 Mud effects on rate of penetration

The following are based on curve fits obtained in laboratory tests as reported by Chilingarian.

A. Determine the effect on the ROP with a change in the Plastic Viscosity:

$$\mathrm{ROP}_2 = \left(\mathrm{ROP}_1 \times 10^{0.003\left(\overline{PV}_1 - \overline{PV}_2\right)}\right) \tag{7.2}$$

where

ROP_2 = new rate of penetration in ft/h
ROP_1 = current rate of penetration in ft/h
PV_1 = current Plastic Viscosity in cP
PV_2 = new Plastic Viscosity in cP

Example: Determine the effect on the ROP with a change in the Plastic Viscosity:

Data: Current rate of penetration = 100 ft/h
Current plastic viscosity = 24 cP
New plastic viscosity = 36 cP

$$\mathrm{ROP}_2 = \left(100 \times 10^{0.003(24-36)}\right) = 92\,\frac{\mathrm{ft}}{\mathrm{h}}$$

Spreadsheet 7.2.A in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 ROP with a change in PV.

B. Determine the effect on the ROP with a change in the bentonite content of the mud:

$$\mathrm{ROP}_2 = \left(\mathrm{ROP}_1 \times 10^{0.051\left(\overline{V}_{\mathrm{B1}} - V_{\mathrm{B2}}\right)}\right) \tag{7.3}$$

where

ROP_2 = new rate of penetration in ft/h
ROP_1 = current rate of penetration in ft/h
V_{B1} = original volume of bentonite in the mud in %
V_{B2} = new volume of bentonite in the mud in %

Example: Determine the effect on the ROP with a change in the volume percent of bentonite in the mud:

Data: Current rate of penetration = 100 ft/h
Current bentonite volume = 2.0% (*v*/v)
New bentonite volume = 4.0% (v/v)

$$\mathrm{ROP}_2 = \left(100 \times 10^{0.051(2.0-4.0)}\right) = 79.1 \text{ ft/h}$$

Spreadsheet 7.2.B in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 ROP with a change in bentonite content.

C. Determine the effect on the ROP with an increase in the Total Solids Content in the mud:

$$\mathrm{ROP}_2 = \left(\mathrm{ROP}_1 \times 10^{0.0066(V_{S1} - V_{S2})}\right) \tag{7.4}$$

where

ROP_2 = new rate of penetration in ft/h
V_{S1} = original volume of total solids in the mud in %
V_{S2} = new volume of total solids in the mud in %

Example: Determine the effect on the ROP with an increase in the Total Solids in the mud:

Data: Current rate of penetration = 100 ft/h
Current total solids volume = 10.0% (*v*/v)
New total solids volume = 12.0% (v/v)

$$\mathrm{ROP}_2 = \left(100 \times 10^{0.0066(10.0-12.0)}\right) = 97.0 \text{ ft/h}$$

Spreadsheet 7.2.C in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 ROP with an increase in the total solids content.

D. Determine the effect on the ROP with a decrease in the filtrate of the mud:

$$\mathrm{ROP}_2 = (\mathrm{ROP}_1)\left(\frac{V_{F2} + 35}{V_{F1} + 35}\right) \tag{7.5}$$

where

ROP_2 = new rate of penetration in ft/h
V_{F1} = original volume of the filtrate in $\mathrm{cm}^3/30\,\mathrm{min}$
V_{F2} = new volume of the filtrate in $\mathrm{cm}^3/30\,\mathrm{min}$

Example: Determine the effect on the ROP with a decrease in the filtrate of the mud:

Data: Current rate of penetration = 100 ft/h
Current filtrate volume = $14.0\,\mathrm{cm}^3/30\,\mathrm{min}$
New filtrate volume = $8.0\,\mathrm{cm}^3/30\,\mathrm{min}$

$$\mathrm{ROP}_2 = (100)\left(\frac{8.0 + 35}{14.0 + 35}\right) = 87.8 \text{ ft/h}$$

Spreadsheet 7.2.D in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 ROP with a decrease in the filtrate.

E. Determine the effect on the ROP with a change in the oil content (volume of oil <30%) of the mud:

$$\mathrm{ROP}_2 = (\mathrm{ROP}_1)\left[\frac{\sin\left((10.6)(V_{O2}) - 48.3\right) + 10.33}{\sin\left((10.6)(V_{O1}) - 48.3\right) + 10.33}\right] \tag{7.6}$$

where

ROP_2 = new rate of penetration in ft/h
V_{O1} = original volume of oil in the mud in %
V_{O2} = new volume of oil in the mud in %

Example: Determine the effect on the ROP with a change in the oil content (volume of oil <30%) of the mud:

Data: Current rate of penetration = 100 ft/h
Current oil volume = 5.0%
New oil volume = 10.0%

$$\mathrm{ROP}_2 = (100)\left[\frac{\sin\left(((10.6)(10.0) - 48.3) + 10.33\right)}{\sin\left(((10.6)(5.0) - 48.3) + 10.33\right)}\right] = 120.5 \text{ ft/h}$$

Spreadsheet 7.2.E in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 ROP with a change in the oil content.

F. Determine the effect on the ROP with a change in total fluid properties of the mud for depths ranging from 8000 to 12,000 ft:

$$\mathrm{ROP}_2 = (\mathrm{ROP}_1)\left(e^{0.382(\rho_1 - \rho_2)}\right) \tag{7.7}$$

where

ROP_2 = new rate of penetration in ft/h
ρ_1 = original mud weight in ppg
ρ_2 = new mud weight in ppg

Example: Determine the effect on the ROP with a change in total fluid properties of the mud:

Data: Current rate of penetration = 100 ft/h
Current mud weight = 12.0 ppg
New mud weight = 14.0 ppg

$$\mathrm{ROP}_2 = (100)\left(e^{0.382(12.0 - 14.0)}\right) = 46.6 \text{ ft/h}$$

Spreadsheet 7.2.F in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 ROP with a change in total fluid properties.

7.3 Cuttings concentration % by volume

A. Determine the Net Transport Efficiency of the Cuttings in the Annulus in %:

$$\mathrm{NT_E} = \frac{(V_a - V_{CS})}{V_a}(100) \tag{7.8}$$

where

$\mathrm{NT_E}$ = transport ratio of the cuttings (dimensionless)
V_a = fluid velocity in the annulus in ft/min
V_{CS} = slip velocity of the cuttings in ft/min

Example: Determine the Net Transport Efficiency of the cuttings with the following conditions:

Data: Annular velocity = 117.5 ft/min
Slip velocity of cuttings = 40.5 ft/min

$$\mathrm{NT_E} = \frac{(117.5 - 40.5)}{117.5}(100) = 65.5\%$$

B. Determine the Volume of Cuttings in the Annulus based on the Rate of Penetration in %:

$$C_a = \frac{(D_h{}^2)(\mathrm{ROP})}{448.4(Q)\left(\frac{\mathrm{NT_E}}{100}\right)}(100) \tag{7.9}$$

where

C_a = concentration of cuttings in annulus in %
D_h = diameter of hole in in. (Normally the Bit Size, but some enlargement can be added if needed)
ROP = rate of penetration in ft/h
Q = pump output in gpm
$\mathrm{NT_E}$ = net transport efficiency in %

Example: Determine the volume of cuttings in the annulus with the following conditions:

Data: Bit size = 17½ in.
Rate of penetration = 25 ft/h
Pump output = 800 gpm
Transport ratio = 65.5%

$$C_a = \frac{(17.5^2)(25)}{448.4(800)\left(\frac{65.5}{100}\right)}(100) = 3.3\%$$

C. Determine the mud weight in the annulus based on the concentration of cuttings:

$$MW_a = ((G_C)(8.34))\left(\frac{C_a}{100}\right) + \left((MW)\left(1 - \left(\frac{C_a}{100}\right)\right)\right) \tag{7.10}$$

where

MW_a = mud weight in the annulus based on the concentration of cuttings in ppg
G_C = specific gravity of the cuttings in gm/cm^3
MW = mud weight in use in ppg

Example: Determine the mud weight in the annulus based on the concentration of cuttings with the following:

Data: Mud weight = 9.5 ppg
Specific gravity of cuttings = 2.0 g/cm^3
Volume of cuttings in annulus = 3.2%

$$MW_a = ((2.0)(8.34))\left(\frac{3.2}{100}\right) + \left((9.5)\left(1 - \left(\frac{3.2}{100}\right)\right)\right) = 9.7 \text{ ppg}$$

D. Determine the Maximum Rate of Penetration needed to limit the Annular Mud Weight and prevent loss of circulation in the top hole section of the well.

$$\text{ROP} = \frac{(Q)(FG - MW_a)}{(D_h{}^2)((0.0057(C_{SG})) - (0.00068(FG)))} \tag{7.11}$$

where

ROP = rate of penetration in ft/h
FG = fracture gradient in ppg
MW = mud weight in ppg
C_{SG} = specific gravity of cuttings in gm/cm^3

Example: Determine the maximum penetration rate to limit the annular mud weight below the fracture gradient in ft/h:

Data:	Bit size	= 17½ in.
	Fracture gradient	= 10.0 ppg
	Mud weight	= 9.5 ppg
	Cutting specific gravity	= 2.0 g/cm^3
	Pump output	= 800 gpm

$$\mathrm{ROP} = \frac{(800)(10.0 - 9.5)}{(17.5^2)((0.0057(2.0)) - (0.00068(10.0)))} = 284 \text{ ft/h}$$

E. Determine the Rate of Penetration required to maintain a certain volume percent of cuttings in the annulus. (Assuming 100% cuttings transport up the annulus).

$$\mathrm{ROP_M} = \frac{(C_a)(Q)}{\left(\frac{(D_h{}^2)}{1470}\right)(1 - C_a)} \tag{7.12}$$

where

$\mathrm{ROP_M}$ = maximum instantaneous rate of penetration in ft/h
C_a = cuttings concentration
Q = pump output in gpm

Example: Determine the Rate of Penetration required to maintain a certain volume percent of cuttings in the annulus.

Data:	Bit size	= 17½ in.
	Volume of cuttings	= 4%
	Pump output	= 800 gpm

$$\mathrm{ROP_M} = \frac{(0.04)(800)}{\left(\frac{(17.5^2)}{1470}\right)(1 - 0.04)} = 160 \text{ ft/h}$$

Spreadsheet 7.3 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Cuttings concentration % by volume.

7.4 "d" Exponent and corrected "d" exponent

Formula 1: The "*d*" exponent is derived from the general drilling equations:

$$R + N = a\left(\frac{W^d}{D}\right) \tag{7.13}$$

where

R = penetration rate in ft/h
N = rotary speed in rpm
a = rock drillability constant (dimensionless)
W = weight on bit in lb
d = exponent in general drilling equation (dimensionless)

Formula 2: The "*d*" exponent equation:

$$d = \frac{\log\left(\frac{R}{60N}\right)}{\log\left(\frac{12W}{1000D}\right)} \tag{7.14}$$

where

d = "*d*" exponent (dimensionless)
R = penetration rate in ft/h
N = rotary speed in rpm
W = weight on bit in 1000 lb
D = bit size in in.

Example:

$$\begin{aligned} \mathrm{R} &= 30\,\mathrm{ft/h} \\ \mathrm{N} &= 120 \\ \mathrm{W} &= 35{,}000\,\mathrm{lb} \\ \mathrm{D} &= 8\tfrac{1}{2}\,\mathrm{in.} \end{aligned}$$

$$d = \frac{\log\left(\frac{30}{60(120)}\right)}{\log\left(\frac{12(35)}{1000(8.5)}\right)} = 1.82$$

Formula 2: The Corrected "*d*" exponent equation:
The "*d*" exponent is influenced by mud weight variations, so modifications have to be made to correct the changes in the mud weight:

$$d_c = d\left(\frac{9.0}{MW_a}\right) \tag{7.15}$$

where

d_c = corrected "d" exponent (dimensionless)
9.0 normal mud weight in ppg
MW_a = actual mud weight in ppg

Example:

$$\begin{aligned} d &= 1.64 \\ MW_a &= 12.7 \text{ ppg} \\ d_c &= 1.64\left(\frac{9.0}{12.7}\right) = 1.16 \end{aligned}$$

Spreadsheet 7.4 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 d Exponent and corrected d exponent.

7.5 Cost per foot

$$C_f = \frac{B + (C_r \times (R_t + T_{rt}))}{F} \tag{7.16}$$

where

C_f = cost per foot in $
B = bit cost in $
C_r = rig cost in $ per hour
R_t = rotating time in hours
T_{rt} = round trip time in hours
F = footage per bit in ft

Example: Determine the drilling cost per foot in $, using the following:

Data: Bit cost = $2500
Rig cost = $900/h
Rotating time = 65 h
Round trip time = 6 h (for depth—10,000 ft)
Footage per bit = 1300 ft

$$C_f = \frac{2500 + (900 \times (65 + 6))}{1300} = 51.08 \text{ \$/ft}$$

Spreadsheet 7.5 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Cost per foot.

7.6 Rig loads

A. Determine the Dead Line Tension for the Drill Line.
Step 1: Determine the total Hook Load on the hoist in lb:
(a) Determine the buoyant weight of the casing in lb:

$$W_{cs} = (W_c)(L_c)(BF) \tag{7.17}$$

where

W_{cs} = buoyant weight of the casing in lb
W_c = casing weight in lb/ft
L_c = length of casing in ft
BF = buoyancy factor $\left[\frac{(65.5-MW)}{65.5}\right]$

(b) Determine the total hook load on the hoist in lb:

$$L_{hoist} = (W_{cs}) + (W_{block}) \tag{7.18}$$

where

L_{hoist} = total hook load on the hoist in lb
W_{block} = traveling block weight in lb

(c) Determine the Dead Line Tension in lb:

$$T_{dl} = \frac{(L_{hoist} + \text{MOP})}{n} \tag{7.19}$$

where

T_{dl} = dead line tension in lb
MOP = maximum overpull in lb
n = number of lines in the derrick

Step 2: Determine the static Fast Line Tension in lb:

$$T_{fls} = T_{dl} \tag{7.20}$$

where

T_{fls} = static fast line tension in lb

Step 3: Determine the dynamic Fast Line Tension in lb:
(a) Method 1 calculation:

$$T_{fld1} = \frac{(L_{hoist} + \text{MOP})}{n(1 - (0.02(n)))} \tag{7.21}$$

where

T_{fld1} = dynamic fast line tension in lb

(b) Method 2 alternate calculation:

$$T_{\text{fld2}} = \frac{(L_{\text{hoist}} + \text{MOP})}{n(0.98^n)} \tag{7.22}$$

where

T_{fld2} = dynamic fast line tension in lb

Step 4: Determine the Safety Factor for the drill line:

(a) Standard calculation using the Dead Line data:

$$SF_{\text{dl}} = \frac{T_{\text{A}}}{T_{\text{dl}}} \tag{7.23}$$

where

SF_{dl} = standard safety factor using the deal line data
T_{A} = allowable hoist line tension in lb

(b) Conservative Safety Factor for the drill line using the Method 1 dynamic fast line tension:

$$SF_{\text{fldl}} = \frac{T_{\text{A}}}{T_{\text{fld1}}} \tag{7.24}$$

where

SF_{fld1} = standard safety factor using the deal line data

(c) Conservative Safety Factor for the drill line using the Method 2 dynamic fast line tension:

$$SF_{\text{fld2}} = \frac{T_{\text{A}}}{T_{\text{fld2}}} \tag{7.25}$$

where

SF_{fld2} = standard safety factor using the dead line data

B. Determine the static load for the derrick in lb:

$$L_{\text{mast}} = \left[\frac{(n+2)}{n}\right](L_{\text{hoist}}) \tag{7.26}$$

where

L_{mast} = buoyed weight of casing string and traveling block in lb

Note: This value should be compared to the rated load of the derrick.

C. Determine the static load of the rig substructure in lb:

(a) Determine the air weight of the drill collars and drill pipe stored on the pipe rack in lb:

$$L_{sb} = [(W_{DC})(L_{DC})] + [(W_{HW})(L_{HW})] + \left[\left(W_{dp}\right)\left(L_{dp}\right)\right] \tag{7.27}$$

where

L_{sb} = air weight of the drill string on the pipe rack in lb
W_{DC} = weight of drill collars in lb/ft
L_{DC} = length of the drill collars in ft
W_{HW} = weight of the heavy weight drill pipe in lb/ft
L_{HW} = length of the heavy weight in ft
W_{dp} = weight of the drill pipe in lb/ft
L_{dp} = length of the drill pipe in ft

(b) Determine the buoyed weight of the casing set in the slips in lb:

$$L_{rt} = W_{cs} \tag{7.28}$$

where

L_{rt} = buoyed weight of the casing set in the slips in lb

(c) Determine the total load on the substructure in lb:

$$L_{SS} = (L_{sb}) + (L_{rt}) \tag{7.29}$$

Example: Determine the Line Tension and Rig Load with the following:

Data:	Mud weight	= 12.0 ppg
	Buoyancy factor	= 0.8168
	Casing weight	= 68 lb/ft (N-80)
	Casing length	= 12,000 ft
	Traveling block	= 50,000 lb
	Maximum overpull	= 100,000 lb
	Allowable cable tension	= 396,000 lb
	Number of lines	= 12
	Drill collar weight	= 147.0 lb/ft
	Drill collar length	= 810 ft
	Heavy weight pipe	= 50.3 lb/ft
	Heavy weight pipe length	= 900 ft
	Drill pipe weight	= 22.61 lb/ft
	Drill pipe length	= 10,290 ft

A. Determine the dead line tension for the drill line.

Step 1: Determine the Total Hook Load on the hoist in lb:

(a) Determine the buoyant weight of the casing in lb:

$$W_{cs} = (68.0)(12{,}000)(0.8168) = 666{,}508.8 \text{ lb}$$

(b) Determine the Total Hook Load on the hoist in lb:

$$L_{hoist} = (666{,}508.8) + (50{,}000) = 716{,}508.8 \text{ lb}$$

(c) Determine the Dead Line Tension in lb:

$$T_{dl} = \frac{(716{,}508.8 + 100{,}000)}{12} = 68{,}042.4 \text{ lb}$$

Step 2: Determine the static Fast Line Tension in lb:

$$T_{fls} = 68{,}042.4 \text{ lb}$$

Step 3: Determine the dynamic Fast Line Tension in lb:

(a) Method 1 calculation:

$$T_{fld1} = \frac{(716{,}508.8 + 100{,}000)}{12(1 - (0.02(12)))} = 89{,}529.5 \text{ lb}$$

(b) Method 2 alternate calculation:

$$T_{fld2} = \frac{(716{,}508.8 + 100{,}000)}{12\left(0.98^{12}\right)} = 86{,}709.5 \text{ lb}$$

Step 4: Determine the Safety Factor for the drill line:

(a) Standard calculation using the Dead Line data:

$$SF_{dl} = \frac{396{,}000}{68{,}042.4} = 5.82$$

(b) Conservative Safety Factor for the drill line using the **Method 1** dynamic fast line tension:

$$SF_{fldl} = \frac{396{,}000}{89{,}529.5} = 4.42$$

(c) Conservative Safety Factor for the drill line using the **Method 2** dynamic fast line tension:

$$SF_{fld2} = \frac{396{,}000}{86{,}709.5} = 4.57$$

B. Determine the Static Load for the Derrick in lb:

$$L_{mast} = \left[\frac{(12 + 2)}{12}\right](716{,}508.8) = 835{,}926.9 \text{ lb}$$

Note: This value should be compared to the rated load of the derrick.

(c) Determine the Static Load of the Rig Substructure in lb:

(a) Determine the air weight of the drill collars and drill pipe stored on the pipe rack in lb:

$$L_{sb} = [(147.0)(810)] + [(50.3)(900)] + [(22.61)(10{,}290)] = 396{,}996.9 \text{ lb}$$

(b) Determine the buoyed weight of the casing set in the slips in lb:

$$L_{rt} = 666{,}508.8 \text{ lb}$$

(c) Determine the total load on the substructure in lb:

$$L_{SS} = (396{,}996.9\ ) + (666{,}508.8) = 1{,}063{,}505.7 \text{ lb}$$

Spreadsheet 7.6 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Rig loads.

7.7 Ton-mile (TM) calculations

All types of ton-mile service should be calculated and recorded in order to obtain a true picture of the total service received from the rotary drilling line. These include:

1. Round trip ton-miles
2. Drilling or "connection" ton-miles
3. Coring ton-miles
4. Ton-miles setting casing
5. Short-trip ton-miles

7.7.1 Round trip ton-miles (RT_{TM})

$$RT_{TM} = \frac{\left((W_{bdp})(D)\left(L_{ps} + D\right) + ((2D)((2W_{tb}) + W_{bdc}))\right)}{(5820)(2000)} \tag{7.30}$$

where

RT_{TM} = round trip in ton-miles
W_{bdp} = buoyed weight of drill pipe in lb/ft
D = depth of hole in ft
L_{ps} = length of one stand of drill pipe (average) in ft
W_{tb} = weight of traveling block or top drive assembly in lb
W_{bdc} = buoyed weight of drill collars in mud minus the buoyed weight of the same length of drill pipe in lb
2000 = number of pounds in one short ton
5280 = number of ft in one mile

Example: Calculate the round trip ton-miles with the following data:

Data:	Mud weight	= 12.6 lb/gal
	Measured depth	= 15,000 ft
	Drill pipe size	= 6⅝ in., (25.20 lb/ft, S-135)
	Drill collar weight	= 147 lb/ft
	Drill collar length	= 500 ft
	Traveling block assembly	= 100,000 lb
	Average length of one stand	= 93 ft (treble)

Step 1: Calculate the buoyancy factor:

$$BF = \left(\frac{65.5 - 12.6}{65.5}\right) = 0.8076$$

Step 2: Calculate the buoyed weight of the drill pipe in mud in lb/ft:

$$W_{dp} = (25.20)(0.8076) = 20.35 \text{ lb/ft}$$

Step 3: Calculate the buoyed weight of the drill collars in the mud minus the buoyed weight of the same length of drill pipe in lb:

$$W_{dc} = ((500)(147.0)(0.8076)) - ((500)(25.20)(0.8076)) = 49{,}183 \text{ lb}$$

Step 4: Calculate the round trip ton-miles:

$$RT_{TM} = \frac{((20.35)(15{,}000)(93 + 15{,}000)) + ((2(15{,}000))((2(100{,}000)) + 49{,}183))}{(5280)(2000)}$$

$$= 1144.2 \text{ ton/miles}$$

Spreadsheet 7.7.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Round trip ton-miles.

7.7.2 Drilling or "connection" ton-miles

The ton-miles calculation used for the drilling operation is expressed in terms of the work performed making round trips. These are the actual ton-miles of work required to drill down the length of a joint of drill pipe (≈31 ft), plus picking up, connecting, and starting to drill with the next joint.

To determine connection or drilling ton-miles, multiply 3 times the difference between the ton-miles for the current round trip minus the ton-miles for the previous round trip:

$$TM_D = 3(TM_2 - TM_1) \tag{7.31}$$

where

TM_D = drilling or "connection" ton-miles

TM_2 = ton-miles for one round trip—depth where drilling stopped before coming out of the hole

TM_1 = ton-miles for one round trip—depth where drilling started

Example:

Ton-miles for trip @ 4600 ft = 64.6
Ton-miles for trip @ 4000 ft = 53.7

$$TM_{\mathrm{D}} = 3(64.6 - 53.7) = 32.7 \text{ ton/miles}$$

Spreadsheet 7.7.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Drilling or connection ton-miles.

7.7.3 Ton-miles during coring operations

The ton-miles of work performed in coring operations, as for drilling operations, is expressed in terms of work performed in making round trips.

To determine ton-miles while coring, take 2 times ton-miles for one round trip at the depth where coring stopped minus ton-miles for one round trip at the depth where coring began:

$$TM_{\mathrm{C}} = 2(TM_4 - TM_3) \tag{7.32}$$

where

TM_{C} = ton-miles while coring
TM_4 = ton-miles for one round trip—depth where coring stopped before coming out of the hole
TM_3 = ton-miles for one round trip—depth where coring started

Example:

Ton-miles to start coring @ 12,000 ft = 1041.0
Ton-miles where coring stopped @ 12,090 ft = 1050.1

$$TM_{\mathrm{C}} = 2(1050.1 - 1041.0) = 18.2 \text{ ton/miles}$$

Spreadsheet 7.7.3 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Ton-miles during coring operations.

7.7.4 Ton-miles setting casing

The calculation of the ton-miles for setting casing is determined just like the one for drill pipe, but with the buoyed weight of the casing being used. The result is multiplied by one-half because setting casing is a one-way (½ round trip) operation. Ton-miles for setting casing can be determined from the following formula:

$$TM_{\mathrm{CSG}} = \frac{(0.5)\left(W_{\mathrm{csgb}}\right)(D)\left(L_{\mathrm{csjt}} + D\right) + ((D)(W_{\mathrm{tb}}))}{(5280)(2000)} \tag{7.33}$$

where

TM_{CSG} = ton-miles setting casing
W_{csgb} = buoyed weight of casing in lb/ft
L_{csjt} = length of one joint of casing in ft
W_{tb} = weight of traveling block assembly in lb

Example:

Mud weight = 12.6 ppg (BF = 0.8076)
Casing = 10¾ × 9.760 (P-110, 55.5 lb/ft)
Casing joint length = 40 ft
Casing length = 12,000 ft
Traveling block = 100,000 lb

$$TM_{CSG} = \frac{(0.5)((55.5)(0.8076))(12{,}000)(40 + 12{,}000) + ((12{,}000)(100{,}000))}{(5280)(2000)}$$
$$= 420.1 \text{ ton/miles}$$

Spreadsheet 7.7.4 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Ton-miles setting casing.

7.7.5 Ton-miles while making short trips

The ton-miles of work performed on short trip operations are also expressed in terms of round trips. Analysis shows that the ton-miles of work done in making a short trip are equal to the difference in round trip ton-miles for the two depths in question.

$$TM_{ST} = (TM_6 - TM_5) \tag{7.34}$$

where

TM_{ST} = ton-miles for short trip
T_6 = ton-miles for one round trip at the deeper depth, the depth of the bit before starting the short trip
T_5 = ton-miles for one round trip at the shallower depth, the depth that the bit is pulled up to at the end of the short trip

Example:

Ton-miles where short trip started = 1144.2
Ton-miles where short trip ended = 846.0

$$TM_{ST} = (1144.2 - 846.0) = 298.2 \text{ ton/miles}$$

Spreadsheet 7.7.5 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Ton-miles while making short trips.

7.7.6 Cutoff practices for rotary drilling line

Refer to Section 6 in API RP 9B for recommended cutoff lengths for drill line based on ton-mile calculations.

7.7.7 Calculate the length of drill line cutoff

$$DL_{CO} = \left(\frac{RT_{TM}}{CO_G}\right) \tag{7.35}$$

where

DL_{CO} = drill line cutoff in ft
CO_G = cutoff goal from Table 7.1

TABLE 7.1 Ton-mile goal per foot of rope[a]

Drum diameter	Rope diameter							
	1 in.	1⅛ in.	1¼ in.	1⅜ in.	1½ in.	1⅝ in.	1¾ in.	2 in.
18 in.	6.0	9.0						
19 in.	6.0	9.0						
20 in.	7.0	9.0						
21 in	7.0	10.0						
22 in.	7.0	10.0						
23 in.	8.0	10.0	13.0					
24 in.	8.0	10.0	13.0	17.0				
25 in.	8.0	10.0	14.0	17.0				
26 in.	9.0	11.0	14.0	17.0				
27 in.	9.0	12.0	15.0	18.0				
28 in.		12.0	15.0	18.0				
29 in.		12.0	15.0	18.0				
30 in.		13.0	16.0	19.0	20.0			
31 in.			16.0	19.0	21.0			
32 in.			17.0	20.0	22.0			
33 in.			17.0	20.0	23.0			
34 in.			18.0	21.0	24.0			

TABLE 7.1 Ton-mile goal per foot of rope[a]—cont'd

Drum diameter	Rope diameter							
	1 in.	1⅛ in.	1¼ in.	1⅜ in.	1½ in.	1⅝ in.	1¾ in.	2 in.
35 in.				21.0	25.0			
36 in.				22.0	25.0	28	30	
42 in.								32
48 in.								
60 in.								

[a] *Premium ropes such as plastic impregnated can provide additional service.*

Using the Ton-Miles calculated in **Step 4** of Section 7.7.1, calculate the amount of drill line to be cut.

Data: Drum circumference 42 in.
Drill line diameter 2 in.
Ton-miles calculated 1144.2

$$DL_{CO} = \left(\frac{114.2}{32}\right) = 35.9 \text{ ft}$$

Spreadsheet 7.7.7 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Calculate the length of drill line cutoff.

7.8 Hydrostatic pressure decrease when pulling pipe out of the hole

7.8.1 When pulling DRY pipe

Step 1: Determine the number of barrels displaced when pulling **DRY** pipe in bbl:

$$V_d = S_n \times L_s \times P_d \tag{7.36}$$

where

V_d = volume of displaced mud when pulling **DRY** pipe in bbl
S_n = number of stands
L_s = average length per stand in ft
P_d = pipe displacement in bbl/ft

Example: Determine volume of barrels displaced when pulling **DRY** pipe out of the hole.

Data: $S_n = 5$ stands
$L_s = 93$ ft/std
$P_d = 0.00807$ bbl/ft (DP = 6⅝ × 5.965 in.)

$$V_d = 5 \times 93 \times 0.00807 = 3.75 \text{ bbl}$$

Step 2: Determine the Hydrostatic Pressure Decrease when pulling **DRY** pipe in psi:

$$HP_d = \left(\frac{V_d}{C_a - P_d}\right) \times 0.052 \times MW \tag{7.37}$$

where

HP_d = hydrostatic pressure decrease when pulling **DRY** pipe in psi
V_d = volume of displaced mud when pulling **DRY** pipe in bbl
C_a = casing annulus volume bbl/ft

Example: Determine hydrostatic pressure decrease when pulling **DRY** pipe out of the hole.

Data: $V_d = 3.75$ bbls from Step 1
$C_a = 0.0773$ bbl/ft (Casing ID = 8.921 in.)
$P_d = 0.00807$ bbl/ft (DP = 6⅝ × 5.965 in.)
$MW = 11.5$ ppg

$$HP_d = \left(\frac{3.75}{0.0773 - 0.00807}\right) \times 0.052 \times 11.5 = 32.4 \text{ psi}$$

Spreadsheet 7.8.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Pulling DRY pipe.

7.8.2 When pulling WET pipe

Step 1: Determine volume of barrels displaced when pulling **WET** pipe out of the hole.

$$V_w = S_n \times L_s \times (P_d + C_p) \tag{7.38}$$

where

V_w = volume of displaced mud when pulling **WET** pipe in bbl
S_n = number of stands pulled

P_d = pipe displacement in bbl/ft
C_p = capacity of pipe in bbl/ft

Example: Determine the volume of barrels displaced when pulling **WET** pipe out of the hole.

Data: $S_n = 5$ stands
$L_s = 93$ ft/std
$P_d = 0.00807$ bbl/ft (DP = 6⅝ × 5.965 in.)
$C_p = 0.0346$ bbl/ft (DP ID = 5.965 in.)

$$V_w = 5 \times 93 \times (0.00807 + 0.0346) = 19.8 \text{ bbl}$$

Step 2: Determine the Hydrostatic Pressure Decrease in psi:

$$HP_w = \frac{V_b}{(C_a - (P_d + C_p))} \times 0.052 \times MW \tag{7.39}$$

where

HP_w = hydrostatic pressure decrease when pulling **WET** pipe in psi
C_a = capacity of casing or open hole in bbl/ft
P_d = pipe displacement in bbl/ft
C_p = Capacity of pipe in bbl/ft

Example: Determine the hydrostatic pressure decrease when pulling DRY pipe out of the hole.

Data: $S_n = 5$ stands
$L_s = 93$ ft/std
$P_d = 0.00807$ bbl/ft (DP = 6⅝ × 5.965 in.)
$C_p = 0.0346$ bbl/ft (DP ID = 5.965 in.)
$C_a = 0.0773$ bbl/ft (Casing ID = 8.921 in.)
$MW = 11.5$ ppg

$$HP_w = \left(\frac{19.8}{0.0773 - (0.00807 + 0.0346)}\right)(0.052)(11.5) = 341.9 \text{ psi}$$

Spreadsheet 7.8.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Pulling WET pipe.

7.9 Loss of overbalance due to falling mud level

7.9.1 Feet of pipe pulled DRY to lose overbalance

$$L_{pd} = \frac{(O_{bL})(C_a - C_{pd} - P_d)}{(MW)(0.052)(C_{pd} + P_d)} \tag{7.40}$$

where

L_{pd} = length of pipe pulled DRY in ft
O_{bL} = amount of overbalance LOST in psi
C_a = capacity of casing or open hole in bbl/ft
P_d = pipe displacement in bbl/ft
C_{pd} = capacity of pipe in bbl/ft (**DRY** pipe)

Example: Determine the **FEET** of **DRY** pipe that must be pulled to lose the overbalance using the following data.

Data: O_{bL} = 150 psi
C_a = 0.0773 bbl/ft (Casing ID = 8.921 in.)
C_{pd} = 0.0346 bbl/ft (DP ID = 5.965 in.) (**DRY** pipe)
P_d = 0.00807 bbl/ft (DP = 6⅝ × 5.965 in.)
MW = 11.5 ppg

$$L_{pd} = \frac{(150)\ (0.0773 - 0.346 - 0.00807)}{(11.5)\ (0.052)\ (0.0346 + 0.00807)} = 203.6 \text{ ft}$$

Spreadsheet 7.9.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Feet of pipe pulled DRY to lose overbalance.

7.9.2 Feet of pipe pulled WET to lose overbalance

$$L_{pw} = \frac{(O_b)(C_a - C_{pw} - P_d)}{(MW)(0.052)(C_{pw} + P_d)} \tag{7.41}$$

where

L_{pw} = length of pipe pulled **WET** in ft

Example: Determine the feet of **WET** pipe that must be pulled to lose the overbalance using the following data.

Data: O_b = 150 psi
C_a = 0.0773 bbl/ft (Casing ID = 8.921 in.)
C_{pw} = 0.0346 bbl/ft (DP ID = 5.965 in.) (Mud filled pipe ≅ **WET**)
P_d = 0.00807 bbl/ft (DP = 6⅝ × 5.965 in.)
MW = 11.5 ppg

$$L_{pw} = \frac{150 \times (0.0773 - 0.0346 - 0.00807)}{11.5 \times 0.052 \times (0.0346 + 0.00807)} = 203.6 \text{ ft}$$

Spreadsheet 7.9.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Feet of pipe pulled WET to lose overbalance.

7.10 Lost circulation

7.10.1 Loss of overbalance

Step 1: Calculate the number of feet of fluid (water for WBM; base oil for OBM/SBM) added to the top of the annulus:

$$L_f = \frac{V_f}{(9.71 \times 10^{-4})\left(D_h^2 - D_p^2\right)} \tag{7.42}$$

where

L_f = length of fluid added to the top of the annulus in ft
V_f = volume of fluid added to the top of the annulus in bbl
D_h = diameter of hole, casing, or pipe in inches
D_p = diameter of pipe in inches

Step 2: Calculate the reduction in bottomhole pressure (BHP) due to the unweighted fluid added to the top of the annulus:

$$P_{BHP} = (L_f)(W_{ma} - W_f)(0.052) \tag{7.43}$$

where

P_{BHP} = decrease in bottom hole pressure in psi
W_{ma} = weight of mud in annulus in ppg
W_f = weight of fluid added to the top of the annulus in ppg

Step 3: Calculate the equivalent mud weight at TVD due to fluid added to the top of the annulus:

$$W_{me} = (W_{ma}) - \left(\frac{\left(\frac{P_{BHP}}{0.052}\right)}{D_{TV}}\right) \tag{7.44}$$

where

W_{me} = equivalent mud weight in lb/gal
D_{TV} = total vertical depth in ft

Example: Calculate the equivalent mud weight due to the addition of unweighted fluid to the top of the annulus with the following data:

Data:	Mud weight	= 12.5 lb/gal
	Water added	= 150 bbl (required to fill the annulus)
	Weight of water	= 8.55 lb/gal (Sea water)
	TVD	= 10,000 ft
	Riser ID	= 18¾ in.
	Drill pipe size	= 6⅝ × 5.965 in.

Step 1: Calculate the length of annulus covered by the volume of water added:

$$L_f = \frac{(150)}{(9.71 \times 10^{-4})(18.75^2 - 6.625^2)} = 501.8 \text{ ft}$$

Step 2: Calculate the reduction in bottom hole pressure:

$$P_{BHP} = (501.8)(12.5 - 8.55)(0.052) = 103.1 \text{ psi}$$

Step 3: Calculate the equivalent mud weight at the TVD:

$$W_{mc} = (12.5) - \left(\frac{\left(\frac{103.1}{0.052}\right)}{10{,}000}\right) = 12.3 \text{ ppg}$$

Spreadsheet 7.10.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Loss of overbalance.

7.10.2 Determine the equivalent mud weight in ppg that will balance the formation losing mud volume

Step 1: Determine the level of the mud in the annulus in ft by adding water to fill up the casing:

$$L_w = \left(\frac{V_w}{C_a}\right) \tag{7.45}$$

where

L_w = length of water added to annulus in ft
V_w = volume of water added to the annulus in bbl
C_a = capacity of the annulus in bbl/ft

Step 2: Determine the mud weight equivalent at the depth of the loss of circulation in ppg:

$$MW_e = \frac{((L_w)(G_w)) + ((D_m - L_w)((MW)(0.052)))}{((D_m)(0.052))} \tag{7.46}$$

where

MW_e = mud weight equivalent at depth of lost circulation in ppg
G_w = gradient of water added to annulus (Fresh = 0.433 psi/ft; Sea Water = 0.455 psi/ft)
D_m = measured depth of interest (normally the casing shoe) in ft
MW = mud weight in use when loss occurred in ppg

Example: Determine the equivalent mud weight in ppg with the following conditions:

Data:	Casing size	= 9⅝ × 8.755 in. (N-80, 43.5 lb/ft)
	Casing length	= 3500 ft
	Annulus capacity	= 0.0502 bbl/ft
	Drill pipe size	= 5.0 in. (19.5 lb/ft)
	Fresh water added	= 25 bbl (0.433 psi/ft)
	Mud weight	= 12.2 ppg

Step 1: Determine the level of the mud in the annulus in ft by adding water to fill up the casing:

$$L_w = \left(\frac{25}{0.0502}\right) = 498 \text{ ft}$$

Step 2: Determine the mud weight equivalent at the depth of the loss of circulation in ppg:

$$MW_e = \frac{((498)(0.433)) + ((3500 - 498)((12.2)(0.052)))}{((3500)(0.052))} = 11.7 \text{ ppg}$$

Spreadsheet 7.10.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 MW to balance the formation losing mud volume.

7.10.3 Determine the depth of the fluid level with loss of circulation in natural fractured formations

$$D_{fl} = \frac{(\Delta W_{ds})}{[(DP_{aw})(1 - BF)]} \tag{7.47}$$

where

D_{fl} = depth of fluid level in ft
ΔW_{ds} = increase in string weight caused by a loss of buoyancy in lb
DP_{aw} = adjusted weight of the drill pipe including tool joints in lb/ft
BF = buoyancy factor

Example: Determine the depth of the fluid level with the following:

Data:	Fractured formation depth	= 7575 ft
	Mud weight	= 11.8 ppg
	String weight increase	= 5000 lb
	Drill pipe weight	= 19.50 lb/ft
	Buoyancy factor	= 0.8198

$$D_{fl} = \frac{(5000)}{[(19.50)(1 - 0.8198)]} = 1422.9 \text{ ft}$$

Spreadsheet 7.10.3 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Depth of fluid level with loss of circulation.

7.10.4 Determine the amount of mud loss that can occur before the well begins to flow from a gas-bearing formation

Step 1: Determine the length of fluid drop before taking a kick in ft:

$$L_m = \frac{(P_{ob})}{(G_m)} \tag{7.48}$$

where

L_m = maximum length of fluid drop before taking a gas kick in ft
P_{ob} = pressure of overbalance in psi
G_m = mud gradient in psi/ft

Step 2: Determine the volume of mud that can be lost before the well kicks in bbl:

$$V_m = (L_m)(C_a) \tag{7.49}$$

where

V_m = volume of mud that can be lost before the well kicks in bbl
C_a = capacity of the annulus in bbl/ft

Example: Determine how much mud can be lost with the following conditions:

Data:	Planned overbalance	= 250 psi
	Casing size	= 9⅝ × 8.681 in. (N-80, 47 lb/ft,)
	Annulus capacity	= 0.04892 bbl/ft
	Drill pipe size	= 5.0 in.
	Mud weight	= 12.0 ppg
	Mud gradient	= 0.624 psi/ft

Step 1: Determine the length of fluid drop before taking a kick in ft:

$$L_m = \frac{(250)}{(0.624)} = 400.6 \text{ ft}$$

Step 2: Determine the volume of mud that can be lost before the well kicks in bbl:

$$V_m = (400.6)(0.04892) = 19.6 \text{ bbl}$$

Spreadsheet 7.10.4 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Mud loss before the well begins to flow.

7.11 Core analysis technique

7.11.1 Extraction and saturation determinations (dean stark analysis)

Step 1: Determine the pore volume and porosity.

(a) Determine the pore volume in cm^3:

$$V_p = \frac{(W_{SC} - W_e)}{\rho_w} \tag{7.50}$$

where

V_p = volume of pore in cm^3
W_{SC} = weight of saturated core in gm
W_e = weight of dried core in gm
ρ_w = specific gravity of water in gm/cm^3

(b) Determine the porosity of the core in %:

$$\varnothing = \frac{(V_p)}{(V_b)}(100) \tag{7.51}$$

where

$\varnothing$ = core porosity in %
V_b = bulk volume of the core in cm^3

Step 2: Determine the water saturation in the core in %:

$$S_w = \frac{(V_w)}{(V_p)}(100) \tag{7.52}$$

where

S_w = water saturation in %
V_w = volume of water extracted from the core in cm^3

Step 3: Determine the oil saturation in %:

$$S_O = \frac{(W_{sc} - W_{dc} - (V_w(\rho_w)))}{(V_p)}(100) \tag{7.53}$$

where

S_O = oil saturation in %
W_{sc} = weight of the saturated sample in gm
W_{dc} = weight of the dry core in gm

Step 4: Determine the gas saturation in %:

$$S_g = (100 - S_w - S_O) \tag{7.54}$$

where

S_g = gas saturation in %

Example: Determine the Dean-Stark Analysis with the following data:

Data:	Weight of saturated core	$= 74.0\,g$
	Weight of dried core	$= 65.0\,g$
	Specific gravity of water	$= 1.0\,g/cm^3$
	Bulk volume of core	$= 25.75\,cm^3$
	Volume of water extracted from core	$= 1.8\,cm^3$

Step 1: Determine the pore volume and porosity.

(a) Determine the pore volume in cm^3:

$$V_p = \frac{(74.0 - 65.0)}{1.0} = 9.0\ cm^3$$

(b) Determine the porosity of the core in %:

$$\varnothing = \frac{(9.0)}{(25.75)}(100) = 35.0\%$$

Step 2: Determine the water saturation in the core in %:

$$S_w = \frac{(1.8)}{(9.0)}(100) = 20\%$$

Step 3: Determine the oil saturation in %:

$$S_O = \frac{(74.0 - 65.0 - (1.8(1.0)))}{(9.0)}(100) = 80\%$$

Step 4: Determine the gas saturation in %:

$$S_g = (100 - 80.0 - 20.0) = 0\%$$

Spreadsheet 7.11 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Dean stark analysis.

7.12 Temperature correction for brines

7.12.1 Determine the clear brine fluid weight to be mixed at the surface to balance the required bottomhole pressure at the bottomhole temperature conditions

Step 1: Select the appropriate weight loss factor (F_{wl}) in ppg/°F (Table 7.2):

TABLE 7.2 Brine weight loss factor.

Brine weight, ppg	Weight loss, ppg/°F
8.4–9.0	0.0017
9.1–11.0	0.0025
11.1–14.5	0.0033
14.6–17.0	0.0040
17.1–19.2	0.0045

Step 2: Determine the brine weight to be mixed at surface conditions in ppg:

$$MW_{SB} = (MW_f) + [(T_f - T_S)(F_{WL})] \tag{7.55}$$

where

MW_{SB} = brine weight at surface temperature in ppg
MW_f = formation mud weight in ppg
T_f = formation temperature at total depth in °F
T_S = surface temperature in °F
F_{WL} = brine weight loss factor in ppg/°F

Example: Determine the brine weight to be mixed at surface conditions in ppg:

Data: $MW_f = 11.4$ ppg
$T_f = 180$°F
$T_S = 80$°F
$F_{WL} = 0.0033$ ppg/°F

$$MW_{SB} = (11.4) + [(180 - 80)(0.0033)] = 11.7 \text{ ppg}$$

Spreadsheet 7.12.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 CBF weight mixed at surface to balance BHP.

7.13 Tubing stretch

A. Determine the Tubing Stretch from the Weight of the String:

$$S_W = \frac{(W_T)(D_{TVD})\left(\frac{L_{tbg}}{2}\right)}{(A_p)(3.0 \times 10^7)} \tag{7.56}$$

where

S_W = tubing stretch due to the weight of the string in ft
W_T = weight of the tubing (plus connection) in lb/ft
D_{TVD} = total vertical depth in ft
L_{tbg} = measured length of the tubing in ft
A_p = cross sectional area of the tubing in in.^{2}

B. Determine the Tubing Stretch from the Expansion of the Tube due to Internal Pressure:

$$S_E = \frac{(0.6)\left[\left(\frac{(A_{OD})((MW)(0.052)(D_{TVD}))}{2}\right) - \left(\frac{(A_{ID})((MW)(0.052)(D_{TVD}))}{2}\right)\right](L_{tbg})}{(A_{pw})(3.0 \times 10^7)} \tag{7.57}$$

where

S_E = tubing stretch from the expansion due to internal pressure in ft
A_{OD} = cross sectional area of the OD of the tubing in in.^{2}
MW = mud weight in ppg
A_{ID} = cross sectional area of the ID of the tubing in in.^{2}
A_{pw} = cross sectional area of the tubing wall in in.^{2} (OD – ID)

C. Determine the Tubing Stretch from Buoyancy of the tubing in ft:

$$S_B = \frac{(-(MW)(0.052)(D_{TVD}))(A_{Pw})(L_{tbg})}{(A_{Pw})(3.0 \times 10^7)} \tag{7.58}$$

where

S_B = tubing stretch from the buoyancy in the completion fluid in ft

D. Determine the Tubing Stretch from Temperature changes in the well in ft:

(a) Determine the Δ Temperature in the well:

$$\Delta T = \frac{(T_{BH} - T_{surface})}{2} \tag{7.59}$$

where

ΔT = temperature change in the well in °F
T_{BH} = bottom hole temperature in °F
$T_{surface}$ = surface temperature in °F

(b) Determine the tubing stretch from the temperature change in the well in ft:

$$S_T = \left(6.9 \times 10^{-6}\right)(\Delta T)\left(L_{tbg}\right) \tag{7.60}$$

where

S_T = tubing stretch from temperature changes in the well in ft

E. Determine the total tubing stretch in the well:

$$S_{total} = (S_W) + (S_E) + (S_B) + (S_T) \tag{7.61}$$

Example: Determine the Tubing Stretch with the following conditions:

Data:	Depth TVD	= 10,000 ft
	Measured depth	= 12,000 ft
	Mud weight	= 8.34 ppg
	Tubing size	= 2 ⅞ in. (8.7 lb/ft, ID = 2.259 in.)
	Cross sectional area of the tubing OD	= 6.49 $in.^2$
	Cross sectional area of the tubing ID	= 4.01 $in.^2$
	Cross sectional area of the tubing wall	= 2.48 $in.^2$
	Surface temperature	= 80°F
	Bottom hole temperature	= 275°F

A. Determine the Tubing Stretch from the Weight of the String:

$$S_W = \frac{(8.7)(10{,}000)\left(\frac{12{,}000}{2}\right)}{(2.48)\left(3.0 \times 10^7\right)} = 7.02 \text{ ft}$$

B. Determine the Tubing Stretch from the Expansion of the Tube due to Internal Pressure:

$$S_E = \frac{(0.6)\left[\left(\frac{(6.49)((8.34)(0.052)(10{,}000))}{2}\right) - \left(\frac{(4.01)((8.34)(0.052)(10{,}000))}{2}\right)\right](12{,}000)}{(2.48)\left(3.0 \times 10^7\right)}$$

$$= 0.52 \text{ ft}$$

C. Determine the Tubing Stretch from Buoyancy of the tubing in ft:

$$S_B = \frac{(-(8.34)(0.052)(10{,}000))(2.48)(12{,}000)}{(2.48)\left(3.0 \times 10^7\right)} = -1.73 \text{ ft}$$

D. Determine the Tubing Stretch from Temperature changes in the well in ft:

(a) Determine the Δ Temperature in the well:

$$\Delta T = \frac{(275 - 80)}{2} = 97.5^{\circ}\text{F}$$

(b) Determine the tubing stretch from the temperature change in the well in ft:

$$S_{\text{T}} = \left(6.9 \times 10^{-6}\right)(97.5)(12{,}000) = 8.07 \text{ ft}$$

E. Determine the total tubing stretch in the well:

$$S_{\text{total}} = (7.02) + (0.52) + (-1.73) + (8.07) = 13.9 \text{ ft}$$

Spreadsheet 7.13 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Tubing stretch.

7.14 Directional drilling calculations

7.14.1 Directional survey calculations

The following are the two most commonly used methods to calculate directional surveys:

1. Angle averaging method

$$\text{North} = (MD)\left(\sin\left(\frac{(I_1 + I_2)}{2}\right)\right)\left(\cos\left(\frac{(A_1 + A_2)}{2}\right)\right) \tag{7.62}$$

$$\text{East} = (MD)\left(\sin\left(\frac{(I_1 + I_2)}{2}\right)\right)\left(\sin\left(\frac{(A_1 + A_2)}{2}\right)\right) \tag{7.63}$$

$$\text{Vertical} = (MD)\left(\cos\left(\frac{(I_1 + I_2)}{2}\right)\right) \tag{7.64}$$

2. Radius of curvature method

$$\text{North} = \left(\frac{(MD)((\cos I_1) - (\cos I_2))((\sin A_2) - (\sin A_1))}{(I_2 - I_1)(A_2 - A_1)}\right)\left(\left(\frac{180}{\pi}\right)^2\right) \tag{7.65}$$

$$\text{East} = \left(\frac{(MD)((\cos I_1) - (\cos I_2))((\cos A_1) - (\cos A_2))}{(I_2 - I_1)(A_2 - A_1)}\right)\left(\left(\frac{180}{\pi}\right)^2\right) \tag{7.66}$$

$$\text{Vertical} = \left(\frac{(MD)((\sin I_2) - (\sin I_1))}{(I_2 - I_1)}\right)\left(\frac{180}{\pi}\right) \tag{7.67}$$

where

MD = course length between surveys in measured depth in ft
I_1 = inclination (angle) at upper survey in degrees (°)
I_2 = inclination (angle) at lower survey in degrees (°)
A_1 = direction at upper survey
A_2 = direction at lower survey

Example: Use the averaging method and the radius of curvature method to calculate the following surveys:

	Survey 1	**Survey 2**
Depth (ft)	7482	7782
Inclination (°)	4	8
Azimuth (°)	10	35

1. Angle averaging method

$$\text{North} = (300)\left(\sin\left(\frac{(4+8)}{2}\right)\right)\left(\cos\left(\frac{(10+35)}{2}\right)\right) = 29 \text{ ft}$$

$$\text{East} = (300)\left(\sin\left(\frac{(4+8)}{2}\right)\right)\left(\sin\left(\frac{(10+35)}{2}\right)\right) = 12.0 \text{ ft}$$

$$\text{Vertical} = (300)\left(\cos\left(\frac{(4+8)}{2}\right)\right) = 297.1 \text{ ft}$$

2. Radius of curvature method

$$\text{North} = \left(\frac{(300)((\cos 4) - (\cos 8))((\sin 35) - (\sin 10))}{(8-4)(35-10)}\right)\left(\left(\frac{180}{3.1416}\right)^2\right)$$
$$= 28.8 \text{ ft}$$

$$\text{East} = \left(\frac{(300)((\cos 4) - (\cos 8))((\cos 10) - (\cos 35))}{(8-4)(35-10)}\right)\left(\left(\frac{180}{3.1416}\right)^2\right)$$
$$= 11.9 \text{ ft}$$

$$\text{Vertical} = \left(\frac{(300)((\sin 8) - (\sin 4))}{(8-4)}\right)\left(\frac{180}{3.1416}\right) = 300.5 \text{ ft}$$

Spreadsheet 7.14.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Directional survey calculations.

7.14.2 Deviation/departure calculation

Deviation is defined as departure of the wellbore from the vertical, measured by the horizontal distance from the rotary table to the target. The amount of departure is a function of the drift angle (inclination) and hole depth.

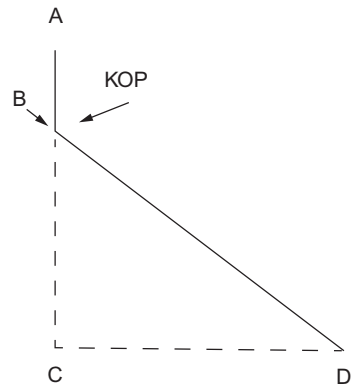
Fig. 7.1 illustrates how to determine the deviation/departure:

To calculate the deviation/departure (CD) in ft:

$$CD = (\sin I)(BD) \tag{7.68}$$

where

CD = deviation/departure of the wellbore from vertical
I = inclination (angle) of wellbore
BD = distance from KOP to the bottom of the hole (MD)



Data :

AB = Distance from the surface location to the KOP

BC = Distance from KOP to the true vertical depth (TVD)

BD = Distance from KOP to the bottom of the hole (MD)

CD = Deviation /departure—departure of the wellbore from the vertical

AC = True vertical depth

AD = Measured depth

FIG. 7.1 Determine deviation/departure.

Example: Kick off point (KOP) is a distance of 2000 ft from the surface. MD is 8000 ft. Hole angle (inclination) is 20°. Therefore, the distance from KOP to MD = 6000 ft (BD).

$$CD = (\sin 20)(6000) = 2052 \text{ ft}$$

From the calculation, the measured depth (MD) is 2052 ft away from the vertical wellbore.

Spreadsheet 7.14.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Deviation—Departure calculation.

7.14.3 Dogleg severity calculation

Method 1:

Dogleg Severity (DLS) is usually given in degrees per 100 ft. The following formula provides dogleg severity in degrees/100 ft and is based on the radius of curvature method:

$$DLS_r = \left[\cos^{-1}\left((\cos I_1)\left(\cos I^2\right) + ((\sin I_1)(\sin I_2))(\cos(A_2 - A_1))\right)\right]\left(\frac{100}{CL}\right) \tag{7.69}$$

where

DLS_r = Dogleg severity based on radius of curvature in degrees/100 ft
CL = course length in ft (distance between survey points in ft)
I_1 = inclination (angle) at upper survey in degrees (°)
I_2 = inclination (angle) at lower survey in degrees (°)
A_1 = direction at upper survey in degrees (°)
A_2 = direction at lower survey in degrees (°)

Example:

	Survey 1	Survey 2
Depth, ft	4231	4262
Inclination, degrees	13.5	14.7
Azimuth, degrees	N 10 E	N 19 E

$$DLS_r = \left[\cos^{-1}((\cos 13.5)(\cos 14.7) + ((\sin 13.5)(\sin 14.7))(\cos(19 - 10)))\right]\left(\frac{100}{31}\right) = 7.8 \text{ per } 100 \text{ ft}$$

Note: Conversion factors used in the spreadsheet solution include the following:

$\frac{3.1416}{180} = 0.01745$ to convert degrees into radians and $\frac{180}{3.1416} = 57.3$ to convert radians into degrees.

Method 2:

This method of calculating dogleg severity is based on the tangential method:

$$DLS_t = \frac{100}{CL(((\sin I_1)(\sin I_2))(((\sin A_1)(\sin A_2)) + ((\cos A_1)(\cos A_2))) + ((\cos I_1)(\cos I_2)))} \tag{7.70}$$

where

DLS_t = Dogleg severity based on the tangential method in degrees/100 ft
CL = course length in ft
I_1 = inclination (angle) at upper survey in degrees (°)
I_2 = inclination (angle) at lower survey in degrees (°)
A_1 = direction at upper survey in degrees (°)
A_2 = direction at lower survey in degrees (°)

Example:

	Survey 1	Survey 2
Depth, ft	4231	4262
Inclination, degrees	13.5	14.7
Azimuth, degrees	N 10 E	N 19 E

$$DLS_t = \frac{100}{31(((\sin 13.5)(\sin 14.7))(((\sin 10)(\sin 19)) + ((\cos 10)(\cos 19))) + ((\cos 13.5)(\cos 14.7)))} = 3.2 \text{ per } 100 \text{ ft} \tag{7.71}$$

Spreadsheet 7.14.3 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 DLS.

7.14.4 Available weight on bit in directional wells

A directionally drilled well requires that a correction be made in total drill collar weight because only a portion of the total weight will be available to the bit:

$$P = (W)(\cos I)(BF) \tag{7.72}$$

where

P = partial weight available for the bit in lb
W = total weight of drill collars in lb
cos = cosine

I = inclination (angle) in degrees (°)
BF = buoyancy factor

Example:

Mud weight	= 15.0 ppg
Drill collar weight	= 45,000 lb
Inclination	= 25°
Buoyancy factor	= 0.7710

$$P = (45{,}000)(0.7710)(0.9063) = 31{,}444 \text{ lb}$$

Spreadsheet 7.14.4 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Available WOB in directional wells.

7.14.5 Determining true vertical depth

The following is a simple method of correcting for the TVD on directional wells. This calculation will provide the approximate TVD interval corresponding to the measured interval and is generally accurate enough for any pressure calculations. At the next survey, the TVD should be corrected to correspond to the directional driller's calculated true vertical depth.

$$TVD_2 = (\cos I)(CL) + (TVD_1) \tag{7.73}$$

where

TVD_2 = new true vertical depth in ft
cos = cosine
I = inclination (angle) in degrees (°)
CL = course length (number of feet since last survey)
TVD_1 = last true vertical depth in ft

Example:

TVD (last survey)	= 8500 ft
Inclination	= 40°
Course length	= 30 ft

$$TVD_2 = (0.7660)(30) + 8500 = 8523 \text{ ft}$$

Spreadsheet 7.14.5 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 TVD.

7.15 Hole washout

Washing out the wellbore while drilling can cause other problems such as poor cement jobs, stuck pipe due to poor hole cleaning, etc. Zeidler conducted a study of the enlargement problem and identified three forces acting on the borehole wall. They were the action of fluid movement, solids (sand blasting effect) and nozzle velocity impacting the wall of the hole. Of those forces, bit nozzle erosion is the most dominate form of erosion. In shallow unconsolidated sands, fluids can effectively remove the formation but as the formations gain compressive strength, the resultant force is less. This calculation is an estimation of the risk of hole enlargement due to nozzle velocity.

Step 1: Calculate the nozzle velocity:

$$V_{\mathrm{N}} = \frac{(417.2)\left(\frac{Q}{N_{\mathrm{n}}}\right)}{N^2} \tag{7.74}$$

where

V_{N} = nozzle velocity in ft/s
Q = pump output in gpm
N_{n} = number of nozzles
N^2 = nozzle size in 32nds

Step 2: Calculate the nozzle shear rate in s^{-1}:

$$\gamma_{\mathrm{N}} = \frac{(96)(V_{\mathrm{N}})}{\left(\frac{N}{32}\right)} \tag{7.75}$$

where

γ_N = nozzle shear rate in s^{-1}

Example: Calculate the nozzle shear rate with the following data:

Q = 460 gpm
Nozzle size = 12/32nds
Number of nozzles = 3

Step 1: Calculate the nozzle velocity:

$$V_{\mathrm{N}} = \frac{(417.2)\left(\frac{460}{3}\right)}{12^2} = 444.2 \text{ ft/s}$$

Step 2: Calculate the nozzle shear rate in s^{-1}:

$$\gamma_{\mathrm{N}} = \frac{(96)(444.2)}{\left(\frac{12}{32}\right)} = 113{,}715.2 \text{ s}^{-1}$$

Note: The risk for hole washout is high when the nozzle shear rate is above 100,000 s^{-1}

Spreadsheet 7.15 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Hole washout.

Appendix: Supplementary material

Supplementary material related to this chapter can be found online at https://www.elsevier.com/books-and-journals/book-companion/9780323905497.

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CHAPTER

8

Air and gas calculations

Note 1: This chapter has unit system consistent equations, not field equations. The equations may be worked using either the USCS or SI. If USCS is used, pressure is (lb/ft.^{2} abs), specific weight (lb/ft.^{3}) or specific gravity, temperature (°Rankin), and depth (ft.). If SI is used, pressure is (N/m^2 abs), specific weight (N/m^3) or specific gravity, temperature (Kelvin), and depth (m). API Standard Conditions are 14.696 psia (~14.7 psia), 59°F, and 0% humidity.

Note 2: Flow problems with gas require that the calculations be carried out from a flow state that is known (usually at the exit). This requires that the calculations proceed from that known position in the flow to subsequent upstream positions until the injection pressure and temperature are determined.

8.1 Static gas column

$$P_{bh} = P_{wh} e^{\left(\frac{S_g}{R}\right)\frac{H}{T_{av}}} \tag{8.1}$$

where

P_{bh} = bottomhole pressure (lb/ft.^{2} abs)
P_{wh} = wellhead pressure at surface (lb/ft.^{2} abs)
H = TVD of the well (ft.)
S_g = specific gravity of the gas
R = engineering gas constant at API standard conditions (53.36 ft. lb/lb °R)

Example: Determine the approximate bottomhole shut-in pressure of a well filled with natural gas with an $S_g = 0.7$ and a TVD = 10,000 ft. The wellhead pressure is reading 1800 psig, and the average temperature in the well is determined to be 139°F. The wellhead surface location is approximately sea level.

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$$P_{wh} = 1800 + 14.7 = 1814.7 \text{ psia}$$

$$P_{wh} = 1814.7(144) = 261{,}317 \text{ lbl/ft.}^2\text{abs}$$

$$T_{av} = 139^\circ + 460^\circ = 599^\circ\text{R}$$

$$P_{bh} = (261{,}317)e^{\frac{\left(\frac{0.7}{53.36}\right)(10{,}000)}{(599)}} = (261{,}317)(1.244) = 325{,}078\frac{\text{lb}}{\text{ft.}^2}\text{abs}$$

$$P_{bh} = \frac{325{,}078}{144} = 2257 \text{ psia}$$

or

$$P_{bh} = 2257 - 14.7 = 2242 \text{ psig}$$

This section may be found in Spreadsheet 8.1.

Spreadsheets 8.1–8.5 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Air and gas calculations.

8.2 Direct circulation: Flow up the annulus (from annulus bottomhole to exit)

Unlike incompressible hydraulic flow calculations, compressible gas flow calculations must begin with a known pressure and temperature. This is usually at the exit to the system (top of annulus). In this case, the exit conditions are known, since the flow exits the annulus into the atmosphere at the surface. In essence, the calculation will proceed from the exit, then upstream to the bottom of the annulus. The bottomhole pressure in the annulus is

$$P_{bh} = \left\lfloor \left(P_{ex}^2 + b_a T_{av}^2\right)e^{\frac{2a_a H}{T_{av}}} b_a T_{av}^2 \right\rfloor \tag{8.2}$$

where

P_{ex} = exit pressure at the top of the annulus at the surface (lb/ft.^{2} abs) and

$$a_a = \left(\frac{S_g}{R}\right)\left\lfloor 1 + \left(\frac{\dot{w}_s}{\dot{w}_g}\right)\right\rfloor$$

$$b_a = \frac{f_a}{2g\left(D_h - D_p\right)}\left(\frac{R}{S_g}\right)^2 \frac{\dot{w}_g^2}{\left(\frac{\pi}{4}\right)^2\left(D_h^2 - D_p^2\right)^2}$$

$$f_a = \left[\frac{1}{2\,log_{10}\left(\frac{D_h - D_p}{\varepsilon}\right) + 1.14}\right]^2$$

$\varepsilon = 0.0005$ ft (absolute roughness of inside of casing and outside of pipe)
$g = 32.2\,\text{ft./s}^2$
$D_h =$ inside diameter of annulus (ft.)
$D_p =$ pipe outside diameter (ft.)

$$Q_g = \frac{f_g}{60}$$

$$\dot{w}_g = \gamma_g Q_g$$

where

$$\gamma_g = \frac{P_g S_g}{RT}$$

$$\dot{w}_s = \left(\frac{\pi}{4}\right) D_h^2 (62.4) 2.7 \left(\frac{\text{ROP}}{3600}\right)$$

Example: Determine the approximate bottomhole annulus pressure in a well being drilled with a 6⅛ in. drill bit on a drill string made of API 5.0 in., 19.50 lb/ft. nominal weight (i.d. = 4.276 in.) drill string run inside an API 7⅝ in. casing, 39.00 lb/ft. nominal weight (i.d. = 6.625 in.). The well is being drilled at an ROP of 60 ft./h, and the drill fluid is inert air with a volumetric flow of 2000 SCFM that is produced by the nitrogen generator ($S_g = 0.97$). The well is vertical with a depth of 10,000 ft., and the surface location is near sea level at mid-latitudes in North America. The geothermal gradient at this drilling location is approximately 0.016° per ft.

$$t_{ex} = t_{at}\ (\text{use average temperature})$$

$$t_{ex} = 59^\circ\text{F}\ (\text{API standard temperature})$$

$$t_{bh} = 59^\circ + 0.016^\circ\ (10{,}000)$$

$$t_{bh} = 219^\circ\text{F}$$

$$t_{av} = \left(\frac{59^\circ + 219^\circ}{2}\right) = 139^\circ\text{F}$$

$$t_{av} = 139^\circ + 460^\circ = 599^\circ\text{R}$$

$$p_{ex} = p_{at}$$

$$p_{ex} = 14.7\,\text{psia}\ (\text{API standard pressure})$$

$$P_{ex} = 14.7(144) = 2116.8\,\text{lb/ft.}^2\,\text{abs}$$

$$t_{ex} = t_{at}$$

$$t_{ex} = 59^\circ\text{F}$$

$$T_{ex} = 59^\circ + 460^\circ = 519^\circ\text{R}$$

$$\gamma_g = \frac{(2116.8)(0.97)}{(53.36)(519)} = 0.0741\,\text{lb/ft.}^3$$

$$q_g = 2000\,\text{SCFM}$$

$$Q_g = \frac{q_g}{60} = \frac{2000}{60} = 33.3\,\text{ft.}^3/\text{s}$$

$$\dot{w}_g = (0.0741)(33.3) = 2.471\,\text{lb/s}$$

$$\dot{w}_s = \left(\frac{\pi}{4}\right)\left(\frac{6.125}{12}\right)^2(62.4)(2.7)\left(\frac{60}{3600}\right) = 0.575\,\text{lb/s}$$

$$a_a = \left(\frac{0.97}{53.36}\right)\left[1 + \left(\frac{0.575}{2.471}\right)\right] = 0.0224$$

$$D_c = \frac{6.625}{12} = 0.552\,\text{ft.}$$

$$D_p = \frac{5.0}{12} = 0.417\,\text{ft.}$$

$$f_a = \left[\frac{1}{2log_{10}\left(\frac{0.552-0.417}{0.0005}\right) + 1.14}\right]^2 = 0.0277$$

$$b_a = \frac{0.0278}{2(32.2)(0.552-0.417)}\left(\frac{53.36}{0.97}\right)^2\frac{(2.471)^2}{\left(\frac{\pi}{4}\right)^2\left((0.552)^2-(0.417)^2\right)^2} = 5535.95$$

$$P_{abh} = \left[(2116.8)^2 - (5535.95)(599)^2\ e^{\frac{2(0.0224)(10{,}000)}{599}} - (5535.95)(599)^2\right]^{0.5}$$

$$= 47{,}117.4\,\text{psia}$$

$$P_{abh} = \left(\frac{47{,}117.4}{144}\right) = 327.2\,\text{psia}$$

or

$$P_{abh} = 327.2 - 14.7 = 312.5\,\text{psig}$$

The data input for this section is found in Spreadsheet "Tables 8.2–8.3 INPUT." The results may be found in Spreadsheet 8.2.

Spreadsheets 8.1–8.5 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Air and gas calculations.

8.3 Direct circulation: Flow down the inside of the drill pipe (from the bottom of the inside of the drill string to the injection at the top of the drill string)

In nearly all air and gas drilling operations, the drill bit nozzles are not jetted. Therefore, there is little or no pressure loss through the drill bit open orifices, and it can be assumed that the pressure and temperature at the bottom of the annulus will be approximately the same as the pressure and temperature at the bottom of the inside of the drill pipe just above the drill bit. The injection pressure into the inside of the drill string is

$$P_{\text{in}} = \left[\frac{P_{\text{ai}}^2 + b_{\text{i}} T_{\text{av}}^2 \left(e^{\frac{2a_{\text{i}}H}{T_{\text{av}}}} - 1 \right)}{e^{\frac{2_{a}H}{T_{\text{av}}}}} \right]^{0.5} \tag{8.3}$$

where

P_{ai} = pressure above the drill bit inside the drill string at the bottom of the well (lb/ft.^{2} abs) and

$$a_{\text{i}} = \left(\frac{S_{\text{g}}}{R} \right)$$

$$b_{\text{i}} = \frac{f_{\text{i}}}{2gD_{\text{i}}} \left(\frac{R}{S_{\text{g}}} \right)^2 \frac{\dot{w}_{\text{g}}^2}{\left(\frac{\pi}{4}\right)^2 D_{\text{i}}^4}$$

$$f_{\text{i}} = \left[\frac{1}{2\,log_{\,10}\left(\frac{D_{\text{i}}}{\varepsilon}\right) + 1.14} \right]^2$$

$\varepsilon = 0.0005$ ft (absolute roughness of inside of the drill pipe)
$g = 32.2$ ft./s^2
D_{i} = inside drill pipe diameter (ft.)

Example: For the previous example, determine the approximate pressure of the nitrogen generator-produced inert air injected into the top of the inside of the drill string (the inside diameter of the drill string is 4.276 in.).

$$P_{\text{abi}} = 327.2 \text{ psia}$$

or

$$P_{\text{abi}} = 327.2(144) = 48{,}116.8 \text{ lb/ft.}^2\text{abs}$$

$$a_i = \left(\frac{0.97}{53.36}\right) = 0.0182$$

$$D_i = \frac{4.276}{12} = 0.356 \text{ ft.}$$

$$f_i = \left[\frac{1}{2\,log_{10}\left(\frac{0.356}{0.0005}\right) + 1.14}\right]^2 = 0.0213$$

$$b_i = \frac{0.0213}{2(32.2)(0.356)}\left(\frac{53.36}{0.97}\right)^2 \frac{(2.471)^2}{\left(\frac{\pi}{4}\right)^2(0.356)^4} = 1728.2$$

$$P_{in} = \left[\frac{(47{,}116.8)^2 + (1728.2)(599)^2\left(e^{\frac{2(0.0182)(10{,}000)}{599}} - 1\right)}{e^{\frac{2(0.0182)(10{,}000)}{599}}}\right]^{0.5}$$
$$= 38{,}627.3 \text{ lb/ft.}^2\text{abs}$$

or

$$P_{in} = 268.2 \text{ psia}$$

or

$$P_{in} = 268.2 - 14.7 = 253.5 \text{ psig}$$

This section may be found in Spreadsheet 8.3.

Spreadsheets 8.1–8.5 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Air and gas calculations.

8.4 Reverse circulation: Flow up the inside of tubing string

Reverse circulation is often used in gas and condensate well workover operations. In such operations, it is necessary to flow nitrogen-generated inert air down the annulus between the inside of the casing and the outside of the production tubing and up through the inside of the production tubing. In this manner, the pressure at the bottom of the well is reduced, which in turn allows the natural gas or condensate to flow from the formation and intermix with the injected inert air and proceed up the tubing to the surface. As the flow from the formation increases, the inert air flow can be reduced as the formation begins to flow naturally through the production tubing and the tubing head choke. The flow from the bottom of the inside of the tubing to the surface is

$$P_{bt} = \left[\left(P_{th}^2 + b_{it} T_{av}^2 \right) e^{\frac{2a_{ti}H}{T_{av}}} - b_{it} T_{av}^2 \right]^{0.5} \tag{8.4}$$

where

P_{bt} = pressure above the drill bit inside the drill string at the bottom of the tubing (lb/ft.^{2} abs) and

$$a_{ti} = \left(\frac{S_g}{R} \right)$$

$$b_{ti} = \frac{f_i}{2gD_{ti}} \left(\frac{R}{S_g} \right)^2 \frac{\dot{w}_{tg}^2}{\left(\frac{\pi}{4}\right)^2 D_{ti}^4}$$

$$\dot{w}_{tg} = \dot{w}_{g1} + \dot{w}_{g2}$$

$$f_{ti} = \left[\frac{1}{2\,log_{10}\left(\frac{D_{ti}}{\varepsilon}\right) + 1.14} \right]^2$$

$\varepsilon = 0.0005$ ft (absolute roughness of inside of the drill pipe)
$g = 32.2$ ft./s^2
D_{ti} = inside tubing diameter (ft.)

Example: Determine the approximate inside bottom pressure at the bottom of the tubing string. The production tubing string is API 2⅞ in., 6.50 lb/ft. nominal weight (i.d. = 2.441 in.), and is hung in a 10,000 ft vertical well inside API 7⅝ in. casing, 39.00 lb/ft. nominal weight (i.d. = 6.625 in.). A flow of 500 SCFM nitrogen-generated inert air ($S_g = 0.97$) is injected into the top of the annulus between the inside of the casing and the outside of the tubing. The flow continues to the bottom of the annulus and then flows up the inside of the tubing to the tubing head and to the choke at the surface. The tubing head pressure is to be kept at a constant 100 psig via the choke as the natural gas production is initiated with the reverse circulation operation. The temperature at the wellhead during circulation is estimated to be the surface ambient (standard API) temperature (59°F). The natural gas ($S_g = 0.7$) producing formation has the potential to flow at a rate of up to 700 SCFM (or 1,008,000 SCFD). This illustrative example will show the calculations for a 200 SCFM of natural gas production rate. The geothermal gradient at this drilling location is approximately 0.016° per ft. The well is located at sea level at midlatitudes in North America.

$$q_{g1} = 500 \text{ SCFM (inert air)}$$

$$q_{g2} = 200 \text{ SCFM (natural gas)}$$

$$S_{g1} = 0.97$$

$$P_{at} = 14.7 \text{ psia}$$

$$P_{th} = 100 \text{ psia}$$

$$P_{th} = 100 + 14.7 = 114.7 \text{ psia}$$

$$P_{th} = 114.7(144) = 16{,}516.8 \text{ lb/ft.}^2\text{abs}$$

$$T_{th} = t_{at}\text{(use average atomshperic temperature)}$$

$$t_{bh} = t_{at} + 0.016^{\circ}(10{,}000) = 219^{\circ}\text{F}$$

$$t_{av} = \left(\frac{59^{\circ} + 219^{\circ}}{2}\right) = 139^{\circ}\text{F}$$

$$T_{av} = 139^{\circ} + 460^{\circ} = 519^{\circ}\text{R}$$

$$\gamma_{g1} = \frac{(2116.8)(0.97)}{(53.36)(519)} = 0.0741 \text{ lb/ft.}^3$$

$$q_{g1} = 500 \text{ SCFM}$$

$$Q_g = \frac{q_{g1}}{60} = \frac{500}{60} = 8.33 \text{ ft.}^3\text{/s}$$

$$\dot{w}_{g1} = (0.0741)(8.33) = 0.618 \text{ lb/s}$$

$$\gamma_{g2} = \frac{(2116.8)(0.7)}{(53.36)(519)} = 0.0535 \text{ lb/ft.}^3$$

$$Q_{g2} = \frac{q_{g1}}{60} = \frac{200}{60} = 3.33 \text{ ft.}^3\text{/s}$$

$$\dot{w}_{g2} = (0.0535)(3.33) = 0.178 \text{ lb/s}$$

$$\dot{w}_{tg} = (0.618) + (0.178) = 0.796 \text{ lb/s}$$

$$a_{ti} = \left(\frac{0.97}{53.36}\right) = 0.0182$$

$$d_{ti} = 2.441 \text{ in}$$

$$D_{ti} = \frac{2.441}{12} = 0.203 \text{ ft.}$$

$$f_{ti} = \left[\frac{1}{2\,log_{10}\left(\frac{0.203}{0.0005}\right) + 1.14}\right]^2 = 0.0247$$

$$b_{ti} = \frac{0.0247}{2(32.2)(0.203)}\left(\frac{53.36}{0.97}\right)^2 \frac{(0.796)^2}{\left(\frac{\pi}{4}\right)^2(0.203)^4} = 3427.3$$

$$P_{bt} = \left[\left((16{,}516.8)^2 + (3427.3)(599)^2\right)e^{\frac{2(0.182)(10{,}000)}{599}} - (3427.3)(599)^2\right]^{0.5}$$
$$= 39{,}078.5 \text{ lb/ft.}^2\text{abs}$$

$$P_{bt} = \left(\frac{39{,}078.5}{144}\right) = 271.8 \text{ psia}$$

Or

$$P_{bt} = 271.8 - 14.7 = 256.7 \text{ psig}$$

This section may be found in Spreadsheet 8.4.

Spreadsheets 8.1–8.5 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Air and gas calculations.

8.5 Reverse circulation: Flow down the annulus

The pressure at the bottom of the tubing is known and is approximately the pressure at the bottomhole pressure in the annulus. In essence, the calculation will proceed from the bottom of the annulus upstream to determine the injection pressure at the top of the annulus. The injection pressure at the top of the annulus is

$$P_{in} = \left[\frac{P_{ba}^2 + b_a T_{av}^2\left(e^{\frac{2a_a H}{T_{av}}} - 1\right)}{e^{\frac{2a_a H}{T_{av}}}}\right]^{0.5} \tag{8.5}$$

where

P_{ba} = pressure at bottom of annulus (lb/ft.^{2} abs) and

$$a_a = \left(\frac{S_g}{R}\right)$$

$$b_a = \frac{f_a}{2g(D_c - D_{to})}\left(\frac{R}{S_g}\right)^2 \frac{\dot{w}_{g1}^2}{\left(\frac{\pi}{4}\right)^2\left(D_c^2 - D_{to}^2\right)^2}$$

$$f_a = \left[\frac{1}{2\,log_{10}\left(\frac{D_c - D_{to}}{\varepsilon}\right) + 1.14}\right]^2$$

$\varepsilon = 0.0005$ ft. (absolute roughness of inside of the drill pipe)
$g = 32.2$ ft./s^2
$D_c =$ inside diameter of annulus casing in ft.
$D_{to} =$ tubing outside diameter in ft.

$$\dot{w}_{g1} = \gamma_{g1} Q_{g1}$$

where

$Q_g =$ gas flow rate (ft.^{3}/s)

$$\gamma_g = \frac{P_g S_g}{R T_g}$$

Example: Using the data from the above example, determine the approximate annulus injection pressure into the well that has been worked over and is being put back into production.

$$d_{to} = 2.875 \text{ in.}$$

$$D_{to} = \frac{2.875}{12} = 0.240 \text{ ft.}$$

$$D_c = \frac{6.625}{12} = 0.552 \text{ ft.}$$

$$t_{in} = t_{at}$$

$$t_{in} = 59^\circ \text{F}$$

$$T_{ex} = 59^\circ + 460^\circ = 519^\circ$$

$$\gamma_{g1} = \frac{(2116.8)(0.97)}{(53.36)(519)} = 0.0741 \text{ lb/ft.}^3$$

$$Q_g = \frac{500}{60} = 8.33 \text{ ft.}^3/\text{s}$$

TABLE 8.1 Natural gas flow versus injection pressure.

q_{ng} (SCFM)	p_{in} (psig)
0	157.6
100	171.6
200	186.3

$$\dot{w}_{g1} = (0.0741)(8.33) = 0.618 \text{ lb/s}$$

$$a_a = \left(\frac{0.97}{53.36}\right) = 0.0182$$

$$f_a = \left[\frac{1}{2\,log_{10}\left(\frac{0.552-0.240}{0.0005}\right) + 1.14}\right]^2 = 0.0221$$

$$b_a = \frac{0.0221}{2(32.2)(0.552-0.240)}\left(\frac{53.36}{0.97}\right)^2 \frac{(0.618)^2}{\left(\frac{\pi}{4}\right)^2\left[(0.552)^2-(0.240)^{2^2}\right]} = 33.8$$

$$P_{in} = \left[\frac{(39{,}078.5)^2 + (33.8)(599)^2\left(e^{\frac{2(0.0182)(10{,}000)}{599}} - 1\right)}{e^{\frac{2(0.0182)(10{,}000)}{599}}}\right]^{0.5} = 28{,}949.9 \text{ lb/ft.}^2\text{abs}$$

$$P_{in} = \left(\frac{28{,}949.9}{144}\right) = 201.0 \text{ psia}$$

or

$$P_{in} = 201.0 - 14.7 = 186.3 \text{ psig}$$

Table 8.1 gives the results of the above calculations for a natural gas flow of 0 SCFM, 100 SCFM, and 200 SCFM.

This section may be found in Spreadsheet 8.5.

Spreadsheets 8.1–8.5 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Air and gas calculations.

8.6 Reverse circulation: Adjusting for reservoir pressure

If the inert air is injected into the top of a well annulus that is under pressure from the reservoir, then the compressor (and nitrogen generator) system will have to overcome the static pressure at the top of the annulus in order to initiate flow.

Example: Let us assume that the static annulus pressure is given by the example in number 1 above. Fig. 8.1 shows an example Inflow Performance Relationship (IPR) of the static well given in number 1. With no flow from the reservoir ($q_{ng}=0$), the compressor system would have to inject inert air at

$$P_{inr} = 157.6 + 1800.0 = 1957.6 \text{ psia}$$

where the injection pressure is taken from Table 8.2.

Table 8.2 gives the approximate bottomhole pressures and tubing head pressures for the natural gas flow rates given in Table 8.1. From Fig. 8.1, the flowing reservoir pressure and its associated tubing head pressure can be estimated.

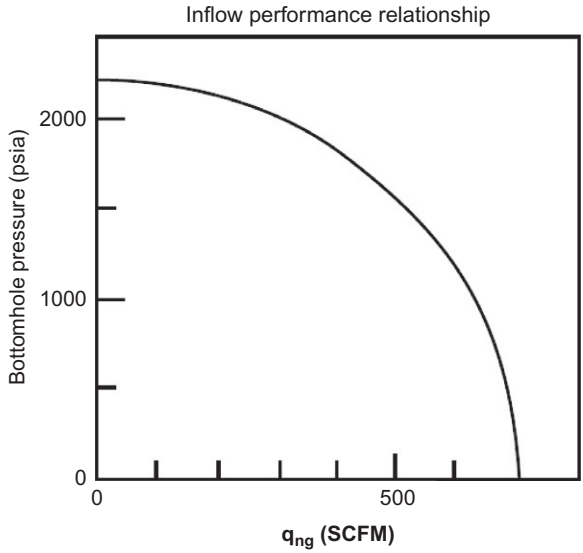


FIG. 8.1 IPR for example well.

TABLE 8.2 Approximate compressor injection pressures.

q_{ng} (SCFM)	p_{res} (psia)	p_{th} (psia)	p_{th} (psig)	p_{com} (psig)
0	2259	1814.7	1800.0	1957.6
100	2241.6	1800.7	1786.0	1957.6
200	2151.5	1728.3	1713.6	1899.9

Table 8.2 shows that the inert air compressor injection pressures will decrease as the natural gas begins to flow up the production tubing to the surface as the well is brought into production. These injection pressures will require a 1000 SCFM primary compressor (either helical screw type or reciprocating piston) and nitrogen generator filter unit and a booster compressor (which must be a reciprocating piston). The nitrogen generator "rule of thumb" is that only about 50% of the primary compressed air will be available after the 1000 SCFM of compressed air flows through the nitrogen generator filter unit. This leaves just 500 SCFM to be injected by the booster compressor into the well.

Appendix: Supplementary material

Supplementary material related to this chapter can be found online at https://www.elsevier.com/books-and-journals/book-companion/9780323905497.

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Appendix A

A.1 Tank capacity determinations

A.1.1 Rectangular tanks with flat bottom

(a) Determine the volume of a Full rectangular tank in bbl:

$$V_{RTf} = ((L_T)(W_T)(D_T))(0.17825) \tag{A.1}$$

where

V_{RTf} = volume of a **Full** rectangular tank with a flat bottom in bbl
L_T = length of tank in ft
W_T = width of tank in ft
D_T = depth of tank in ft
0.17825 = conversion from ft^3 to barrels

(b) Determine the volume of a **Partially** filled rectangular tank with a flat bottom in bbl:

$$V_{RTp} = ((L_T)(W_T)(D_L))(0.17825) \tag{A.2}$$

where

V_{RTp} = volume of a **Partially** full rectangular tank with a flat bottom in bbl
D_L = depth of liquid in tank in ft

Example 1: Determine the total capacity of a rectangular tank with a flat bottom using the following data:

Length = 30 ft
Width = 10 ft
Depth = 8 ft

(a) Determine the volume of a **Full** rectangular tank in bbl:

$$V_{RT} = ((30)(10)(8))(0.17825) = 427.8 \text{ bbl}$$

(b) Determine the volume of a rectangular tank with a flat bottom filled with only 5½ ft of fluid in bbl:

$$V_{RTp} = ((30)(10)(5.5))(0.17825) = 294.1 \text{ bbl}$$

Spreadsheet A.1.1 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Rectangular tanks with flat bottom.

A.1.2 Rectangular tanks with sloping sides

(a) Determine the volume of a **Full** rectangular tank with sloping sides in bbl:

$$V_{RTS} = \left(((L_T)(W_T)(D_T)) - \left(2\left(\left(0.5\left(D\left(\frac{W_t - W_b}{2}\right)\right)\right)(L)\right)\right)\right)(0.17825) \tag{A.3}$$

W_b = width of tank bottom in ft

Example: Determine the total tank capacity using the following data:

Length	= 30.0 ft
Width (top)	= 10.0 ft
Width (bottom)	= 6.0 ft
Depth	= 8.0 ft

$$V_{RTS} = \left(((30)(10)(8)) - \left(2\left(\left(0.5\left(8\left(\frac{10-6}{2}\right)\right)\right)(30)\right)\right)\right)(0.17825)$$
$$= 342.2 \text{ bbl}$$

Spreadsheet A.1.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Rectangular tanks with sloping sides.

A.1.3 Horizontal cylindrical (lay down) tank

(a) Determine the total volume of a Full horizontal cylindrical tank in bbl:

$$V_{HCTf} = (\pi)(r^2)(L_c)(0.17825) \tag{A.4}$$

where

V_{HCTf} = volume of a **Full** horizontal circular tank in bbl
$\pi = 3.14$
r = radius in ft
L_c = length of horizontal cylindrical tank in ft

(b) Determine the **Partially** filled volume of horizontal cylindrical tank in bbl:

$$V_{HCTp} = (L_c)\left[(0.017543)(r^2)\left(cos^{-1}\left(\frac{r - D_L}{r}\right)\right)\right. \quad \text{(A.5)}$$

$$\left. - \left(\left(\sqrt{(2(D_L)(r)) - (D_L^2)}\right)\right)(r - D_L)\right](0.17825)$$

where

V_{HCTp} = volume of **Partially** filled horizontal cylindrical tank in bbl
D_L = depth of liquid in tank in ft

Example: Determine the capacity of a cylindrical tank with the following dimensions:

Length = 15.0 ft
Diameter = 10.0 ft

(a) Determine the total volume of a **Full** horizontal cylindrical tank in bbl:

$$V_{CT} = (3.1416)(5^2)(30)(0.17825) = 420\ bbl$$

(b) Determine the **Partially** filled volume of horizontal cylindrical tank filled with 4 feet of liquid in bbl:

$$V_{HCTp} = (30)\left[(0.017543)(5^2)\left(cos^{-1}\left(\frac{5-4}{5}\right)\right)\right.$$

$$\left. - \left(\left(\sqrt{(2(4)(5)) - (4^2)}\right)\right)(5-4)\right](0.17825)$$

$$= 156.9 \text{ bbl}$$

Spreadsheet A.1.3 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Horizontal cylindrical (lay down) tank.

A.1.4 Tapered vertical cylindrical tanks

(a) Volume of **Full** vertical cylindrical tank section in bbl:

$$V_{HCTf} = (\pi)\left(r^2\right)(D_t)(0.17825) \tag{A.6}$$

where

V_{HCTf} = volume of **Full** vertical cylindrical tank section in bbl
D_t = depth of vertical cylindrical tank section in ft

(b) Volume of **Full** tapered section in bbl:

$$V_t = (0.059)(\pi)(H_t)\left(r_t^2 + r_b^2 + ((r_b)(r_t))\right) \tag{A.6}$$

where

V_t = volume of tapered section, bbl
H_t = height of tapered section in ft
r_c = radius of cylindrical section in ft
r_b = radius at bottom ft
0.059 = conversion of ft^3 to bbl

(c) Calculate the total volume of the tapered cylindrical tank:

$$V_{TC} = V_{CT} + V_t \tag{A.7}$$

where

V_{TC} = total volume of tapered cylindrical tank in bbl

(d) Determine the volume of a Partially filled vertical cylindrical tank in bbl:

Note: If the liquid is filled up inside the vertical cylindrical portion of the tank, use Eq. (A.3) and change L_c to D_t for that section and add to that volume the full volume of the tapered section of the tank.

If the liquid is contained in the tapered section, use Eq. (A.6) and change H_t to the depth of the liquid (D_L) and calculate the volume only in the tapered section.

Example: Determine the volume of a vertical cylindrical tank with the following dimensions:

Height of vertical cylindrical section = 5.0 ft

Height of tapered section = 10.0 ft
Radius of vertical cylindrical section = 6.0 ft
Radius at bottom of tapered section = 1.0 ft

Solution:

(a) Volume of the vertical cylindrical section:

$$V_{CT} = (3.14)\left(6.0^2\right)(5.0)(0.17825) = 100.7 \text{ bbl}$$

(b) Volume of tapered section:

$$V_t = (0.059)(3.14)(10.0)\left(6.0^2 + 1.0^2 + ((6.0)(1.0))\right) = 79.7 \text{ bbl}$$

(c) Total volume: 100.7 + 79.7 = 180.4 bbl

(d) Calculate the volume of a **Partially** filled vertical cylindrical tank with the dimensions listed above in bbl:

Vertical section filled with 2 ft of liquid:

$$V_{VTp} = (3.1416)\left(6^2\right)(2)(0.17825) = 40.3 \text{ bbl}$$

where

V_{VTp} = volume of **Partially** filled vertical section of cylindrical tank in bbl

Tapered section full:

$$V_t = (0.059)(3.14)(10.0)\left(6.0^2 + 1.0^2 + ((6.0)(1.0))\right) = 79.7 \text{ bbl}$$

Total volume = 40.3 + 79.7 = 120 bbl

Spreadsheet A.1.4 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Tapered vertical cylindrical tanks.

A.1.5 Vertical cylindrical tank

(a) Full vertical cylindrical tank capacity:

$$V_{VC} = (\pi)\left(r_c^2\right)(H_T)(0.17825) \tag{A.8}$$

where

V_{VC} = volume of vertical cylindrical tank in bbl
r_c = radius of vertical cylindrical tank in ft
H_T = height of vertical cylindrical tank in ft

(b) Partial volume:

$$V_{VC} = (\pi)\left(r_c^2\right)(D_L)(0.17825) \tag{A.9}$$

Example: Determine the total volume of the following tank:

Height = 20 ft
Radius = 6 ft

(a) Full vertical cylindrical tank capacity:

$$V_{VC} = (3.1416)\left(6^2\right)(20)(0.17825) = 403.2 \text{ bbl}$$

(b) Partial volume of a vertical cylindrical tank with 8 ft of liquid in it:

$$V_{VC} = (3.1416)\left(6^2\right)(8)(0.17825) = 161.3 \text{ bbl}$$

Spreadsheet A.1.5 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Vertical cylindrical tank.

TABLE A.1 Drill pipe capacity and displacement (English system).

Size OD (in.)	Size ID (in.)	Weight (lb/ft)	Capacity (bbl/ft)	Displacement (bbl/ft)
2⅜	1.815	6.65	0.00320	0.00228
2⅞	2.150	10.40	0.00449	0.00354
3½	2.764	13.30	0.00742	0.00448
3½	2.602	15.50	0.00658	0.00532
4	3.340	14.00	0.01084	0.00471
4½	3.826	16.60	0.01422	0.00545
4½	3.640	20.00	0.01287	0.00680
5	4.276	19.50	0.01766	0.00652
5	4.214	20.50	0.01730	0.00704
5½	4.778	21.90	0.02218	0.00721
5½	4.670	24.70	0.02119	0.00820
5⁹⁄₁₆	4.859	22.20	0.02294	0.00712
6⅝	5.965	25.20	0.03456	0.00807

TABLE A.2 Heavy weight drill pipe and displacement

Size OD (in.)	Size ID (in.)	Weigh (lb/ft)	Capacity (bbl/ft)	Displacement (bbl/ft)
3½	2.0625	25.3	0.00421	0.00921
4	2.25625	29.7	0.00645	0.01082
4½	2.75	41.0	0.00743	0.01493
5	3.0	49.3	0.00883	0.01796

A.2 Pipe capacities, displacements, and weight calculations

(a) Capacity of pipe:

$$C_p = \frac{P_{ID}^2}{1029.4} \tag{A.10}$$

where

C_p = capacity of pipe in bbl/ft
P_{ID} = pipe ID in inches

(b) Displacement of pipe:

$$D_p = \frac{(P_{OD}^2) - (P_{ID}^2)}{1029.4} \tag{A.11}$$

where

D_p = displacement of pipe in bbl/ft
P_{OD} = pipe OD in inches

(c) Weight of pipe (Table A.3):

$$W_p = (D_p)(2747) \tag{A.12}$$

where

W_p = weight of pipe in lb/ft
2747 = weight of steel in lb/bbl

Spreadsheet A.2 in the online version at https://www.elsevier.com/books-and-journals/book-companion/9780323905497 Pipe capacities, displacements, and weight calculations.

TABLE A.3 Drill collar capacity and displacement.

OD (in.)	ID capacity	1½ (in.) .0022	1¾ (in.) .0030	2.0 (in.) .0039	2¼ (in.) .0049	2½ (in.) .0061	2¾ (in.) .0073	3.0 (in.) .0087	3¼ (in.) .0103	3½ (in.) .0119	3¾ (in.) .0137	4.0 (in.) .0155	4¼ (in.) .0175
4.0	#/ft	36.7	34.5	32.0	29.2	–	–	–	–	–	–	–	–
	Disp.	0.0133	0.0125	0.0116	0.0106								
4¼	#/ft	42.2	40.0	37.5	34.7	–	–	–	–	–	–	–	–
	Disp.	0.0153	0.0145	0.0136	0.0126								
4½	#/ft	48.1	45.9	43.4	40.6	–	–	–	–	–	–	–	–
	Disp.	0.0175	0.0167	0.0158	0.0148								
4¾	#/ft	54.3	52.1	49.5	46.8	43.6	–	–	–	–	–	–	–
	Disp.	0.0197	0.0189	0.0180	0.0170	.0159							
5.0	#/ft	60.8	58.6	56.3	53.3	50.1	–	–	–	–	–	–	–
	Disp.	0.0221	0.0213	0.0214	0.0194	0.0182							
5¼	#/ft	67.6	65.4	62.9	60.1	56.9	53.4	–	–	–	–	–	–
	Disp.	0.0246	0.0238	0.0229	0.0219	0.0207	0.0194						
5½	#/ft	74.8	72.6	70.5	67.3	64.1	60.6	56.8	–	–	–	–	–
	Disp.	0.0272	0.0264	0.0255	0.0245	0.0233	0.0221	0.0207					
5¾	#/ft	82.3	80.1	77.6	74.8	71.6	68.1	64.3	–	–	–	–	–
	Disp.	0.0299	0.0291	0.0282	0.0272	0.0261	0.2048	0.0234					

6.0	#/ft	90.1	87.9	85.4	82.6	79.4	75.9	72.1	67.9	63.4	–	–	–
	Disp.	0.0328	0.0320	0.0311	0.0301	0.0289	0.0276	67.9	0.0247	0.0231			
6¼	#/ft	98.0	95.8	93.3	90.5	87.3	83.8	80.0	75.8	71.3	–	–	–
	Disp.	0.0356	0.0349	0.0339	0.0329	0.0318	0.0305	0.0291	0.0276	0.0259			
6½	#/ft	107.0	104.8	102.3	99.5	96.3	92.8	89.0	84.8	80.3	–	–	–
	Disp.	.0389	0.0381	0.0372	0.0362	0.0350	0.0338	0.0324	0.0308	0.0292			
6¾	#/ft	116.0	113.8	111.3	108.5	105.3	101.8	98.0	93.8	89.3	–	–	–
	Disp.	0.0422	0.0414	0.0405	0.0395	0.0383	0.0370	0.0356	0.0341	0.0325			
7.0	#/ft	125.0	122.8	120.3	117.5	114.3	110.8	107.0	102.8	98.3	93.4	88.3	–
	Disp.	0.0455	0.0447	0.0438	0.0427	0.0416	0.0403	0.0389	0.0374	0.0358	0.0340	0.0321	
7¼	#/ft	134.	131.8	129.3	126.5	123.3	119.8	116.0	111.8	107.3	102.4	97.3	–
	Disp.	0.0487	0.0479	0.0470	0.0460	0.0449	0.0436	0.0422	0.0407	0.0390	0.0372	0.0354	
7½	#/ft	144.0	141.8	139.3	136.5	133.3	129.8	126.0	121.8	117.3	112.4	107.3	–
	Disp.	0.0524	0.0516	0.0507	0.0497	0.0485	0.0472	0.0458	0.0443	0.0427	0.0409	0.0390	
7¾	#/ft	154.0	151.8	149.3	146.5	143.3	139.8	136.0	131.8	127.3	122.4	117.3	–
	Disp.	0.0560	0.0552	0.0543	0.0533	0.0521	0.0509	0.0495	0.0479	0.0463	0.0445	0.0427	
8.0	#/ft	165.0	162.8	160.3	157.5	154.3	150.8	147.0	142.8	138.3	133.4	123.3	122.8
	Disp.	0.0600	0.0592	0.0583	0.0573	0.0561	0.0549	0.0535	0.0520	0.0503	0.0485	0.0467	0.0447

Continued

TABLE A.3 Drill collar capacity and displacement—cont'd

OD (in.)	ID capacity	1½ (in.) .0022	1¾ (in.) .0030	2.0 (in.) .0039	2¼ (in.) .0049	2½ (in.) .0061	2¾ (in.) .0073	3.0 (in.) .0087	3¼ (in.) .0103	3½ (in.) .0119	3¾ (in.) .0137	4.0 (in.) .0155	4¼ (in.) .0175
8¼	#/ft	176.0	173.8	171.3	168.3	165.3	161.8	158.0	153.8	149.3	144.4	139.3	133.8
	Disp.	0.0640	0.0632	0.0623	0.0613	0.0601	0.0589	0.0575	0.0560	0.0543	0.0525	0.0507	0.0487
8½	#/ft	187.0	184.8	182.3	179.5	176.3	172.8	169.0	164.8	160.3	155.4	150.3	144.8
	Disp.	0.0680	0.0672	0.0663	0.0653	0.0641	0.0629	0.0615	0.0600	0.0583	0.0565	0.0547	0.0527
8¾	#/ft	199.0	196.8	194.3	191.5	188.3	184.8	181.0	176.8	172.3	167.4	162.3	156.8
	Disp.	0.0724	0.0716	0.0707	0.0697	0.0685	0.0672	0.0658	0.0613	0.0697	0.0609	0.0590	0.0570
9.0	#/ft	210.2	208.0	205.6	202.7	199.6	196.0	192.2	188.0	183.5	178.7	173.5	168.0
	Disp.	0.0765	0.0757	0.0748	0.0738	0.0726	0.0714	0.0700	0.0685	0.0668	0.0651	0.0632	0.0612
10.0	#/ft	260.9	258.8	256.3	253.4	250.3	246.8	242.9	238.8	234.3	229.4	224.2	118.7
	Disp.	0.0950	0.0942	0.0933	0.0923	0.0911	0.0898	0.0084	0.0869	0.0853	0.0835	0.0816	0.0796

Appendix: Supplementary material

Supplementary material related to this chapter can be found online at https://www.elsevier.com/books-and-journals/book-companion/9780323905497.

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Appendix B: Conversion factors

To convert		
From	**To**	**Multiply by**
Area		
Square inches	Square centimeters	6.45
Square inches	Square millimeters	645.2
Square centimeters	Square inches	0.155
Square millimeters	Square inches	1.55×10^{-3}
Circulation rate		
Barrels/min	Gallons/min	42.0
Cubic feet/min	Cubic meters/s	4.72×10^{-4}
Cubic feet/min	Gallons/min	7.48
Cubic feet/min	Liters/min	28.32
Cubic meters/s	Gallons/min	15,850
Cubic meters/s	Cubic feet/min	2118
Cubic meters/s	Liters/min	60,000
Gallons/min	Barrels/min	0.0238
Gallons/mm	Cubic feet/min	0.134
Gallons/min	Liters/min	3.79
Gallons/min	Cubic meters/s	6.309×10^{-5}
Liters/min	Cubic meters/s	1.667×10^{-5}
Liters/min	Cubic feet/min	0.0353
Liters/min	Gallons/min	0.264

Continued

To convert		
From	**To**	**Multiply by**
Impact force		
Pounds	Dynes	4.45×10^{-5}
Pounds	Kilograms	0.454
Pounds	Newtons	4.448
Dynes	Pounds	2.25×10^{-6}
Kilograms	Pounds	2.20
Newtons	Pounds	0.2248
Length		
Feet	Meters	0.305
Inches	Millimeters	25.40
Inches	Centimeters	2.54
Centimeters	Inches	0.394
Millimeters	Inches	0.03937
Meters	Feet	3.281
Mud weight		
Pounds/gallon	Pounds/ft^3	7.48
Pounds/gallon	Specific gravity	0.120
Pounds/gallon	Grams/cm^3	0.1198
Grams/cm^3	Pounds/gallon	8.347
Pounds/ft^3	Pounds/gallon	0.134
Specific gravity	Pounds/gallon	8.34
Power		
Horsepower	Horsepower (metric)	1.014
Horsepower	Kilowatts	0.746
Horsepower	Foot-pounds/s	550
Horsepower (metric)	Horsepower	0.986
Horsepower (metric)	Foot-pounds/s	542.5
Kilowatts	Horsepower	1.341
Foot-pounds/s	Horsepower	0.00181

To convert		
From	**To**	**Multiply by**
Pressure		
Atmospheres	Pounds/sq. in.	14.696
Atmospheres	Kilograms/sq. cm	1.033
Atmospheres	Pascals	1.033×10^{5}
Kilograms/sq. cm	Atmospheres	0.9678
Kilograms/sq. cm	Pounds/sq. in.	14.223
Kilograms/sq. cm	Atmospheres	0.9678
Pounds/sq. in.	Atmospheres	0.0680
Pounds/sq. in.	Kilograms/sq. cm	0.0703
Pounds/sq. in.	Pascals	6.894×10^{3}
Velocity		
Feet/s	Meters/s	0.305
Feet/min	Meters/s	5.08×10^{-3}
Meters/s	Feet/min	196.8
Meters/s	Feet/s	3.28
Volume		
Barrels	Gallons	42
Cubic centimeters	Cubic feet	3.51×10^{-5}
Cubic centimeters	Cubic inches	0.06102
Cubic centimeters	Cubic meters	
Cubic centimeters	Gallons	2.642×10^{-4}
Cubic centimeters	Liters	0.001
Cubic feet	Cubic centimeters	28,320
Cubic feet	Cubic feet	1728
Cubic feet	Cubic meters	0.02832
Cubic feet	Gallons	7.48
Cubic feet	Liters	28.32
Cubic inches	Cubic centimeters	16.39

Continued

To convert		
From	**To**	**Multiply by**
Cubic inches	Cubic feet	5.787×10^{-4}
Cubic inches	Cubic meters	1.639×10^{-5}
Cubic inches	Gallons	4.329×10^{-3}
Cubic inches	Liters	0.01639
Cubic meters	Cubic centimeters	10^{6}
Cubic meters	Cubic feet	35.31
Cubic meters	Gallons	264.2
Gallons	Barrels	0.0238
Gallons	Cubic centimeters	3785
Gallons	Cubic feet	0.1337
Gallons	Cubic inches	231
Gallons	Cubic meters	3.785×10^{-3}
Gallons	Liters	3.785
Weight		
Pounds	Tons (metric)	4.535×10^{-4}
Tons (metric)	Pounds	2205
Tons (metric)	Kilograms	1000

Appendix C: Average annual atmospheric conditions

This appendix gives the graphic representation of the average atmospheric conditions for midlatitudes (30–60°N) of the North American continent. Fig. C.1 gives the average annual atmospheric pressure of air for

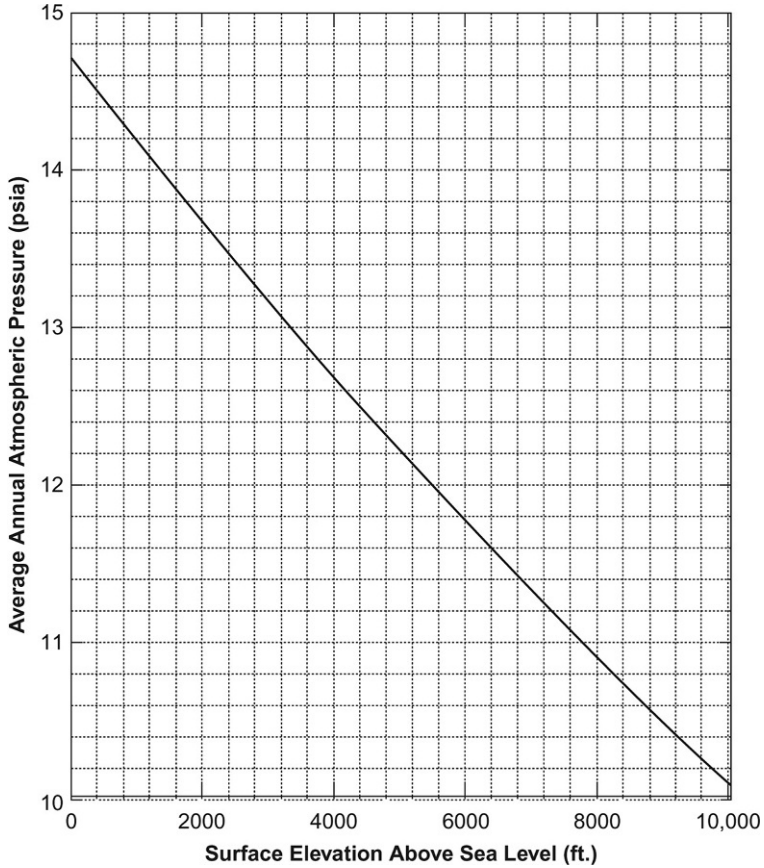


FIG. C.1 Average annual atmospheric pressure versus surface elevation above mean sea level (USCS units).

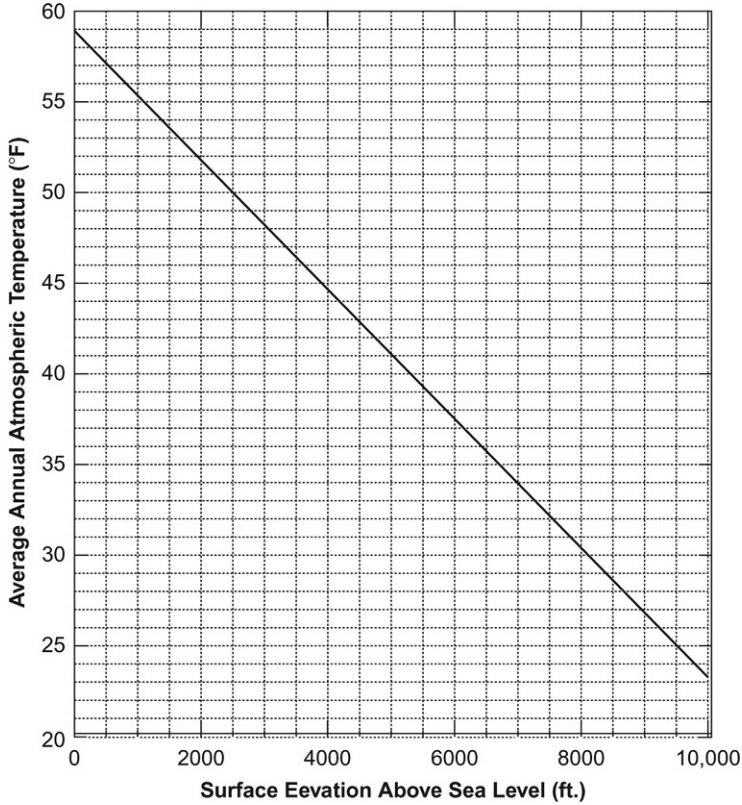


FIG. C.2 Average annual atmospheric temperature versus surface elevation above sea level (USCS units).

midlatitudes of the North American continent as a function of surface elevation location above mean sea level. These average annual atmospheric pressures are of critical importance in predicting the actual weight rate of flow of air (or other gases) at an actual drilling location. Fig. C.1 is given in USCS units.

Fig. C.2 gives the average annual atmospheric temperature of air for midlatitudes of the North American continent as a function of surface elevation location above mean sea level. These average annual atmospheric temperatures are of critical importance in predicting the approximate geothermal temperature at an actual drilling location. Fig. C.2 is given in USCS units.

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